



Electrical, Electromagnetic, and Optical Characterization of the InP/InGaAs Alloy System

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A Thesis Proposal  
Presented to the Faculty of the  
Department of Electronics and Computer Engineering  
Gokongwei College of Engineering  
De La Salle University

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In Partial Fulfillment of the  
Requirements for the Degree of  
Bachelor of Science in Electronics and Communications Engineering

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by  
  
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January, 2025



De La Salle University

## ORAL DEFENSE RECOMMENDATION SHEET

This thesis proposal, entitled **Electrical, Electromagnetic, and Optical Characterization of the InP/InGaAs Alloy System**, prepared and submitted by thesis group, ESG-04, composed of:

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in partial fulfillment of the requirements for the degree of **Bachelor of Science in Electronics and Communications Engineering (BS-ECE)** has been examined and is recommended for acceptance and approval for **ORAL DEFENSE**.

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**Dr. Francisco D. Baltasar**  
*Adviser*

January 22, 2025



## ABSTRACT

Keep your abstract short by giving the gist/nutshell of your thesis proposal. Use the following checklist questions to help you in crafting your abstract.

- ☐ Did you briefly state what you intend to do?
- ☐ Did you concisely discuss the problem statement?
- ☐ Did you tersely mention the objectives in general terms?
- ☐ Did you succinctly describe the methodology for the target audience?
- ☐ Did you strongly describe your significant results and your conclusions?

*Index Terms*—alloy system, characterization, InP, InGaAs (see IEEE Taxonomy and Thesaurus).



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E.1 A quadrilateral image example. . . . . 90

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## ABBREVIATIONS

|     |      |                                 |    |
|-----|------|---------------------------------|----|
| 158 | AC   | Alternating Current.....        | 86 |
| 159 | CSS  | Cascading Style Sheet.....      | 86 |
| 160 | HTML | Hyper-text Markup Language..... | 86 |
| 161 | XML  | eXtensible Markup Language..... | 86 |



## NOTATION

162

|     |                 |                                                       |    |
|-----|-----------------|-------------------------------------------------------|----|
| 163 | $ \mathcal{S} $ | the number of elements in the set $\mathcal{S}$ ..... | 88 |
| 164 | $\emptyset$     | the set with no elements .....                        | 88 |
| 165 | $h(t)$          | impulse response .....                                | 78 |
| 166 | $\mathcal{S}$   | a collection of distinct objects .....                | 88 |
| 167 | $\mathcal{U}$   | the set containing everything .....                   | 88 |
| 168 | $x(t)$          | input signal represented in the time domain .....     | 78 |
| 169 | $y(t)$          | output signal represented in the time domain .....    | 78 |

170 Throughout this thesis proposal, mathematical notations conform to ISO 80000-2 standard,  
171 e.g., variable names are printed in italics, the only exception being acronyms like, e.g., SNR,  
172 which are printed in regular font. Constants are also set in regular font like  $j$ . Standard  
173 functions and operators are also set in regular font, e.g., in  $\sin(\cdot)$ ,  $\max\{\cdot\}$ . Commonly  
174 used notations are  $t$ ,  $f$ ,  $j = \sqrt{-1}$ ,  $n$  and  $\exp(\cdot)$ , which refer to the time variable, frequency  
175 variable, imaginary unit,  $n$ th variable, and exponential function, respectively.



176

## GLOSSARY

177

Functional Analysis

the branch of mathematics concerned with the study of spaces of functions

178

matrix

a concise and useful way of uniquely representing and working with linear transformations; a rectangular table of elements



179

## LISTINGS

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## Chapter 1

197

## INTRODUCTION



## 1.1 Background of the Study

Aside from the usual text descriptions of the background, put here figures that will cast images to your audience about the context of your work.

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232 Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla  
233 tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue  
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235 Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit  
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238 Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec  
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240 placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor.  
241 Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla  
242 tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue  
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244 Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit





amet ipsum. Nunc quis urna dictum turpis accumsan semper.

## 1.2 Prior Studies

Put here a narrative and a summary (not a duplicate) of your literature review chapter. In this section, summarize and highlight the gap(s) found in the literature review in Chapter 2. Preferably, a table showing the summary would be helpful.

Prior Studies or Literature Review<sup>1</sup> (expansion of the Prior Studies) is basically about competition. Competition.

So the suggested goals in writing the narrative of the Prior Studies in summative and highlighted forms are, in no particular order:

1. to mention the problem briefly;
2. to show the features of the existing literature in solving the problem
3. to show the weaknesses of the solutions of existing literature
4. to show how your solution is better (can be better (for proposals))

If the suggested table will be placed, please discuss it in light of the above-mentioned items.

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla

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<sup>1</sup>The main difference between the Prior Studies and Literature Review is that the Prior Studies is done in a concise manner. By the way, this is also an example of a footnote usage.



264 tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue  
 265 a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris.  
 266 Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit  
 267 amet ipsum. Nunc quis urna dictum turpis accumsan semper.

## 268 1.3 Problem Statement

269 The problem statement needs to be very clear and to the point.

270 A persuasive problem statement from a contextualized and intended-audience-awareness  
 271 perspective consists of:

272 1. PS1: description of the ideal scenario for your intended audience

- 273 • Describe the goals, desired state, or the values that your audience considers  
 274 important and that are relevant to the problem.

275 2. PS2: reality of the situation

- 276 • Describe a condition that prevents the goal, state, or value discussed in PS1  
 277 from being achieved or realized at the present time.
- 278 • It is imperative to make the audience feel the pain point.

279 3. PS3: consequences for the audience

- 280 • Using specific details, show how the situation contains a little promise of  
 281 improvement unless something is done.



282 After the above-mentioned items, succinctly describe your solution. Please avoid describing  
 283 your entire solution here since you will articulate and elucidate it by showing what you want  
 284 to achieve through your objectives, and how you will make it through your methodology.  
 285 A well-constructed problem statement will convince your audience that the problem is real  
 286 and worth having you solve it.

287 Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem.  
 288 Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec  
 289 ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus  
 290 placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor.  
 291 Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla  
 292 tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue  
 293 a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris.  
 294 Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit  
 295 amet ipsum. Nunc quis urna dictum turpis accumsan semper.

## 296 1.4 Objectives and Deliverables

297 Your objectives are the states that you desire to achieve in solving the problem. The general  
 298 objective is the main state to be achieved whereas the specific ones are sub-states to be  
 299 achieved.

### 300 1.4.1 General Objective (GO)

301 GO: To Morbi quis dolor. ;



### 1.4.2 Specific Objectives (SOs)

- SO1: To Quisque egestas wisi eget nunc. ;
- SO2: To Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. ;
- SO3: To Nullam cursus pulvinar lectus. ;
- SO4: To Morbi blandit ligula feugiat magna. ;
- SO5: To Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. ;

### 1.4.3 Expected Deliverables

Table 1.1 shows the outputs, products, results, achievements, gains, realizations, and/or yields of the Thesis Proposal.

TABLE 1.1 EXPECTED DELIVERABLES PER OBJECTIVE

| Objectives               | Expected Deliverables |
|--------------------------|-----------------------|
| GO: To Morbi quis dolor. | ⋮                     |

## 1.5 Significance of the Study

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor.



317 Praesent in sapien. Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Duis fringilla  
318 tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue  
319 a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris.  
320 Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit  
321 amet ipsum. Nunc quis urna dictum turpis accumsan semper.

## 322 1.5.1 Technical Benefit

323

324 1. First itemtext

325 2. Second itemtext

326 3. Last itemtext

327 4. First itemtext

328 5. Second itemtext

## 329 1.5.2 Social Impact

330

331 1. First itemtext

332 2. Second itemtext

333 3. Last itemtext

334 4. First itemtext



5. Second itemtext

### 1.5.3 Environmental Welfare

1. First itemtext

2. Second itemtext

3. Last itemtext

4. First itemtext

5. Second itemtext

## 1.6 Assumptions, Scope, and Delimitations

Bulletize your assumptions in one group, and then bulletize the scope in another, and do the same for your delimitations. The assumptions to put here are those major facts or statements that are *key* for your proposed solution to work. Scope refers to the space(s) for the operation of your proposed solution, whereas delimitations are the limits of the operation of your proposed solution.

### 1.6.1 Assumptions

1. ...;

2. ...;



3. ...;

### 1.6.2 Scope

1. ...;

2. ...;

3. ...;

### 1.6.3 Delimitations

1. ...;

2. ...;

3. ...;

## 1.7 Description and Methodology of the Thesis Proposal

A purpose of the description here is to re-steer/remind the panelist/reader again by tersely describing what your thesis is about (i.e. the problem and the main goal you want to achieve) in another way without sounding repetitive.

Your methodology is your means of achieving your stated objectives. What you put here is the summary of your methodology chapter.

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## 1.8 Estimated Work Schedule and Budget

The estimated work schedule can be represented as a Gantt Chart or a combination of  
 Project Network Diagram, Work Breakdown Structure, and Critical Path. The budget can  
 be made into a Bill of Materials, financial plan, or if your Thesis Proposal is funded and  
 part of larger project, the cost, and date for reaching each milestone and/or deliverable for  
 your part of the project.

For ECE Department undergraduate theses, the individual Gantt Chart or Work Break-  
 down Schedule and Bill of Materials will be included in this section and be removed in the  
 final document.

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395 **1.9 Overview of the Thesis Proposal**

396 Provide here a brief summary and what the reader should expect from each succeeding  
397 chapter. Show how each chapter is connected with each other.



De La Salle University

398

## Chapter 2

399

## LITERATURE REVIEW



400 It is to be noted that each subsection in this chapter should discuss in narrative form  
 401 each table that is presented in order to point out to the reader what the author(s) intend to  
 402 convey.

## 403 2.1 Existing Work

404 Cite and summarize here relevant and significant literature (dissertations, theses, journals,  
 405 patents, notable conference papers) through a table and descriptions to prove that no one  
 406 has done your work yet and/or that your work is not a duplication of existing ones. Your  
 407 focus here is what has *been done*.

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## 2.2 Lacking in the Approaches

You can summarize the weaknesses of existing approaches by a tabular comparison of the  
 literature. Your focus here is what has *not been done*, i.e. what features were missed, what  
 solutions were not considered, what the demerits are, etc. Through these items, you then  
 can introduce the necessity for doing your proposed solution.

It is to be noted that the degree of novelty for undergraduate thesis is lower than those  
 for graduate school. If a Ph.D. dissertation/thesis has a high degree of novelty and that for  
 an undergraduate is low, then a master's thesis is somewhere between the two.

Briefly include here the following in order to remind the reader why you are highlighting  
 the weaknesses of the solutions of existing literature.

- mentioning the problem
- showing how your solution is better (can be better (for proposals))

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## 510 2.3 Summary

511 Provide the gist of this chapter such that it reflects the contents and the message.



512

## **Chapter 3**

513

## **THEORETICAL CONSIDERATIONS**





Before starting the first section, provide an overview of the purpose of this chapter and its contents, and how they are relevant to your methodology. Discuss in this chapter the relevant theories and concepts that should support your proposed solutions.

This chapter is for providing the context to your panelist/reader. It is actually an expanded form of the Background of the Study that you have put in Chapter 1.

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Fig. 3.1 A quadrilateral image example.

## 564 3.1 Summary

565 Provide the gist of this chapter such that it reflects the contents and the message.



566

## Chapter 4

567

## DESIGN CONSIDERATIONS



Before starting the first section, provide an overview of the purpose of this chapter and its contents, and how they are relevant to your methodology.

Your primary goal in the Design Considerations chapter is to describe to your panelist/readers the key topics that fall further under Theoretical Considerations, but should be placed here instead since they are geared towards your Methodology. These key topics are those that you have directly adopted in making your solution/methodology. You can think of the connection of the Design Considerations chapter to the Theoretical Considerations chapter in this way: if your Theoretical Considerations chapter serves as the main foundation of a building, then the Design Considerations chapter functions as the columns.

The Design Considerations chapter is an avenue for explaining why you considered the topics here for your proposed methodology. This chapter is different from your methodology, because topics you discuss here are already accepted as part of the body of knowledge, and may have not been developed by you.

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## 4.1 Standards

Standards are essential for successful projects and impactful research. They provide a common framework and ensure consistency, quality, and safety across various disciplines. By adhering to established standards, your work becomes more reliable, interoperable, and valuable in real-world applications. Standards also demonstrate your understanding of industry best practices and enhance the credibility of your research.

To effectively integrate standards into your project, begin by identifying relevant standards related to your specific field. Thoroughly research and understand the requirements and guidelines outlined within these standards. Align your project objectives and methodologies to meet or exceed these standards. Document your use of standards in this section, including how and why specific standards were chosen. Finally, evaluate your results against the established standards, justifying any deviations from the norm with sound



638 reasoning and evidence.

639 **4.2 Summary**

640 Provide the gist of this chapter such that it reflects the contents and message.





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## Chapter 5

642

## METHODOLOGY



Put an overview of the contents of chapter. Mention here your methodology flow through a figure and provide an overview of it and how your methodology achieves your objectives. How your methodology achieves each of your specific objectives is what your panelists/examiners will be looking for. Specify how your methodology achieves your general objective and specific objectives. A point-by-point comparison how your methodology achieves each of your specific objectives is expected in the final Thesis Proposal.

Also make sure that you refer clearly to the chapters on the Literature Review, Theoretical Considerations, and Design Considerations showing how your methodology ties with those that you have discussed in those chapters.

Make an overview of the contents of the chapter. Put here your methodology flow through a figure and provide an overview of it.

In summative form, Table 5.1 indicates the approaches, designs, modes, processes, programs, techniques, and/or ways that the Thesis Proposal reaches the objectives.

TABLE 5.1 SUMMARY OF METHODS FOR REACHING THE OBJECTIVES

| Objectives               | Methods                                                                                                | Locations         |
|--------------------------|--------------------------------------------------------------------------------------------------------|-------------------|
| GO: To Morbi quis dolor. | 1. First itemtext<br>2. Second itemtext<br>3. Last itemtext<br>4. First itemtext<br>5. Second itemtext | Sec. 5.1 on p. 31 |

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| Objectives                                                                                            | Methods                                                                                                                                                                          | Locations         |
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| SO1: To Quisque egestas wisi eget nunc.                                                               | <ol style="list-style-type: none"> <li>1. First itemtext</li> <li>2. Second itemtext</li> <li>3. Last itemtext</li> <li>4. First itemtext</li> <li>5. Second itemtext</li> </ol> | Sec. 5.1 on p. 31 |
| SO2: To Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. | <ol style="list-style-type: none"> <li>1. First itemtext</li> <li>2. Second itemtext</li> <li>3. Last itemtext</li> <li>4. First itemtext</li> <li>5. Second itemtext</li> </ol> | Sec. 5.1 on p. 31 |
| SO3: To Nullam cursus pulvinar lectus.                                                                | <ol style="list-style-type: none"> <li>1. First itemtext</li> <li>2. Second itemtext</li> <li>3. Last itemtext</li> <li>4. First itemtext</li> <li>5. Second itemtext</li> </ol> | Sec. 5.1 on p. 31 |
| SO4: To Morbi blandit ligula feugiat magna.                                                           | <ol style="list-style-type: none"> <li>1. First itemtext</li> <li>2. Second itemtext</li> <li>3. Last itemtext</li> <li>4. First itemtext</li> <li>5. Second itemtext</li> </ol> | Sec. 5.1 on p. 31 |

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| Objectives                                                               | Methods                                                                                                                                                                          | Locations         |
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| SO5: To Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. | <ol style="list-style-type: none"> <li>1. First itemtext</li> <li>2. Second itemtext</li> <li>3. Last itemtext</li> <li>4. First itemtext</li> <li>5. Second itemtext</li> </ol> | Sec. 5.1 on p. 31 |

## 5.1 Implementation

Summarize the process used to create/set-up the work with an explanation of such process, instruments, and materials that you used if any. If the description is lengthy, use condensed bullet points.

*Rule of thumb:* Implementation is how you made your work; (keywords: implemented, created, made, soldered, programmed, etc.).

If you wrote a program or made a simulation, you must state how the program or simulation functions in this section. An algorithm or a pseudocode as shown in Table E.2 is a good example.

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## 711 5.2 Evaluation

712 Describe the procedures for evaluating the correct behavior and outcome of your work,  
 713 including what information you need to gather and how you will obtain or measure it.

714 *Rule of thumb:* Evaluation is how you tested your work; (keywords: measured, tested,  
 715 compared, simulated, etc.).

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### 5.3 Summary

Provide the gist of this chapter such that it reflects the contents and the message.





De La Salle University

763

## **Chapter 6**

764

## **RESULTS AND DISCUSSIONS**



765 Show in this chapter proofs why your proposed solution works. However, presenting  
 766 results ("It worked") without an appropriate explanation does not show thorough under-  
 767 standing. Aside from the data and results that you have obtained, and their explanation,  
 768 the discussion includes why components of your proposed solution work did or did not  
 769 work in accordance to what you described in the evaluation process, and how the proposed  
 770 solution performed and faired. Interpret the results and the reasons why they were obtained.  
 771 If your results are incorrect, apparent discrepancies from theory should be pointed out and  
 772 explained. In essence, what do the results mean? Citing existing publication can help you  
 773 compare your results and your explanations.

774 The next items below is not related to the description of this results and discussions  
 775 chapter, but serves as an opener for the  $\LaTeX$  portion of this template.

776 Here is an example of a citation for ISO 80000-2 standard [ISO, 2009]. Another one  
 777 is [Einstein, 1905] and [Croft, 1978].

778 In using this template, the user is expected to have a working knowledge of  $\LaTeX$ . A  
 779 good introduction is in [Oetiker et al., 2014]. Its latest version can be accessed at [http://](http://www.ctan.org/tex-archive/info/lshort)  
 780 [www.ctan.org/tex-archive/info/lshort](http://www.ctan.org/tex-archive/info/lshort). See the Appendix of `document_guide.pdf` for  
 781 examples.

782 In aggregate form, Table 6.1 shows the outcomes and completions in applying the  
 783 methodology of the Thesis Proposalper objective.

TABLE 6.1 SUMMARY OF RESULTS FOR ACHIEVING THE OBJECTIVES

| Objectives | Results | Locations |
|------------|---------|-----------|
|------------|---------|-----------|

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| Objectives                                                                                            | Results                                                                                                                                                                          | Locations         |
|-------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| GO: To Morbi quis dolor.                                                                              | <ol style="list-style-type: none"> <li>1. First itemtext</li> <li>2. Second itemtext</li> <li>3. Last itemtext</li> <li>4. First itemtext</li> <li>5. Second itemtext</li> </ol> | Sec. 5.1 on p. 31 |
| SO1: To Quisque eget wisi eget nunc.                                                                  | <ol style="list-style-type: none"> <li>1. First itemtext</li> <li>2. Second itemtext</li> <li>3. Last itemtext</li> <li>4. First itemtext</li> <li>5. Second itemtext</li> </ol> | Sec. 5.1 on p. 31 |
| SO2: To Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. | <ol style="list-style-type: none"> <li>1. First itemtext</li> <li>2. Second itemtext</li> <li>3. Last itemtext</li> <li>4. First itemtext</li> <li>5. Second itemtext</li> </ol> | Sec. 5.1 on p. 31 |
| SO3: To Nullam cursus pulvinar lectus.                                                                | <ol style="list-style-type: none"> <li>1. First itemtext</li> <li>2. Second itemtext</li> <li>3. Last itemtext</li> <li>4. First itemtext</li> <li>5. Second itemtext</li> </ol> | Sec. 5.1 on p. 31 |

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| Objectives                                                               | Results                                                                                                                                                                          | Locations         |
|--------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| SO4: To Morbi blandit ligula feugiat magna.                              | <ol style="list-style-type: none"> <li>1. First itemtext</li> <li>2. Second itemtext</li> <li>3. Last itemtext</li> <li>4. First itemtext</li> <li>5. Second itemtext</li> </ol> | Sec. 5.1 on p. 31 |
| SO5: To Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. | <ol style="list-style-type: none"> <li>1. First itemtext</li> <li>2. Second itemtext</li> <li>3. Last itemtext</li> <li>4. First itemtext</li> <li>5. Second itemtext</li> </ol> | Sec. 5.1 on p. 31 |

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**6.1 Summary**

Provide the gist of this chapter such that it reflects the contents and the message.



831

## **Chapter 7**

832

## **CONCLUSIONS, RECOMMENDATIONS, AND**

833

## **FUTURE DIRECTIVES**



## 834 7.1 Concluding Remarks

835 In this Thesis Proposal, . . .

836 Put here the main points that should be known and learned about the work topic.  
837 Summarize or give the gist of the essential principles and inferences drawn from your  
838 results.

## 839 7.2 Contributions

840 The interrelated contributions and supplements that have been developed by the author(s)  
841 in this Thesis Proposal are listed as follows. Only those that are unique to the authors' work  
842 are included.

- 843 • the ;
- 844 • the ;
- 845 • the ;

## 846 7.3 Recommendations

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## 7. Conclusions, Recommendations, and Future Directives



# De La Salle University

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 879 tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue  
 880 a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris.  
 881 Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit  
 882 amet ipsum. Nunc quis urna dictum turpis accumsan semper.

883 Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem.  
 884 Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec  
 885 ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus  
 886 placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor.  
 887 Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla  
 888 tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue  
 889 a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris.  
 890 Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit  
 891 amet ipsum. Nunc quis urna dictum turpis accumsan semper.

## 892 7.4 Future Prospects

893 There are several prospects that may be extended for further studies. . . . So the suggested  
 894 topics are listed in the following.

- 895 1. the . . . .
- 896 2. the . . . .
- 897 3. the . . . .

## 7. Conclusions, Recommendations, and Future Directives



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898 Note that for ECE undergraduate theses, as per the directions of the thesis adviser,  
899 Recommendations and Future Directives will be removed for the hardbound copy but will  
900 be retained for database storage.



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1194  $\LaTeX$ -comment this and the following texts after you have implemented them. See the  
1195 following references for helpful guides for the bibliography and script editing in general.  
1196 Note that the links might be unavailable, but the names can be searched in the Web.

- 1197 1. IEEE Citation Reference: [www.ieee.org/documents/ieeecitationref.pdf](http://www.ieee.org/documents/ieeecitationref.pdf)
- 1198 2. IEEE Editorial Style manual: [www.ieee.org/documents/style\\_manual.pdf](http://www.ieee.org/documents/style_manual.pdf)
- 1199 3. IEEE Abbreviations for Transactions, Journals, Letters, and Magazines: [www.ieee.org/documents/trans\\_journal\\_names.pdf](http://www.ieee.org/documents/trans_journal_names.pdf)

1201 Also in your BibTeX file, enclose letters or words that should all be in uppercase in curly  
1202 brackets. Example: IBM, Philippines, eXtensible Markup Language.



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1204

## **Appendix A**

1205

## **STUDENT RESEARCH ETHICS CLEARANCE**



1206

**RESEARCH ETHICS CLEARANCE FORM<sup>1</sup>**  
**For Thesis Proposals**

Names of Student Researcher(s):

Dela Cruz, Juan Z.

College: **Gokongwei College of Engineering**Department: **Electronics and Communications Engineering**Course: **PhD-ECE**Expected Duration of the Project: from: **April 2015**to: **April 2017**

Ethical considerations

None

(The [Ethics Checklists](#) may be used as guides in determining areas for ethical concern/consideration)

To the best of my knowledge, the ethical issues listed above have been addressed in the research.

Dr. Francisco D. Baltasar

Name and Signature of Adviser/Mentor:

Date: **April 8, 2017**

Noted by:

Dr. Rafael W. Sison

Name and Signature of the Department Chairperson:

Date: **April 8, 2017**

<sup>1</sup> The same form can be used for the reports of completed projects. The appropriate heading need only be used.



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## **Appendix B**

1208

# **ANSWERS TO QUESTIONS TO THIS THESIS PROPOSAL**

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## B1 How important is the problem to practice?

A possible answer to this question is the summary of your Significance of the Study, and that portion of the Problem Statement where you describe the ideal scenario for your intended audience.

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## B2 How will you know if the solution/s that you will achieve would be better than existing ones?

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### B2.1 How will you measure the improvement/s?

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### **B2.1.1 What is/are your basis/bases for the improvement/s?**

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### **B2.1.2 Why did you choose that/those basis/bases?**

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### **B2.1.3 How significant are your measure/s of the improvement/s?**

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### **B3 What is the difference of the solution/s from existing ones?**

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#### **B3.1 How is it different from previous and existing ones?**

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### **B4 What are the assumptions made (that are behind for your proposed solution to work)?**

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#### **B4.1 Will your proposed solution/s be sensitive to these assumptions?**

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#### **B4.2 Can your proposed solution/s be applied to more general cases when some assumptions are eliminated? If so, how?**

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#### **B5 What is the necessity of your approach / proposed solution/s?**

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### **B5.1 What will be the limits of applicability of your proposed solution/s?**

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### **B5.2 What will be the message of the proposed solution to technical people? How about to non-technical managers and business people?**

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### **B6 How will you know if your proposed solution/s is/are correct?**

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### **B6.1 Will your results warrant the level of mathematics used (i.e., will the end justify the means)?**

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### **B7 Is/are there an/\_ alternative way/s to get to the same solution/s?**

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### **B7.1 Can you come up with illustrating examples, or even better, counterexamples to your proposed solution/s?**

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## **B7.2 Is there an approximation that can arrive at essentially the same proposed solution/s more easily?**

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## **B8 If you were the examiner of your Thesis Proposal, how would you present the Thesis Proposal in another way? Give your remarks, especially for your methodology and the results and discussions.**

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## **B8.1 What are the weaknesses of your Thesis Proposal, specifically your methodology and the results and discussions?**

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## **Appendix C**

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## **REVISIONS TO THE PROPOSAL**





Make a table with the following columns for showing the summary of revisions to the proposal based on the comments of the panel of examiners.

1. Examiner

2. Comment

3. Summary of how the comment was addressed

4. Locations in the document where the changes have been reflected

TABLE C.1 SUMMARY OF REVISIONS TO THE PROPOSAL

| Examiner                  | Comment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Summary of how the comment was addressed                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Locations                                                      |
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| Dr. Francisco D. Baltasar | <p>Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.</p> | <p>Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.</p> <p><b>First</b> itemtext</p> <p><b>Second</b> itemtext</p> <p><b>Last</b> itemtext</p> <p><b>First</b> itemtext</p> <p><b>Second</b> itemtext</p> | <p>Sec. 5.1 on p. 31, Sec. 5.2 on p. 33, Fig. 3.1 on p. 22</p> |

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| Examiner               | Comment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Summary of how the comment was addressed                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Locations                                                      |
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| Dr. Amado Z. Hernandez | <p>Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.</p> | <p>Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.</p> <p><b>First</b> itemtext</p> <p><b>Second</b> itemtext</p> <p><b>Last</b> itemtext</p> <p><b>First</b> itemtext</p> <p><b>Second</b> itemtext</p> | <p>Sec. 5.1 on p. 31, Sec. 5.2 on p. 33, Fig. 3.1 on p. 22</p> |

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| Examiner           | Comment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Summary of how the comment was addressed                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Locations                                                      |
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| Dr. Jose Y. Alonzo | <p>Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.</p> | <p>Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.</p> <ul style="list-style-type: none"> <li>• First itemtext</li> <li>• Second itemtext</li> <li>• Last itemtext</li> <li>• First itemtext</li> <li>• Second itemtext</li> </ul> | <p>Sec. 5.1 on p. 31, Sec. 5.2 on p. 33, Fig. 3.1 on p. 22</p> |

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| Examiner               | Comment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Summary of how the comment was addressed                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Locations                                                      |
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| Dr. Mariana X. Mercado | <p>Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.</p> | <p>Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.</p> <ol style="list-style-type: none"> <li>1. First itemtext</li> <li>2. Second itemtext</li> <li>3. Last itemtext</li> <li>4. First itemtext</li> <li>5. Second itemtext</li> </ol> | <p>Sec. 5.1 on p. 31, Sec. 5.2 on p. 33, Fig. 3.1 on p. 22</p> |

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| Examiner            | Comment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Summary of how the comment was addressed                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Locations                                                      |
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| Dr. Rafael W. Sison | <p>Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.</p> | <p>Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.</p> | <p>Sec. 5.1 on p. 31, Sec. 5.2 on p. 33, Fig. 3.1 on p. 22</p> |



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## **Appendix D**

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## **REVISIONS TO THE FINAL**



Make a table with the following columns for showing the summary of revisions to the proposal based on the comments of the panel of examiners.

1. Examiner
2. Comment
3. Summary of how the comment has been addressed
4. Locations in the document where the changes have been reflected

TABLE D.1 SUMMARY OF REVISIONS TO THE THESIS PROPOSAL

| Examiner                  | Comment                                                                                                                                                                          | Summary of how the comment has been addressed                                                                                                                                                                                                                                                                                      | Locations                                                      |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| Dr. Francisco D. Baltasar | <ol style="list-style-type: none"> <li>1. First itemtext</li> <li>2. Second itemtext</li> <li>3. Last itemtext</li> <li>4. First itemtext</li> <li>5. Second itemtext</li> </ol> | <ol style="list-style-type: none"> <li>1. First itemtext</li> <li>2. Second itemtext</li> <li>3. Last itemtext</li> <li>4. First itemtext</li> <li>5. Second itemtext</li> </ol> <p><b>First</b> itemtext</p> <p><b>Second</b> itemtext</p> <p><b>Last</b> itemtext</p> <p><b>First</b> itemtext</p> <p><b>Second</b> itemtext</p> | <p>Sec. 5.1 on p. 31, Sec. 5.2 on p. 33, Fig. 3.1 on p. 22</p> |

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| Examiner               | Comment                                                                                                | Summary of how the comment has been addressed                                                                                                                                                                                            | Locations                                                              |
|------------------------|--------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| Dr. Amado Z. Hernandez | 1. First itemtext<br>2. Second itemtext<br>3. Last itemtext<br>4. First itemtext<br>5. Second itemtext | 1. First itemtext<br>2. Second itemtext<br>3. Last itemtext<br>4. First itemtext<br>5. Second itemtext<br><br><b>First</b> itemtext<br><b>Second</b> itemtext<br><b>Last</b> itemtext<br><b>First</b> itemtext<br><b>Second</b> itemtext | Sec. 5.1<br>on p. 31,<br>Sec. 5.2<br>on p. 33,<br>Fig. 3.1 on<br>p. 22 |
| Dr. Jose Y. Alonzo     | 1. First itemtext<br>2. Second itemtext<br>3. Last itemtext<br>4. First itemtext<br>5. Second itemtext | 1. First itemtext<br>2. Second itemtext<br>3. Last itemtext<br>4. First itemtext<br>5. Second itemtext<br><br>• First itemtext<br>• Second itemtext<br>• Last itemtext<br>• First itemtext<br>• Second itemtext                          | Sec. 5.1<br>on p. 31,<br>Sec. 5.2<br>on p. 33,<br>Fig. 3.1 on<br>p. 22 |

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| Examiner               | Comment                                                                                                | Summary of how the comment has been addressed                                                          | Locations                                                              |
|------------------------|--------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| Dr. Mariana X. Mercado | 1. First itemtext<br>2. Second itemtext<br>3. Last itemtext<br>4. First itemtext<br>5. Second itemtext | 1. First itemtext<br>2. Second itemtext<br>3. Last itemtext<br>4. First itemtext<br>5. Second itemtext | Sec. 5.1<br>on p. 31,<br>Sec. 5.2<br>on p. 33,<br>Fig. 3.1 on<br>p. 22 |
| Dr. Rafael W. Sison    | 1. First itemtext<br>2. Second itemtext<br>3. Last itemtext<br>4. First itemtext<br>5. Second itemtext | 1. First itemtext<br>2. Second itemtext<br>3. Last itemtext<br>4. First itemtext<br>5. Second itemtext | Sec. 5.1<br>on p. 31,<br>Sec. 5.2<br>on p. 33,<br>Fig. 3.1 on<br>p. 22 |



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## **Appendix E**

### **USAGE EXAMPLES**

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The user is expected to have a working knowledge of  $\text{\LaTeX}$ . A good introduction is in [Oetiker et al., 2014]. Its latest version can be accessed at <http://www.ctan.org/tex-archive/info/lshort>.

## E1 Equations

The following examples show how to typeset equations in  $\text{\LaTeX}$ . This section also shows examples of the use of `\gls{ }` commands in conjunction with the items that are in the `notation.tex` file. **Please make sure that the entries in `notation.tex` are those that are referenced in the  $\text{\LaTeX}$  document files used by this Thesis Proposal. Please comment out unused notations and be careful with the commas and brackets in `notation.tex`.**

In (E.1), the output signal  $y(t)$  is the result of the convolution of the input signal  $x(t)$  and the impulse response  $h(t)$ .

$$y(t) = h(t) * x(t) = \int_{-\infty}^{+\infty} h(t - \tau) x(\tau) d\tau \quad (\text{E.1})$$

Other example equations are as follows.

$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} V_2 \\ I_2 \end{bmatrix} \quad (\text{E.2})$$

$$\frac{1}{2} < \left[ \text{mod} \left( \left\lfloor \frac{y}{17} \right\rfloor 2^{-17\lfloor x \rfloor - \text{mod}(\lfloor y \rfloor, 17)}, 2 \right) \right], \quad (\text{E.3})$$

$$|\zeta(x)^3 \zeta(x + iy)^4 \zeta(x + 2iy)| = \exp \sum_{n,p} \frac{3 + 4 \cos(ny \log p) + \cos(2ny \log p)}{np^{nx}} \geq 1 \quad (\text{E.4})$$



The verbatim L<sup>A</sup>T<sub>E</sub>X code of Sec. E1 is in List. E.1.

```

1 The following examples show how to typeset equations in \LaTeX. This
  section also shows examples of the use of \verb| \gls{ } | commands
  in conjunction with the items that are in the \verb| notation.tex |
  file. \textbf{Please make sure that the entries in} \verb| notation.
  tex |\textbf{ are those that are referenced in the \LaTeX \
  document files used by this \documentType. Please comment out
  unused notations and be careful with the commas and brackets in} \
  verb| notation.tex |.
2
3 In~\eqref{eq:conv}, the output signal \gls{not:output_sigt} is the
  result of the convolution of the input signal \gls{not:input_sigt}
  and the impulse response \gls{not:ir}.
4
5 \begin{eqnarray}
6   y\left( t \right) = h\left( t \right) * x\left( t \right)=\int_{-\infty}^{\infty}h\left( t-\tau \right)x\left( \tau \right) \,
   \mathrm{d}\tau
7   \label{eq:conv}
8 \end{eqnarray}
9
10 Other example equations are as follows.
11
12 \begin{eqnarray}
13   \left[ \begin{matrix} \frac{V_1}{I_1} \end{matrix} \right] =
14   \begin{bmatrix} A & B \\ C & D \end{bmatrix}
15   \label{eq:ABCD}
16 \end{eqnarray}
17
18 \begin{eqnarray}
19   \left[ \begin{matrix} \frac{V_2}{I_2} \end{matrix} \right]
20   \label{eq:ABCD}
21 \end{eqnarray}
22
23 \begin{eqnarray}
24   \frac{1}{2} < \left\lfloor \mathrm{mod} \left( \left\lfloor \frac{y}{17} \right\rfloor 2^{-17} \lfloor x \rfloor - \mathrm{mod}(\lfloor y \rfloor, 17) \right), 2 \right\rfloor \right\rfloor,
25 \end{eqnarray}
26
27 \begin{eqnarray}
28   | \zeta(x)^3 \zeta(x + iy)^4 \zeta(x + 2iy) | =
29   \exp \sum_{n,p} \frac{3 + 4 \cos( ny \log p) + \cos( 2ny \log p)}{n^p}
30 \end{eqnarray}

```



## E2 Notations

In order to use the standardized notation, the user is highly suggested to see the ISO 80000-2 standard [ISO, 2009].

See [https://en.wikipedia.org/wiki/Help:Displaying\\_a\\_formula](https://en.wikipedia.org/wiki/Help:Displaying_a_formula) and [https://en.wikipedia.org/wiki/List\\_of\\_mathematical\\_symbols](https://en.wikipedia.org/wiki/List_of_mathematical_symbols) for L<sup>A</sup>T<sub>E</sub>X maths and other notations, respectively.

The following were taken from `isomath-test.tex`.

### E2.1 Math alphabets

If there are other symbols in place of Greek letters in a math alphabet, it uses T1 or OT1 font encoding instead of OML.

|                         |                                                                                                                                                                                                                                                             |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>mathnormal</code> | $A, B, \Gamma, \Delta, \Theta, \Lambda, \Xi, \Pi, \Sigma, \Phi, \Psi, \Omega, \alpha, \beta, \pi, \nu, \omega, v, w, 0, 1, 9$                                                                                                                               |
| <code>mathit</code>     | $A, B, \Gamma, \Delta, \Theta, \Lambda, \Xi, \Pi, \Sigma, \Phi, \Psi, \Omega, \mathfrak{f}, \mathfrak{f}, \beta, ^\circ, !, v, w, 0, 1, 9$                                                                                                                  |
| <code>mathrm</code>     | $A, B, \Gamma, \Delta, \Theta, \Lambda, \Xi, \Pi, \Sigma, \Phi, \Psi, \Omega, \mathfrak{f}, \mathfrak{f}, \beta, ^\circ, !, v, w, 0, 1, 9$                                                                                                                  |
| <code>mathbf</code>     | $\mathbf{A}, \mathbf{B}, \mathbf{\Gamma}, \mathbf{\Delta}, \mathbf{\Theta}, \mathbf{\Lambda}, \mathbf{\Xi}, \mathbf{\Pi}, \mathbf{\Sigma}, \mathbf{\Phi}, \mathbf{\Psi}, \mathbf{\Omega}, \mathbf{f}, \mathbf{f}, \mathbf{\beta}, ^\circ, !, v, w, 0, 1, 9$ |
| <code>mathsf</code>     | $A, B, \Gamma, \Delta, \Theta, \Lambda, \Xi, \Pi, \Sigma, \Phi, \Psi, \Omega, \mathfrak{f}, \mathfrak{f}, \beta, ^\circ, !, v, w, 0, 1, 9$                                                                                                                  |
| <code>mathtt</code>     | $A, B, \Gamma, \Delta, \Theta, \Lambda, \Xi, \Pi, \Sigma, \Phi, \Psi, \Omega, \uparrow, \downarrow, \beta, ^\circ, !, v, w, 0, 1, 9$                                                                                                                        |

New alphabets bold-italic, sans-serif-italic, and sans-serif-bold-italic.

|                        |                                                                                                                                                                                                                                           |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>mathbfit</code>  | $\mathbf{A}, \mathbf{B}, \mathbf{\Gamma}, \mathbf{\Delta}, \mathbf{\Theta}, \mathbf{\Lambda}, \mathbf{\Xi}, \mathbf{\Pi}, \mathbf{\Sigma}, \mathbf{\Phi}, \mathbf{\Psi}, \mathbf{\Omega}, \alpha, \beta, \pi, \nu, \omega, v, w, 0, 1, 9$ |
| <code>mathsf</code>    | $A, B, \Gamma, \Delta, \Theta, \Lambda, \Xi, \Pi, \Sigma, \Phi, \Psi, \Omega, \alpha, \beta, \pi, \nu, \omega, v, w, 0, 1, 9$                                                                                                             |
| <code>mathsfbit</code> | $\mathbf{A}, \mathbf{B}, \mathbf{\Gamma}, \mathbf{\Delta}, \mathbf{\Theta}, \mathbf{\Lambda}, \mathbf{\Xi}, \mathbf{\Pi}, \mathbf{\Sigma}, \mathbf{\Phi}, \mathbf{\Psi}, \mathbf{\Omega}, \alpha, \beta, \pi, \nu, \omega, v, w, 0, 1, 9$ |

Do the math alphabets match?

$\alpha x \alpha \omega \alpha x \alpha \omega \alpha x \alpha \omega \quad TC\Theta\Gamma TC\Theta\Gamma TC\Theta\Gamma$

### E2.2 Vector symbols

Alphabetic symbols for vectors are boldface italic,  $\lambda = e_1 \cdot a$ , while numeric ones (e.g. the zero vector) are bold upright,  $a + 0 = a$ .

### E2.3 Matrix symbols

Symbols for matrices are boldface italic, too:<sup>1</sup>  $\mathbf{A} = \mathbf{E} \cdot \mathbf{A}$ .

<sup>1</sup>However, matrix symbols are usually capital letters whereas vectors are small ones. Exceptions are physical quantities like the force vector  $\mathbf{F}$  or the electrical field  $\mathbf{E}$ .



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**E2.4 Tensor symbols**

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Symbols for tensors are sans-serif bold italic,

$$\boldsymbol{\alpha} = \boldsymbol{e} \cdot \boldsymbol{a} \quad \Longleftrightarrow \quad \alpha_{ijl} = e_{ijk} \cdot a_{kl}.$$

1492

The permittivity tensor describes the coupling of electric field and displacement:

$$\boldsymbol{D} = \epsilon_0 \boldsymbol{\epsilon}_r \boldsymbol{E}$$



## E2.5 Bold math version

The “bold” math version is selected with the commands `\boldmath` or `\mathversion{bold}`

|                         |                                                                                                                                                    |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>mathnormal</code> | $A, B, \Gamma, \Delta, \Theta, \Lambda, \Xi, \Pi, \Sigma, \Phi, \Psi, \Omega, \alpha, \beta, \pi, \nu, \omega, v, w, 0, 1, 9$                      |
| <code>mathit</code>     | $A, B, \Gamma, \Delta, \Theta, \Lambda, \Xi, \Pi, \Sigma, \Phi, \Psi, \Omega, \mathfrak{f}, \mathfrak{fi}, \mathfrak{B}, ^\circ, !, v, w, 0, 1, 9$ |
| <code>mathrm</code>     | $A, B, \Gamma, \Delta, \Theta, \Lambda, \Xi, \Pi, \Sigma, \Phi, \Psi, \Omega, \mathfrak{f}, \mathfrak{fi}, \mathfrak{B}, ^\circ, !, v, w, 0, 1, 9$ |
| <code>mathbf</code>     | $A, B, \Gamma, \Delta, \Theta, \Lambda, \Xi, \Pi, \Sigma, \Phi, \Psi, \Omega, \mathfrak{f}, \mathfrak{fi}, \mathfrak{B}, ^\circ, !, v, w, 0, 1, 9$ |
| <code>mathsf</code>     | $A, B, \Gamma, \Delta, \Theta, \Lambda, \Xi, \Pi, \Sigma, \Phi, \Psi, \Omega, \mathfrak{f}, \mathfrak{fi}, \mathfrak{B}, ^\circ, !, v, w, 0, 1, 9$ |
| <code>mathtt</code>     | $A, B, \Gamma, \Delta, \Theta, \Lambda, \Xi, \Pi, \Sigma, \Phi, \Psi, \Omega, \uparrow, \downarrow, \mathfrak{B}, ^\circ, !, v, w, 0, 1, 9$        |

New alphabets bold-italic, sans-serif-italic, and sans-serif-bold-italic.

|                       |                                                                                                                               |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <code>mathbf</code>   | $A, B, \Gamma, \Delta, \Theta, \Lambda, \Xi, \Pi, \Sigma, \Phi, \Psi, \Omega, \alpha, \beta, \pi, \nu, \omega, v, w, 0, 1, 9$ |
| <code>mathsf</code>   | $A, B, \Gamma, \Delta, \Theta, \Lambda, \Xi, \Pi, \Sigma, \Phi, \Psi, \Omega, \alpha, \beta, \pi, \nu, \omega, v, w, 0, 1, 9$ |
| <code>mathsfbf</code> | $A, B, \Gamma, \Delta, \Theta, \Lambda, \Xi, \Pi, \Sigma, \Phi, \Psi, \Omega, \alpha, \beta, \pi, \nu, \omega, v, w, 0, 1, 9$ |

Do the math alphabets match?

$\alpha x \alpha \omega \alpha x \alpha \omega \alpha x \alpha \omega \quad TC\Theta\Gamma TC\Theta\Gamma TC\Theta\Gamma$

### E2.5.1 Vector symbols

Alphabetic symbols for vectors are boldface italic,  $\lambda = e_1 \cdot a$ , while numeric ones (e.g. the zero vector) are bold upright,  $a + 0 = a$ .

### E2.5.2 Matrix symbols

Symbols for matrices are boldface italic, too:<sup>2</sup>  $\Lambda = E \cdot A$ .

### E2.5.3 Tensor symbols

Symbols for tensors are sans-serif bold italic,

$$\alpha = e \cdot a \quad \Longleftrightarrow \quad \alpha_{ijl} = e_{ijk} \cdot a_{kl}.$$

The permittivity tensor describes the coupling of electric field and displacement:

$$D = \epsilon_0 \epsilon_r E$$

<sup>2</sup>However, matrix symbols are usually capital letters whereas vectors are small ones. Exceptions are physical quantities like the force vector  $F$  or the electrical field  $E$ .



1506 The verbatim  $\text{\LaTeX}$  code of Sec. E2 is in List. E.2.

Listing E.2: Sample  $\text{\LaTeX}$  code for notations usage

```

1507 1 % A teststring with Latin and Greek letters::
1508 2 \newcommand{\teststring}{%
1509 3 % capital Latin letters
1510 4 % A,B,C,
1511 5 A,B,
1512 6 % capital Greek letters
1513 7 %\Gamma,\Delta,\Theta,\Lambda,\Xi,\Pi,\Sigma,\Upsilon,\Phi,\Psi,
1514 8 \Gamma,\Delta,\Theta,\Lambda,\Xi,\Pi,\Sigma,\Phi,\Psi,\Omega,
1515 9 % small Greek letters
1516 10 \alpha,\beta,\pi,\nu,\omega,
1517 11 % small Latin letters:
1518 12 % compare \nu, \omega, v, and w
1519 13 v,w,
1520 14 % digits
1521 15 0,1,9
1522 16 }
1523 17
1524 18
1525 19 \subsection{Math alphabets}
1526 20
1527 21 If there are other symbols in place of Greek letters in a math
1528 22 alphabet, it uses T1 or OT1 font encoding instead of OML.
1529 23
1530 24 \begin{eqnarray*}
1531 25 \mbox{\mathnormal} & & \mbox{\teststring} \\
1532 26 \mbox{\mathit} & & \mbox{\mathit{\teststring}} \\
1533 27 \mbox{\mathrm} & & \mbox{\mathrm{\teststring}} \\
1534 28 \mbox{\mathbf} & & \mbox{\mathbf{\teststring}} \\
1535 29 \mbox{\mathsf} & & \mbox{\mathsf{\teststring}} \\
1536 30 \mbox{\mathtt} & & \mbox{\mathtt{\teststring}} \\
1537 31 \end{eqnarray*}
1538 32 New alphabets bold-italic, sans-serif-italic, and sans-serif-bold-
1539 33 italic.
1540 34 \begin{eqnarray*}
1541 35 \mbox{\mathbfit} & & \mbox{\mathbfit{\teststring}} \\
1542 36 \mbox{\mathsf fit} & & \mbox{\mathsf fit{\teststring}} \\
1543 37 \mbox{\mathsfbfit} & & \mbox{\mathsfbfit{\teststring}} \\
1544 38 \end{eqnarray*}
1545 39 %
1546 40 Do the math alphabets match?
1547 41 $
1548 42 \mathnormal {a x \alpha \omega}
1549 43 \mathbfit {a x \alpha \omega}
1550 44 \mathsfbfit{a x \alpha \omega}
1551 45 \quad
1552 46 \mathsfbfit{T C \Theta \Gamma}
1553 47 \mathbfit {T C \Theta \Gamma}
1554 48 \mathnormal {T C \Theta \Gamma}
1555 49 $
1556 50
1557 51 \subsection{Vector symbols}
1558 52

```





```

1561      Alphabetic symbols for vectors are boldface italic,
1562       $\vec{\lambda} = \vec{e}_1 \cdot \vec{a}$ ,
1563      while numeric ones (e.g. the zero vector) are bold upright,
1564       $\vec{a} + \vec{0} = \vec{a}$ .
1565
1566      \subsection{Matrix symbols}
1567
1568      Symbols for matrices are boldface italic, too:%
1569      \footnote{However, matrix symbols are usually capital letters whereas
1570      vectors
1571      are small ones. Exceptions are physical quantities like the force
1572      vector  $\vec{F}$  or the electrical field  $\vec{E}$ .%
1573    }
1574       $\Lambda = E \cdot A$ .
1575
1576
1577      \subsection{Tensor symbols}
1578
1579      Symbols for tensors are sans-serif bold italic,
1580
1581      [
1582        \tensorsym{\alpha} = \tensorsym{e} \cdot \tensorsym{a}
1583        \quad \Longleftrightarrow \quad
1584        \alpha_{ijl} = e_{ijk} \cdot a_{kl}.
1585      ]
1586
1587
1588      The permittivity tensor describes the coupling of electric field and
1589      displacement: [
1590         $D = \epsilon_0 \text{\tensorsym{\epsilon}}_{\text{\mathrm{r}}} \vec{E}$ ]
1591
1592
1593
1594      \newpage
1595      \subsection{Bold math version}
1596
1597      The ‘‘bold’’ math version is selected with the commands
1598      \verb+\boldmath+ or \verb+\mathversion{bold}+
1599
1600      {\boldmath
1601        \begin{eqnarray*}
1602          \mbox{\mathnormal} & & \teststring \\
1603          \mbox{\mathit} & & \mathit{\teststring} \\
1604          \mbox{\mathrm} & & \mathrm{\teststring} \\
1605          \mbox{\mathbf} & & \mathbf{\teststring} \\
1606          \mbox{\mathsf} & & \mathsf{\teststring} \\
1607          \mbox{\mathtt} & & \mathtt{\teststring}
1608        \end{eqnarray*}
1609        New alphabets bold-italic, sans-serif-italic, and sans-serif-bold-
1610        italic.
1611        \begin{eqnarray*}
1612          \mbox{\mathbfit} & & \mathbfit{\teststring} \\
1613          \mbox{\mathsfit} & & \mathsfit{\teststring} \\
1614          \mbox{\mathsfbfit} & & \mathsfbfit{\teststring}
1615        \end{eqnarray*}
1616        %
1617        Do the math alphabets match?

```



```

1618 108
1619 109 $
1620 110 \mathnormal {a x \alpha \omega}
1621 111 \mathbfit {a x \alpha \omega}
1622 112 \mathsfbfit{a x \alpha \omega}
1623 113 \quad
1624 114 \mathsfbfit{T C \Theta \Gamma}
1625 115 \mathbfit {T C \Theta \Gamma}
1626 116 \mathnormal {T C \Theta \Gamma}
1627 117 $
1628 118
1629 119 \subsection{Vector symbols}
1630 120
1631 121 Alphabetic symbols for vectors are boldface italic,
1632 122 $\vec{\lambda}=\vec{e}_{1}\cdot\vec{a}$,
1633 123 while numeric ones (e.g. the zero vector) are bold upright,
1634 124 $\vec{a} + \vec{0} = \vec{a}$.
1635 125
1636 126
1637 127
1638 128
1639 129 \subsection{Matrix symbols}
1640 130
1641 131 Symbols for matrices are boldface italic, too:%
1642 132 \footnote{However, matrix symbols are usually capital letters whereas
1643 133 vectors
1644 133 are small ones. Exceptions are physical quantities like the force
1645 134 vector $\vec{F}$ or the electrical field $\vec{E}$.%
1646 135 }
1647 136 $\matrixsym{\Lambda}=\matrixsym{E}\cdot\matrixsym{A}$.
1648 137
1649 138
1650 139 \subsection{Tensor symbols}
1651 140
1652 141 Symbols for tensors are sans-serif bold italic,
1653 142
1654 143 \[
1655 144 \tensorsym{\alpha} = \tensorsym{e}\cdot\tensorsym{a}
1656 145 \quad \Longleftarrow \quad
1657 146 \alpha_{ijl} = e_{ijk}\cdot a_{kl}.
1658 147 \]
1659 148
1660 149 The permittivity tensor describes the coupling of electric field and
1661 150 displacement: \[
1662 151 \vec{D}=\epsilon_{0}\tensorsym{\epsilon}_{\mathrm{r}}\vec{E}\]
1663 152 }

```



## E3 Abbreviation

This section shows examples of the use of  $\LaTeX$  commands in conjunction with the items that are in the `abbreviation.tex` and in the `glossary.tex` files. Please see List. E.3. **To lessen the  $\LaTeX$  parsing time, it is suggested that you use `\acr{ }` only for the first occurrence of the word to be abbreviated.**

Again please see List. E.3. Here is an example of first use: alternating current (ac). Next use: ac. Full: alternating current (ac). Here's an acronym referenced using `\acr` : hyper-text markup language (html). And here it is again: html. If you are used to the glossaries package, note the difference in using `\gls` : hyper-text markup language (html). And again (no difference): hyper-text markup language (html). For plural use `\glspl` . Here are some more entries:

- extensible markup language (xml) and cascading style sheet (css).
- Next use: xml and css.
- Full form: extensible markup language (xml) and cascading style sheet (css).
- Reset again.
- Start with a capital. Hyper-text markup language (html).
- Next: Html. Full: Hyper-text markup language (html).
- Prefer capitals? Extensible markup language (XML). Next: XML. Full: extensible markup language (XML).
- Prefer small-caps? Cascading style sheet (CSS). Next: CSS. Full: cascading style sheet (CSS).
- Resetting all acronyms.
- Here are the acronyms again:
- Hyper-text markup language (HTML), extensible markup language (XML) and cascading style sheet (CSS).
- Next use: HTML, XML and CSS.
- Full form: Hyper-text markup language (HTML), extensible markup language (XML) and cascading style sheet (CSS).



- 1694 • Provide your own link text: style sheet.

1695 The verbatim  $\text{\LaTeX}$  code of Sec. E3 is in List. E.3.

### Listing E.3: Sample $\text{\LaTeX}$ code for abbreviations usage

```

1 Again please see List.~\ref{lst:abbrv}. Here is an example of first use:
  \acr{ac}. Next use: \acr{ac}. Full: \gls{ac}. Here's an acronym
  referenced using \verb| \acr |: \acr{html}. And here it is again: \
  acr{html}. If you are used to the \texttt{glossaries} package, note
  the difference in using \verb| \gls |: \gls{html}. And again (no
  difference): \gls{html}. Here are some more entries:
2
3 \begin{itemize}
4
5   \item \acr{xml} and \acr{css}.
6
7   \item Next use: \acr{xml} and \acr{css}.
8
9   \item Full form: \gls{xml} and \gls{css}.
10
11  \item Reset again. \glsresetall{abbreviation}
12
13  \item Start with a capital. \Acr{html}.
14
15  \item Next: \Acr{html}. Full: \Gls{html}.
16
17  \item Prefer capitals? \renewcommand{\acronymfont}[1]{\
    MakeTextUppercase{#1}} \Acr{xml}. Next: \acr{xml}. Full: \gls{xml}
    }.
18
19  \item Prefer small-caps? \renewcommand{\acronymfont}[1]{\textsc{#1}}
    \Acr{css}. Next: \acr{css}. Full: \gls{css}.
20
21  \item Resetting all acronyms.\glsresetall{abbreviation}
22
23  \item Here are the acronyms again:
24
25  \item \Acr{html}, \acr{xml} and \acr{css}.
26
27  \item Next use: \Acr{html}, \acr{xml} and \acr{css}.
28
29  \item Full form: \Gls{html}, \gls{xml} and \gls{css}.
30
31  \item Provide your own link text: \glslink{[textbf]css}{style}
32
33 \end{itemize}

```



## E4 Glossary

This section shows examples of the use of `\gls{ }` commands in conjunction with the items that are in the `glossary.tex` and `notation.tex` files. Note that entries in `notation.tex` are prefixed with “not:” label (see List. E.4).

**Please make sure that the entries in `notation.tex` are those that are referenced in the  $\LaTeX$  document files used by this Thesis Proposal. Please comment out unused notations and be careful with the commas and brackets in `notation.tex` .**

- Matrices are usually denoted by a bold capital letter, such as  $\mathbf{A}$ . The matrix’s  $(i, j)$ th element is usually denoted  $a_{ij}$ . Matrix  $\mathbf{I}$  is the identity matrix.
- A set, denoted as  $\mathcal{S}$ , is a collection of objects.
- The universal set, denoted as  $\mathcal{U}$ , is the set of everything.
- The empty set, denoted as  $\emptyset$ , contains no elements.
- Functional Analysis is seen as the study of complete normed vector spaces, i.e., Banach spaces.
- The cardinality of a set, denoted as  $|\mathcal{S}|$ , is the number of elements in the set.

The verbatim  $\LaTeX$  code for the part of Sec. E4 is in List. E.4.

Listing E.4: Sample  $\LaTeX$  code for glossary and notations usage

```

1 \begin{itemize}
2
3   \item \Glspl{matrix} are usually denoted by a bold capital letter,
      such as  $\mathbf{A}$ . The  $\mathbf{A}$ 's  $(i,j)$ th element is
      usually denoted  $a_{ij}$ .  $\mathbf{I}$  is the
      identity  $\mathbf{I}$ .
4
5   \item A set, denoted as  $\mathbf{S}$ , is a collection of objects.
6
7   \item The universal set, denoted as  $\mathbf{U}$ , is the
      set of everything.
8
9   \item The empty set, denoted as  $\emptyset$ , contains no
      elements.
10
11  \item  $\mathbf{FA}$  is seen as the study of complete
      normed vector spaces, i.e., Banach spaces.
12
13  \item The cardinality of a set, denoted as  $\mathbf{C}$ , is
      the number of elements in the set.
14
15 \end{itemize}

```



## E5 Figure

This section shows several ways of placing figures. PDFL<sup>A</sup>T<sub>E</sub>X compatible files are PDF, PNG, and JPG. Please see the `figure` subdirectory.



Fig. E.1 A quadrilateral image example.



1715 Fig. E.1 is a gray box enclosed by a dark border. List. E.5 shows the corresponding  
1716  $\text{\LaTeX}$  code.

Listing E.5: Sample  $\text{\LaTeX}$  code for a single figure

```
1 \begin{figure}[!htbp]
2   \centering
3   \includegraphics[width=0.5\textwidth]{example}
4   \caption{A quadrilateral image example.}
5   \label{fig:example}
6 \end{figure}
7 \cleardoublepage
8
9 Fig.~\ref{fig:example} is a gray box enclosed by a dark border. List.~\
  ref{lst:onefig} shows the corresponding  $\text{\LaTeX}$  \ code.
10 \end{figure}
```





(a) A sub-figure in the top row.



(b) A sub-figure in the middle row.



(c) A sub-figure in the bottom row.

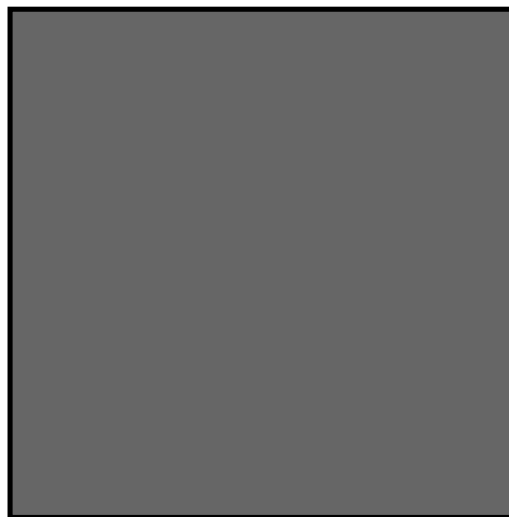
Fig. E.2 Figures on top of each other. See List. E.6 for the corresponding  $\text{\LaTeX}$  code.

Listing E.6: Sample L<sup>A</sup>T<sub>E</sub>X code for three figures on top of each other

```
1 \begin{figure}[!htbp]
2 \centering
3 \subbottom[A sub-figure in the top row.]{
4 \includegraphics[width=0.35\textwidth]{example_gray_box}
5 \label{fig:top}
6 }
7 \vfill
8 \subbottom[A sub-figure in the middle row.]{
9 \includegraphics[width=0.35\textwidth]{example_gray_box}
10 \label{fig:mid}
11 }
12 \vfill
13 \subbottom[A sub-figure in the bottom row.]{
14 \includegraphics[width=0.35\textwidth]{example_gray_box}
15 \label{fig:botm}
16 }
17 \caption{Figures on top of each other}
18 \label{fig:tmb}
19 \end{figure}
```



(a) A sub-figure in the upper-left corner.



(b) A sub-figure in the upper-right corner.



(c) A sub-figure in the lower-left corner.



(d) A sub-figure in the lower-right corner

Fig. E.3 Four figures in each corner. See List. E.7 for the corresponding  $\text{\LaTeX}$  code.

Listing E.7: Sample  $\text{\LaTeX}$  code for the four figures

```

1 \begin{figure}[!htbp]
2 \centering
3 \subbottom[A sub-figure in the upper-left corner.]{
4 \includegraphics[width=0.45\textwidth]{example_gray_box}
5 \label{fig:upprleft}
6 }
7 \hfill
8 \subbottom[A sub-figure in the upper-right corner.]{
9 \includegraphics[width=0.45\textwidth]{example_gray_box}
10 \label{fig:uppright}
11 }
12 \vfill
13 \subbottom[A sub-figure in the lower-left corner.]{
14 \includegraphics[width=0.45\textwidth]{example_gray_box}
15 \label{fig:lowerleft}
16 }
17 \hfill
18 \subbottom[A sub-figure in the lower-right corner]{
19 \includegraphics[width=0.45\textwidth]{example_gray_box}
20 \label{fig:lowright}
21 }
22 \caption{Four figures in each corner. See List.\ref{lst:fourfigs} for
23 the corresponding \LaTeX \ code.}
24 \label{fig:fourfig}
25 \end{figure}

```



## E6 Table

This section shows an example of placing a table (a long one). Table E.1 are the triples.

TABLE E.1 FEASIBLE TRIPLES FOR HIGHLY VARIABLE GRID

| Time (s) | Triple chosen  | Other feasible triples                              |
|----------|----------------|-----------------------------------------------------|
| 0        | (1, 11, 13725) | (1, 12, 10980), (1, 13, 8235), (2, 2, 0), (3, 1, 0) |
| 2745     | (1, 12, 10980) | (1, 13, 8235), (2, 2, 0), (2, 3, 0), (3, 1, 0)      |
| 5490     | (1, 12, 13725) | (2, 2, 2745), (2, 3, 0), (3, 1, 0)                  |
| 8235     | (1, 12, 16470) | (1, 13, 13725), (2, 2, 2745), (2, 3, 0), (3, 1, 0)  |
| 10980    | (1, 12, 16470) | (1, 13, 13725), (2, 2, 2745), (2, 3, 0), (3, 1, 0)  |
| 13725    | (1, 12, 16470) | (1, 13, 13725), (2, 2, 2745), (2, 3, 0), (3, 1, 0)  |
| 16470    | (1, 13, 16470) | (2, 2, 2745), (2, 3, 0), (3, 1, 0)                  |
| 19215    | (1, 12, 16470) | (1, 13, 13725), (2, 2, 2745), (2, 3, 0), (3, 1, 0)  |
| 21960    | (1, 12, 16470) | (1, 13, 13725), (2, 2, 2745), (2, 3, 0), (3, 1, 0)  |
| 24705    | (1, 12, 16470) | (1, 13, 13725), (2, 2, 2745), (2, 3, 0), (3, 1, 0)  |
| 27450    | (1, 12, 16470) | (1, 13, 13725), (2, 2, 2745), (2, 3, 0), (3, 1, 0)  |
| 30195    | (2, 2, 2745)   | (2, 3, 0), (3, 1, 0)                                |
| 32940    | (1, 13, 16470) | (2, 2, 2745), (2, 3, 0), (3, 1, 0)                  |
| 35685    | (1, 13, 13725) | (2, 2, 2745), (2, 3, 0), (3, 1, 0)                  |
| 38430    | (1, 13, 10980) | (2, 2, 2745), (2, 3, 0), (3, 1, 0)                  |
| 41175    | (1, 12, 13725) | (1, 13, 10980), (2, 2, 2745), (2, 3, 0), (3, 1, 0)  |
| 43920    | (1, 13, 10980) | (2, 2, 2745), (2, 3, 0), (3, 1, 0)                  |
| 46665    | (2, 2, 2745)   | (2, 3, 0), (3, 1, 0)                                |
| 49410    | (2, 2, 2745)   | (2, 3, 0), (3, 1, 0)                                |
| 52155    | (1, 12, 16470) | (1, 13, 13725), (2, 2, 2745), (2, 3, 0), (3, 1, 0)  |
| 54900    | (1, 13, 13725) | (2, 2, 2745), (2, 3, 0), (3, 1, 0)                  |
| 57645    | (1, 13, 13725) | (2, 2, 2745), (2, 3, 0), (3, 1, 0)                  |
| 60390    | (1, 12, 13725) | (2, 2, 2745), (2, 3, 0), (3, 1, 0)                  |
| 63135    | (1, 13, 16470) | (2, 2, 2745), (2, 3, 0), (3, 1, 0)                  |
| 65880    | (1, 13, 16470) | (2, 2, 2745), (2, 3, 0), (3, 1, 0)                  |
| 68625    | (2, 2, 2745)   | (2, 3, 0), (3, 1, 0)                                |
| 71370    | (1, 13, 13725) | (2, 2, 2745), (2, 3, 0), (3, 1, 0)                  |
| 74115    | (1, 12, 13725) | (2, 2, 2745), (2, 3, 0), (3, 1, 0)                  |
| 76860    | (1, 13, 13725) | (2, 2, 2745), (2, 3, 0), (3, 1, 0)                  |
| 79605    | (1, 13, 13725) | (2, 2, 2745), (2, 3, 0), (3, 1, 0)                  |
| 82350    | (1, 12, 13725) | (2, 2, 2745), (2, 3, 0), (3, 1, 0)                  |
| 85095    | (1, 12, 13725) | (1, 13, 10980), (2, 2, 2745), (2, 3, 0), (3, 1, 0)  |
| 87840    | (1, 13, 16470) | (2, 2, 2745), (2, 3, 0), (3, 1, 0)                  |
| 90585    | (1, 13, 16470) | (2, 2, 2745), (2, 3, 0), (3, 1, 0)                  |
| 93330    | (1, 13, 13725) | (2, 2, 2745), (2, 3, 0), (3, 1, 0)                  |
| 96075    | (1, 13, 16470) | (2, 2, 2745), (2, 3, 0), (3, 1, 0)                  |
| 98820    | (1, 13, 16470) | (2, 2, 2745), (2, 3, 0), (3, 1, 0)                  |
| 101565   | (1, 13, 13725) | (2, 2, 2745), (2, 3, 0), (3, 1, 0)                  |
| 104310   | (1, 13, 16470) | (2, 2, 2745), (2, 3, 0), (3, 1, 0)                  |
| 107055   | (1, 13, 13725) | (2, 2, 2745), (2, 3, 0), (3, 1, 0)                  |
| 109800   | (1, 13, 13725) | (2, 2, 2745), (2, 3, 0), (3, 1, 0)                  |
| 112545   | (1, 12, 16470) | (1, 13, 13725), (2, 2, 2745), (2, 3, 0), (3, 1, 0)  |
| 115290   | (1, 13, 16470) | (2, 2, 2745), (2, 3, 0), (3, 1, 0)                  |
| 118035   | (1, 13, 13725) | (2, 2, 2745), (2, 3, 0), (3, 1, 0)                  |
| 120780   | (1, 13, 16470) | (2, 2, 2745), (2, 3, 0), (3, 1, 0)                  |
| 123525   | (1, 13, 13725) | (2, 2, 2745), (2, 3, 0), (3, 1, 0)                  |

*Continued on next page*



Continued from previous page

| Time (s) | Triple chosen  | Other feasible triples                             |
|----------|----------------|----------------------------------------------------|
| 126270   | (1, 12, 16470) | (1, 13, 13725), (2, 2, 2745), (2, 3, 0), (3, 1, 0) |
| 129015   | (2, 2, 2745)   | (2, 3, 0), (3, 1, 0)                               |
| 131760   | (2, 2, 2745)   | (2, 3, 0), (3, 1, 0)                               |
| 134505   | (1, 13, 16470) | (2, 2, 2745), (2, 3, 0), (3, 1, 0)                 |
| 137250   | (1, 13, 13725) | (2, 2, 2745), (2, 3, 0), (3, 1, 0)                 |
| 139995   | (2, 2, 2745)   | (2, 3, 0), (3, 1, 0)                               |
| 142740   | (2, 2, 2745)   | (2, 3, 0), (3, 1, 0)                               |
| 145485   | (1, 12, 16470) | (1, 13, 13725), (2, 2, 2745), (2, 3, 0), (3, 1, 0) |
| 148230   | (2, 2, 2745)   | (2, 3, 0), (3, 1, 0)                               |
| 150975   | (1, 13, 16470) | (2, 2, 2745), (2, 3, 0), (3, 1, 0)                 |
| 153720   | (1, 12, 13725) | (2, 2, 2745), (2, 3, 0), (3, 1, 0)                 |
| 156465   | (1, 13, 13725) | (2, 2, 2745), (2, 3, 0), (3, 1, 0)                 |
| 159210   | (1, 13, 13725) | (2, 2, 2745), (2, 3, 0), (3, 1, 0)                 |
| 161955   | (1, 13, 16470) | (2, 2, 2745), (2, 3, 0), (3, 1, 0)                 |
| 164700   | (1, 13, 13725) | (2, 2, 2745), (2, 3, 0), (3, 1, 0)                 |



1720 List. E.8 shows the corresponding  $\text{\LaTeX}$  code.

Listing E.8: Sample  $\text{\LaTeX}$  code for making typical table environment

```

1721 1 \begin{center}
1722 2 {\scriptsize
1723 3 \begin{tabularx}{\textwidth}{p{0.1\textwidth}|p{0.2\textwidth}|p{0.5\textwidth}}
1724 4 \caption{Feasible triples for highly variable grid} \label{tab:triple_
1725 5 grid} \\
1726 6 \hline
1727 7 \textbf{Time (s)} &
1728 8 \textbf{Triple chosen} &
1729 9 \textbf{Other feasible triples} \\
1730 10 \hline
1731 11 \endfirsthead
1732 12 \multicolumn{3}{c}{\textit{Continued from previous page}} \\
1733 13 \hline
1734 14 \hline
1735 15 \textbf{Time (s)} &
1736 16 \textbf{Triple chosen} &
1737 17 \textbf{Other feasible triples} \\
1738 18 \hline
1739 19 \endhead
1740 20 \hline
1741 21 \multicolumn{3}{r}{\textit{Continued on next page}} \\
1742 22 \endfoot
1743 23 \hline
1744 24 \endlastfoot
1745 25 \hline
1746 26
1747 27
1748 28 0 & (1, 11, 13725) & (1, 12, 10980), (1, 13, 8235), (2, 2, 0), (3, 1, 0) \\
1749 29 & \\
1750 30 2745 & (1, 12, 10980) & (1, 13, 8235), (2, 2, 0), (2, 3, 0), (3, 1, 0) \\
1751 31 & \\
1752 32 5490 & (1, 12, 13725) & (2, 2, 2745), (2, 3, 0), (3, 1, 0) \\
1753 33 8235 & (1, 12, 16470) & (1, 13, 13725), (2, 2, 2745), (2, 3, 0), (3, 1, 0) \\
1754 34 10980 & (1, 12, 16470) & (1, 13, 13725), (2, 2, 2745), (2, 3, 0), (3, 1, 0) \\
1755 35 13725 & (1, 12, 16470) & (1, 13, 13725), (2, 2, 2745), (2, 3, 0), (3, 1, 0) \\
1756 36 16470 & (1, 13, 16470) & (2, 2, 2745), (2, 3, 0), (3, 1, 0) \\
1757 37 19215 & (1, 12, 16470) & (1, 13, 13725), (2, 2, 2745), (2, 3, 0), (3, 1, 0) \\
1758 38 21960 & (1, 12, 16470) & (1, 13, 13725), (2, 2, 2745), (2, 3, 0), (3, 1, 0) \\
1759 39 24705 & (1, 12, 16470) & (1, 13, 13725), (2, 2, 2745), (2, 3, 0), (3, 1, 0) \\
1760 40 27450 & (1, 12, 16470) & (1, 13, 13725), (2, 2, 2745), (2, 3, 0), (3, 1, 0) \\
1761 41 30195 & (2, 2, 2745) & (2, 3, 0), (3, 1, 0) \\
1762 42 32940 & (1, 13, 16470) & (2, 2, 2745), (2, 3, 0), (3, 1, 0) \\
1763 43 35685 & (1, 13, 13725) & (2, 2, 2745), (2, 3, 0), (3, 1, 0) \\
1764 44 38430 & (1, 13, 10980) & (2, 2, 2745), (2, 3, 0), (3, 1, 0)

```



```

1775 43 41175 & (1, 12, 13725) & (1, 13, 10980), (2, 2, 2745), (2, 3, 0), (3, 1,
1776      0) \\
1777 44 43920 & (1, 13, 10980) & (2, 2, 2745), (2, 3, 0), (3, 1, 0) \\
1778 45 46665 & (2, 2, 2745) & (2, 3, 0), (3, 1, 0) \\
1779 46 49410 & (2, 2, 2745) & (2, 3, 0), (3, 1, 0) \\
1780 47 52155 & (1, 12, 16470) & (1, 13, 13725), (2, 2, 2745), (2, 3, 0), (3, 1,
1781      0) \\
1782 48 54900 & (1, 13, 13725) & (2, 2, 2745), (2, 3, 0), (3, 1, 0) \\
1783 49 57645 & (1, 13, 13725) & (2, 2, 2745), (2, 3, 0), (3, 1, 0) \\
1784 50 60390 & (1, 12, 13725) & (2, 2, 2745), (2, 3, 0), (3, 1, 0) \\
1785 51 63135 & (1, 13, 16470) & (2, 2, 2745), (2, 3, 0), (3, 1, 0) \\
1786 52 65880 & (1, 13, 16470) & (2, 2, 2745), (2, 3, 0), (3, 1, 0) \\
1787 53 68625 & (2, 2, 2745) & (2, 3, 0), (3, 1, 0) \\
1788 54 71370 & (1, 13, 13725) & (2, 2, 2745), (2, 3, 0), (3, 1, 0) \\
1789 55 74115 & (1, 12, 13725) & (2, 2, 2745), (2, 3, 0), (3, 1, 0) \\
1790 56 76860 & (1, 13, 13725) & (2, 2, 2745), (2, 3, 0), (3, 1, 0) \\
1791 57 79605 & (1, 13, 13725) & (2, 2, 2745), (2, 3, 0), (3, 1, 0) \\
1792 58 82350 & (1, 12, 13725) & (2, 2, 2745), (2, 3, 0), (3, 1, 0) \\
1793 59 85095 & (1, 12, 13725) & (1, 13, 10980), (2, 2, 2745), (2, 3, 0), (3, 1,
1794      0) \\
1795 60 87840 & (1, 13, 16470) & (2, 2, 2745), (2, 3, 0), (3, 1, 0) \\
1796 61 90585 & (1, 13, 16470) & (2, 2, 2745), (2, 3, 0), (3, 1, 0) \\
1797 62 93330 & (1, 13, 13725) & (2, 2, 2745), (2, 3, 0), (3, 1, 0) \\
1798 63 96075 & (1, 13, 16470) & (2, 2, 2745), (2, 3, 0), (3, 1, 0) \\
1799 64 98820 & (1, 13, 16470) & (2, 2, 2745), (2, 3, 0), (3, 1, 0) \\
1800 65 101565 & (1, 13, 13725) & (2, 2, 2745), (2, 3, 0), (3, 1, 0) \\
1801 66 104310 & (1, 13, 16470) & (2, 2, 2745), (2, 3, 0), (3, 1, 0) \\
1802 67 107055 & (1, 13, 13725) & (2, 2, 2745), (2, 3, 0), (3, 1, 0) \\
1803 68 109800 & (1, 13, 13725) & (2, 2, 2745), (2, 3, 0), (3, 1, 0) \\
1804 69 112545 & (1, 12, 16470) & (1, 13, 13725), (2, 2, 2745), (2, 3, 0), (3,
1805      1, 0) \\
1806 70 115290 & (1, 13, 16470) & (2, 2, 2745), (2, 3, 0), (3, 1, 0) \\
1807 71 118035 & (1, 13, 13725) & (2, 2, 2745), (2, 3, 0), (3, 1, 0) \\
1808 72 120780 & (1, 13, 16470) & (2, 2, 2745), (2, 3, 0), (3, 1, 0) \\
1809 73 123525 & (1, 13, 13725) & (2, 2, 2745), (2, 3, 0), (3, 1, 0) \\
1810 74 126270 & (1, 12, 16470) & (1, 13, 13725), (2, 2, 2745), (2, 3, 0), (3,
1811      1, 0) \\
1812 75 129015 & (2, 2, 2745) & (2, 3, 0), (3, 1, 0) \\
1813 76 131760 & (2, 2, 2745) & (2, 3, 0), (3, 1, 0) \\
1814 77 134505 & (1, 13, 16470) & (2, 2, 2745), (2, 3, 0), (3, 1, 0) \\
1815 78 137250 & (1, 13, 13725) & (2, 2, 2745), (2, 3, 0), (3, 1, 0) \\
1816 79 139995 & (2, 2, 2745) & (2, 3, 0), (3, 1, 0) \\
1817 80 142740 & (2, 2, 2745) & (2, 3, 0), (3, 1, 0) \\
1818 81 145485 & (1, 12, 16470) & (1, 13, 13725), (2, 2, 2745), (2, 3, 0), (3,
1819      1, 0) \\
1820 82 148230 & (2, 2, 2745) & (2, 3, 0), (3, 1, 0) \\
1821 83 150975 & (1, 13, 16470) & (2, 2, 2745), (2, 3, 0), (3, 1, 0) \\
1822 84 153720 & (1, 12, 13725) & (2, 2, 2745), (2, 3, 0), (3, 1, 0) \\
1823 85 156465 & (1, 13, 13725) & (2, 2, 2745), (2, 3, 0), (3, 1, 0) \\
1824 86 159210 & (1, 13, 13725) & (2, 2, 2745), (2, 3, 0), (3, 1, 0) \\
1825 87 161955 & (1, 13, 16470) & (2, 2, 2745), (2, 3, 0), (3, 1, 0) \\
1826 88 164700 & (1, 13, 13725) & (2, 2, 2745), (2, 3, 0), (3, 1, 0) \\
1827 89 \end{tabularx}
1828 90 }
1830 91 \end{center}

```





## E7 Algorithm or Pseudocode Listing

Table E.2 shows an example pseudocode. Note that if the pseudocode exceeds one page, it can mean that its implementation is not modular. List. E.9 shows the corresponding  $\text{\LaTeX}$  code.

TABLE E.2 CALCULATION OF  $y = x^n$

|                   |                                      |
|-------------------|--------------------------------------|
| <b>Input(s):</b>  |                                      |
| $n$               | : $n$ th power; $n \in \mathbb{Z}^+$ |
| $x$               | : base value; $x \in \mathbb{R}^+$   |
| <b>Output(s):</b> |                                      |
| $y$               | : result; $y \in \mathbb{R}^+$       |

**Require:**  $n \geq 0 \vee x \neq 0$

**Ensure:**  $y = x^n$

```

1:  $y \leftarrow 1$ 
2: if  $n < 0$  then
3:    $X \leftarrow 1/x$ 
4:    $N \leftarrow -n$ 
5: else
6:    $X \leftarrow x$ 
7:    $N \leftarrow n$ 
8: end if
9: while  $N \neq 0$  do
10:  if  $N$  is even then
11:     $X \leftarrow X \times X$ 
12:     $N \leftarrow N/2$ 
13:  else  $\{N \text{ is odd}\}$ 
14:     $y \leftarrow y \times X$ 
15:     $N \leftarrow N - 1$ 
16:  end if
17: end while

```

Listing E.9: Sample L<sup>A</sup>T<sub>E</sub>X code for algorithm or pseudocode listing usage

```

1 \begin{table}[!htbp]
2   \caption{Calculation of  $y = x^n$ }
3   \label{tab:calcxn}
4   {\footnotesize
5     \begin{tabular}{lll}
6       \hline
7       \hline
8       {\bfseries Input(s):} & & \\
9       $n$ & : & $n$th power; $n$ \in \mathbb{Z}^{+}$ \\
10      $x$ & : & base value; $x$ \in \mathbb{R}^{+}$ \\
11      \hline
12      {\bfseries Output(s):} & & \\
13      $y$ & : & result; $y$ \in \mathbb{R}^{+}$ \\
14      \hline
15      \hline
16      \\
17    \end{tabular}
18  }
19  \begin{algorithmic}[1]
20    {\footnotesize
21      \REQUIRE $n \geq 0$ \vee $x \neq 0$
22      \ENSURE $y = x^n$
23      \STATE $y \leftarrow 1$
24      \IF{$n < 0$}
25        \STATE $X \leftarrow 1 / x$
26        \STATE $N \leftarrow -n$
27      \ELSE
28        \STATE $X \leftarrow x$
29        \STATE $N \leftarrow n$
30      \ENDIF
31      \WHILE{$N \neq 0$}
32        \IF{$N$ is even}
33          \STATE $X \leftarrow X \times X$
34          \STATE $N \leftarrow N / 2$
35        \ELSE[$N$ is odd]
36          \STATE $y \leftarrow y \times X$
37          \STATE $N \leftarrow N - 1$
38        \ENDIF
39      \ENDWHILE
40    }
41  \end{algorithmic}
42 \end{table}

```



## E8 Program/Code Listing

List. E.10 is a program listing of a C code for computing Fibonacci numbers by calling the actual code. Please see the `code` subdirectory.

Listing E.10: Computing Fibonacci numbers in C (./code/fibo.c)

```

1  /* fibo.c -- It prints out the first N Fibonacci
2  *              numbers.
3  */
4
5  #include <stdio.h>
6
7  int main(void) {
8      int n;          /* Number of fibonacci numbers we will print */
9      int i;          /* Index of fibonacci number to be printed next */
10     int current;     /* Value of the (i)th fibonacci number */
11     int next;        /* Value of the (i+1)th fibonacci number */
12     int twoaway;     /* Value of the (i+2)th fibonacci number */
13
14     printf("How many Fibonacci numbers do you want to compute? ");
15     scanf("%d", &n);
16     if (n<=0)
17         printf("The number should be positive.\n");
18     else {
19         printf("\n\n\tI\t\tFibonacci(I)\t\n\t===== \n");
20         next = current = 1;
21         for (i=1; i<=n; i++) {
22             printf("\t%d\t\t\t%d\n", i, current);
23             twoaway = current+next;
24             current = next;
25             next = twoaway;
26         }
27     }
28 }
29
30 /* The output from a run of this program was:
31
32 How many Fibonacci numbers do you want to compute? 9
33
34     I      Fibonacci(I)
35     =====
36     1      1
37     2      1
38     3      2
39     4      3
40     5      5
41     6      8
42     7     13
43     8     21
44     9     34
45
46 */

```



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List. E.11 shows the corresponding  $\text{\LaTeX}$  code.

Listing E.11: Sample  $\text{\LaTeX}$  code for program listing

```
1 List.\ref{lst:fib_c} is a program listing of a C code for computing  
   Fibonacci numbers by calling the actual code. Please see the \verb|  
   code | subdirectory.
```



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## E9 Referencing

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Referencing chapters: This appendix is in Appendix E, which is about examples in using various  $\LaTeX$  commands.

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Referencing sections: This section is Sec. E9, which shows how to refer to the locations of various labels that have been placed in the  $\LaTeX$  files. List. E.12 shows the corresponding  $\LaTeX$  code.

### Listing E.12: Sample $\LaTeX$ code for referencing sections

```
1 Referencing sections: This section is Sec.~\ref{sec:ref}, which shows
  how to refer to the locations of various labels that have been
  placed in the \LaTeX \ files. List.~\ref{lst:refsec} shows the
  corresponding \LaTeX \ code.
```

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Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.



## E9.1 A subsection

Referencing subsections: This section is Sec. E9.1, which shows how to refer to a subsection. List. E.13 shows the corresponding  $\LaTeX$  code.

### Listing E.13: Sample $\LaTeX$ code for referencing subsections

```
1 Referencing subsections: This section is Sec.~\ref{sec:subsec}, which
  shows how to refer to a subsection. List.~\ref{lst:refsub} shows the
  corresponding \LaTeX \ code.
```

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.



### E9.1.1 A sub-subsection

Referencing sub-subsections: This section is Sec. E9.1.1, which shows how to refer to a sub-subsection. List. E.14 shows the corresponding  $\LaTeX$  code.

#### Listing E.14: Sample $\LaTeX$ code for referencing sub-subsections

```
1 Referencing sub-subsections: This section is Sec.\ref{sec:subsubsec},
   which shows how to refer to a sub-subsection. List.\ref{lst:
   refsubsub} shows the corresponding \LaTeX \ code.
```

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.



## E10 Citing

Citing bibliography content is done using BibTeX. It requires the creation of a BibTeX file (.bib extension name), and then added in the argument of `\bibliography{ }`. For each .bib file, separate them by a comma in the argument of `\bibliography{ }` without the extension name. Building your BibTeX file (references.bib) can be done easily with a tool called JabRef ([www.jabref.org](http://www.jabref.org)).

The following subsections are examples of citations.

### E10.1 Books

- ['Chicago', 1982]
- [Aristotle, 1877]
- [Aristotle, 1907]
- [Aristotle, 1968]
- [Aristotle, 1929]
- [ABCM, 1959]
- [Augustine, 1995]
- [Averroes, 1982]
- [Butcher, 1981]
- [Chapman, 1975]
- [Cicero, 1995]
- [Coleridge, 1983]
- [Cotton et al., 1999]
- [van Gennep, 1909a]
- [van Gennep, 1909b]
- [van Gennep, 1960]
- [Gerhardt, 2000]
- [Gonzalez, 2001]





- 1904 • [Goossens et al., 1994]
- 1905 • [Hammond, 1997]
- 1906 • [HersHKovitz, 1962]
- 1907 • [Hoel, 1971a]
- 1908 • [Homer, 2004]
- 1909 • [Knuth, 1981a]
- 1910 • [Knuth, 1981b]
- 1911 • [Knuth, 1973a]
- 1912 • [Kullback, 1997a]
- 1913 • [Kullback, 1997b]
- 1914 • [Kullback, 1959]
- 1915 • [Malinowski, 1972]
- 1916 • [Maron, 2000]
- 1917 • [Massa, 2004]
- 1918 • [McColvin, 2004]
- 1919 • [Nietzsche, 1988b]
- 1920 • [Nietzsche, 1988a]
- 1921 • [Oetiker et al., 2014]
- 1922 • [Piccato, 2001]
- 1923 • [Smart, 1976]
- 1924 • [Vázquez de Parga et al., 1993]
- 1925 • [Wilde, 1899]
- 1926 • [Wood, 1961]
- 1927 • [Worman, 2002]
- 1928 • [Wright, 1978a]
- 1929 • [Lipcoll et al., 1977]

**E10.2 Booklets**

- [Knvth, 1988]

**E10.3 Proceedings**

- [Oz and Yannakakis, 1983]

**E10.4 In books**

- [von Brandt and Hoffmann, 1987]
- [BSI, 1973a]
- [Eckstein and Zuckermann, 1960]
- [Feigl, 1958]
- [Gordon, 1975]
- [Hanson, 1967]
- [Hoel, 1971b]
- [Hyman, 1981]
- [Kant, 1968a]
- [Kant, 1968b]
- [Knuth, 1973b]
- [Knuth, 1973c]
- [Lincoll, 1977a]
- [Lincoll, 2004]
- [Lincoll, 1977b]
- [McNeill, 1963]
- [Milton, 1924]
- [Nietzsche, 1988c]



- 1953 • [Ogilvy, 1965]
- 1954 • [Pines, 1979]
- 1955 • [Ramsbottom, 1931]
- 1956 • [Ranganthan, 1951]
- 1957 • [Thomson, 1971]
- 1958 • [Westfahl, 2004]
- 1959 • [Wright, 1963]
- 1960 • [Wright, 1978b]

### E10.5 In proceedings

- 1962 • [Chave, 1964]
- 1963 • [Chomsky, 1973]
- 1964 • [Moraux, 1979]
- 1965 • [Oaho et al., 1983a]
- 1966 • [Oaho et al., 2004]
- 1967 • [Oaho et al., 1983b]
- 1968 • [Salam, 1968]

### E10.6 Journals

- 1970 • [Aamport, 2004]
- 1971 • [Aamport, 1986a]
- 1972 • [Aamport, 1986b]
- 1973 • [Aksin et al., 2006]
- 1974 • [Angenendt, 2002]
- 1975 • [Aslin, 1949]



- |      |                                 |
|------|---------------------------------|
| 1976 | • [Baez and Lauda, 2004a]       |
| 1977 | • [Bertram and Wentworth, 1996] |
| 1978 | • [Bry and Afflerbach, 1968]    |
| 1979 | • [Doody, 1974]                 |
| 1980 | • [Einstein, 1905]              |
| 1981 | • [Fletcher and Hopkins, 1907]  |
| 1982 | • [Gillies, 1933]               |
| 1983 | • [Glashow, 1961]               |
| 1984 | • [Godfrey, 1959]               |
| 1985 | • [Hanlon, 1972]                |
| 1986 | • [Heller and Lederis, 1958]    |
| 1987 | • [Herrmann et al., 2006]       |
| 1988 | • [Hostetler et al., 1998]      |
| 1989 | • [Howells, 1966a]              |
| 1990 | • [Howells, 1966b]              |
| 1991 | • [Howells, 1951]               |
| 1992 | • [ISO, 2009]                   |
| 1993 | • [Jackson, 1979]               |
| 1994 | • [Johnson, 1974]               |
| 1995 | • [Moore, 1998]                 |
| 1996 | • [Moore, 1965]                 |
| 1997 | • [Prufer, 1964]                |
| 1998 | • [Reese, 1958]                 |
| 1999 | • [Sarfraz and Razzak, 2002]    |



- 2000 • [Shore, 1991]
- 2001 • [Sigfridsson and Ryde, 1998]
- 2002 • [Weinberg, 1967]
- 2003 • [Yoon et al., 2006]
- 2004 • [GAJ, 1986]

### E10.7 Theses/dissertations

- 2006 • [Croft, 1978]
- 2007 • [Maguire, 1976]
- 2008 • [Mann, 1968]
- 2009 • [Masterly, 1988a]
- 2010 • [Masterly, 1988b]
- 2011 • [Phony-Baloney, 1988a]
- 2012 • [Phony-Baloney, 1988b]

### E10.8 Technical Reports and Others

- 2014 • ['Brunswick', 1985]
- 2015 • [BSI, 1983]
- 2016 • [BSI, 1978]
- 2017 • [BSI, 1976]
- 2018 • [BSI, 1973b]
- 2019 • [Ellis and Walton, 1971]
- 2020 • [Terrific, 1988]
- 2021 • [Terrific, 1988]
- 2022 • [Winget Ltd., 1967]



2023 • [Ünderwood et al., 2004]

2024 • [Ünderwood et al., 1988]

2025 • [Downes, 1974]

2026 • [Exchequer, 1639]

2027 • [Pym, 1624]

2028 • [Traquair, 1638]

## 2029 **E10.9 Miscellaneous**

2030 • [Almendro et al., 1998]

2031 • [Baez and Lauda, 2004b]

2032 • [Chiu and Chow, 1978]

2033 • [Itzhaki, 1996]

2034 • [Kowalik and Isard, 1995]

2035 • [Laufenberg et al., 2006]

2036 • [Loh, 1992]

2037 • [Markey, 2005]

2038 • [Missilany, 1984]

2039 • [Padhye et al., 1999]

2040 • [Sorace et al., 1997]

2041 • [Wassenberg and Sanders, 2010]

2042 • [Missilany, 2004]



## E11 Index

For key words or topics that are expected (or the user would like) to appear in the Index, use `\index{key}`, where `key` is an example keyword to appear in the Index. For example, Fredholm integral and Fourier operator of the following paragraph are in the Index.

If we make a very large matrix with complex exponentials in the rows (i.e., cosine real parts and sine imaginary parts), and increase the resolution without bound, we approach the kernel of the Fredholm integral equation of the 2nd kind, namely the Fourier operator that defines the continuous Fourier transform.

List. E.15 is a program listing of the above-mentioned paragraph.

Listing E.15: Sample  $\text{\LaTeX}$  code for Index usage

```
1 If we make a very large matrix with complex exponentials in the rows (i.
  e., cosine real parts and sine imaginary parts), and increase the
  resolution without bound, we approach the kernel of the \index{
  Fredholm integral} Fredholm integral equation of the 2nd kind,
  namely the \index{Fourier} Fourier operator that defines the
  continuous Fourier transform.
```



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## E12 Adding Relevant PDF Pages

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Examples of such PDF pages are Standards, Datasheets, Specification Sheets, Application Notes, etc. Selected PDF pages can be added (see List. E.16), but note that the options must be tweaked. See the manual of `pdfpages` for other options.

Listing E.16: Sample  $\text{\LaTeX}$  code for including PDF pages

```
1 \includepdf[pages={8-10},%  
2 offset=3.5mm -10mm,%  
3 scale=0.73,%  
4 frame,%  
5 pagecommand={},]  
6 {./reference/Xilinx2015-UltraScale-Architecture-Overview.pdf}
```





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UltraScale Architecture and Product Overview

## Virtex UltraScale FPGA Feature Summary

Table 6: Virtex UltraScale FPGA Feature Summary

|                                  | VU065   | VU080   | VU095     | VU125     | VU160     | VU190     | VU440     |
|----------------------------------|---------|---------|-----------|-----------|-----------|-----------|-----------|
| Logic Cells                      | 626,640 | 780,000 | 940,800   | 1,253,280 | 1,621,200 | 1,879,920 | 4,432,680 |
| CLB Flip-Flops                   | 716,160 | 891,424 | 1,075,200 | 1,432,320 | 1,852,800 | 2,148,480 | 5,065,920 |
| CLB LUTs                         | 358,080 | 445,712 | 537,600   | 716,160   | 926,400   | 1,074,240 | 2,532,960 |
| Maximum Distributed RAM (Mb)     | 4.8     | 3.9     | 4.8       | 9.7       | 12.7      | 14.5      | 28.7      |
| Block RAM/FIFO w/ECC (36Kb each) | 1,260   | 1,421   | 1,728     | 2,520     | 3,276     | 3,780     | 2,520     |
| Total Block RAM (Mb)             | 44.3    | 50.0    | 60.8      | 88.6      | 115.2     | 132.9     | 88.6      |
| CMT (1 MMCM, 2 PLLs)             | 10      | 16      | 16        | 20        | 30        | 30        | 30        |
| I/O DLLs                         | 40      | 64      | 64        | 80        | 120       | 120       | 120       |
| Fractional PLLs                  | 5       | 8       | 8         | 10        | 15        | 15        | 0         |
| Maximum HP I/Os <sup>(1)</sup>   | 468     | 780     | 780       | 780       | 650       | 650       | 1,404     |
| Maximum HR I/Os <sup>(2)</sup>   | 52      | 52      | 52        | 104       | 52        | 52        | 52        |
| DSP Slices                       | 600     | 672     | 768       | 1,200     | 1,560     | 1,800     | 2,880     |
| System Monitor                   | 1       | 1       | 1         | 2         | 3         | 3         | 3         |
| PCIe Gen3 x8                     | 2       | 4       | 4         | 4         | 5         | 6         | 6         |
| 150G Interlaken                  | 3       | 6       | 6         | 6         | 8         | 9         | 0         |
| 100G Ethernet                    | 3       | 4       | 4         | 6         | 9         | 9         | 3         |
| GTH 16.3Gb/s Transceivers        | 20      | 32      | 32        | 40        | 52        | 60        | 48        |
| GTY 30.5Gb/s Transceivers        | 20      | 32      | 32        | 40        | 52        | 60        | 0         |

**Notes:**

1. HP = High-performance I/O with support for I/O voltage from 1.0V to 1.8V.
2. HR = High-range I/O with support for I/O voltage from 1.2V to 3.3V.



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## UltraScale Architecture and Product Overview

## Virtex UltraScale Device-Package Combinations and Maximum I/Os

Table 7: Virtex UltraScale Device-Package Combinations and Maximum I/Os

| Package <sup>(1)(2)(3)</sup> | Package Dimensions (mm) | VU065              | VU080              | VU095              | VU125              | VU160              | VU190              | VU440              |
|------------------------------|-------------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|
|                              |                         | HR, HP<br>GTH, GTY | HR, HP<br>GTH, GTY | HR, HP<br>GTH, GTY | HR, HP<br>GTH, GTY | HR, HP<br>GTH, GTY | HR, HP<br>GTH, GTY | HR, HP<br>GTH, GTY |
| FFVC1517                     | 40x40                   | 52, 468<br>20, 20  | 52, 468<br>20, 20  | 52, 468<br>20, 20  |                    |                    |                    |                    |
| FFVD1517                     | 40x40                   |                    | 52, 286<br>32, 32  | 52, 286<br>32, 32  |                    |                    |                    |                    |
| FLVD1517                     | 40x40                   |                    |                    |                    | 52, 286<br>40, 32  |                    |                    |                    |
| FFVB1760                     | 42.5x42.5               |                    | 52, 650<br>32, 16  | 52, 650<br>32, 16  |                    |                    |                    |                    |
| FLVB1760                     | 42.5x42.5               |                    |                    |                    | 52, 650<br>36, 16  |                    |                    |                    |
| FFVA2104                     | 47.5x47.5               |                    | 52, 780<br>28, 24  | 52, 780<br>28, 24  |                    |                    |                    |                    |
| FLVA2104                     | 47.5x47.5               |                    |                    |                    | 52, 780<br>28, 24  |                    |                    |                    |
| FFVB2104                     | 47.5x47.5               |                    | 52, 650<br>32, 32  | 52, 650<br>32, 32  |                    |                    |                    |                    |
| FLVB2104                     | 47.5x47.5               |                    |                    |                    | 52, 650<br>40, 36  |                    |                    |                    |
| FLGB2104                     | 47.5x47.5               |                    |                    |                    |                    | 52, 650<br>40, 36  | 52, 650<br>40, 36  |                    |
| FFVC2104                     | 47.5x47.5               |                    |                    | 52, 364<br>32, 32  |                    |                    |                    |                    |
| FLVC2104                     | 47.5x47.5               |                    |                    |                    | 52, 364<br>40, 40  |                    |                    |                    |
| FLGC2104                     | 47.5x47.5               |                    |                    |                    |                    | 52, 364<br>52, 52  | 52, 364<br>52, 52  |                    |
| FLGB2377                     | 50x50                   |                    |                    |                    |                    |                    |                    | 52, 1248<br>36, 0  |
| FLGA2577                     | 52.5x52.5               |                    |                    |                    |                    |                    | 0, 448<br>60, 60   |                    |
| FLGA2892                     | 55x55                   |                    |                    |                    |                    |                    |                    | 52, 1404<br>48, 0  |

## Notes:

1. Go to [Ordering Information](#) for package designation details.
2. All packages have 1.0mm ball pitch.
3. Packages with the same last letter and number sequence, e.g., A2104, are footprint compatible with all other UltraScale architecture-based devices with the same sequence. The footprint compatible devices within this family are outlined. See the [UltraScale Architecture Product Selection Guide](#) for details on inter-family migration.



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UltraScale Architecture and Product Overview

## Virtex UltraScale+ FPGA Feature Summary

Table 8: Virtex UltraScale+ FPGA Feature Summary

|                                  | VU3P    | VU5P      | VU7P      | VU9P      | VU11P     | VU13P     |
|----------------------------------|---------|-----------|-----------|-----------|-----------|-----------|
| Logic Cells                      | 689,640 | 1,051,010 | 1,379,280 | 2,068,920 | 2,147,040 | 2,862,720 |
| CLB Flip-Flops                   | 788,160 | 1,201,154 | 1,576,320 | 2,364,480 | 2,453,760 | 3,271,680 |
| CLB LUTs                         | 394,080 | 600,577   | 788,160   | 1,182,240 | 1,226,880 | 1,635,840 |
| Max. Distributed RAM (Mb)        | 12.0    | 18.3      | 24.1      | 36.1      | 34.8      | 46.4      |
| Block RAM/FIFO w/ECC (36Kb each) | 720     | 1,024     | 1,440     | 2,160     | 2,016     | 2,688     |
| Block RAM (Mb)                   | 25.3    | 36.0      | 50.6      | 75.9      | 70.9      | 94.5      |
| UltraRAM Blocks                  | 320     | 470       | 640       | 960       | 1,152     | 1,536     |
| UltraRAM (Mb)                    | 90.0    | 132.2     | 180.0     | 270.0     | 324.0     | 432.0     |
| CMTs (1 MMCM and 2 PLLs)         | 10      | 20        | 20        | 30        | 12        | 16        |
| Max. HP I/O <sup>(1)</sup>       | 520     | 832       | 832       | 832       | 624       | 832       |
| DSP Slices                       | 2,280   | 3,474     | 4,560     | 6,840     | 8,928     | 11,904    |
| System Monitor                   | 1       | 2         | 2         | 3         | 3         | 4         |
| GTY Transceivers 32.75Gb/s       | 40      | 80        | 80        | 120       | 96        | 128       |
| PCIe Gen3 x16 and Gen4 x8        | 2       | 4         | 4         | 6         | 3         | 4         |
| 150G Interlaken                  | 3       | 4         | 6         | 9         | 9         | 12        |
| 100G Ethernet w/RS-FEC           | 3       | 4         | 6         | 9         | 6         | 8         |

**Notes:**

1. HP = High-performance I/O with support for I/O voltage from 1.0V to 1.8V.

## Virtex UltraScale+ Device-Package Combinations and Maximum I/Os

Table 9: Virtex UltraScale+ Device-Package Combinations and Maximum I/Os

| Package<br><sup>(1)(2)(3)</sup> | Package<br>Dimensions<br>(mm) | VU3P    | VU5P    | VU7P    | VU9P     | VU11P   | VU13P    |
|---------------------------------|-------------------------------|---------|---------|---------|----------|---------|----------|
|                                 |                               | HP, GTY | HP, GTY | HP, GTY | HP, GTY  | HP, GTY | HP, GTY  |
| FFVC1517                        | 40x40                         | 520, 40 |         |         |          |         |          |
| FLVF1924                        | 45x45                         |         |         |         |          | 624, 64 |          |
| FLVA2104                        | 47.5x47.5                     |         | 832, 52 | 832, 52 | 832, 52  |         |          |
| FHVA2104                        | 52.5x52.5 <sup>(4)</sup>      |         |         |         |          |         | 832, 52  |
| FLVB2104                        | 47.5x47.5                     |         | 702, 76 | 702, 76 | 702, 76  | 624, 76 |          |
| FHVB2104                        | 52.5x52.5 <sup>(4)</sup>      |         |         |         |          |         | 702, 76  |
| FLVC2104                        | 47.5x47.5                     |         | 416, 80 | 416, 80 | 416, 104 | 416, 96 |          |
| FHVC2104                        | 52.5x52.5 <sup>(4)</sup>      |         |         |         |          |         | 416, 104 |
| FLVA2577                        | 52.5x52.5                     |         |         |         | 448, 120 | 448, 96 | 448, 128 |

**Notes:**

- Go to [Ordering Information](#) for package designation details.
- All packages have 1.0mm ball pitch.
- Packages with the same last letter and number sequence, e.g., A2104, are footprint compatible with all other UltraScale devices with the same sequence. The footprint compatible devices within this family are outlined.
- These 52.5x52.5mm overhang packages have the same PCB ball footprint as the corresponding 47.5x47.5mm packages (i.e., the same last letter and number sequence) and are footprint compatible.



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## Appendix F

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## SOME LIST OF MATH SYMBOLS



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List of mathematical symbols

From Wikipedia, the free encyclopedia

This is a list of symbols found within all branches of mathematics to express a formula or to represent a constant.

When reading the list, it is important to recognize that a mathematical concept is independent of the symbol chosen to represent it. For many of the symbols below, the symbol is usually synonymous with the corresponding concept (ultimately an arbitrary choice made as a result of the cumulative history of mathematics), but in some situations a different convention may be used. For example, depending on context, the triple bar "≡" may represent congruence or a definition. Further, in mathematical logic, numerical equality is sometimes represented by "≐" instead of "=", with the latter representing equality of well-formed formulas. In short, convention dictates the meaning.

Each symbol is shown both in HTML, whose display depends on the browser's access to an appropriate font installed on the particular device, and typeset as an image using TeX.

Contents

- 1 Guide
- 2 Basic symbols
- 3 Symbols based on equality
- 4 Symbols that point left or right
- 5 Brackets
- 6 Other non-letter symbols
- 7 Letter-based symbols
  - 7.1 Letter modifiers
  - 7.2 Symbols based on Latin letters
  - 7.3 Symbols based on Hebrew or Greek letters
- 8 Variations
- 9 See also
- 10 References
- 11 External links

Guide

This list is organized by symbol type and is intended to facilitate finding an unfamiliar symbol by its visual appearance. For a related list organized by mathematical topic, see List of mathematical symbols by subject. That list also includes LaTeX and HTML markup and Unicode code points for each symbol.

- **Basic symbols:** Symbols widely used in mathematics, roughly through first-year calculus. More advanced meanings are included with some symbols listed here.
- **Symbols based on equality "≡":** Symbols derived from or similar to the equal sign, including double-headed arrows. Not surprisingly these symbols are often associated with an equivalence relation.
- **Symbols that point left or right:** Symbols, such as < and >, that appear to point to one side or another.
- **Brackets:** Symbols that are placed on either side of a variable or expression, such as [x].
- **Other non-letter symbols:** Symbols that do not fall in any of the other categories.
- **Letter-based symbols:** Many mathematical symbols are based on, or closely resemble, a letter in some alphabet. This section includes such symbols, including symbols that resemble upside-down letters. Many letters have conventional meanings in various branches of mathematics and physics. These are not listed here. The See also section, below, has several lists of such usages.
  - **Letter modifiers:** Symbols that can be placed on or next to any letter to modify the letter's meaning.
  - **Symbols based on Latin letters,** including those symbols that resemble or contain an X
  - **Symbols based on Hebrew or Greek letters** e.g. ℵ, 2, ̂, Δ, π, II, σ, Z, Φ. *Note:* symbols resembling Λ are grouped with "V" under Latin letters.
- **Variations:** Usage in languages written right-to-left

Basic symbols

## F. Some List of Math Symbols



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| Symbol in HTML | Symbol in TeX | Name<br>Read as<br>Category            | Explanation                                                                                                                                  | Examples                                                                                                                                        |
|----------------|---------------|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| +              | +             | addition                               |                                                                                                                                              |                                                                                                                                                 |
|                |               | plus;<br>add                           | $4 + 6$ means the sum of 4 and 6.                                                                                                            | $2 + 7 = 9$                                                                                                                                     |
|                |               | arithmetic                             |                                                                                                                                              |                                                                                                                                                 |
|                |               | disjoint union                         |                                                                                                                                              |                                                                                                                                                 |
|                |               | the disjoint union of ... and ...      | $A_1 + A_2$ means the disjoint union of sets $A_1$ and $A_2$ .                                                                               | $A_1 = \{3, 4, 5, 6\} \wedge A_2 = \{7, 8, 9, 10\} \Rightarrow A_1 + A_2 = \{(3, 1), (4, 1), (5, 1), (6, 1), (7, 2), (8, 2), (9, 2), (10, 2)\}$ |
| −              | −             | subtraction                            |                                                                                                                                              |                                                                                                                                                 |
|                |               | minus;<br>take;<br>subtract            | $9 - 4$ means the subtraction of 4 from 9.                                                                                                   | $8 - 3 = 5$                                                                                                                                     |
|                |               | arithmetic                             |                                                                                                                                              |                                                                                                                                                 |
|                |               | negative sign                          |                                                                                                                                              |                                                                                                                                                 |
|                |               | negative;<br>minus;<br>the opposite of | $-3$ means the additive inverse of the number 3.                                                                                             | $-(-5) = 5$                                                                                                                                     |
| ±              | ±             | arithmetic                             |                                                                                                                                              |                                                                                                                                                 |
|                |               | set-theoretic complement               | $A - B$ means the set that contains all the elements of $A$ that are not in $B$ .                                                            | $\{1, 2, 4\} - \{1, 3, 4\} = \{2\}$                                                                                                             |
|                |               | minus;<br>without                      | ( $\setminus$ can also be used for set-theoretic complement as described below.)                                                             |                                                                                                                                                 |
|                |               | set theory                             |                                                                                                                                              |                                                                                                                                                 |
|                |               | plus-minus                             |                                                                                                                                              |                                                                                                                                                 |
| ±              | ±             | plus or minus                          | $6 \pm 3$ means both $6 + 3$ and $6 - 3$ .                                                                                                   | The equation $x = 5 \pm \sqrt{4}$ , has two solutions, $x = 7$ and $x = 3$ .                                                                    |
|                |               | arithmetic                             |                                                                                                                                              | Note: $\{\{\text{sgn}(\pm)\}\}$ was used to get $\sqrt{4}$ .                                                                                    |
|                |               | plus-minus                             |                                                                                                                                              |                                                                                                                                                 |
|                |               | plus or minus measurement              | $10 \pm 2$ or equivalently $10 \pm 20\%$ means the range from $10 - 2$ to $10 + 2$ .                                                         | If $a = 100 \pm 1$ mm, then $a \geq 99$ mm and $a \leq 101$ mm.                                                                                 |
|                |               | minus-plus                             |                                                                                                                                              |                                                                                                                                                 |
| ∓              | ∓             | minus or plus                          | $6 \pm (3 \mp 5)$ means $6 + (3 - 5)$ and $6 - (3 + 5)$ .                                                                                    | $\cos(x \pm y) = \cos(x) \cos(y) \mp \sin(x) \sin(y)$ .                                                                                         |
|                |               | arithmetic                             |                                                                                                                                              |                                                                                                                                                 |
|                |               | multiplication                         |                                                                                                                                              |                                                                                                                                                 |
|                |               | times;<br>multiplied by                | $3 \times 4$ or $3 \cdot 4$ means the multiplication of 3 by 4.                                                                              | $7 \cdot 8 = 56$                                                                                                                                |
|                |               | arithmetic                             |                                                                                                                                              |                                                                                                                                                 |
| ×              | ×             | dot product                            |                                                                                                                                              |                                                                                                                                                 |
|                |               | scalar product                         |                                                                                                                                              |                                                                                                                                                 |
|                |               | dot                                    | $\mathbf{u} \cdot \mathbf{v}$ means the dot product of vectors $\mathbf{u}$ and $\mathbf{v}$                                                 | $(1, 2, 5) \cdot (3, 4, -1) = 6$                                                                                                                |
|                |               | linear algebra                         |                                                                                                                                              |                                                                                                                                                 |
|                |               | vector algebra                         |                                                                                                                                              |                                                                                                                                                 |
| ·              | ·             | cross product                          |                                                                                                                                              |                                                                                                                                                 |
|                |               | vector product                         |                                                                                                                                              |                                                                                                                                                 |
|                |               | cross                                  | $\mathbf{u} \times \mathbf{v}$ means the cross product of vectors $\mathbf{u}$ and $\mathbf{v}$                                              | $(1, 2, 5) \times (3, 4, -1) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 2 & 5 \\ 3 & 4 & -1 \end{vmatrix} = (-22, 16, -2)$   |
|                |               | linear algebra                         |                                                                                                                                              |                                                                                                                                                 |
|                |               | vector algebra                         |                                                                                                                                              |                                                                                                                                                 |
| ·              | ·             | placeholder                            |                                                                                                                                              |                                                                                                                                                 |
|                |               | (silent)                               | $A \cdot$ means a placeholder for an argument of a function.                                                                                 | $ \cdot $                                                                                                                                       |
|                |               | functional analysis                    | Indicates the functional nature of an expression without assigning a specific symbol for an argument.                                        |                                                                                                                                                 |
|                |               |                                        |                                                                                                                                              |                                                                                                                                                 |
|                |               |                                        |                                                                                                                                              |                                                                                                                                                 |
| ÷              | ÷             | division (Obelus)                      |                                                                                                                                              |                                                                                                                                                 |
|                |               | divided by;<br>over                    | $6 \div 3$ or $6/3$ means the division of 6 by 3.                                                                                            | $2 \div 4 = 0.5$                                                                                                                                |
|                |               | arithmetic                             |                                                                                                                                              | $12/4 = 3$                                                                                                                                      |
|                |               | quotient group                         |                                                                                                                                              |                                                                                                                                                 |
|                |               | mod                                    | $G/H$ means the quotient of group $G$ modulo its subgroup $H$ .                                                                              | $\{0, a, 2a, b, b+a, b+2a\} / \{0, b\} = \{\{0, b\}, \{a, b+a\}, \{2a, b+2a\}\}$                                                                |
| /              | /             | group theory                           |                                                                                                                                              |                                                                                                                                                 |
|                |               | quotient set                           |                                                                                                                                              |                                                                                                                                                 |
|                |               | mod                                    | $A/\sim$ means the set of all $\sim$ equivalence classes in $A$ .                                                                            | If we define $\sim$ by $x \sim y \Leftrightarrow x - y \in \mathbb{Z}$ , then $\mathbb{R}/\sim = \{x + n : n \in \mathbb{Z}, x \in [0, 1)\}$ .  |
|                |               | set theory                             |                                                                                                                                              |                                                                                                                                                 |
|                |               |                                        |                                                                                                                                              |                                                                                                                                                 |
| √              | √             | square root                            |                                                                                                                                              |                                                                                                                                                 |
|                |               | the (principal) square root of         | $\sqrt{x}$ means the nonnegative number whose square is $x$ .                                                                                | $\sqrt{4} = 2$                                                                                                                                  |
|                |               | real numbers                           |                                                                                                                                              |                                                                                                                                                 |
|                |               | complex square root                    |                                                                                                                                              |                                                                                                                                                 |
|                |               | the (complex) square root of           | If $z = r \exp(i\varphi)$ is represented in polar coordinates with $-\pi < \varphi \leq \pi$ , then $\sqrt{z} = \sqrt{r} \exp(i\varphi/2)$ . | $\sqrt{-1} = i$                                                                                                                                 |
| Σ              | Σ             | complex numbers                        |                                                                                                                                              |                                                                                                                                                 |
|                |               | summation                              |                                                                                                                                              |                                                                                                                                                 |
|                |               | sum over ... from ... to ... of        | $\sum_{k=1}^n a_k$ means $a_1 + a_2 + \dots + a_n$ .                                                                                         | $\sum_{k=1}^4 k^2 = 1^2 + 2^2 + 3^2 + 4^2 = 1 + 4 + 9 + 16 = 30$                                                                                |
|                |               | arithmetic                             |                                                                                                                                              |                                                                                                                                                 |
|                |               |                                        |                                                                                                                                              |                                                                                                                                                 |
| ∫              | ∫             | indefinite integral or antiderivative  |                                                                                                                                              |                                                                                                                                                 |
|                |               | indefinite integral of                 | $\int f(x) dx$ means a function whose derivative is $f$ .                                                                                    | $\int x^2 dx = \frac{x^3}{3} + C$                                                                                                               |
|                |               |                                        |                                                                                                                                              |                                                                                                                                                 |
|                |               |                                        |                                                                                                                                              |                                                                                                                                                 |
|                |               |                                        |                                                                                                                                              |                                                                                                                                                 |

## F. Some List of Math Symbols



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|                                                                                     |                                                      |                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------------------------------------------------------------------------------------|------------------------------------------------------|---------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                     |                                                      | - OR -<br>the<br>antiderivative of<br>calculus                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                                                                     |                                                      | definite integral                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                                                                     |                                                      | integral from ...<br>to ... of ... with<br>respect to<br>calculus                     | $\int_a^b f(x) dx$ means the signed area between the $x$ -axis and the graph of the function $f$ between $x = a$ and $x = b$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | $\int_a^b x^2 dx = \frac{b^3 - a^3}{3}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                                                                     |                                                      | line integral<br>line/ path/ curve/<br>integral of ...<br>along ...<br>calculus       | $\int_C f ds$ means the integral of $f$ along the curve $C$ ,<br>$\int_a^b f(\mathbf{r}(t))  \mathbf{r}'(t)  dt$ , where $\mathbf{r}$ is a parametrization of $C$ . (If the curve is closed, the symbol $\oint$ may be used instead, as described below.)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| $\oint$                                                                             | $\oint$                                              | Contour<br>integral;<br>closed line<br>integral<br>contour integral<br>of<br>calculus | Similar to the integral, but used to denote a single integration over a closed curve or loop. It is sometimes used in physics texts involving equations regarding Gauss's Law, and while these formulas involve a closed surface integral, the representations describe only the first integration of the volume over the enclosing surface. Instances where the latter requires simultaneous double integration, the symbol $\oint$ would be more appropriate. A third related symbol is the closed volume integral, denoted by the symbol $\oint_V$ .<br>The contour integral can also frequently be found with a subscript capital letter $C$ , $\oint_C$ , denoting that a closed loop integral is, in fact, around a contour $C$ , or sometimes dually appropriately, a circle $C$ . In representations of Gauss's Law, a subscript capital $S$ , $\oint_S$ , is used to denote that the integration is over a closed surface. | If $C$ is a Jordan curve about 0, then $\oint_C \frac{1}{z} dz = 2\pi i$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| $\therefore$                                                                        | $\therefore$                                         | therefore<br>because;<br>so;<br>hence<br>everywhere                                   | Sometimes used in proofs before logical consequences.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | All humans are mortal. Socrates is a human. $\therefore$ Socrates is mortal.                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| $\because$                                                                          | $\because$                                           | because<br>because;<br>since<br>everywhere                                            | Sometimes used in proofs before reasoning.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 11 is prime $\because$ it has no positive integer factors other than itself and one.                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| $!$                                                                                 | $!$                                                  | factorial<br>factorial<br>combinatorics                                               | $n!$ means the product $1 \times 2 \times \dots \times n$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b><math>4! = 1 \times 2 \times 3 \times 4 = 24</math></b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                                                                                     |                                                      | logical negation<br>not<br>propositional<br>logic                                     | The statement $\neg A$ is true if and only if $A$ is false.<br>A slash placed through another operator is the same as " $\neg$ " placed in front.<br>(The symbol $\neg$ is primarily from computer science. It is avoided in mathematical texts, where the notation $\neg A$ is preferred.)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | $\neg(\neg A) \Leftrightarrow A$<br>$x \neq y \Leftrightarrow \neg(x = y)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| $\neg$                                                                              | $\neg$                                               | logical negation<br>not<br>propositional<br>logic                                     | The statement $\neg A$ is true if and only if $A$ is false.<br>A slash placed through another operator is the same as " $\neg$ " placed in front.<br>(The symbol $\sim$ has many other uses, so $\neg$ or the slash notation is preferred. Computer scientists will often use $\neg$ but this is avoided in mathematical texts.)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | $\neg(\neg A) \Leftrightarrow A$<br>$x \neq y \Leftrightarrow \neg(x = y)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| $\propto$                                                                           | $\propto$                                            | proportionality<br>is proportional<br>to;<br>varies as<br>everywhere                  | $y \propto x$ means that $y = kx$ for some constant $k$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | if $y = 2x$ , then $y \propto x$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| $\infty$                                                                            | $\infty$                                             | infinity<br>infinity<br>numbers                                                       | $\infty$ is an element of the extended number line that is greater than all real numbers; it often occurs in limits.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | $\lim_{x \rightarrow \infty} \frac{1}{ x } = 0$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| $\blacksquare$<br>$\square$<br>$\blacksquare$<br>$\square$<br>$\blacktriangleright$ | $\blacksquare$<br>$\square$<br>$\blacktriangleright$ | end of proof<br>QED;<br>tombstone;<br>Halmos finality<br>symbol<br>everywhere         | Used to mark the end of a proof.<br>(May also be written Q.E.D.)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | (1) $a + 0 := a$ (def.)<br>(2) $a + \text{succ}(b) := \text{succ}(a + b)$ (def.)<br>Proposition. $3 + 2 = 5$ .<br>Proof.<br>$3 + 2 = 3 + \text{succ}(1)$ (definition of succ)<br>$3 + \text{succ}(1) = \text{succ}(3 + 1)$ (2)<br>$\text{succ}(3 + 1) = \text{succ}(3 + \text{succ}(0))$ (definition of succ)<br>$\text{succ}(3 + \text{succ}(0)) = \text{succ}(\text{succ}(3 + 0))$ (2)<br>$\text{succ}(\text{succ}(3 + 0)) = \text{succ}(\text{succ}(3))$ (1)<br>$\text{succ}(\text{succ}(3)) = \text{succ}(4) = 5$ (definition of succ) $\blacksquare$ |

Symbols based on equality

## F. Some List of Math Symbols



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| Symbol in HTML                    | Symbol in TeX                                                                                     | Name                                                     | Explanation                                                                                                                                   | Examples                                                                                         |
|-----------------------------------|---------------------------------------------------------------------------------------------------|----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
|                                   |                                                                                                   | Read as                                                  |                                                                                                                                               |                                                                                                  |
|                                   |                                                                                                   | Category                                                 |                                                                                                                                               |                                                                                                  |
| =                                 | =                                                                                                 | equality                                                 |                                                                                                                                               |                                                                                                  |
|                                   |                                                                                                   | is equal to; equals                                      | $\mathbf{x = y}$ means $\mathbf{x}$ and $\mathbf{y}$ represent the same math object (Both symbols have the same value).                       | $\mathbf{2 = 2}$<br>$\mathbf{1 + 1 = 2}$                                                         |
|                                   |                                                                                                   | everywhere                                               |                                                                                                                                               |                                                                                                  |
| ≠                                 | ≠                                                                                                 | inequality                                               | $\mathbf{x \neq y}$ means that $\mathbf{x}$ and $\mathbf{y}$ do not represent the same math object (Both symbols do not have the same value). |                                                                                                  |
|                                   |                                                                                                   | is not equal to; does not equal                          | <i>(The forms !=, /= or &lt;&gt; are generally used in programming languages where ease of typing and use of ASCII text is preferred.)</i>    | $\mathbf{2 + 2 \neq 5}$                                                                          |
|                                   |                                                                                                   | everywhere                                               |                                                                                                                                               |                                                                                                  |
| ≈                                 | ≈                                                                                                 | approximately equal                                      | $x \approx y$ means $x$ is approximately equal to $y$ .                                                                                       |                                                                                                  |
|                                   |                                                                                                   | is approximately equal to                                | <i>This may also be written ≈, ≐, ∼, ≒ (Libra Symbol), or ≐.</i>                                                                              | $\pi \approx 3.14159$                                                                            |
|                                   |                                                                                                   | everywhere                                               |                                                                                                                                               |                                                                                                  |
|                                   |                                                                                                   | isomorphism                                              | $G \cong H$ means that group $G$ is isomorphic (structurally identical) to group $H$ .                                                        |                                                                                                  |
|                                   |                                                                                                   | is isomorphic to                                         |                                                                                                                                               | $\mathbb{Q}_8 / \mathbb{C}_2 \approx \mathbb{V}$                                                 |
|                                   |                                                                                                   | group theory                                             | (≐ can also be used for isomorphic, as described below.)                                                                                      |                                                                                                  |
| ∼                                 | ∼                                                                                                 | probability distribution                                 |                                                                                                                                               |                                                                                                  |
|                                   |                                                                                                   | has distribution                                         | $X \sim D$ , means the random variable $X$ has the probability distribution $D$ .                                                             | $X \sim \mathcal{N}(0,1)$ , the standard normal distribution                                     |
|                                   |                                                                                                   | statistics                                               |                                                                                                                                               |                                                                                                  |
|                                   |                                                                                                   | row equivalence                                          |                                                                                                                                               |                                                                                                  |
|                                   |                                                                                                   | is row equivalent to                                     | $A \sim B$ means that $B$ can be generated by using a series of elementary row operations on $A$                                              | $\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}$ |
|                                   |                                                                                                   | matrix theory                                            |                                                                                                                                               |                                                                                                  |
|                                   |                                                                                                   | same order of magnitude                                  | $m \sim n$ means the quantities $m$ and $n$ have the same order of magnitude, or general size.                                                | $2 \sim 5$                                                                                       |
|                                   |                                                                                                   | roughly similar; poorly approximates; is on the order of | <i>(Note that <math>\sim</math> is used for an approximation that is poor, otherwise use <math>\approx</math>.)</i>                           | $8 \times 9 \sim 100$                                                                            |
|                                   |                                                                                                   | approximation theory                                     |                                                                                                                                               | but $\pi^2 \not\sim 10$                                                                          |
|                                   |                                                                                                   | similarity                                               |                                                                                                                                               |                                                                                                  |
|                                   |                                                                                                   | is similar to <sup>[1]</sup>                             | $\triangle ABC \sim \triangle DEF$ means triangle $ABC$ is similar to (has the same shape) triangle $DEF$ .                                   |                                                                                                  |
|                                   |                                                                                                   | geometry                                                 |                                                                                                                                               |                                                                                                  |
|                                   |                                                                                                   | asymptotically equivalent                                |                                                                                                                                               |                                                                                                  |
| is asymptotically equivalent to   | $f \sim g$ means $\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = 1$ .                            | $x \sim x+1$                                             |                                                                                                                                               |                                                                                                  |
| asymptotic analysis               |                                                                                                   |                                                          |                                                                                                                                               |                                                                                                  |
| equivalence relation              |                                                                                                   |                                                          |                                                                                                                                               |                                                                                                  |
| are in the same equivalence class | $a \sim b$ means $\mathbf{b} \in [\mathbf{a}]$ (and equivalently $\mathbf{a} \in [\mathbf{b}]$ ). | $1 \sim 5 \bmod 4$                                       |                                                                                                                                               |                                                                                                  |
| everywhere                        |                                                                                                   |                                                          |                                                                                                                                               |                                                                                                  |
| ≡                                 | ≡                                                                                                 |                                                          |                                                                                                                                               |                                                                                                  |
|                                   |                                                                                                   |                                                          |                                                                                                                                               |                                                                                                  |
|                                   |                                                                                                   |                                                          |                                                                                                                                               |                                                                                                  |
|                                   |                                                                                                   | definition                                               | $x := y, y := x$ or $x \equiv y$ means $x$ is defined to be another name for $y$ , under certain assumptions taken in context.                | $\cosh x := \frac{e^x + e^{-x}}{2}$                                                              |
|                                   |                                                                                                   | is defined as; is equal by definition to                 | <i>(Some writers use <math>\equiv</math> to mean congruence).</i>                                                                             | $[\mathbf{a}, \mathbf{b}] := \mathbf{a} \cdot \mathbf{b} - \mathbf{b} \cdot \mathbf{a}$          |
|                                   |                                                                                                   | everywhere                                               | $P \equiv Q$ means $P$ is defined to be logically equivalent to $Q$ . $P \equiv Q$ means if and only if (iff)                                 |                                                                                                  |
|                                   |                                                                                                   |                                                          |                                                                                                                                               |                                                                                                  |
|                                   |                                                                                                   |                                                          |                                                                                                                                               |                                                                                                  |
|                                   |                                                                                                   |                                                          |                                                                                                                                               |                                                                                                  |
|                                   |                                                                                                   |                                                          |                                                                                                                                               |                                                                                                  |
|                                   |                                                                                                   |                                                          |                                                                                                                                               |                                                                                                  |
|                                   |                                                                                                   |                                                          |                                                                                                                                               |                                                                                                  |
|                                   |                                                                                                   |                                                          |                                                                                                                                               |                                                                                                  |
|                                   |                                                                                                   |                                                          |                                                                                                                                               |                                                                                                  |
| ℜ                                 | ℜ                                                                                                 | congruence                                               |                                                                                                                                               |                                                                                                  |
|                                   |                                                                                                   | is congruent to                                          | $\triangle ABC \cong \triangle DEF$ means triangle $ABC$ is congruent to (has the same measurements as) triangle $DEF$ .                      |                                                                                                  |
|                                   |                                                                                                   | geometry                                                 |                                                                                                                                               |                                                                                                  |
|                                   |                                                                                                   | isomorphic                                               | $G \cong H$ means that group $G$ is isomorphic (structurally identical) to group $H$ .                                                        |                                                                                                  |
|                                   |                                                                                                   | is isomorphic to                                         |                                                                                                                                               | $\mathbb{V} \cong \mathbb{C}_2 \times \mathbb{C}_2$                                              |
|                                   |                                                                                                   | abstract algebra                                         | (≐ can also be used for isomorphic, as described above.)                                                                                      |                                                                                                  |
| ≡                                 | ≡                                                                                                 | congruence relation                                      |                                                                                                                                               |                                                                                                  |
|                                   |                                                                                                   | ... is congruent to ... modulo ...                       | $a \equiv b \pmod{n}$ means $a - b$ is divisible by $n$                                                                                       | $5 \equiv 2 \pmod{3}$                                                                            |
|                                   |                                                                                                   | modular arithmetic                                       |                                                                                                                                               |                                                                                                  |
| ⇔                                 | ⇔                                                                                                 | material equivalence                                     |                                                                                                                                               |                                                                                                  |
|                                   |                                                                                                   | if and only if; iff                                      | $A \Leftrightarrow B$ means $A$ is true if $B$ is true and $A$ is false if $B$ is false.                                                      | $x + 5 = y + 2 \Leftrightarrow x + 3 = y$                                                        |
|                                   |                                                                                                   | propositional logic                                      |                                                                                                                                               |                                                                                                  |

### Symbols that point left or right



## F. Some List of Math Symbols



# De La Salle University

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| Symbol in HTML | Symbol in TeX | Name<br>Read as<br>Category                                    | Explanation                                                                                                                                                                     | Examples                                                                                                               |
|----------------|---------------|----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| <              | <             | strict inequality                                              |                                                                                                                                                                                 |                                                                                                                        |
|                |               | is less than,                                                  | $x < y$ means $x$ is less than $y$ .                                                                                                                                            | $3 < 4$                                                                                                                |
|                |               | is greater than                                                | $x > y$ means $x$ is greater than $y$ .                                                                                                                                         | $5 > 4$                                                                                                                |
|                |               | order theory                                                   |                                                                                                                                                                                 |                                                                                                                        |
| >              | >             | proper subgroup                                                |                                                                                                                                                                                 | $5\mathbb{Z} < \mathbb{Z}$                                                                                             |
|                |               | is a proper subgroup of                                        | $H < G$ means $H$ is a proper subgroup of $G$ .                                                                                                                                 | $A_3 < S_3$                                                                                                            |
|                |               | group theory                                                   |                                                                                                                                                                                 |                                                                                                                        |
|                |               |                                                                |                                                                                                                                                                                 |                                                                                                                        |
| ≪              | ≪             | significant (strict) inequality                                | $x \ll y$ means $x$ is much less than $y$ .                                                                                                                                     | $0.003 \ll 1000000$                                                                                                    |
|                |               | is much less than,                                             | $x \gg y$ means $x$ is much greater than $y$ .                                                                                                                                  |                                                                                                                        |
|                |               | is much greater than                                           |                                                                                                                                                                                 |                                                                                                                        |
|                |               | order theory                                                   |                                                                                                                                                                                 |                                                                                                                        |
| ≫              | ≫             | asymptotic comparison                                          | $f \ll g$ means the growth of $f$ is asymptotically bounded by $g$ .                                                                                                            | $x \ll e^x$                                                                                                            |
|                |               | is of smaller order than,                                      | (This is I. M. Vinogradov's notation. Another notation is the Big $O$ notation, which looks like $f = O(g)$ .)                                                                  |                                                                                                                        |
|                |               | is of greater order than                                       |                                                                                                                                                                                 |                                                                                                                        |
|                |               | analytic number theory                                         |                                                                                                                                                                                 |                                                                                                                        |
| ≍              | ≍             | absolute continuity                                            |                                                                                                                                                                                 |                                                                                                                        |
|                |               | is absolutely continuous with respect to                       | $\mu \ll \nu$ means that $\mu$ is absolutely continuous with respect to $\nu$ , i.e., whenever $\nu(A) = 0$ , we have $\mu(A) = 0$ .                                            | If $\mathfrak{c}$ is the counting measure on $[0, 1]$ and $\mu$ is the Lebesgue measure, then $\mu \ll \mathfrak{c}$ . |
|                |               | measure theory                                                 |                                                                                                                                                                                 |                                                                                                                        |
|                |               |                                                                |                                                                                                                                                                                 |                                                                                                                        |
| ≤              | ≤             | inequality                                                     | $x \leq y$ means $x$ is less than or equal to $y$ .                                                                                                                             |                                                                                                                        |
|                |               | is less than or equal to,                                      | $x \geq y$ means $x$ is greater than or equal to $y$ .                                                                                                                          | $3 \leq 4$ and $5 \leq 5$                                                                                              |
|                |               | is greater than or equal to                                    | (The forms $\leq$ and $\geq$ are generally used in programming languages, where ease of typing and use of ASCII text is preferred.)                                             | $5 \geq 4$ and $5 \geq 5$                                                                                              |
|                |               | order theory                                                   |                                                                                                                                                                                 |                                                                                                                        |
| ⊆              | ⊆             | subgroup                                                       |                                                                                                                                                                                 | $\mathbb{Z} \subseteq \mathbb{Z}$                                                                                      |
|                |               | is a subgroup of                                               | $H \subseteq G$ means $H$ is a subgroup of $G$ .                                                                                                                                | $A_3 \subseteq S_3$                                                                                                    |
|                |               | group theory                                                   |                                                                                                                                                                                 |                                                                                                                        |
|                |               |                                                                |                                                                                                                                                                                 |                                                                                                                        |
| ≲              | ≲             | reduction                                                      |                                                                                                                                                                                 | If                                                                                                                     |
|                |               | is reducible to                                                | $A \leq B$ means the problem $A$ can be reduced to the problem $B$ .                                                                                                            | $\exists f \in F. \forall a \in \mathbb{N}. a \in A \leftrightarrow f(a) \in B$                                        |
|                |               | computational complexity theory                                | Subscripts can be added to the $\leq$ to indicate what kind of reduction.                                                                                                       | then                                                                                                                   |
|                |               |                                                                |                                                                                                                                                                                 | $A \leq_P B$                                                                                                           |
| ≡              | ≡             | congruence relation                                            |                                                                                                                                                                                 |                                                                                                                        |
|                |               | ... is less than ... is greater than ...                       | $7k \equiv 28 \pmod{2}$ is only true if $k$ is an even integer. Assume that the problem requires $k$ to be non-negative; the domain is defined as $0 \leq k < \infty$ .         | $10a \equiv 5 \pmod{5}$ for $1 \leq a \leq 10$                                                                         |
|                |               | modular arithmetic                                             |                                                                                                                                                                                 |                                                                                                                        |
|                |               |                                                                |                                                                                                                                                                                 |                                                                                                                        |
| ≦              | ≦             | vector inequality                                              | $x \leq y$ means that each component of vector $x$ is less than or equal to each corresponding component of vector $y$ .                                                        |                                                                                                                        |
|                |               | ... is less than or equal... is greater than or equal...       | $x \geq y$ means that each component of vector $x$ is greater than or equal to each corresponding component of vector $y$ .                                                     |                                                                                                                        |
|                |               | order theory                                                   | It is important to note that $x \leq y$ remains true if every element is equal. However, if the operator is changed, $x \leq y$ is true if and only if $x \neq y$ is also true. |                                                                                                                        |
|                |               |                                                                |                                                                                                                                                                                 |                                                                                                                        |
| ≲              | ≲             | Karp reduction                                                 |                                                                                                                                                                                 |                                                                                                                        |
|                |               | is Karp reducible to; is polynomial-time many-one reducible to | $L_1 < L_2$ means that the problem $L_1$ is Karp reducible to $L_2$ . <sup>[2]</sup>                                                                                            | If $L_1 < L_2$ and $L_2 \in \mathbf{P}$ , then $L_1 \in \mathbf{P}$ .                                                  |
|                |               | computational complexity theory                                |                                                                                                                                                                                 |                                                                                                                        |
|                |               |                                                                |                                                                                                                                                                                 |                                                                                                                        |
| ≲              | ≲             | Nondominated order                                             |                                                                                                                                                                                 |                                                                                                                        |
|                |               | is nondominated by                                             | $P < Q$ means that the element $P$ is nondominated by element $Q$ . <sup>[3]</sup>                                                                                              | If $P_1 < Q_2$ then $\forall_i P_i \leq Q_i \wedge \exists P_i < Q_i$                                                  |
|                |               | Multi-objective optimization                                   |                                                                                                                                                                                 |                                                                                                                        |
|                |               |                                                                |                                                                                                                                                                                 |                                                                                                                        |
| ⊴              | ⊴             | normal subgroup                                                |                                                                                                                                                                                 |                                                                                                                        |
|                |               | is a normal subgroup of                                        | $N \triangleleft G$ means that $N$ is a normal subgroup of group $G$ .                                                                                                          | $Z(G) \triangleleft G$                                                                                                 |
|                |               | group theory                                                   |                                                                                                                                                                                 |                                                                                                                        |
|                |               |                                                                |                                                                                                                                                                                 |                                                                                                                        |
| ⊵              | ⊵             | ideal                                                          |                                                                                                                                                                                 |                                                                                                                        |
|                |               | is an ideal of                                                 | $I \triangleleft R$ means that $I$ is an ideal of ring $R$ .                                                                                                                    | $(2) \triangleleft \mathbb{Z}$                                                                                         |
|                |               | ring theory                                                    |                                                                                                                                                                                 |                                                                                                                        |
|                |               |                                                                |                                                                                                                                                                                 |                                                                                                                        |
| ⊶              | ⊶             | antijoin                                                       |                                                                                                                                                                                 |                                                                                                                        |
|                |               | the antijoin of                                                | $R \bowtie S$ means the antijoin of the relations $R$ and $S$ , the tuples in $R$ for which there is not a tuple in $S$ that is equal on their common attribute names.          | $R \bowtie S = R - R \times S$                                                                                         |
|                |               | relational algebra                                             |                                                                                                                                                                                 |                                                                                                                        |
|                |               |                                                                |                                                                                                                                                                                 |                                                                                                                        |
| ⇒              | ⇒             | material implication                                           | $A \Rightarrow B$ means if $A$ is true then $B$ is also true; if $A$ is false then nothing is said about $B$ .                                                                  |                                                                                                                        |
|                |               | implies; if ... then                                           | ( $\Rightarrow$ may mean the same as $\Rightarrow$ , or it may have the meaning for functions given below.)                                                                     | $x = 2 \Rightarrow x^2 = 4$ is true, but $x^2 = 4 \Rightarrow x = 2$ is in general false (since $x$ could be $-2$ ).   |
|                |               | propositional logic; Heyting algebra                           | ( $\Rightarrow$ may mean the same as $\Rightarrow$ . <sup>[4]</sup> or it may have the meaning for superset given below.)                                                       |                                                                                                                        |
|                |               |                                                                |                                                                                                                                                                                 |                                                                                                                        |
| ⊂              | ⊂             | subset                                                         | (subset) $A \subseteq B$ means every element of $A$ is also an element of $B$ . <sup>[5]</sup>                                                                                  | $(A \cap B) \subseteq A$                                                                                               |
|                |               | is a subset of                                                 | (proper subset) $A \subset B$ means $A \subseteq B$ but $A \neq B$ .                                                                                                            | $\mathbb{N} \subset \mathbb{Q}$                                                                                        |
|                |               | set theory                                                     |                                                                                                                                                                                 | $\mathbb{Q} \subset \mathbb{R}$                                                                                        |
|                |               |                                                                | (Some writers use the symbol $\subset$ as if it were the same as $\subseteq$ .)                                                                                                 |                                                                                                                        |



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|               |               |                                                                                    |                                                                                                                                                                           |                                                                                                                                                               |
|---------------|---------------|------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\supseteq$   | $\supseteq$   | superset                                                                           | $A \supseteq B$ means every element of $B$ is also an element of $A$ .                                                                                                    | $(A \cup B) \supseteq B$                                                                                                                                      |
| $\supset$     | $\supset$     | is a superset of<br>set theory                                                     | $A \supset B$ means $A \supseteq B$ but $A \neq B$ .<br>(Some writers use the symbol $\supset$ as if it were the same as $\supseteq$ .)                                   | $\mathbb{R} \supset \mathbb{Q}$                                                                                                                               |
| $\rightarrow$ | $\rightarrow$ | function arrow<br>from ... to<br>set theory, type theory                           | $f: X \rightarrow Y$ means the function $f$ maps the set $X$ into the set $Y$ .                                                                                           | Let $f: \mathbb{Z} \rightarrow \mathbb{N} \cup \{0\}$ be defined by $f(x) := x^2$ .                                                                           |
| $\mapsto$     | $\mapsto$     | function arrow<br>maps to<br>set theory                                            | $f: a \mapsto b$ means the function $f$ maps the element $a$ to the element $b$ .                                                                                         | Let $f: x \mapsto x + 1$ (the successor function).                                                                                                            |
| $<:$          | $<:$          | subtype<br>is a subtype of<br>type theory                                          | $T_1 <: T_2$ means that $T_1$ is a subtype of $T_2$ .                                                                                                                     | If $S <: T$ and $T <: U$ then $S <: U$ (transitivity).                                                                                                        |
| $<\cdot$      | $<\cdot$      | cover<br>is covered by<br>order theory                                             | $x <\cdot y$ means that $x$ is covered by $y$ .                                                                                                                           | $\{1, 8\} <\cdot \{1, 3, 8\}$ among the subsets of $\{1, 2, \dots, 10\}$ ordered by containment.                                                              |
| $\vDash$      | $\vDash$      | entailment<br>entails<br>model theory                                              | $A \vDash B$ means the sentence $A$ entails the sentence $B$ , that is in every model in which $A$ is true, $B$ is also true.                                             | $A \vDash A \vee \neg A$                                                                                                                                      |
| $\vdash$      | $\vdash$      | inference<br>infers;<br>is derived from<br>propositional logic,<br>predicate logic | $x \vdash y$ means $y$ is derivable from $x$ .                                                                                                                            | $A \rightarrow B \vdash \neg B \rightarrow \neg A$                                                                                                            |
|               |               | partition<br>is a partition of<br>number theory                                    | $p \vdash n$ means that $p$ is a partition of $n$ .                                                                                                                       | $(4, 3, 1, 1) \vdash 9, \sum_{\lambda \vdash n} (f_{\lambda})^2 = n!$                                                                                         |
| $\langle  $   | $\langle  $   | bra vector<br>the bra ...;<br>the dual of ...<br>Dirac notation                    | $\langle \phi  $ means the dual of the vector $ \phi\rangle$ , a linear functional which maps a ket $ \psi\rangle$ onto the inner product $\langle \phi   \psi \rangle$ . |                                                                                                                                                               |
| $  \rangle$   | $  \rangle$   | ket vector<br>the ket ...;<br>the vector ...<br>Dirac notation                     | $ \phi\rangle$ means the vector with label $\phi$ , which is in a Hilbert space.                                                                                          | A qubit's state can be represented as $\alpha 0\rangle + \beta 1\rangle$ , where $\alpha$ and $\beta$ are complex numbers s.t. $ \alpha ^2 +  \beta ^2 = 1$ . |

Brackets

## F. Some List of Math Symbols



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| Symbol in HTML                      | Symbol in TeX                       | Name<br>Read as<br>Category                                                                                                                                                                                                                                                                           | Explanation                                                                                                                                                                                                                                                                                                                           | Examples                                                                                                                                                                                                                                                                                                 |
|-------------------------------------|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                     | $\binom{n}{k}$                      | combination;<br>binomial coefficient<br>$n$ choose $k$<br>combinatorics                                                                                                                                                                                                                               | $\binom{n}{k} = \frac{n!/(n-k)!}{k!} = \frac{(n-k+1) \cdots (n-2) \cdot (n-1) \cdot n}{k!}$<br>means (in the case of $n =$ positive integer) the number of combinations of $k$ elements drawn from a set of $n$ elements.<br>(This may also be written as $C(n, k)$ , $C(n; k)$ , ${}_nC_k$ , ${}_nC_k$ , or $\langle n \rangle_k$ .) | $\binom{73}{5} = \frac{73!/(73-5)!}{5!} = \frac{60 \cdot 70 \cdot 71 \cdot 72 \cdot 73}{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5} = 15020394$<br>$\binom{.5}{7} = \frac{-.5 \cdot -.4 \cdot -.3 \cdot -.2 \cdot -.1 \cdot -.5 \cdot -.5}{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 \cdot 7} = \frac{.33}{2048}$ |
|                                     | $\binom{u}{k}$                      | multiset coefficient<br>$u$ multichoose $k$<br>combinatorics                                                                                                                                                                                                                                          | $\binom{u}{k} = \binom{u+k-1}{k} = \frac{(u+k-1)!}{k!}$<br>(when $u$ is positive integer)<br>means reverse or rising binomial coefficient.                                                                                                                                                                                            | $\binom{-5.5}{7} = \frac{-.5 \cdot -.4 \cdot -.3 \cdot -.2 \cdot -.1 \cdot -.5 \cdot -.5}{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 \cdot 7} = \binom{.5}{7} = \frac{.33}{2048}$                                                                                                                         |
|                                     | $  \dots  $                         | absolute value;<br>modulus<br><br>absolute value of;<br>modulus of<br>numbers<br><br>Euclidean norm or<br>Euclidean length or<br>magnitude<br>Euclidean norm of<br>geometry<br>determinant<br>determinant of<br>matrix theory<br>cardinality<br>cardinality of;<br>size of;<br>order of<br>set theory | $ x $ means the distance along the real line (or across the complex plane) between $x$ and zero.<br><br>$ x $ means the (Euclidean) length of vector $x$ .<br><br>$ A $ means the determinant of the matrix $A$ .<br><br>$ X $ means the cardinality of the set $X$ .<br>(# may be used instead as described below.)                  | $ 3  = 3$<br>$ -5  =  5  = 5$<br>$ i  = 1$<br>$ 3 + 4i  = 5$<br><br>For $x = (3, -4)$<br>$ x  = \sqrt{3^2 + (-4)^2} = 5$<br><br>$\begin{vmatrix} 1 & 2 \\ 2 & 9 \end{vmatrix} = 5$<br><br>$  \{3, 5, 7, 9\}   = 4$                                                                                       |
| $\  \dots \ $                       | $\  \dots \ $                       | norm<br>norm of;<br>length of<br>linear algebra<br><br>nearest integer<br>function<br><br>nearest integer to<br>numbers                                                                                                                                                                               | $\ x\ $ means the norm of the element $x$ of a normed vector space. <sup>[6]</sup><br><br>$\lfloor x \rfloor$ means the nearest integer to $x$ .<br>(This may also be written $\lfloor x \rfloor$ , $\text{L}x$ , $\text{nint}(x)$ or $\text{Round}(x)$ .)                                                                            | $\ x + y\  \leq \ x\  + \ y\ $<br><br>$\ 1\  = 1$ , $\ 1.6\  = 2$ , $\  -2.4\  = -2$ , $\ 3.49\  = 3$                                                                                                                                                                                                    |
| $\{ \dots \}$                       | $\{ \dots \}$                       | set brackets<br>the set of ...<br>set theory                                                                                                                                                                                                                                                          | $\{a, b, c\}$ means the set consisting of $a$ , $b$ , and $c$ . <sup>[7]</sup>                                                                                                                                                                                                                                                        | $\mathbb{N} = \{ 1, 2, 3, \dots \}$                                                                                                                                                                                                                                                                      |
| $\{ : \}$<br>$\{   \}$<br>$\{ ; \}$ | $\{ : \}$<br>$\{   \}$<br>$\{ ; \}$ | set builder notation<br>the set of ...<br>such that<br>set theory                                                                                                                                                                                                                                     | $\{x : P(x)\}$ means the set of all $x$ for which $P(x)$ is true. <sup>[7]</sup> $\{x   P(x)\}$ is the same as $\{x : P(x)\}$ .                                                                                                                                                                                                       | $\{n \in \mathbb{N} : n^2 < 20\} = \{ 1, 2, 3, 4 \}$                                                                                                                                                                                                                                                     |
| $\lfloor \dots \rfloor$             | $\lfloor \dots \rfloor$             | floor<br>floor;<br>greatest integer;<br>entier<br>numbers                                                                                                                                                                                                                                             | $\lfloor x \rfloor$ means the floor of $x$ , i.e. the largest integer less than or equal to $x$ .<br>(This may also be written $\lfloor x \rfloor$ , $\text{floor}(x)$ or $\text{int}(x)$ .)                                                                                                                                          | $\lfloor 4 \rfloor = 4$ , $\lfloor 2.1 \rfloor = 2$ , $\lfloor 2.9 \rfloor = 2$ , $\lfloor -2.6 \rfloor = -3$                                                                                                                                                                                            |
| $\lceil \dots \rceil$               | $\lceil \dots \rceil$               | ceiling<br>ceiling<br>numbers                                                                                                                                                                                                                                                                         | $\lceil x \rceil$ means the ceiling of $x$ , i.e. the smallest integer greater than or equal to $x$ .<br>(This may also be written $\text{ceil}(x)$ or $\text{ceiling}(x)$ .)                                                                                                                                                         | $\lceil 4 \rceil = 4$ , $\lceil 2.1 \rceil = 3$ , $\lceil 2.9 \rceil = 3$ , $\lceil -2.6 \rceil = -2$                                                                                                                                                                                                    |
| $\lfloor \dots \rfloor$             | $\lfloor \dots \rfloor$             | nearest integer<br>function<br><br>nearest integer to<br>numbers                                                                                                                                                                                                                                      | $\lfloor x \rfloor$ means the nearest integer to $x$ .<br>(This may also be written $\lfloor x \rfloor$ , $\lfloor x \rfloor$ , $\text{nint}(x)$ or $\text{Round}(x)$ .)                                                                                                                                                              | $\lfloor 2 \rfloor = 2$ , $\lfloor 2.6 \rfloor = 3$ , $\lfloor -3.4 \rfloor = -3$ , $\lfloor 4.49 \rfloor = 4$                                                                                                                                                                                           |
| $[ : ]$                             | $[ : ]$                             | degree of a field extension<br>the degree of field theory                                                                                                                                                                                                                                             | $[K : F]$ means the degree of the extension $K : F$ .                                                                                                                                                                                                                                                                                 | $[Q(\sqrt{2}) : Q] = 2$<br>$[C : R] = 2$<br>$[R : Q] = \infty$                                                                                                                                                                                                                                           |
| $[ ]$<br>$[ , ]$                    | $[ ]$<br>$[ , ]$<br>$[ , ]$         | equivalence class<br>the equivalence class of                                                                                                                                                                                                                                                         | $[a]$ means the equivalence class of $a$ , i.e. $\{x : x \sim a\}$ , where $\sim$ is an equivalence relation.<br>$[a]_R$ means the same, but with $R$ as the equivalence relation.                                                                                                                                                    | Let $a \sim b$ be true iff $a \equiv b \pmod{5}$ .<br>Then $[2] = \{\dots, -8, -3, 2, 7, \dots\}$ .                                                                                                                                                                                                      |

## F. Some List of Math Symbols



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|                         |                                                                                            |                                                                                                                                                                                                                                                              |                                                                                                                                                                                                  |
|-------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [ , , ]                 | abstract algebra                                                                           |                                                                                                                                                                                                                                                              |                                                                                                                                                                                                  |
|                         | floor                                                                                      |                                                                                                                                                                                                                                                              |                                                                                                                                                                                                  |
|                         | floor;<br>greatest integer;<br>entier<br>numbers                                           | $[x]$ means the floor of $x$ , i.e. the largest integer less than or equal to $x$ .<br><br>(This may also be written $\lfloor x \rfloor$ , $\text{floor}(x)$ or $\text{int}(x)$ . Not to be confused with the nearest integer function, as described below.) | $[3] = 3, [3.5] = 3, [3.99] = 3, [-3.7] = -4$                                                                                                                                                    |
|                         | nearest integer function                                                                   | $\lceil x \rceil$ means the nearest integer to $x$ .                                                                                                                                                                                                         |                                                                                                                                                                                                  |
|                         | nearest integer to<br>numbers                                                              | (This may also be written $\lceil x \rceil$ , $\lceil x \rceil$ , $\text{nint}(x)$ or $\text{Round}(x)$ . Not to be confused with the floor function, as described above.)                                                                                   | $[2] = 2, [2.6] = 3, [-3.4] = -3, [4.49] = 4$                                                                                                                                                    |
|                         | Iverson bracket                                                                            |                                                                                                                                                                                                                                                              |                                                                                                                                                                                                  |
|                         | 1 if true, 0 otherwise                                                                     | $[S]$ maps a true statement $S$ to 1 and a false statement $S$ to 0.                                                                                                                                                                                         | $[0=5]=0, [7>0]=1, [2 \in \{2,3,4\}]=1, [5 \in \{2,3,4\}]=0$                                                                                                                                     |
|                         | propositional logic                                                                        |                                                                                                                                                                                                                                                              |                                                                                                                                                                                                  |
|                         | image                                                                                      | $f[X]$ means $\{f(x) : x \in X\}$ , the image of the function $f$ under the set $X \subseteq \text{dom}(f)$ .                                                                                                                                                |                                                                                                                                                                                                  |
|                         | image of ...<br>under ...<br>everywhere                                                    | (This may also be written as $f[X]$ if there is no risk of confusing the image of $f$ under $X$ with the function application $f$ of $X$ . Another notation is $\text{Im } f$ , the image of $f$ under its domain.)                                          | $\sin(\mathbb{R}) = [-1, 1]$                                                                                                                                                                     |
| ( )<br>( )<br>( )       | closed interval                                                                            | $[a, b] = \{x \in \mathbb{R} : a \leq x \leq b\}$ .                                                                                                                                                                                                          | 0 and 1/2 are in the interval $[0, 1]$ .                                                                                                                                                         |
|                         | closed interval                                                                            |                                                                                                                                                                                                                                                              |                                                                                                                                                                                                  |
|                         | order theory                                                                               |                                                                                                                                                                                                                                                              |                                                                                                                                                                                                  |
|                         | commutator                                                                                 |                                                                                                                                                                                                                                                              |                                                                                                                                                                                                  |
|                         | the commutator of                                                                          | $[g, h] = g^{-1}h^{-1}gh$ (or $ghg^{-1}h^{-1}$ ), if $g, h \in G$ (a group).                                                                                                                                                                                 | $x^y = x[x, y]$ (group theory).                                                                                                                                                                  |
|                         | group theory,<br>ring theory                                                               | $[a, b] = ab - ba$ , if $a, b \in R$ (a ring or commutative algebra).                                                                                                                                                                                        | $[AB, C] = A[B, C] + [A, C]B$ (ring theory).                                                                                                                                                     |
|                         | triple scalar product                                                                      |                                                                                                                                                                                                                                                              |                                                                                                                                                                                                  |
|                         | the triple scalar product of                                                               | $[a, b, c] = a \times b \cdot c$ , the scalar product of $a \times b$ with $c$ .                                                                                                                                                                             | $[a, b, c] = [b, c, a] = [c, a, b]$ .                                                                                                                                                            |
|                         | vector calculus                                                                            |                                                                                                                                                                                                                                                              |                                                                                                                                                                                                  |
|                         | function application of<br>set theory                                                      | $f(x)$ means the value of the function $f$ at the element $x$ .                                                                                                                                                                                              | If $f(x) = x^2$ , then $f(3) = 3^2 = 9$ .                                                                                                                                                        |
| ( )<br>( )<br>( )       | image                                                                                      | $f[X]$ means $\{f(x) : x \in X\}$ , the image of the function $f$ under the set $X \subseteq \text{dom}(f)$ .                                                                                                                                                |                                                                                                                                                                                                  |
|                         | image of ...<br>under ...<br>everywhere                                                    | (This may also be written as $f[X]$ if there is a risk of confusing the image of $f$ under $X$ with the function application $f$ of $X$ . Another notation is $\text{Im } f$ , the image of $f$ under its domain.)                                           | $\sin(\mathbb{R}) = [-1, 1]$                                                                                                                                                                     |
|                         | precedence grouping<br>parentheses<br>everywhere                                           | Perform the operations inside the parentheses first.                                                                                                                                                                                                         | $(8/4)2 = 2/2 = 1$ , but $8/(4/2) = 8/2 = 4$ .                                                                                                                                                   |
|                         | tuple                                                                                      | An ordered list (or sequence, or horizontal vector, or row vector) of values.                                                                                                                                                                                | $(a, b)$ is an ordered pair (or 2-tuple).                                                                                                                                                        |
|                         | tuple; $n$ -tuple;<br>ordered<br>pair/triple/etc;<br>row vector;<br>sequence<br>everywhere | (Note that the notation $(a, b)$ is ambiguous: it could be an ordered pair or an open interval. Set theorists and computer scientists often use angle brackets $\langle \rangle$ instead of parentheses.)                                                    | $(a, b, c)$ is an ordered triple (or 3-tuple).<br><br>$()$ is the empty tuple (or 0-tuple).                                                                                                      |
|                         | highest common factor                                                                      |                                                                                                                                                                                                                                                              |                                                                                                                                                                                                  |
|                         | highest common factor;<br>greatest common divisor; hcf; gcd                                | $(a, b)$ means the highest common factor of $a$ and $b$ .<br><br>(This may also be written $\text{hcf}(a, b)$ or $\text{gcd}(a, b)$ .)                                                                                                                       | $(3, 7) = 1$ (they are coprime); $(15, 25) = 5$ .                                                                                                                                                |
|                         | number theory                                                                              |                                                                                                                                                                                                                                                              |                                                                                                                                                                                                  |
|                         | ( , )<br>( , )<br>( , )                                                                    | open interval<br>open interval<br>order theory                                                                                                                                                                                                               | $(a, b) = \{x \in \mathbb{R} : a < x < b\}$ .<br><br>(Note that the notation $(a, b)$ is ambiguous: it could be an ordered pair or an open interval. The notation $[a, b]$ can be used instead.) |
|                         | left-open interval                                                                         |                                                                                                                                                                                                                                                              | 4 is not in the interval $(4, 18)$ .<br><br>$(0, +\infty)$ equals the set of positive real numbers.                                                                                              |
| ( , ]<br>( , ]<br>( , ] | half-open interval;<br>left-open interval                                                  | $[a, b] = \{x \in \mathbb{R} : a \leq x \leq b\}$ .                                                                                                                                                                                                          | $(-1, 7]$ and $(-\infty, -1]$                                                                                                                                                                    |
|                         | order theory                                                                               |                                                                                                                                                                                                                                                              |                                                                                                                                                                                                  |

## F. Some List of Math Symbols



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|                                         |                                |                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                         |
|-----------------------------------------|--------------------------------|-----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $[, )$<br>$[, [$                        | $(, )$<br>$(, [$               | right-open interval                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                         |
|                                         |                                | half-open interval;<br>right-open interval                                                          | $[a, b) = \{x \in \mathbb{R} : a \leq x < b\}$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | $[4, 18)$ and $[1, +\infty)$                                                                                                                                                                                            |
| $\langle \rangle$<br>$\langle, \rangle$ | $( )$<br>$(, )$                | order theory                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                         |
|                                         |                                | inner product<br>inner product of<br>linear algebra                                                 | <p><math>\langle u, v \rangle</math> means the inner product of <math>u</math> and <math>v</math>, where <math>u</math> and <math>v</math> are members of an inner product space.</p> <p><i>Note that the notation <math>\langle u, v \rangle</math> may be ambiguous: it could mean the inner product or the linear span.</i></p> <p><i>There are many variants of the notation, such as <math>\langle u   v \rangle</math> and <math>\langle u   v \rangle</math>, which are described below. For spatial vectors, the dot product notation, <math>x \cdot y</math> is common. For matrices, the colon notation <math>A : B</math> may be used. As <math>\langle</math> and <math>\rangle</math> can be hard to type, the more "keyboard friendly" forms <math>&lt;</math> and <math>&gt;</math> are sometimes seen. These are avoided in mathematical texts.</i></p> | <p>The standard inner product between two vectors <math>x = (2, 3)</math> and <math>y = (-1, 5)</math> is:<br/> <math>\langle x, y \rangle = 2 \times -1 + 3 \times 5 = 13</math></p>                                   |
|                                         |                                | average<br>average of<br>statistics                                                                 | <p>let <math>S</math> be a subset of <math>N</math> for example, <math>\langle S \rangle</math> represents the average of all the element in <math>S</math>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <p>for a time series : <math>g(t)</math> (<math>t = 1, 2, \dots</math>)</p> <p>we can define the structure functions <math>S_g(\tau)</math>:</p> $S_g = \langle  g(t+\tau) - g(t) ^p \rangle_t$                         |
|                                         |                                | linear span<br>(linear) span of,<br>linear hull of<br>linear algebra                                | <p><math>\langle S \rangle</math> means the span of <math>S \subseteq V</math>. That is, it is the intersection of all subspaces of <math>V</math> which contain <math>S</math>.</p> <p><math>\langle u_1, u_2, \dots \rangle</math> is shorthand for <math>\langle \{u_1, u_2, \dots\} \rangle</math>.</p> <p><i>Note that the notation <math>\langle u, v \rangle</math> may be ambiguous: it could mean the inner product or the linear span.</i></p> <p><i>The span of <math>S</math> may also be written as <math>\text{Sp}(S)</math>.</i></p>                                                                                                                                                                                                                                                                                                                     | $\left\langle \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\rangle = \mathbb{R}^3$                                             |
|                                         |                                | subgroup generated by a set<br>the subgroup generated by<br>group theory                            | <p><math>\langle S \rangle</math> means the smallest subgroup of <math>G</math> (where <math>S \subseteq G</math>, a group) containing every element of <math>S</math>.</p> <p><math>\langle g_1, g_2, \dots \rangle</math> is shorthand for <math>\langle g_1, g_2, \dots \rangle</math>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <p>In <math>S_3</math>, <math>\langle (1\ 2) \rangle = \{id, (1\ 2)\}</math> and <math>\langle (1\ 2\ 3) \rangle = \{id, (1\ 2\ 3), (1\ 3\ 2)\}</math>.</p>                                                             |
|                                         |                                | tuple<br>tuple; $n$ -tuple;<br>ordered<br>pair/triple/etc;<br>row vector;<br>sequence<br>everywhere | <p>An ordered list (or sequence, or horizontal vector, or row vector) of values.</p> <p><i>(The notation <math>(a, b)</math> is often used as well.)</i></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <p><math>\langle a, b \rangle</math> is an ordered pair (or 2-tuple).</p> <p><math>\langle a, b, c \rangle</math> is an ordered triple (or 3-tuple).</p> <p><math>\emptyset</math> is the empty tuple (or 0-tuple).</p> |
| $\langle   \rangle$<br>$(   )$          | $\langle   \rangle$<br>$(   )$ | inner product<br>inner product of<br>linear algebra                                                 | <p><math>\langle u   v \rangle</math> means the inner product of <math>u</math> and <math>v</math>, where <math>u</math> and <math>v</math> are members of an inner product space.<sup>[8]</sup> <math>\langle u   v \rangle</math> means the same.</p> <p><i>Another variant of the notation is <math>\langle u, v \rangle</math> which is described above. For spatial vectors, the dot product notation, <math>x \cdot y</math> is common. For matrices, the colon notation <math>A : B</math> may be used. As <math>\langle</math> and <math>\rangle</math> can be hard to type, the more "keyboard friendly" forms <math>&lt;</math> and <math>&gt;</math> are sometimes seen. These are avoided in mathematical texts.</i></p>                                                                                                                                    |                                                                                                                                                                                                                         |

Other non-letter symbols

## F. Some List of Math Symbols



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| Symbol in HTML | Symbol in TeX | Name<br>Read as<br>Category                                                                          | Explanation                                                                                                                                                                 | Examples                                                                                                                            |
|----------------|---------------|------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| *              | *             | convolution                                                                                          |                                                                                                                                                                             |                                                                                                                                     |
|                |               | convolution;<br>convolved with                                                                       | $f * g$ means the convolution of $f$ and $g$ .                                                                                                                              | $(f * g)(t) = \int_0^t f(\tau)g(t - \tau) d\tau$ .                                                                                  |
|                |               | functional analysis                                                                                  |                                                                                                                                                                             |                                                                                                                                     |
|                |               | complex conjugate                                                                                    | $z^*$ means the complex conjugate of $z$ .                                                                                                                                  | $(3 + 4i)^* = 3 - 4i$ .                                                                                                             |
|                |               | conjugate                                                                                            |                                                                                                                                                                             |                                                                                                                                     |
|                |               | complex numbers                                                                                      | $(\bar{z})$ can also be used for the conjugate of $z$ , as described below.)                                                                                                |                                                                                                                                     |
|                |               | group of units                                                                                       | $R^*$ consists of the set of units of the ring $R$ , along with the operation of multiplication.                                                                            | $(\mathbb{Z}/6\mathbb{Z})^* = \{[1], [2], [3], [4]\}$<br>$\cong C_4$                                                                |
|                |               | the group of units of                                                                                |                                                                                                                                                                             |                                                                                                                                     |
|                |               | ring theory                                                                                          | This may also be written $R^*$ as described above, or $U(R)$ .                                                                                                              |                                                                                                                                     |
|                |               | hyperreal numbers                                                                                    |                                                                                                                                                                             |                                                                                                                                     |
| $\propto$      | $\propto$     | the (set of) hyperreals                                                                              | ${}^*\mathbf{R}$ means the set of hyperreal numbers. Other sets can be used in place of $\mathbf{R}$ .                                                                      | ${}^*\mathbf{N}$ is the hypernatural numbers.                                                                                       |
|                |               | non-standard analysis                                                                                |                                                                                                                                                                             |                                                                                                                                     |
|                |               | Hodge dual                                                                                           |                                                                                                                                                                             |                                                                                                                                     |
|                |               | Hodge dual;<br>Hodge star<br>linear algebra                                                          | $\star v$ means the Hodge dual of a vector $v$ . If $v$ is a $k$ -vector within an $n$ -dimensional oriented inner product space, then $\star v$ is an $(n-k)$ -vector.     | If $\{e_i\}$ are the standard basis vectors of $\mathbf{R}^n$ ,<br>$\star(e_1 \wedge e_2 \wedge e_3) = e_4 \wedge e_5$              |
| $\propto$      | $\propto$     | proportionality                                                                                      |                                                                                                                                                                             |                                                                                                                                     |
|                |               | is proportional to;<br>varies as<br>everywhere                                                       | $y \propto x$ means that $y = kx$ for some constant $k$ .                                                                                                                   | if $y = 2x$ , then $y \propto x$ .                                                                                                  |
|                |               | Karp reduction <sup>[9]</sup>                                                                        |                                                                                                                                                                             |                                                                                                                                     |
|                |               | is Karp reducible to;<br>is polynomial-time many-one reducible to<br>computational complexity theory | $A \propto B$ means the problem $A$ can be polynomially reduced to the problem $B$ .                                                                                        | If $L_1 \propto L_2$ and $L_2 \in \mathbf{P}$ , then $L_1 \in \mathbf{P}$ .                                                         |
| \              | \             | set-theoretic complement                                                                             | $A \setminus B$ means the set that contains all those elements of $A$ that are not in $B$ . <sup>[5]</sup>                                                                  | $\{1,2,3,4\} \setminus \{3,4,5,6\} = \{1,2\}$                                                                                       |
|                |               | minus;<br>without;<br>throw out;<br>not                                                              | ( $\setminus$ can also be used for set-theoretic complement as described above.)                                                                                            |                                                                                                                                     |
|                |               | set theory                                                                                           |                                                                                                                                                                             |                                                                                                                                     |
|                |               |                                                                                                      |                                                                                                                                                                             |                                                                                                                                     |
|                |               | conditional event given                                                                              | $P(A B)$ means the probability of the event $A$ occurring given that $B$ occurs.                                                                                            | if $X$ is a uniformly random day of the year $P(X \text{ is May } 25   X \text{ is in May}) = 1/31$                                 |
|                |               | probability                                                                                          |                                                                                                                                                                             |                                                                                                                                     |
|                |               | restriction of ... to ...;<br>restricted to<br>set theory                                            | $f _A$ means the function $f$ is restricted to the set $A$ , that is, it is the function with domain $A \cap \text{dom}(f)$ that agrees with $f$ .                          | The function $f: \mathbf{R} \rightarrow \mathbf{R}$ defined by $f(x) = x^2$ is not injective, but $f _{\mathbf{R}^+}$ is injective. |
|                |               | such that<br>such that;<br>so that<br>everywhere                                                     | $ $ means "such that", see $\bullet$ , $\circ$ (described below).                                                                                                           | $S = \{(x,y)   0 < y < f(x)\}$<br>The set of $(x,y)$ such that $y$ is greater than 0 and less than $f(x)$ .                         |
|                |               | divisor, divides                                                                                     | $a   b$ means $a$ divides $b$ .                                                                                                                                             |                                                                                                                                     |
|                |               | divides                                                                                              | $a \nmid b$ means $a$ does not divide $b$ .                                                                                                                                 | Since $15 = 3 \times 5$ , it is true that $3   15$ and $5   15$ .                                                                   |
|                |               | number theory                                                                                        | (The symbol $ $ can be difficult to type, and its negation is rare, so a regular but slightly shorter vertical bar $ $ character is often used instead.)                    |                                                                                                                                     |
|                |               |                                                                                                      |                                                                                                                                                                             |                                                                                                                                     |
|                |               | exact divisibility                                                                                   | $p^a    n$ means $p^a$ exactly divides $n$ (i.e. $p^a$ divides $n$ but $p^{a+1}$ does not).                                                                                 | $2^2    360$ .                                                                                                                      |
|                |               | exactly divides<br>number theory                                                                     |                                                                                                                                                                             |                                                                                                                                     |
|                |               | parallel                                                                                             | $x \parallel y$ means $x$ is parallel to $y$ .<br>$x \nparallel y$ means $x$ is not parallel to $y$ .<br>$x \# y$ means $x$ is equal and parallel to $y$ .                  |                                                                                                                                     |
|                |               | is parallel to                                                                                       |                                                                                                                                                                             | If $l \parallel m$ and $m \perp n$ then $l \perp n$ .                                                                               |
|                |               | geometry                                                                                             | (The symbol $\parallel$ can be difficult to type, and its negation is rare, so two regular but slightly longer vertical bar $\parallel$ characters are often used instead.) |                                                                                                                                     |
|                |               | incomparability                                                                                      |                                                                                                                                                                             |                                                                                                                                     |
| #              | #             | is incomparable to                                                                                   | $x \parallel y$ means $x$ is incomparable to $y$ .                                                                                                                          | $\{1,2\} \parallel \{2,3\}$ under set containment.                                                                                  |
|                |               | order theory                                                                                         |                                                                                                                                                                             |                                                                                                                                     |
|                |               | cardinality                                                                                          |                                                                                                                                                                             |                                                                                                                                     |
|                |               | cardinality of;<br>size of;<br>order of<br>set theory                                                | $\#X$ means the cardinality of the set $X$ .<br>( $\#$ may be used instead as described above.)                                                                             | $\#\{4, 6, 8\} = 3$                                                                                                                 |
| #              | #             | connected sum                                                                                        |                                                                                                                                                                             |                                                                                                                                     |
|                |               | connected sum of;<br>knot sum of;<br>knot composition of                                             | $A \# B$ is the connected sum of the manifolds $A$ and $B$ . If $A$ and $B$ are knots, then this denotes the knot sum, which has a slightly stronger condition.             | $A \# S^n$ is homeomorphic to $A$ , for any manifold $A$ , and the sphere $S^n$ .                                                   |
|                |               | topology, knot theory                                                                                |                                                                                                                                                                             |                                                                                                                                     |
|                |               |                                                                                                      |                                                                                                                                                                             |                                                                                                                                     |

## F. Some List of Math Symbols



# De La Salle University

2071

|  |  |                                            |                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--|--|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  |  | primorial                                  |                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|  |  | primorial                                  | $n\#$ is product of all prime numbers less than or equal to $n$ .                                                                                                                                                                                          | $12\# = 2 \times 3 \times 5 \times 7 \times 11 = 2310$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|  |  | number theory                              |                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|  |  | such that                                  |                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|  |  | such that;<br>so that                      | $\therefore$ means "such that", and is used in proofs and the set-builder notation ( <i>described below</i> ).                                                                                                                                             | $\exists n \in \mathbb{N}$ : $n$ is even.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|  |  | everywhere                                 |                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|  |  | field extension                            | $K : F$ means the field $K$ extends the field $F$ .                                                                                                                                                                                                        | $\mathbb{R} : \mathbb{Q}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|  |  | extends;<br>over                           | <i>This may also be written as <math>K \supseteq F</math>.</i>                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|  |  | field theory                               |                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|  |  | inner product of<br>matrices               | $A : B$ means the Frobenius inner product of the matrices $A$ and $B$ .                                                                                                                                                                                    | $A : B = \sum_{ij} A_{ij} B_{ij}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|  |  | inner product of<br>linear algebra         | The general inner product is denoted by $\langle u, v \rangle$ , $\langle u   v \rangle$ or $(u   v)$ , as described below: For spatial vectors, the dot product notation, $x \cdot y$ is common. See also bra–ket notation.                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|  |  | index of a<br>subgroup                     | The index of a subgroup $H$ in a group $G$ is the "relative size" of $H$ in $G$ : equivalently, the number of "copies" (cosets) of $H$ that fill up $G$                                                                                                    | $ G : H  = \frac{ G }{ H }$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|  |  | index of subgroup                          |                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|  |  | group theory                               |                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|  |  | division                                   |                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|  |  | divided by<br>over                         | $A : B$ means the division of $A$ with $B$ (dividing $A$ by $B$ )                                                                                                                                                                                          | $10 : 2 = 5$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|  |  | everywhere                                 |                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|  |  | vertical ellipsis                          |                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|  |  | vertical ellipsis                          | Denotes that certain constants and terms are missing out (e.g. for clarity) and that only the important terms are being listed.                                                                                                                            | $P(\mathbf{r}, t) = \chi E(\mathbf{r}, t_1) E(\mathbf{r}, t_2) E(\mathbf{r}, t_3)$                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|  |  | everywhere                                 |                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|  |  | wreath product                             | $A \wr H$ means the wreath product of the group $A$ by the group $H$ .                                                                                                                                                                                     | $S_n \wr \mathbb{Z}_2$ is isomorphic to the automorphism group of the complete bipartite graph on $(n, n)$ vertices.                                                                                                                                                                                                                                                                                                                                                                                                                           |
|  |  | wreath product of<br>... by ...            | <i>This may also be written <math>A \wr H</math>.</i>                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|  |  | group theory                               |                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|  |  | downwards zigzag<br>arrow                  |                                                                                                                                                                                                                                                            | $x + 4 = x - 3 \times$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|  |  | contradiction; this<br>contradicts that    | Denotes that contradictory statements have been inferred. For clarity, the exact point of contradiction can be appended.                                                                                                                                   | Statement: Every finite, non-empty, ordered set has a largest element. Otherwise, let's assume that $\mathbf{X}$ is a finite, non-empty, ordered set with no largest element. Then, for some $\mathbf{x}_1 \in \mathbf{X}$ , there exists an $\mathbf{x}_2 \in \mathbf{X}$ with $\mathbf{x}_1 < \mathbf{x}_2$ , but then there's also an $\mathbf{x}_3 \in \mathbf{X}$ with $\mathbf{x}_2 < \mathbf{x}_3$ , and so on. Thus, $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \dots$ are distinct elements in $\mathbf{X}$ . $\mathbf{x}$ is finite. |
|  |  | everywhere                                 |                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|  |  | exclusive or                               |                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|  |  | xor                                        | The statement $A \oplus B$ is true when either A or B, but not both, are true. $A \vee B$ means the same.                                                                                                                                                  | $(\neg A) \oplus A$ is always true, $A \oplus A$ is always false.                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|  |  | propositional<br>logic, Boolean<br>algebra |                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|  |  | direct sum                                 | The direct sum is a special way of combining several objects into one general object.                                                                                                                                                                      | Most commonly, for vector spaces $U$ , $V$ , and $W$ , the following consequence is used:                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|  |  | direct sum of                              | <i>(The bun symbol <math>\oplus</math>, or the coproduct symbol <math>\sqcup</math>, is used; <math>\vee</math> is only for logic.)</i>                                                                                                                    | $U = V \oplus W \Leftrightarrow (U = V + W) \wedge (V \cap W = \{0\})$                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|  |  | abstract algebra                           |                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|  |  | Kulkarni–Nomizu<br>product                 | Derived from the tensor product of two symmetric type (0,2) tensors; it has the algebraic symmetries of the Riemann tensor.                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|  |  | Kulkarni–Nomizu<br>product                 | $f = g \oslash h$ has components                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|  |  | tensor algebra                             | $f_{\alpha\beta\gamma\delta} = g_{\alpha\gamma} h_{\beta\delta} + g_{\beta\delta} h_{\alpha\gamma} - g_{\alpha\delta} h_{\beta\gamma} - g_{\beta\gamma} h_{\alpha\delta}$ .                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|  |  | D'Alembertian;<br>wave operator            | It is the generalisation of the Laplace operator in the sense that it is the differential operator which is invariant under the isometry group of the underlying space and it reduces to the Laplace operator if restricted to time independent functions. | $\square = \frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \frac{\partial^2}{\partial x^2} - \frac{\partial^2}{\partial y^2} - \frac{\partial^2}{\partial z^2}$                                                                                                                                                                                                                                                                                                                                                                                |
|  |  | non-Euclidean<br>Laplacian                 |                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|  |  | vector calculus                            |                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

### Letter-based symbols

Includes upside-down letters.

#### Letter modifiers

Also called diacritics.



2072

| Symbol<br>in<br>HTML | Symbol<br>in TeX | Name                                               | Explanation                                                                                                        | Examples                                                                                                                                             |
|----------------------|------------------|----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
|                      |                  | Read as<br>Category                                |                                                                                                                    |                                                                                                                                                      |
| $\bar{a}$            | $\bar{a}$        | mean                                               |                                                                                                                    |                                                                                                                                                      |
|                      |                  | overbar;<br>... bar                                | $\bar{x}$ (often read as "x bar") is the mean (average value of $x_i$ ).                                           | $x = \{1, 2, 3, 4, 5\}; \bar{x} = 3$                                                                                                                 |
|                      |                  | statistics                                         |                                                                                                                    |                                                                                                                                                      |
|                      |                  | finite sequence,<br>tuple                          |                                                                                                                    |                                                                                                                                                      |
|                      |                  | finite sequence,<br>tuple                          | $\bar{a}$ means the finite sequence/tuple $(a_1, a_2, \dots, a_n)$ .                                               | $\bar{a} := (a_1, a_2, \dots, a_n)$                                                                                                                  |
|                      |                  | model theory                                       |                                                                                                                    |                                                                                                                                                      |
|                      |                  | algebraic closure                                  |                                                                                                                    |                                                                                                                                                      |
|                      |                  | algebraic closure<br>of                            | $\bar{F}$ is the algebraic closure of the field $F$ .                                                              | The field of algebraic numbers is sometimes denoted as $\bar{\mathbb{Q}}$ because it is the algebraic closure of the rational numbers $\mathbb{Q}$ . |
|                      |                  | field theory                                       |                                                                                                                    |                                                                                                                                                      |
| $\hat{a}$            | $\hat{a}$        | complex conjugate<br>conjugate                     | $\bar{z}$ means the complex conjugate of $z$ .                                                                     | $\overline{3 + 4i} = 3 - 4i$                                                                                                                         |
|                      |                  | complex numbers                                    | ( $z^*$ can also be used for the conjugate of $z$ , as described above.)                                           |                                                                                                                                                      |
|                      |                  | topological closure<br>(topological)<br>closure of | $\bar{S}$ is the topological closure of the set $S$ .                                                              | In the space of the real numbers, $\bar{\mathbb{Q}} = \mathbb{R}$ (the rational numbers are dense in the real numbers).                              |
|                      |                  | topology                                           | This may also be denoted as $\text{cl}(S)$ or $\text{Cl}(S)$ .                                                     |                                                                                                                                                      |
| $\hat{a}$            | $\hat{a}$        | unit vector                                        |                                                                                                                    |                                                                                                                                                      |
|                      |                  | hat                                                | $\hat{\mathbf{a}}$ (pronounced "a hat") is the normalized version of vector $\mathbf{a}$ , having length 1.        |                                                                                                                                                      |
|                      |                  | geometry                                           |                                                                                                                    |                                                                                                                                                      |
| $\hat{a}$            | $\hat{a}$        | estimator                                          |                                                                                                                    |                                                                                                                                                      |
|                      |                  | estimator for                                      | $\hat{\theta}$ is the estimator or the estimate for the parameter $\theta$ .                                       | The estimator $\hat{\mu} = \frac{\sum x_i}{n}$ produces a sample estimate $\hat{\mu}(x)$ for the mean $\mu$ .                                        |
|                      |                  | statistics                                         |                                                                                                                    |                                                                                                                                                      |
| $'$                  | $'$              | derivative                                         | $f'(x)$ means the derivative of the function $f$ at the point $x$ , i.e., the slope of the tangent to $f$ at $x$ . | If $f(x) := x^2$ , then $f'(x) = 2x$ .                                                                                                               |
|                      |                  | ... prime;<br>derivative of                        | (The single-quote character ' is sometimes used instead, especially in ASCII text.)                                |                                                                                                                                                      |
|                      |                  | calculus                                           |                                                                                                                    |                                                                                                                                                      |
| $\dot{a}$            | $\dot{a}$        | derivative                                         | $\dot{x}$ means the derivative of $x$ with respect to time. That is                                                | If $x(t) := t^2$ , then $\dot{x}(t) = 2t$ .                                                                                                          |
|                      |                  | ... dot;<br>time derivative of                     | $\dot{x}(t) = \frac{d}{dt}x(t)$ .                                                                                  |                                                                                                                                                      |
|                      |                  | calculus                                           |                                                                                                                    |                                                                                                                                                      |

Symbols based on Latin letters



## F. Some List of Math Symbols



# De La Salle University

2073

| Symbol in HTML               | Symbol in TeX                | Name<br>Read as<br>Category                                                                                                              | Explanation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Examples                                                                                                                                         |
|------------------------------|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| $\forall$                    | $\forall$                    | universal quantification<br>for all;<br>for any;<br>for each;<br>for every<br>predicate logic                                            | $\forall x: P(x)$ means $P(x)$ is true for all $x$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | $\forall n \in \mathbb{N}: n^2 \geq n$ .                                                                                                         |
| $\mathbb{C}$<br>$\mathbb{C}$ | $\mathbb{C}$<br>$\mathbb{C}$ | complex numbers<br>C;<br>the (set of) complex numbers<br>numbers                                                                         | $\mathbb{C}$ means $\{a + bi : a, b \in \mathbb{R}\}$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | $i = \sqrt{-1} \in \mathbb{C}$                                                                                                                   |
| $\mathfrak{c}$               | $\mathfrak{c}$               | cardinality of the continuum<br>cardinality of the continuum;<br>c;<br>cardinality of the real numbers<br>set theory                     | The cardinality of $\mathbb{R}$ is denoted by $ \mathbb{R} $ or by the symbol $\mathfrak{c}$ (a lowercase Fraktur letter C).                                                                                                                                                                                                                                                                                                                                                                                                                                           | $\mathfrak{c} = 2^{\aleph_1}$                                                                                                                    |
| $\partial$                   | $\partial$                   | partial derivative<br>partial;<br>d<br>calculus<br>boundary<br>boundary of<br>topology<br>degree of a polynomial<br>degree of<br>algebra | $\partial f / \partial x_i$ means the partial derivative of $f$ with respect to $x_i$ , where $f$ is a function on $(x_1, \dots, x_n)$ .<br>$\partial M$ means the boundary of $M$ .<br>$\partial f$ means the degree of the polynomial $f$ .<br>(This may also be written $\deg f$ .)                                                                                                                                                                                                                                                                                 | If $f(x, y) := x^2y$ , then $\partial f / \partial x = 2xy$ .<br>$\partial\{x : \ x\  \leq 2\} = \{x : \ x\  = 2\}$<br>$\partial(x^2 - 1) = 2$   |
| $\mathbb{E}$<br>$\mathbb{E}$ | $\mathbb{E}$<br>$\mathbb{E}$ | expected value<br>expected value<br>probability theory                                                                                   | the value of a random variable one would "expect" to find if one could repeat the random variable process an infinite number of times and take the average of the values obtained                                                                                                                                                                                                                                                                                                                                                                                      | $\mathbb{E}[X] = \frac{x_1p_1 + x_2p_2 + \dots + x_kp_k}{p_1 + p_2 + \dots + p_k}$                                                               |
| $\exists$                    | $\exists$                    | existential quantification<br>there exists;<br>there is;<br>there are<br>predicate logic                                                 | $\exists x: P(x)$ means there is at least one $x$ such that $P(x)$ is true.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | $\exists n \in \mathbb{N}: n$ is even.                                                                                                           |
| $\exists!$                   | $\exists!$                   | uniqueness quantification<br>there exists exactly one<br>predicate logic                                                                 | $\exists! x: P(x)$ means there is exactly one $x$ such that $P(x)$ is true.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | $\exists! n \in \mathbb{N}: n + 5 = 2n$ .                                                                                                        |
| $\in$<br>$\notin$            | $\in$<br>$\notin$            | set membership<br>is an element of;<br>is not an element of<br>everywhere, set theory                                                    | $a \in S$ means $a$ is an element of the set $S$ . <sup>[7]</sup> $a \notin S$ means $a$ is not an element of $S$ . <sup>[7]</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                     | $(1/2)^{-1} \in \mathbb{N}$<br>$2^{-1} \notin \mathbb{N}$                                                                                        |
| $\ni$                        | $\ni$                        | set membership<br>does not contain as an element<br>set theory                                                                           | $S \ni e$ means the same thing as $e \in S$ , where $S$ is a set and $e$ is not an element of $S$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                  |
| $\ni$                        | $\ni$                        | such that symbol<br>such that<br>mathematical logic<br>set membership<br>contains as an element<br>set theory                            | often abbreviated as "s.t."; $:$ and $ $ are also used to abbreviate "such that". The use of $\ni$ goes back to early mathematical logic and its usage in this sense is declining.<br>$S \ni e$ means the same thing as $e \in S$ , where $S$ is a set and $e$ is an element of $S$ .                                                                                                                                                                                                                                                                                  | Choose $2 \ni 2$ and $3 \ni 2$ . (Here $ $ is used in the sense of "divides".)                                                                   |
| $\mathbb{H}$<br>$\mathbb{H}$ | $\mathbb{H}$<br>$\mathbb{H}$ | quaternions or Hamiltonian quaternions<br>H;<br>the (set of) quaternions<br>numbers                                                      | $\mathbb{H}$ means $\{a + bi + cj + dk : a, b, c, d \in \mathbb{R}\}$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                  |
| $\mathbb{N}$<br>$\mathbb{N}$ | $\mathbb{N}$<br>$\mathbb{N}$ | natural numbers<br>the (set of) natural numbers<br>numbers                                                                               | $\mathbb{N}$ means either $\{0, 1, 2, 3, \dots\}$ or $\{1, 2, 3, \dots\}$ .<br><i>The choice depends on the area of mathematics being studied; e.g. number theorists prefer the latter; analysts, set theorists and computer scientists prefer the former. To avoid confusion, always check an author's definition of <math>\mathbb{N}</math>.</i><br><i>Set theorists often use the notation <math>\omega</math> (for least infinite ordinal) to denote the set of natural numbers (including zero), along with the standard ordering relation <math>\leq</math>.</i> | $\mathbb{N} = \{ a  : a \in \mathbb{Z}\}$ <b>or</b> $\mathbb{N} = \{ a  > 0 : a \in \mathbb{Z}\}$                                                |
| $\circ$                      | $\circ$                      | Hadamard product<br>entrywise product<br>linear algebra                                                                                  | For two matrices (or vectors) of the same dimensions $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{m \times n}$ the Hadamard product is a matrix of the same dimensions $\mathbf{A} \circ \mathbf{B} \in \mathbb{R}^{m \times n}$ with elements given by $(\mathbf{A} \circ \mathbf{B})_{ij} = (\mathbf{A})_{ij} \cdot (\mathbf{B})_{ij}$ .<br>In MATLAB this operation is expressed by $\mathbf{A}.*\mathbf{B}$ .                                                                                                                                                           | $\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \circ \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 4 \\ 0 & 0 \end{bmatrix}$ |
| $\circ$                      | $\circ$                      | function composition<br>composed with<br>set theory                                                                                      | $f \circ g$ is the function such that $(f \circ g)(x) = f(g(x))$ . <sup>[10]</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | if $f(x) := 2x$ , and $g(x) := x + 3$ , then $(f \circ g)(x) = 2(x + 3)$ .                                                                       |
| $\mathcal{O}$                | $\mathcal{O}$                | Big O notation<br>big-oh of<br>Computational complexity theory                                                                           | The Big O notation describes the limiting behavior of a function, when the argument tends towards a particular value or infinity.                                                                                                                                                                                                                                                                                                                                                                                                                                      | If $f(x) = 6x^4 - 2x^3 + 5$ and $g(x) = x^4$ , then $f(x) = \mathcal{O}(g(x))$ as $x \rightarrow \infty$                                         |

## F. Some List of Math Symbols



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|              |              |                                                                                                              |                                                                                                                                                                                                                                                       |                                                                                                                                                                                        |
|--------------|--------------|--------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\mathbb{P}$ | $\mathbb{P}$ | projective space                                                                                             | $\mathbb{P}$ means a space with a point at infinity.                                                                                                                                                                                                  | $\mathbb{P}^1, \mathbb{P}^2$                                                                                                                                                           |
|              |              | $\mathbb{P}$ ;<br>the projective space;<br>the projective line;<br>the projective plane                      |                                                                                                                                                                                                                                                       |                                                                                                                                                                                        |
| $\mathbb{P}$ | $\mathbb{P}$ | topology                                                                                                     | $P(X)$ means the probability of the event $X$ occurring.<br><i>This may also be written as <math>P(X)</math>, <math>\Pr(X)</math>, <math>P[X]</math> or <math>\Pr[X]</math>.</i>                                                                      | If a fair coin is flipped, $P(\text{Heads}) = P(\text{Tails}) = 0.5$ .                                                                                                                 |
|              |              | probability                                                                                                  |                                                                                                                                                                                                                                                       |                                                                                                                                                                                        |
| $\mathbb{P}$ | $\mathbb{P}$ | the probability of                                                                                           | Given a set $S$ , the power set of $S$ is the set of all subsets of the set $S$ . The power set of $S$ is                                                                                                                                             | The power set $P(\{0, 1, 2\})$ is the set of all subsets of $\{0, 1, 2\}$ . Hence,<br>$P(\{0, 1, 2\}) = \{\emptyset, \{0\}, \{1\}, \{2\}, \{0, 1\}, \{0, 2\}, \{1, 2\}, \{0, 1, 2\}\}$ |
|              |              | probability theory                                                                                           |                                                                                                                                                                                                                                                       |                                                                                                                                                                                        |
| $\mathbb{P}$ | $\mathbb{P}$ | Power set                                                                                                    | denoted by $P(S)$ .                                                                                                                                                                                                                                   |                                                                                                                                                                                        |
|              |              | the Power set of                                                                                             |                                                                                                                                                                                                                                                       |                                                                                                                                                                                        |
| $\mathbb{Q}$ | $\mathbb{Q}$ | numbers                                                                                                      | $\mathbb{Q}$ means $\{p/q : p \in \mathbb{Z}, q \in \mathbb{N}\}$ .                                                                                                                                                                                   | $3.14000... \in \mathbb{Q}$<br>$\pi \notin \mathbb{Q}$                                                                                                                                 |
|              |              | $\mathbb{Q}$ ;<br>the (set of) rational numbers;<br>the rationals                                            |                                                                                                                                                                                                                                                       |                                                                                                                                                                                        |
| $\mathbb{R}$ | $\mathbb{R}$ | real numbers                                                                                                 | $\mathbb{R}$ means the set of real numbers.                                                                                                                                                                                                           | $\pi \in \mathbb{R}$<br>$\sqrt{-1} \notin \mathbb{R}$                                                                                                                                  |
|              |              | $\mathbb{R}$ ;<br>the (set of) real numbers;<br>the reals                                                    |                                                                                                                                                                                                                                                       |                                                                                                                                                                                        |
| $\dagger$    | $\dagger$    | conjugate transpose                                                                                          | $A^\dagger$ means the transpose of the complex conjugate of $A$ . <sup>[11]</sup><br><i>This may also be written <math>A^{*\top}</math>, <math>A^{1*}</math>, <math>A^*</math>, <math>\overline{A}^\top</math> or <math>\overline{A}^\top</math>.</i> | If $A = (a_{ij})$ then $A^\dagger = (\overline{a_{ji}})$ .                                                                                                                             |
|              |              | conjugate transpose;<br>adjoint;<br>Hermitian<br>adjoint/conjugate<br>/transpose/dagger<br>matrix operations |                                                                                                                                                                                                                                                       |                                                                                                                                                                                        |
| $\top$       | $\top$       | transpose                                                                                                    | $A^\top$ means $A$ , but with its rows swapped for columns.<br><i>This may also be written <math>A^t</math>, <math>A^o</math> or <math>A^u</math>.</i>                                                                                                | If $A = (a_{ij})$ then $A^\top = (a_{ji})$ .                                                                                                                                           |
|              |              | transpose<br>matrix operations                                                                               |                                                                                                                                                                                                                                                       |                                                                                                                                                                                        |
| $\top$       | $\top$       | top element                                                                                                  | $\top$ means the largest element of a lattice.                                                                                                                                                                                                        | $\forall x : x \vee \top = \top$                                                                                                                                                       |
|              |              | the top element                                                                                              |                                                                                                                                                                                                                                                       |                                                                                                                                                                                        |
| $\top$       | $\top$       | lattice theory                                                                                               | $\top$ means the top or universal type; every type in the type system of interest is a subtype of top.                                                                                                                                                | $\forall$ types $T, T' : T < \top$                                                                                                                                                     |
|              |              | top type                                                                                                     |                                                                                                                                                                                                                                                       |                                                                                                                                                                                        |
| $\perp$      | $\perp$      | the top type; top                                                                                            |                                                                                                                                                                                                                                                       |                                                                                                                                                                                        |
|              |              | type theory                                                                                                  |                                                                                                                                                                                                                                                       |                                                                                                                                                                                        |
| $\perp$      | $\perp$      | perpendicular                                                                                                | $x \perp y$ means $x$ is perpendicular to $y$ ; or more generally $x$ is orthogonal to $y$ .                                                                                                                                                          | If $l \perp m$ and $m \perp n$ in the plane, then $l \parallel n$ .                                                                                                                    |
|              |              | is perpendicular to                                                                                          |                                                                                                                                                                                                                                                       |                                                                                                                                                                                        |
| $\perp$      | $\perp$      | geometry                                                                                                     | $W^\perp$ means the orthogonal complement of $W$ (where $W$ is a subspace of the inner product space $V$ ), the set of all vectors in $V$ orthogonal to every vector in $W$ .                                                                         | Within $\mathbb{R}^3$ , $(\mathbb{R}^2)^\perp \cong \mathbb{R}$                                                                                                                        |
|              |              | orthogonal complement                                                                                        |                                                                                                                                                                                                                                                       |                                                                                                                                                                                        |
| $\perp$      | $\perp$      | orthogonal/<br>perpendicular<br>complement of;<br>perp                                                       | $x \perp y$ means $x$ has no factor greater than 1 in common with $y$ .                                                                                                                                                                               | $34 \perp 55$                                                                                                                                                                          |
|              |              | linear algebra                                                                                               |                                                                                                                                                                                                                                                       |                                                                                                                                                                                        |
| $\perp$      | $\perp$      | coprime                                                                                                      | $A \perp B$ means $A$ is an event whose probability is independent of event $B$ .                                                                                                                                                                     | If $A \perp B$ , then $P(A B) = P(A)$ .                                                                                                                                                |
|              |              | is coprime to                                                                                                |                                                                                                                                                                                                                                                       |                                                                                                                                                                                        |
| $\perp$      | $\perp$      | number theory                                                                                                | $\perp$ means the smallest element of a lattice.                                                                                                                                                                                                      | $\forall x : x \wedge \perp = \perp$                                                                                                                                                   |
|              |              | independent                                                                                                  |                                                                                                                                                                                                                                                       |                                                                                                                                                                                        |
| $\perp$      | $\perp$      | is independent of                                                                                            | $\perp$ means the bottom type (a.k.a. the zero type or empty type); bottom is the subtype of every type in the type system.                                                                                                                           | $\forall$ types $T, \perp : T$                                                                                                                                                         |
|              |              | probability                                                                                                  |                                                                                                                                                                                                                                                       |                                                                                                                                                                                        |
| $\perp$      | $\perp$      | bottom element                                                                                               | $x \perp y$ means that $x$ is comparable to $y$ .                                                                                                                                                                                                     | $\{e, \pi\} \perp \{1, 2, e, 3, \pi\}$ under set containment.                                                                                                                          |
|              |              | the bottom element                                                                                           |                                                                                                                                                                                                                                                       |                                                                                                                                                                                        |
| $\perp$      | $\perp$      | lattice theory                                                                                               |                                                                                                                                                                                                                                                       |                                                                                                                                                                                        |
|              |              | bottom type                                                                                                  |                                                                                                                                                                                                                                                       |                                                                                                                                                                                        |
| $\perp$      | $\perp$      | the bottom type;<br>bot                                                                                      |                                                                                                                                                                                                                                                       |                                                                                                                                                                                        |
|              |              | type theory                                                                                                  |                                                                                                                                                                                                                                                       |                                                                                                                                                                                        |
| $\perp$      | $\perp$      | comparability                                                                                                | $A \cup B$ means the set of those elements which are either in $A$ , or in $B$ , or in both. <sup>[5]</sup>                                                                                                                                           | $A \subseteq B \Leftrightarrow (A \cup B) = B$                                                                                                                                         |
|              |              | is comparable to                                                                                             |                                                                                                                                                                                                                                                       |                                                                                                                                                                                        |
| $\cup$       | $\cup$       | order theory                                                                                                 | $A \cap B$ means the set that contains all those elements that $A$ and $B$ have in common. <sup>[5]</sup>                                                                                                                                             | $\{x \in \mathbb{R} : x^2 = 1\} \cap \mathbb{N} = \{1\}$                                                                                                                               |
|              |              | set-theoretic union                                                                                          |                                                                                                                                                                                                                                                       |                                                                                                                                                                                        |
| $\cup$       | $\cup$       | the union of ... or ...;<br>union                                                                            | The statement $A \vee B$ is true if $A$ or $B$ (or both) are true; if both are false, the statement is false.<br>For functions $A(x)$ and $B(x)$ , $A(x) \vee B(x)$ is used to mean $\max(A(x), B(x))$ .                                              | $n \geq 4 \vee n \leq 2 \Leftrightarrow n \neq 3$ when $n$ is a natural number.                                                                                                        |
|              |              | set theory                                                                                                   |                                                                                                                                                                                                                                                       |                                                                                                                                                                                        |
| $\cap$       | $\cap$       | set-theoretic intersection                                                                                   | The statement $A \wedge B$ is true if $A$ and $B$ are both true; else it is false.<br>For functions $A(x)$ and $B(x)$ , $A(x) \wedge B(x)$ is used to mean $\min(A(x), B(x))$ .                                                                       | $n < 4 \wedge n > 2 \Leftrightarrow n = 3$ when $n$ is a natural number.                                                                                                               |
|              |              | intersected with;<br>intersect                                                                               |                                                                                                                                                                                                                                                       |                                                                                                                                                                                        |
| $\vee$       | $\vee$       | set theory                                                                                                   |                                                                                                                                                                                                                                                       |                                                                                                                                                                                        |
|              |              | logical disjunction or<br>join in a lattice                                                                  |                                                                                                                                                                                                                                                       |                                                                                                                                                                                        |
| $\vee$       | $\vee$       | or;<br>max;<br>join                                                                                          |                                                                                                                                                                                                                                                       |                                                                                                                                                                                        |
|              |              | propositional logic,<br>lattice theory                                                                       |                                                                                                                                                                                                                                                       |                                                                                                                                                                                        |
| $\wedge$     | $\wedge$     | logical conjunction or<br>meet in a lattice                                                                  |                                                                                                                                                                                                                                                       |                                                                                                                                                                                        |
|              |              | and;                                                                                                         |                                                                                                                                                                                                                                                       |                                                                                                                                                                                        |

## F. Some List of Math Symbols



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|  |  |  |                                                                                  |                                                                                                                                                                                                 |                                                                                       |
|--|--|--|----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
|  |  |  | min;<br>meet                                                                     |                                                                                                                                                                                                 |                                                                                       |
|  |  |  | propositional logic;<br>lattice theory                                           |                                                                                                                                                                                                 |                                                                                       |
|  |  |  | wedge product                                                                    | $u \wedge v$ means the wedge product of any multivectors $u$ and $v$ . In three-dimensional Euclidean space the wedge product and the cross product of two vectors are each other's Hodge dual. | $u \wedge v = \ast(u \times v)$ if $u, v \in \mathbb{R}^3$                            |
|  |  |  | wedge product;<br>exterior product                                               |                                                                                                                                                                                                 |                                                                                       |
|  |  |  | exterior algebra                                                                 |                                                                                                                                                                                                 |                                                                                       |
|  |  |  | exponentiation                                                                   | $a^b$ means $a$ raised to the power of $b$                                                                                                                                                      | $2^3 = 2^3 = 8$                                                                       |
|  |  |  | ... (raised) to the power<br>of ...<br>everywhere                                | $(a^b)^c$ is more commonly written $a^{bc}$ . The symbol $^$ is generally used in programming languages where ease of typing and use of plain ASCII text is preferred.)                         |                                                                                       |
|  |  |  | multiplication                                                                   | $3 \times 4$ means the multiplication of 3 by 4.                                                                                                                                                | $7 \times 8 = 56$                                                                     |
|  |  |  | times;<br>multiplied by                                                          | (The symbol $\times$ is generally used in programming languages, where ease of typing and use of ASCII text is preferred.)                                                                      |                                                                                       |
|  |  |  | arithmetic                                                                       |                                                                                                                                                                                                 |                                                                                       |
|  |  |  | Cartesian product                                                                |                                                                                                                                                                                                 |                                                                                       |
|  |  |  | the Cartesian product of<br>... and ...;<br>the direct product of ...<br>and ... | $X \times Y$ means the set of all ordered pairs with the first element of each pair selected from $X$ and the second element selected from $Y$ .                                                | $\{1,2\} \times \{3,4\} = \{(1,3), (1,4), (2,3), (2,4)\}$                             |
|  |  |  | set theory                                                                       |                                                                                                                                                                                                 |                                                                                       |
|  |  |  | cross product                                                                    |                                                                                                                                                                                                 |                                                                                       |
|  |  |  | cross                                                                            | $u \times v$ means the cross product of vectors $u$ and $v$                                                                                                                                     | $(1,2,5) \times (3,4,-1) = (-22, 16, -2)$                                             |
|  |  |  | linear algebra                                                                   |                                                                                                                                                                                                 |                                                                                       |
|  |  |  | group of units                                                                   | $R^\times$ consists of the set of units of the ring $R$ , along with the operation of multiplication.                                                                                           | $(\mathbb{Z}/6\mathbb{Z})^\times = \{[1], [2], [3], [4]\} \cong \mathbb{C}_4$         |
|  |  |  | the group of units of                                                            |                                                                                                                                                                                                 |                                                                                       |
|  |  |  | ring theory                                                                      | This may also be written $R^\times$ as described below, or $U(R)$ .                                                                                                                             |                                                                                       |
|  |  |  | tensor product, tensor<br>product of modules                                     | $V \otimes U$ means the tensor product of $V$ and $U$ . $V \otimes_R U$ means the tensor product of modules $V$ and $U$ over the ring $R$ .                                                     | $\{1, 2, 3, 4\} \otimes \{1, 1, 2\} = \{(1, 1, 2), (2, 2, 4), (3, 3, 6), (4, 4, 8)\}$ |
|  |  |  | tensor product of                                                                |                                                                                                                                                                                                 |                                                                                       |
|  |  |  | linear algebra                                                                   |                                                                                                                                                                                                 |                                                                                       |
|  |  |  | semidirect product                                                               | $N \rtimes H$ is the semidirect product of $N$ (a normal subgroup) and $H$ (a subgroup), with respect to $\phi$ . Also, if $G = N \rtimes H$ , then $G$ is said to split over $N$ .             | $D_{2n} \cong \mathbb{C}_n \rtimes \mathbb{C}_2$                                      |
|  |  |  | the semidirect product of                                                        |                                                                                                                                                                                                 |                                                                                       |
|  |  |  | group theory                                                                     | ( $\rtimes$ may also be written the other way round, as $\ltimes$ , or as $\ltimes$ .)                                                                                                          |                                                                                       |
|  |  |  | semijoin                                                                         | $R \ltimes S$ is the semijoin of the relations $R$ and $S$ , the set of all tuples in $R$ for which there is a tuple in $S$ that is equal on their common attribute names.                      | $R \ltimes S = \bigcup_{i_1, \dots, i_n} (R \bowtie_i S)$                             |
|  |  |  | the semijoin of                                                                  |                                                                                                                                                                                                 |                                                                                       |
|  |  |  | relational algebra                                                               |                                                                                                                                                                                                 |                                                                                       |
|  |  |  | natural join                                                                     | $R \bowtie S$ is the natural join of the relations $R$ and $S$ , the set of all combinations of tuples in $R$ and $S$ that are equal on their common attribute names.                           |                                                                                       |
|  |  |  | the natural join of                                                              |                                                                                                                                                                                                 |                                                                                       |
|  |  |  | relational algebra                                                               |                                                                                                                                                                                                 |                                                                                       |
|  |  |  | integers                                                                         | $\mathbb{Z}$ means $\{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$ .                                                                                                                                 | $\mathbb{Z} = \{p, -p : p \in \mathbb{N} \cup \{0\}\}$                                |
|  |  |  | the (set of) integers                                                            | $\mathbb{Z}^+$ or $\mathbb{Z}^{\geq}$ means $\{1, 2, 3, \dots\}$ .                                                                                                                              |                                                                                       |
|  |  |  | numbers                                                                          | $\mathbb{Z}^*$ or $\mathbb{Z}^{\neq}$ means $\{0, 1, 2, 3, \dots\}$ .                                                                                                                           |                                                                                       |
|  |  |  | integers mod $n$                                                                 | $\mathbb{Z}_n$ means $\{[0], [1], [2], \dots, [n-1]\}$ with addition and multiplication modulo $n$ .                                                                                            | $\mathbb{Z}_3 = \{[0], [1], [2]\}$                                                    |
|  |  |  | the (set of) integers<br>modulo $n$                                              | Note that any letter may be used instead of $n$ , such as $p$ . To avoid confusion with $p$ -adic numbers, use $\mathbb{Z}/p\mathbb{Z}$ or $\mathbb{Z}/(p)$ instead.                            |                                                                                       |
|  |  |  | numbers                                                                          |                                                                                                                                                                                                 |                                                                                       |
|  |  |  | $p$ -adic integers                                                               |                                                                                                                                                                                                 |                                                                                       |
|  |  |  | the (set of) $p$ -adic<br>integers                                               |                                                                                                                                                                                                 |                                                                                       |
|  |  |  | numbers                                                                          | Note that any letter may be used instead of $p$ , such as $n$ or $l$ .                                                                                                                          |                                                                                       |

Symbols based on Hebrew or Greek letters

## F. Some List of Math Symbols



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| Symbol in HTML | Symbol in TeX | Name<br>Read as<br>Category                     | Explanation                                                                                                                                                                                                                                                                                                                  | Examples                                                                                                                                       |
|----------------|---------------|-------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| $\aleph$       | $\aleph$      | aleph number<br>aleph<br>set theory             | $\aleph_\alpha$ represents an infinite cardinality (specifically, the $\alpha$ -th one, where $\alpha$ is an ordinal).                                                                                                                                                                                                       | $ \mathbb{N}  = \aleph_0$ , which is called aleph-null.                                                                                        |
| $\beth$        | $\beth$       | beth number<br>beth<br>set theory               | $\beth_\alpha$ represents an infinite cardinality (similar to $\aleph$ , but $\beth$ does not necessarily index all of the numbers indexed by $\aleph$ ).                                                                                                                                                                    | $\beth_1 =  P(\mathbb{N})  = 2^{\aleph_0}$ .                                                                                                   |
| $\delta$       | $\delta$      | Dirac delta function                            | $\delta(x) = \begin{cases} \infty, & x = 0 \\ 0, & x \neq 0 \end{cases}$                                                                                                                                                                                                                                                     | $\delta(x)$                                                                                                                                    |
|                |               | Dirac delta of hyperfunction                    |                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                |
|                |               | Kronecker delta                                 | $\delta_{ij} = \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases}$                                                                                                                                                                                                                                                        | $\delta_{ij}$                                                                                                                                  |
|                |               | Kronecker delta of hyperfunction                |                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                |
|                |               | Functional derivative                           | $\left\langle \frac{\delta F[\varphi(x)]}{\delta \varphi(x)}, f(x) \right\rangle = \int \frac{\delta F[\varphi(x)]}{\delta \varphi(x')} f(x') dx'$<br>$= \lim_{\epsilon \rightarrow 0} \frac{F[\varphi(x) + \epsilon f(x)] - F[\varphi(x)]}{\epsilon}$<br>$= \frac{d}{d\epsilon} F[\varphi + \epsilon f] \Big _{\epsilon=0}$ | $\frac{\delta V(r)}{\delta \rho(r')} = \frac{1}{4\pi\epsilon_0  r - r' }$                                                                      |
| $\Delta$       | $\Delta$      | symmetric difference                            | $A \Delta B$ (or $A \ominus B$ ) means the set of elements in exactly one of $A$ or $B$ .                                                                                                                                                                                                                                    | $\{1,5,6,8\} \Delta \{2,5,8\} = \{1,2,6\}$                                                                                                     |
| $\Theta$       | $\Theta$      | symmetric difference<br>set theory              | (Not to be confused with delta, $\Delta$ , described below.)                                                                                                                                                                                                                                                                 | $\{3,4,5,6\} \Theta \{1,2,5,6\} = \{1,2,3,4\}$                                                                                                 |
| $\Delta$       | $\Delta$      | delta                                           | $\Delta x$ means a (non-infinitesimal) change in $x$ .                                                                                                                                                                                                                                                                       |                                                                                                                                                |
|                |               | delta; change in calculus                       | (If the change becomes infinitesimal, $\delta$ and even $d$ are used instead. Not to be confused with the symmetric difference, written $\Delta$ , above.)                                                                                                                                                                   | $\frac{\Delta y}{\Delta x}$ is the gradient of a straight line.                                                                                |
|                |               | Laplacian                                       |                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                |
| $\nabla$       | $\nabla$      | Laplace operator                                | The Laplace operator is a second order differential operator in n-dimensional Euclidean space                                                                                                                                                                                                                                | If $f$ is a twice-differentiable real-valued function, then the Laplacian of $f$ is defined by $\Delta f = \nabla^2 f = \nabla \cdot \nabla f$ |
|                |               | vector calculus                                 |                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                |
|                |               | gradient                                        |                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                |
|                |               | del; nabla; gradient of                         | $\nabla f(x_1, \dots, x_n)$ is the vector of partial derivatives $(\partial f / \partial x_1, \dots, \partial f / \partial x_n)$ .                                                                                                                                                                                           | If $f(x,y,z) := 3xy + z^2$ , then $\nabla f = (3y, 3x, 2z)$                                                                                    |
|                |               | vector calculus                                 |                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                |
| $\nabla$       | $\nabla$      | divergence                                      |                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                |
|                |               | del dot; divergence of vector calculus          | $\nabla \cdot \vec{v} = \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z}$                                                                                                                                                                                                 | If $\vec{v} := 3xy\mathbf{i} + y^2xz\mathbf{j} + 5\mathbf{k}$ , then $\nabla \cdot \vec{v} = 3y + 2yz$ .                                       |
|                |               | curl                                            | $\nabla \times \vec{v} = \left( \frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z} \right) \mathbf{i} + \left( \frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x} \right) \mathbf{j} + \left( \frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} \right) \mathbf{k}$            | If $\vec{v} := 3xy\mathbf{i} + y^2xz\mathbf{j} + 5\mathbf{k}$ , then $\nabla \times \vec{v} = -y^2\mathbf{i} - 3xz\mathbf{k}$ .                |
| $\pi$          | $\pi$         | Pi                                              | Used in various formulas involving circles; $\pi$ is equivalent to the amount of area a circle would take up in a square of equal width with an area of 4 square units, roughly 3.14159. It is also the ratio of the circumference to the diameter of a circle.                                                              | $A = \pi R^2 = 314.16 \rightarrow R = 10$                                                                                                      |
|                |               | projection                                      |                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                |
|                |               | Projection of relational algebra                | $\pi_{a_1, \dots, a_n}(R)$ restricts $R$ to the $\{a_1, \dots, a_n\}$ attribute set.                                                                                                                                                                                                                                         | $\pi_{Age, Weight}(Person)$                                                                                                                    |
|                |               | Homotopy group                                  |                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                |
| $\Pi$          | $\Pi$         | the $n$ th Homotopy group of                    | $\pi_n(X)$ consists of homotopy equivalence classes of base point preserving maps from an n-dimensional sphere (with base point) into the pointed space X.                                                                                                                                                                   | $\pi_1(\mathcal{S}^1) = \pi_1(\mathcal{S}^1) \oplus \pi_{1-1}(\mathcal{S}^1)$                                                                  |
|                |               | Homotopy theory                                 |                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                |
|                |               | product                                         |                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                |
| $\Pi$          | $\Pi$         | product over ... from ... to ... of             | $\prod_{k=1}^n a_k$ means $a_1 a_2 \dots a_n$ .                                                                                                                                                                                                                                                                              | $\prod_{k=1}^4 (k+2) = (1+2)(2+2)(3+2)(4+2) = 3 \times 4 \times 5 \times 6 = 360$                                                              |
|                |               | arithmetic                                      |                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                |
|                |               | Cartesian product                               | $\prod_{i=1}^n X_i$ means the set of all (n+1)-tuples $(y_0, \dots, y_n)$ .                                                                                                                                                                                                                                                  | $\prod_{i=1}^3 \mathbb{R} = \mathbb{R} \times \mathbb{R} \times \mathbb{R} = \mathbb{R}^3$                                                     |
| $\sqcup$       | $\sqcup$      | the Cartesian product of; the direct product of |                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                |
|                |               | set theory                                      |                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                |
| $\sqcup$       | $\sqcup$      | coproduct                                       | A general construction which subsumes the disjoint union of sets and of topological spaces, the free product of groups, and the direct sum of modules and vector spaces. The coproduct of a family of objects is                                                                                                             |                                                                                                                                                |



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|             |             |                                                                  |                                                                                                                                                                                                                                                                                  |                                                                                        |
|-------------|-------------|------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
|             |             | to ... of<br>category<br>theory                                  | essentially the "least specific" object to which each object in the family admits a morphism.                                                                                                                                                                                    |                                                                                        |
| $\sigma$    | $\sigma$    | selection<br>Selection of<br>relational<br>algebra               | The selection $\sigma_{ab}(R)$ selects all those tuples in $R$ for which $\theta$ holds between the $a$ and the $b$ attribute. The selection $\sigma_{a\theta b}(R)$ selects all those tuples in $R$ for which $\theta$ holds between the $a$ attribute and the value $\theta$ . | $\sigma_{Age \geq 34}(\text{Person})$<br>$\sigma_{Age = \text{Weight}}(\text{Person})$ |
| $\Sigma$    | $\Sigma$    | summation<br>sum over ...<br>from ... to ...<br>of<br>arithmetic | $\sum_{i=1}^n a_i$ means $a_1 + a_2 + \dots + a_n$ .                                                                                                                                                                                                                             | $\sum_{k=1}^4 k^2 = 1^2 + 2^2 + 3^2 + 4^2 = 1 + 4 + 9 + 16 = 30$                       |
| $\emptyset$ | $\emptyset$ | empty set                                                        |                                                                                                                                                                                                                                                                                  |                                                                                        |
| $\{\}$      | $\{\}$      | the empty set<br>set theory                                      | $\emptyset$ means the set with no elements. <sup>[7]</sup> $\{\}$ means the same.                                                                                                                                                                                                | $\{n \in \mathbb{N} : 1 < n^2 < 4\} = \emptyset$                                       |

### Variations

In mathematics written in Arabic, some symbols may be reversed to make right-to-left writing and reading easier.<sup>[13]</sup>

### See also

- Greek letters used in mathematics, science, and engineering
- Diacritic
- ISO 31-11 (Mathematical signs and symbols for use in physical sciences and technology)
- Latin letters used in mathematics
- List of mathematical abbreviations
- List of mathematical symbols by subject
- Mathematical Alphanumeric Symbols (Unicode block)
- Mathematical constants and functions
- Mathematical notation
- Mathematical operators and symbols in Unicode
- Notation in probability and statistics
- Physical constants
- Table of logic symbols
- Table of mathematical symbols by introduction date
- Typographical conventions in mathematical formulae

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### External links

- The complete set of mathematics Unicode characters (<http://krestavilis.com/math.php>)
- Jeff Miller: *Earliest Uses of Various Mathematical Symbols* (<http://jeff560.tripod.com/mathsym.html>)
- Numericana: *Scientific Symbols and Icons* (<http://www.numericana.com/answer/symbol.htm>)
- GIF and PNG Images for Math Symbols (<http://us.metamath.org/symbols/symbols.html>)
- Mathematical Symbols in Unicode (<http://tit.psu.edu/suggestions/international/bylanguage/math.html#browsers>)
- Using Greek and special characters from Symbol font in HTML (<http://www.alanwood.net/demos/symbol.html>)
- Unicode Math Symbols (<http://mathsymbols.net/>) - a quick form for using unicode math symbols.
- DeTeXify handwritten symbol recognition (<http://detexify.kirelabs.org/classify.html>) — doodle a symbol in the box, and the program will tell you what its name is
- Handbook for Spoken Mathematics ([http://web.efzg.hr/dok/MAT/vkojic/Larrys\\_speakeasy.pdf](http://web.efzg.hr/dok/MAT/vkojic/Larrys_speakeasy.pdf)) — pronunciation guide to many commonly used symbols

Some Unicode charts of mathematical operators:

- Index of Unicode symbols (<http://www.unicode.org/charts/#symbols>)
- Range 2100–214F: Unicode Letterlike Symbols (<http://www.unicode.org/charts/PDF/U2100.pdf>)
- Range 2190–21FF: Unicode Arrows (<http://www.unicode.org/charts/PDF/U2190.pdf>)
- Range 2200–22FF: Unicode Mathematical Operators (<http://www.unicode.org/charts/PDF/U2200.pdf>)
- Range 27C0–27EF: Unicode Miscellaneous Mathematical Symbols–A (<http://www.unicode.org/charts/PDF/U27C0.pdf>)
- Range 2980–29FF: Unicode Miscellaneous Mathematical Symbols–B (<http://www.unicode.org/charts/PDF/U2980.pdf>)
- Range 2A00–2AFF: Unicode Supplementary Mathematical Operators (<http://www.unicode.org/charts/PDF/U2A00.pdf>)

Some Unicode cross-references:

- Short list of commonly used LaTeX symbols ([http://www.artofproblemsolving.com/wiki/index.php/LaTeX\\_Symbols](http://www.artofproblemsolving.com/wiki/index.php/LaTeX_Symbols)) and Comprehensive LaTeX Symbol List (<http://mirrors.med.harvard.edu/ctan/info/symbols/comprehensive/>)
- MathML Characters (<http://www.robinlionheart.com/stds/html4/entities-mathml>) - sorts out Unicode, HTML and MathML/TeX names on one page
- Unicode values and MathML names (<http://www.w3.org/TR/REC-MathML/chap6/bycodes.html>)
- Unicode values and Postscript names (<http://svn.ghostscript.com/ghostscript/branches/gs-db/Resource/Decoding/Unicode>) from the source code for Ghostscript

Retrieved from "https://en.wikipedia.org/w/index.php?title=List\_of\_mathematical\_symbols&oldid=730409871"

Categories: Mathematical notation | Mathematics-related lists | Mathematical symbols | Mathematical tables | Mathematical logic | Lists of symbols

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## Appendix G

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## DISLAYING MATH EXPRESSIONS



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## Help:Displaying a formula

From Wikipedia, the free encyclopedia

*"WP:MATH" and "WP:MATHS" redirect here. For the WikiProject on mathematics, see Wikipedia:WikiProject Mathematics. For Wikipedia's mathematics style manual, see Wikipedia:Manual of Style/Mathematics. For the mathematics reference desk, see Wikipedia:Reference desk/Mathematics.*

MediaWiki renders mathematical equations using a combination of html markup and a variant of LaTeX.

The version of LaTeX used is a subset of AMS-LaTeX markup, a superset of LaTeX markup which is in turn a superset of TeX markup, for mathematical formulae. Only a limited part of the full TeX language is supported; see below for details.<sup>[a]</sup>

By default SVG images with non-visible MathML are generated. The older PNG images can be set via user preferences.<sup>[b]</sup> On some browsers like Firefox, it is possible to use MathML for display via extensions; see the main extension page at mw:Extension:Math for details. Client side MathJax is no longer supported.

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## Basics

Math markup goes inside `<math>...</math>`. Chemistry markup goes inside `<math chem>...</math chem>` or `<ce>...</ce>`. All these tags use TeX.

The TeX code has to be put literally: MediaWiki templates, predefined templates, and parameters cannot be used within math tags: pairs of double braces are ignored and `"#"` gives an error message. However, math tags work in the then and else part of `#if`, etc. See `m:Template:Demo_of_attempt_to_use_parameters_within_TeX` (backlinks edit ([https://meta.wikimedia.org/wiki/Template:Demo\\_of\\_attempt\\_to\\_use\\_parameters\\_within\\_TeX?action=edit](https://meta.wikimedia.org/wiki/Template:Demo_of_attempt_to_use_parameters_within_TeX?action=edit))) for more information.





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### LaTeX commands

LaTeX commands are case-sensitive, and take one of the following two formats:

- They start with a backslash `\` and then have a name consisting of letters only. Command names are terminated by a space, a number or any other "non-letter".
- They consist of a backslash `\` and exactly one non-letter.

Some commands need an argument, which has to be given between curly braces `{ }` after the command name. Some commands support optional parameters, which are added after the command name in square brackets `[]`. The general syntax is:

```
\commandname[option1,option2,...](argument1)(argument2)...
```

### Special characters

The following symbols are reserved characters that either have a special meaning under LaTeX or are unavailable in all the fonts. If you enter them directly in your text, they will normally not render, but rather do things you did not intend.

```
# $ % ^ & _ { } ~ \
```

These characters can be entered by adding a prefix backslash or using special sequences:

```
\# \$ \% ^\wedge \& \_ \{ \} \sim \backslash
```

yielding

```
\# \$ \% ^\wedge \& \_ \{ \} \sim \backslash
```

The backslash character `\` can *not* be entered by adding another backslash in front of it (`\\`); this sequence is used for line breaking. For introducing a backslash in math mode, you can use `\backslash` instead which gives `\`.

The command `\tilde` produces a tilde which is placed over the next letter. For example, `\tilde{a}` gives  $\tilde{a}$ . To produce just a tilde character `~`, use `\tilde{ }` which gives `~`, placing a `~` over an empty box. Alternatively `\sim` produces  $\sim$ , a large centred `~` which may be more appropriate in some situations.

The command `\hat` produces a hat over the next character, for example `\hat{o}` produces  $\hat{o}$ . For a stretchable version use `\widehat{abc}` giving  $\widehat{abc}$ . The wedge `\wedge` is normally used as a mathematical operator  $\wedge$  the sequence `^\wedge` produces  $\wedge$  the best equivalent to the ascii caret `^` character.

### Spaces

"Whitespace" characters, such as blank or tab, are treated uniformly as "space" by LaTeX. Several consecutive whitespace characters are treated as one "space". See below for commands that produces spaces of different size.

### LaTeX environments

*Environments* in LaTeX have a role that is quite similar to commands, but they usually have effect on a wider part of formula. Their syntax is:

```
\begin{environmentname}
  text to be influenced
\end{environmentname}
```

Environments supported by Wikipedia include *matrix*, *align*, etc. See below.

### Rendering

By default, the PNG images are rendered black on white, with a transparent background. On darker backgrounds, the characters may show white edges. To remove these, match the PNG background color with the background color of the page using `\pagecolor`. However, black text on a dark background is hard to read and should be avoided altogether where possible.

$$e^{i\pi} + 1 = 0 \quad e^{i\pi} + 1 = 0 \quad e^{i\pi} + 1 = 0$$

The colors, as well as font sizes and types, are independent of browser settings or CSS. Font sizes and types will often deviate from what HTML renders. Vertical alignment with the surrounding text can also be a problem; a work-around is described in the "Alignment with normal text flow" section below. The css selector of the images is `img.tex`.



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The alt text of the PNG images, which is displayed to visually impaired and other readers who cannot see the images, and is also used when the text is selected and copied, defaults to the wikitext that produced the image, excluding the `<math>` and `</math>`. You can override this by explicitly specifying an `alt` attribute for the `math` element. For example, `<math alt="Square root of pi">\sqrt{\pi}</math>` generates an image  $\sqrt{\pi}$  whose alt text is "Square root of pi". This should not be confused with the `title` attribute that produces popup text when the hovering over the PNG image, for example `<math title="pi">\pi</math>` generates an image  $\pi$  whose popup text is "pi".

Apart from function and operator names, as is customary in mathematics, variables and letters are in italics; digits are not. For other text, (like variable labels) to avoid being rendered in italics like variables, use `\text`, `\mbox`, or `\mathrm`. You can also define new function names using `\operatorname{...}`. For example, `\text{abc}` gives *abc*. `\operatorname{...}` provides spacing before and after the operator name when appropriate, as when `a\operatorname{sn}b` is rendered as *a sn b* (with space to the left and right of "sn") and `a\operatorname{sn}(b+c)` as *a sn(b + c)* (with space to the left and not to the right).

Latex does not have full support for Unicode characters and not all characters render. Most Latin characters with accents render correctly. However some do not, in particular those that include multiple diacritics (e.g. with Latin letters used in Vietnamese) or that cannot be precomposed into a single character (such as the uppercase Latin letter W with ring), or that use other diacritics (like the ogonek or the double grave accent, used in Central European languages like Polish, or the horn attached above some vowels in Vietnamese), or other modified letter forms (used in IPA notations, or African languages, or in medieval texts), some digram ligatures (like IJ in Dutch), or Latin letters borrowed from Greek, or small capitals, as well as superscripts and subscript letters. For example, `\text{ø}` or `\mbox{ø}`, and `\text{þ}` or `\mbox{þ}` (used in Icelandic) will give errors.

Force-rerendering of formulas

MediaWiki stores rendered formulas in a cache so that the images of those formulas do not need to be created each time the page is opened by a user. To force the rerendering of all formulas of a page, you must open it with the getter variables `action=purge&mathpurge=true`. Imagine for example there is a wrong rendered formula in the article Integral. To force the rerendering of this formula you need to open the URL <https://en.wikipedia.org/w/index.php?title=Integral&action=purge&mathpurge=true>. Afterwards you need to bypass your browser cache so that the new created images of the formulas are actually downloaded. See also mw:Extension:Math#Purging pages that contain equations for more details.

TeX vs HTML

Main page: *Wikipedia:Rendering math*

Before using TeX markup for producing special characters, it should be noted that, as this comparison table shows, sometimes similar results can be achieved in HTML using Template:Math. See also Help:Special characters.

| TeX syntax                | TeX rendering  | HTML syntax                                                                   | HTML rendering |
|---------------------------|----------------|-------------------------------------------------------------------------------|----------------|
| <code>\alpha</code>       | $\alpha$       | <code>{{math ''\&amp;alpha;''}}</code>                                        | $\alpha$       |
| <code>f(x) = x^2</code>   | $f(x) = x^2$   | <code>{{math ''f''(''x'') { {=} } ''x''&lt;sup&gt;2&lt;/sup&gt;}}</code>      | $f(x) = x^2$   |
| <code>\sqrt{2}</code>     | $\sqrt{2}$     | <code>{{math {{radical} 2}}}</code>                                           | $\sqrt{2}$     |
| <code>\sqrt{1-e^2}</code> | $\sqrt{1-e^2}$ | <code>{{math {{radical} 1 &amp;minus; ''e''&lt;sup&gt;2&lt;/sup&gt;}}}</code> | $\sqrt{1-e^2}$ |

The codes on the left produce the symbols on the right, but the latter can also be put directly in the wikitext, except for ‘=’.



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| HTML syntax                                                                                                                                                                                                                                                                                    | Rendering                                                                                                                                                                                                                                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\alpha$<br>$\beta$<br>$\gamma$<br>$\delta$<br>$\epsilon$<br>$\zeta$<br>$\eta$<br>$\theta$<br>$\iota$<br>$\kappa$<br>$\lambda$<br>$\mu$<br>$\nu$<br>$\xi$<br>$\omicron$<br>$\pi$<br>$\rho$<br>$\sigma$<br>$\varsigma$<br>$\tau$<br>$\upsilon$<br>$\phi$<br>$\chi$<br>$\psi$<br>$\omega$        | $\alpha$ $\beta$ $\gamma$ $\delta$ $\epsilon$ $\zeta$<br>$\eta$ $\theta$ $\iota$ $\kappa$ $\lambda$ $\mu$ $\nu$<br>$\xi$ $\omicron$ $\pi$ $\rho$ $\sigma$ $\varsigma$<br>$\tau$ $\upsilon$ $\phi$ $\chi$ $\psi$ $\omega$                 |
| $\Gamma$<br>$\Delta$<br>$\Theta$<br>$\Lambda$<br>$\Xi$<br>$\Pi$<br>$\Sigma$<br>$\Phi$<br>$\Psi$<br>$\Omega$                                                                                                                                                                                    | $\Gamma$ $\Delta$ $\Theta$ $\Lambda$ $\Xi$ $\Pi$<br>$\Sigma$ $\Phi$ $\Psi$ $\Omega$                                                                                                                                                      |
| $\int$<br>$\sum$<br>$\prod$<br>$\sqrt{\phantom{x}}$<br>$\pm$<br>$\infty$<br>$\approx$<br>$\propto$<br>$\equiv$<br>$\neq$<br>$\leq$<br>$\geq$<br>$\times$<br>$\cdot$<br>$\div$<br>$\partial$<br>$\nabla$<br>$\%$<br>$^\circ$<br>$\therefore$<br>$\emptyset$                                     | $\int$ $\sum$ $\prod$ $\sqrt{\phantom{x}}$ $\pm$ $\infty$<br>$\approx$ $\propto$ $\equiv$ $\neq$ $\leq$ $\geq$<br>$\times$ $\cdot$ $\div$ $\partial$ $\nabla$<br>$\%$ $^\circ$ $\therefore$ $\emptyset$                                  |
| $\in$<br>$\notin$<br>$\cap$<br>$\cup$<br>$\subset$<br>$\supset$<br>$\subseteq$<br>$\supseteq$<br>$\neg$<br>$\wedge$<br>$\vee$<br>$\exists$<br>$\forall$<br>$\Rightarrow$<br>$\Leftrightarrow$<br>$\rightarrow$<br>$\leftrightarrow$<br>$\uparrow$<br>$\downarrow$<br>$\propto$<br>$-$<br>$---$ | $\in$ $\notin$ $\cap$ $\cup$ $\subset$ $\supset$ $\subseteq$ $\supseteq$<br>$\neg$ $\wedge$ $\vee$ $\exists$ $\forall$<br>$\Rightarrow$ $\Leftrightarrow$ $\rightarrow$ $\leftrightarrow$ $\uparrow$ $\downarrow$<br>$\propto$ $-$ $---$ |

The project has settled on using both HTML and TeX because each has advantages in some situations.

### Pros of HTML

1. Formulas in HTML behave more like regular text. In-line HTML formulae always align properly with the rest of the HTML text and, to some degree, can be copied-and-pasted (this is not a problem if TeX is rendered using MathJax, and the alignment should not be a problem for PNG rendering once bug 32694 is fixed).
2. The formula's background and font size match the rest of HTML contents (this can be fixed on TeX formulas by using the commands `\pagecolor` and `\definecolor`) and the appearance respects CSS and browser settings while the typeface is conveniently altered to help you identify formulae.
3. Pages using HTML code for formulae will load faster and they will create less clutter on your hard disk.
4. Formulae typeset with HTML code will be accessible to client-side script links (a.k.a. scriptlets).
5. The display of a formula entered using mathematical templates can be conveniently altered by modifying the templates involved; this modification will affect all relevant formulae without any manual intervention.
6. The HTML code, if entered diligently, will contain all semantic information to transform the equation back to TeX or any other code as needed. It can even contain differences TeX does not normally catch, e.g. `{\math|i' i'}` for the imaginary unit and `{\math|<var>i</var>}` for an arbitrary index variable.
7. Unlike generated bitmaps, HTML is not sensitive to dots per inch variances between viewing platforms.

### Pros of TeX

1. TeX is semantically more precise than HTML.
  1. In TeX, "x" means "mathematical variable  $x$ ", whereas in HTML "x" is generic and somewhat ambiguous.
  2. On the other hand, if you encode the same formula as `{\math|<var>x</var>}`, you get the same visual result  $x$  and no information is lost. This requires diligence and more typing that could make the formula harder to understand as you type it. However, since there are far more readers than editors, this effort is worth considering if no other rendering options are available (such as MathJax, which is available to logged-in users as a preferences opt-in).
2. One consequence of point 1 is that TeX code can be transformed into HTML, but not vice versa.<sup>[1]</sup> This means that on the server side we can always transform a formula, based on its complexity and location within the text, user preferences, type of browser, etc. Therefore, where possible, all the benefits of HTML can be retained, together with the benefits of TeX. It is true that the current situation is not ideal, but that is not a good reason to drop information or contents. It is more a reason to help improve the situation.
3. Another consequence of point 1 is that TeX can be converted to MathML (e.g. by MathJax) for browsers which support it, thus keeping its semantics and allowing the rendering to be better suited for the reader's graphic device.
4. TeX is the preferred text formatting language of most professional mathematicians, scientists, and engineers. It is easier to persuade them to contribute if they can write in TeX.
5. TeX has been specifically designed for typesetting formulae, so input is easier and more natural if you are accustomed to it, and output is more aesthetically pleasing if you focus on a single formula rather than on the whole containing page.
6. Once a formula is done correctly in TeX, it will render reliably, whereas the success of HTML formulae is somewhat dependent on browsers or versions of browsers. Another aspect of this dependency is fonts: the serif font used for rendering formulae is browser-dependent and it may be missing some important glyphs. While the browser is generally capable to substitute a matching glyph from a different font family, it need not be the case for combined glyphs (compare



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" $\bar{a}$ " and " $\bar{a}$ ").

7. When writing in TeX, editors need not worry about whether this or that version of this or that browser supports this or that HTML entity. The burden of these decisions is put on the software. This does not hold for HTML formulae, which can easily end up being rendered wrongly or differently from the editor's intentions on a different browser.<sup>[2]</sup>
8. TeX formulae, by default, render larger and are usually more readable than HTML formulae and are not dependent on client-side browser resources, such as fonts, and so the results are more reliably WYSIWYG.
9. While TeX does not assist you in finding HTML codes or Unicode values (which you can obtain by viewing the HTML source in your browser), copying and pasting from a TeX PNG image in Wikipedia into simple text will return the LaTeX source.

<sup>^</sup> Unless your wikitext follows the style of point 1.2

<sup>^</sup> The entity support problem is not limited to mathematical formulae though; it can be easily solved by using the corresponding characters instead of entities, as the character repertoire links do, except for cases where the corresponding glyphs are visually indiscernible (e.g. `&ndash;` for `'-'` and `&minus;` for `'-'`).

In some cases it may be the best choice to use neither TeX nor the HTML substitutes, but instead the simple ASCII symbols of a standard keyboard (see hereafter, for an example).

### Using MathML

The default MathML/SVG renderer option, selectable through My Preferences - Appearance - Math generate hidden MathML code. This code can be used by screen readers and other assistive technology. To actually display the MathML in Firefox the Native MathML (<https://addons.mozilla.org/en-US/firefox/addon/native-mathml/>) extension and the MathML fonts ([https://developer.mozilla.org/en-US/docs/Mozilla/MathML\\_Project/Fonts](https://developer.mozilla.org/en-US/docs/Mozilla/MathML_Project/Fonts)) must be installed. Details on using MathML in other systems can be found at mw:Extension:Math.

### Formatting using TeX

#### Functions, symbols, special characters



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| Accents/diacritics                                                                                                                                                                       |                                                                                        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| <code>\dot{a}</code> , <code>\ddot{a}</code> , <code>\acute{a}</code> , <code>\grave{a}</code>                                                                                           | $\dot{a}, \ddot{a}, \acute{a}, \grave{a}$                                              |
| <code>\check{a}</code> , <code>\breve{a}</code> , <code>\tilde{a}</code> , <code>\bar{a}</code>                                                                                          | $\check{a}, \breve{a}, \tilde{a}, \bar{a}$                                             |
| <code>\hat{a}</code> , <code>\widehat{a}</code> , <code>\vec{a}</code>                                                                                                                   | $\hat{a}, \widehat{a}, \vec{a}$                                                        |
| Standard numerical functions                                                                                                                                                             |                                                                                        |
| <code>\exp_a b = a^b</code> , <code>\exp b = e^b</code> , <code>10^m</code>                                                                                                              | $\exp_a b = a^b, \exp b = e^b, 10^m$                                                   |
| <code>\ln c</code> , <code>\lg d = \log e</code> , <code>\log_{10} f</code>                                                                                                              | $\ln c, \lg d = \log e, \log_{10} f$                                                   |
| <code>\sin a</code> , <code>\cos b</code> , <code>\tan c</code> , <code>\cot d</code> , <code>\sec e</code> , <code>\csc f</code>                                                        | $\sin a, \cos b, \tan c, \cot d, \sec e, \csc f$                                       |
| <code>\arcsin h</code> , <code>\arccos i</code> , <code>\arctan j</code>                                                                                                                 | $\arcsin h, \arccos i, \arctan j$                                                      |
| <code>\sinh k</code> , <code>\cosh l</code> , <code>\tanh m</code> , <code>\coth n</code>                                                                                                | $\sinh k, \cosh l, \tanh m, \coth n$                                                   |
| <code>\operatorname{sh} l</code> , <code>\operatorname{ch} l</code> , <code>\operatorname{th} m</code> , <code>\operatorname{coth} n</code>                                              | $\operatorname{sh} k, \operatorname{ch} l, \operatorname{th} m, \operatorname{coth} n$ |
| <code>\operatorname{argsh} o</code> , <code>\operatorname{argch} p</code> , <code>\operatorname{argth} q</code>                                                                          | $\operatorname{argsh} o, \operatorname{argch} p, \operatorname{argth} q$               |
| <code>\sgn r</code> , <code>\left s\right </code>                                                                                                                                        | $\operatorname{sgn} r,  s $                                                            |
| <code>\min(x,y)</code> , <code>\max(x,y)</code>                                                                                                                                          | $\min(x,y), \max(x,y)$                                                                 |
| Bounds                                                                                                                                                                                   |                                                                                        |
| <code>\min x</code> , <code>\max y</code> , <code>\inf s</code> , <code>\sup t</code>                                                                                                    | $\min x, \max y, \inf s, \sup t$                                                       |
| <code>\lim u</code> , <code>\liminf v</code> , <code>\limsup w</code>                                                                                                                    | $\lim u, \liminf v, \limsup w$                                                         |
| <code>\dim p</code> , <code>\deg q</code> , <code>\det m</code> , <code>\ker \phi</code>                                                                                                 | $\dim p, \deg q, \det m, \ker \phi$                                                    |
| Projections                                                                                                                                                                              |                                                                                        |
| <code>\Pr j</code> , <code>\hom l</code> , <code>\lVert z \rVert</code> , <code>\arg z</code>                                                                                            | $\Pr j, \operatorname{hom} l, \ z\ , \arg z$                                           |
| Differentials and derivatives                                                                                                                                                            |                                                                                        |
| <code>dt</code> , <code>\operatorname{d}t</code> , <code>\partial t</code> , <code>\nabla \psi</code>                                                                                    | $dt, dt, \partial t, \nabla \psi$                                                      |
| <code>dy/dx</code> , <code>\operatorname{d}y/\operatorname{d}x</code> , <code>\frac{dy}{dx}</code> , <code>\frac{\partial^2}{\partial x_1 \partial x_2} y</code>                         | $dy/dx, dy/dx, \frac{dy}{dx}, \frac{\partial^2}{\partial x_1 \partial x_2} y$          |
| <code>\prime</code> , <code>\backprime</code> , <code>f^{\prime}</code> , <code>f''</code> , <code>f^{(3)}</code> , <code>\dot{y}</code> , <code>\ddot{y}</code>                         | $\prime, \backprime, f', f'', f^{(3)}, \dot{y}, \ddot{y}$                              |
| Letter-like symbols or constants                                                                                                                                                         |                                                                                        |
| <code>\infty</code> , <code>\aleph</code> , <code>\complement</code> , <code>\backepsilon</code> , <code>\eth</code> , <code>\Finv</code> , <code>\hbar</code>                           | $\infty, \aleph, \complement, \backepsilon, \eth, \Finv, \hbar$                        |
| <code>\Im</code> , <code>\imath</code> , <code>\jmath</code> , <code>\Bbbk</code> , <code>\ell</code> , <code>\mho</code> , <code>\wp</code> , <code>\Re</code> , <code>\circledS</code> | $\Im, \imath, \jmath, \Bbbk, \ell, \mho, \wp, \Re, \circledS$                          |
| Modular arithmetic                                                                                                                                                                       |                                                                                        |
| <code>s_k \equiv 0 \pmod{m}</code>                                                                                                                                                       | $s_k \equiv 0 \pmod{m}$                                                                |
| <code>a \bmod b</code>                                                                                                                                                                   | $a \bmod b$                                                                            |
| <code>\gcd(m, n)</code> , <code>\operatorname{lcm}(m, n)</code>                                                                                                                          | $\gcd(m, n), \operatorname{lcm}(m, n)$                                                 |
| <code>\mid</code> , <code>\nmid</code> , <code>\shortmid</code> , <code>\nshortmid</code>                                                                                                | $\mid, \nmid, \shortmid, \nshortmid$                                                   |
| Radicals                                                                                                                                                                                 |                                                                                        |



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For a little more semantics on these symbols, see the brief TeX Cookbook (<http://www.math.upenn.edu/tex-stuff/cookbook.pdf>).

**Larger expressions**

**Subscripts, superscripts, integrals**



2088

| Feature                                                      | Syntax                                                                            | How it looks rendered                                      |
|--------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------|
| Superscript                                                  | <code>a^2</code>                                                                  | $a^2$                                                      |
| Subscript                                                    | <code>a_2</code>                                                                  | $a_2$                                                      |
| Grouping                                                     | <code>10^{30} a^{2+2}</code>                                                      | $10^{30} a^{2+2}$                                          |
|                                                              | <code>a_{i,j} b_{f'}</code>                                                       | $a_{i,j} b_{f'}$                                           |
| Combining sub & super without and with horizontal separation | <code>x_2^3</code>                                                                | $x_2^3$                                                    |
|                                                              | <code>{x_2}^3</code>                                                              | $x_2^3$                                                    |
| Super super                                                  | <code>10^{10^{8}}</code>                                                          | $10^{10^8}$                                                |
| Preceding and/or additional sub & super                      | <code>\sideset{_{1^2}}{_{3^4}}\prod a^b</code>                                    | ${}^2_1 \prod_3 a^b$                                       |
|                                                              | <code>{_{1^2}}!\Omega_{3^4}</code>                                                | ${}_{1^2}!\Omega_3$                                        |
| Stacking                                                     | <code>\overset{\alpha}{\omega}</code>                                             | $\overset{\alpha}{\omega}$                                 |
|                                                              | <code>\underset{\alpha}{\omega}</code>                                            | $\underset{\alpha}{\omega}$                                |
|                                                              | <code>\overset{\alpha}{\underset{\gamma}{\omega}}</code>                          | $\overset{\alpha}{\underset{\gamma}{\omega}}$              |
|                                                              | <code>\stackrel{\alpha}{\omega}</code>                                            | $\overset{\alpha}{\omega}$                                 |
| Derivatives                                                  | <code>x', y'', f', f''</code>                                                     | $x', y'', f', f''$                                         |
|                                                              | <code>x^{\prime}, y^{\prime\prime}</code>                                         | $x', y''$                                                  |
| Derivative dots                                              | <code>\dot{x}, \ddot{x}</code>                                                    | $\dot{x}, \ddot{x}$                                        |
| Underlines, overlines, vectors                               | <code>\hat{a} \ \bar{b} \ \vec{c}</code>                                          | $\hat{a} \ \bar{b} \ \vec{c}$                              |
|                                                              | <code>\overrightarrow{a \ b} \ \overleftarrow{c \ d} \ \widehat{d \ e \ f}</code> | $\overrightarrow{ab} \ \overleftarrow{cd} \ \widehat{def}$ |
|                                                              | <code>\overline{g \ h \ i} \ \underline{j \ k \ l}</code>                         | $\overline{ghi} \ \underline{jkl}$                         |
| Arc (workaround)                                             | <code>\overset{\frown}{AB}</code>                                                 | $\overset{\frown}{AB}$                                     |
| Arrows                                                       | <code>A \xleftarrow{n+\mu-1} B \xrightarrow[n\pm i-1]{T} C</code>                 | $A \xleftarrow{n+\mu-1} B \xrightarrow[n\pm i-1]{T} C$     |
| Overbraces                                                   | <code>\overbrace{1+2+\cdots+100}^{5050}</code>                                    | $\overbrace{1+2+\cdots+100}^{5050}$                        |
| Underbraces                                                  | <code>\underbrace{a+b+\cdots+z}_{26}</code>                                       | $\underbrace{a+b+\cdots+z}_{26}$                           |
| Sum                                                          | <code>\sum_{k=1}^N k^2</code>                                                     | $\sum_{k=1}^N k^2$                                         |
| Sum (force <code>\textstyle</code> )                         | <code>\textstyle \sum_{k=1}^N k^2</code>                                          | $\sum_{k=1}^N k^2$                                         |
| Sum in a fraction (default <code>\textstyle</code> )         | <code>\frac{\sum_{k=1}^N k^2}{a}</code>                                           | $\frac{\sum_{k=1}^N k^2}{a}$                               |
| Sum in a fraction (force <code>\displaystyle</code> )        | <code>\frac{\displaystyle \sum_{k=1}^N k^2}{a}</code>                             | $\frac{\sum_{k=1}^N k^2}{a}$                               |
| Sum in a fraction (alternative limits style)                 | <code>\frac{\sum\limits^{^N}_{k=1} k^2}{a}</code>                                 | $\frac{\sum_{k=1}^N k^2}{a}$                               |
| Product                                                      | <code>\prod_{i=1}^N x_i</code>                                                    | $\prod_{i=1}^N x_i$                                        |
| Product (force <code>\textstyle</code> )                     | <code>\textstyle \prod_{i=1}^N x_i</code>                                         | $\prod_{i=1}^N x_i$                                        |
| Coproduct                                                    | <code>\coprod_{i=1}^N x_i</code>                                                  | $\prod_{i=1}^N x_i$                                        |
| Coproduct (force <code>\textstyle</code> )                   | <code>\textstyle \coprod_{i=1}^N x_i</code>                                       | $\prod_{i=1}^N x_i$                                        |



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|                                                                     |                                                         |                                        |
|---------------------------------------------------------------------|---------------------------------------------------------|----------------------------------------|
| Limit                                                               | <code>\lim_{n \to \infty} x_n</code>                    | $\lim_{n \rightarrow \infty} x_n$      |
| Limit (force <code>\textstyle</code> )                              | <code>\textstyle \lim_{n \to \infty} x_n</code>         | $\lim_{n \rightarrow \infty} x_n$      |
| Integral                                                            | <code>\int\limits_1^3 \frac{e^3/x}{x^2} \, dx</code>    | $\int_1^3 \frac{e^3/x}{x^2} dx$        |
| Integral (alternative limits style)                                 | <code>\int_1^3 \frac{e^3/x}{x^2} \, dx</code>           | $\int_1^3 \frac{e^3/x}{x^2} dx$        |
| Integral (force <code>\textstyle</code> )                           | <code>\textstyle \int_{-N}^N e^x \, dx</code>           | $\int_{-N}^N e^x dx$                   |
| Integral (force <code>\textstyle</code> , alternative limits style) | <code>\textstyle \int_{-N}^N e^x \, dx</code>           | $\int_{-N}^N e^x dx$                   |
| Double integral                                                     | <code>\iint\limits_D \, dx \, dy</code>                 | $\iint_D dx dy$                        |
| Triple integral                                                     | <code>\iiint\limits_E \, dx \, dy \, dz</code>          | $\iiint_E dx dy dz$                    |
| Quadruple integral                                                  | <code>\iiint\limits_F \, dx \, dy \, dz \, dt</code>    | $\iiint_F dx dy dz dt$                 |
| Line or path integral                                               | <code>\int_{(x,y) \in C} x^3 \, dx + 4y^2 \, dy</code>  | $\int_{(x,y) \in C} x^3 dx + 4y^2 dy$  |
| Closed line or path integral                                        | <code>\oint_{(x,y) \in C} x^3 \, dx + 4y^2 \, dy</code> | $\oint_{(x,y) \in C} x^3 dx + 4y^2 dy$ |
| Intersections                                                       | <code>\bigcap_{i=1}^n E_i</code>                        | $\bigcap_{i=1}^n E_i$                  |
| Unions                                                              | <code>\bigcup_{i=1}^n E_i</code>                        | $\bigcup_{i=1}^n E_i$                  |

Display attribute

The `<math>` tag can take a `display` attribute with possible values of `inline` and `block`.

Inline

If the value of the display attribute is `inline`, the contents will be rendered in inline mode; i.e., there will be no new paragraph for the equation and the operators will be rendered to consume only a small amount of vertical space.

Example

The sum  $\sum_{i=0}^{\infty} 2^{-i}$  converges to 2.

The next line-width is not disturbed by large operators.

The code for the math example reads:

```
<math display="inline">\sum_{i=0}^{\infty} 2^{-i}</math>
```

Technical implementation

Technically the command `\textstyle` will be added to the user input before the `tex` command is passed to the renderer. The result will be displayed without further formatting by outputting the image or `MathMLElement` to the page.

Block

In block-style the equation is rendered in its own paragraph and the operators are rendered consuming less horizontal space.

Example

VisualEditor

Cancel

Formula

Done

Formula

E=mc^2

⌵

Display

Default

Inline

Block

Link ID (optional)

This screenshot shows the formula  $E = mc^2$  being edited using VisualEditor. The visual editor shows a button that allows to choose one of three offered modes to display a formula.





2090

The equation

**geometric series:**  $\sum_{i=0}^{\infty} 2^{-i} = 2$

It was entered as

```
<math display="block">\text{geometric series:}\quad \sum_{i=0}^{\infty} 2^{-i}=2 </math>
```

**Technical implementation**

Technically it will add the command `\displaystyle` will be added to the user input, if the user input does not contain the string `\displaystyle` or `\align` before the tex command is passed to the renderer. The result will be displayed in a new paragraph. Therefore, the style of the MathImage is altered i.e. the style attribute "display:block;margin:auto" is added. For MathML it is ensured that display=inline is replaced by display block which produces a new paragraph

**Not specified**

If nothing is specified the current behavior is preserved. That means all equations are rendered in display style but not using a new paragraph.

**Example**

The sum  $\sum_{i=0}^{\infty} 2^{-i}$  converges to 2.

The next line-width is disturbed by large operators.

The code for the math example reads:

```
<math>\sum_{i=0}^{\infty} 2^{-i}</math>
```

The equation

**geometric series:**  $\sum_{i=0}^{\infty} 2^{-i} = 2$

It was entered as

```
<math>\text{geometric series:}\quad \sum_{i=0}^{\infty} 2^{-i}=2 </math>
```

**Fractions, matrices, multilines**



2091

| Feature                                                                  | Syntax                                                                                                                        | How it looks rendered                                                                        |
|--------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| Fractions                                                                | <code>\frac{2}{4}=0.5</code> or <code>{2 \over 4}=0.5</code>                                                                  | $\frac{2}{4} = 0.5$                                                                          |
| Small fractions (force <code>\textstyle</code> )                         | <code>\tfrac{2}{4} = 0.5</code>                                                                                               | $\tfrac{2}{4} = 0.5$                                                                         |
| Large (normal) fractions (force <code>\displaystyle</code> )             | <code>\dfrac{2}{4} = 0.5 \quad \dfrac{2}{c + \dfrac{2}{d + \dfrac{2}{4}}} = a</code>                                          | $\frac{2}{4} = 0.5 \quad \frac{2}{c + \frac{2}{d + \frac{2}{4}}} = a$                        |
| Large (nested) fractions                                                 | <code>\cfrac{2}{c + \cfrac{2}{d + \cfrac{2}{4}}} = a</code>                                                                   | $\frac{2}{c + \frac{2}{d + \frac{2}{4}}} = a$                                                |
| Cancellations in fractions                                               | <code>\cfrac{x}{1 + \cfrac{\cancel{y}}{\cancel{y}}} = \cfrac{x}{2}</code>                                                     | $\frac{x}{1 + \frac{\cancel{y}}{\cancel{y}}} = \frac{x}{2}$                                  |
| Binomial coefficients                                                    | <code>\binom{n}{k}</code>                                                                                                     | $\binom{n}{k}$                                                                               |
| Small binomial coefficients (force <code>\textstyle</code> )             | <code>\tbinom{n}{k}</code>                                                                                                    | $\tbinom{n}{k}$                                                                              |
| Large (normal) binomial coefficients (force <code>\displaystyle</code> ) | <code>\dbinom{n}{k}</code>                                                                                                    | $\dbinom{n}{k}$                                                                              |
| Matrices                                                                 | <pre>\begin{matrix} x &amp; y \\ z &amp; v \end{matrix}</pre>                                                                 | $\begin{matrix} x & y \\ z & v \end{matrix}$                                                 |
|                                                                          | <pre>\begin{vmatrix} x &amp; y \\ z &amp; v \end{vmatrix}</pre>                                                               | $\begin{vmatrix} x & y \\ z & v \end{vmatrix}$                                               |
|                                                                          | <pre>\begin{Vmatrix} x &amp; y \\ z &amp; v \end{Vmatrix}</pre>                                                               | $\begin{Vmatrix} x & y \\ z & v \end{Vmatrix}$                                               |
|                                                                          | <pre>\begin{bmatrix} 0 &amp; \cdots &amp; 0 \\ \vdots &amp; \ddots &amp; \vdots \\ 0 &amp; \cdots &amp; 0 \end{bmatrix}</pre> | $\begin{bmatrix} 0 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 0 \end{bmatrix}$ |
|                                                                          | <pre>\begin{Bmatrix} x &amp; y \\ z &amp; v \end{Bmatrix}</pre>                                                               | $\begin{Bmatrix} x & y \\ z & v \end{Bmatrix}$                                               |
|                                                                          | <pre>\begin{pmatrix} x &amp; y \\ z &amp; v \end{pmatrix}</pre>                                                               | $\begin{pmatrix} x & y \\ z & v \end{pmatrix}$                                               |
|                                                                          | <pre>\bigl( \begin{smallmatrix} a &amp; b \\ c &amp; d \end{smallmatrix} \bigr)</pre>                                         | $\bigl( \begin{smallmatrix} a & b \\ c & d \end{smallmatrix} \bigr)$                         |



2092

|                                                                                                             |                                                                                                                                       |                                                                                                                                                                                                                                                                                                |     |     |     |   |   |   |   |   |   |   |   |   |   |   |   |
|-------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|-----|---|---|---|---|---|---|---|---|---|---|---|---|
| Case distinctions                                                                                           | <pre>\f(n) = \begin{cases} n/2, &amp; \text{if } n \text{ is even} \\ 3n+1, &amp; \text{if } n \text{ is odd} \end{cases}</pre>       | $f(n) = \begin{cases} n/2, & \text{if } n \text{ is even} \\ 3n+1, & \text{if } n \text{ is odd} \end{cases}$                                                                                                                                                                                  |     |     |     |   |   |   |   |   |   |   |   |   |   |   |   |
| Multiline equations                                                                                         | <pre>\begin{align} f(x) &amp;= (a+b)^2 \\ &amp;= a^2 + 2ab + b^2 \end{align}</pre>                                                    | $f(x) = (a+b)^2 = a^2 + 2ab + b^2$                                                                                                                                                                                                                                                             |     |     |     |   |   |   |   |   |   |   |   |   |   |   |   |
|                                                                                                             | <pre>\begin{alignat}{2} f(x) &amp;= (a-b)^2 \\ &amp;= a^2 - 2ab + b^2 \end{alignat}</pre>                                             | $f(x) = (a-b)^2 = a^2 - 2ab + b^2$                                                                                                                                                                                                                                                             |     |     |     |   |   |   |   |   |   |   |   |   |   |   |   |
| Multiline equations (must define number of columns used ({lcl})) (should not be used unless needed)         | <pre>\begin{array}{lcl} z &amp; = &amp; a \\ f(x,y,z) &amp; = &amp; x + y + z \end{array}</pre>                                       | $\begin{array}{lcl} z & = & a \\ f(x,y,z) & = & x + y + z \end{array}$                                                                                                                                                                                                                         |     |     |     |   |   |   |   |   |   |   |   |   |   |   |   |
| Multiline equations (more)                                                                                  | <pre>\begin{array}{lcr} z &amp; = &amp; a \\ f(x,y,z) &amp; = &amp; x + y + z \end{array}</pre>                                       | $\begin{array}{lcr} z & = & a \\ f(x,y,z) & = & x + y + z \end{array}$                                                                                                                                                                                                                         |     |     |     |   |   |   |   |   |   |   |   |   |   |   |   |
| Breaking up a long expression so that it wraps when necessary, at the expense of destroying correct spacing | <pre>f(x) = \sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + \cdots</pre>                                                        | $f(x) = \sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + \cdots$                                                                                                                                                                                                                          |     |     |     |   |   |   |   |   |   |   |   |   |   |   |   |
| Simultaneous equations                                                                                      | <pre>\begin{cases} 3x + 5y + z \\ 7x - 2y + 4z \\ -6x + 3y + 2z \end{cases}</pre>                                                     | $\begin{cases} 3x + 5y + z \\ 7x - 2y + 4z \\ -6x + 3y + 2z \end{cases}$                                                                                                                                                                                                                       |     |     |     |   |   |   |   |   |   |   |   |   |   |   |   |
| Arrays                                                                                                      | <pre>\begin{array}{ c c c } \hline 0 &amp; 0 &amp; 1 \\ 0 &amp; 1 &amp; 1 \\ 1 &amp; 0 &amp; 1 \\ 1 &amp; 1 &amp; 0 \end{array}</pre> | <table border="1"> <tr> <td><math>a</math></td> <td><math>b</math></td> <td><math>S</math></td> </tr> <tr> <td>0</td> <td>0</td> <td>1</td> </tr> <tr> <td>0</td> <td>1</td> <td>1</td> </tr> <tr> <td>1</td> <td>0</td> <td>1</td> </tr> <tr> <td>1</td> <td>1</td> <td>0</td> </tr> </table> | $a$ | $b$ | $S$ | 0 | 0 | 1 | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 1 | 0 |
| $a$                                                                                                         | $b$                                                                                                                                   | $S$                                                                                                                                                                                                                                                                                            |     |     |     |   |   |   |   |   |   |   |   |   |   |   |   |
| 0                                                                                                           | 0                                                                                                                                     | 1                                                                                                                                                                                                                                                                                              |     |     |     |   |   |   |   |   |   |   |   |   |   |   |   |
| 0                                                                                                           | 1                                                                                                                                     | 1                                                                                                                                                                                                                                                                                              |     |     |     |   |   |   |   |   |   |   |   |   |   |   |   |
| 1                                                                                                           | 0                                                                                                                                     | 1                                                                                                                                                                                                                                                                                              |     |     |     |   |   |   |   |   |   |   |   |   |   |   |   |
| 1                                                                                                           | 1                                                                                                                                     | 0                                                                                                                                                                                                                                                                                              |     |     |     |   |   |   |   |   |   |   |   |   |   |   |   |

Parenthesizing big expressions, brackets, bars

| Feature | Syntax                                    | How it looks rendered      |
|---------|-------------------------------------------|----------------------------|
| Bad     | <code>( \frac{1}{2} )</code>              | $(\frac{1}{2})$            |
| Good    | <code>\left ( \frac{1}{2} \right )</code> | $\left(\frac{1}{2}\right)$ |

You can use various delimiters with `\left` and `\right`:



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| Feature                                                                         | Syntax                                                                                                                                                                | How it looks rendered                                                                                         |
|---------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| Parentheses                                                                     | <code>\left ( \frac{a}{b} \right )</code>                                                                                                                             | $\left(\frac{a}{b}\right)$                                                                                    |
| Brackets                                                                        | <code>\left [ \frac{a}{b} \right ] \quad \left \lceil \frac{a}{b} \right \rceil</code>                                                                                | $\left[\frac{a}{b}\right] \quad \left\lceil\frac{a}{b}\right\rceil$                                           |
| Braces                                                                          | <code>\left \{ \frac{a}{b} \right \} \quad \left \{ \frac{a}{b} \right \}</code>                                                                                      | $\left\{\frac{a}{b}\right\} \quad \left\{\frac{a}{b}\right\}$                                                 |
| Angle brackets                                                                  | <code>\left \langle \frac{a}{b} \right \rangle</code>                                                                                                                 | $\left\langle\frac{a}{b}\right\rangle$                                                                        |
| Bars and double bars                                                            | <code>\left   \frac{a}{b} \right   \quad \left \lVert \frac{c}{d} \right \rVert</code>                                                                                | $\left \frac{a}{b}\right  \quad \left\lVert\frac{c}{d}\right\rVert$                                           |
| Floor and ceiling functions:                                                    | <code>\left \lfloor \frac{a}{b} \right \rfloor \quad \left \lceil \frac{c}{d} \right \rceil</code>                                                                    | $\left\lfloor\frac{a}{b}\right\rfloor \quad \left\lceil\frac{c}{d}\right\rceil$                               |
| Slashes and backslashes                                                         | <code>\left / \frac{a}{b} \right \backslash</code>                                                                                                                    | $\left/\frac{a}{b}\right\backslash$                                                                           |
| Up, down, and up-down arrows                                                    | <code>\left \uparrow \frac{a}{b} \right \downarrow \quad \left \Uparrow \frac{a}{b} \right \Downarrow \quad \left \updownarrow \frac{a}{b} \right \Updownarrow</code> | $\uparrow\frac{a}{b}\downarrow \quad \Uparrow\frac{a}{b}\Downarrow \quad \updownarrow\frac{a}{b}\Updownarrow$ |
| Delimiters can be mixed, as long as \left and \right match                      | <code>\left [ 0,1 \right ) \quad \left \langle \psi \right  </code>                                                                                                   | $[0,1) \quad \langle\psi $                                                                                    |
| Use \left. and \right. if you do not want a delimiter to appear                 | <code>\left. \frac{A}{B} \right. \rightarrow X</code>                                                                                                                 | $\left.\frac{A}{B}\right.\rightarrow X$                                                                       |
| Size of the delimiters (add "l" or "r" to indicate the side for proper spacing) | <code>( \bigl ( \Bigl ( \biggl ( \Biggl ( \dots \Biggr ) \biggr ) \Bigr ) \bigr )</code>                                                                              | $(((((\dots])))$                                                                                              |
|                                                                                 | <code>{ \bigl \{ \Bigl \{ \biggl \{ \Biggl \{ \dots \Biggr \} \biggr \} \Bigr \} \bigr \} \rangle \rangle \rangle</code>                                              | $\{\{\{\{\dots\}\}\}\}\rangle\rangle\rangle$                                                                  |
|                                                                                 | <code>  \big  \Big  \bigg  \Bigg  \dots \Bigg  \bigg  \Big  \big </code>                                                                                              | $    \dots    $                                                                                               |
|                                                                                 | <code>\lfloor \bigl \lfloor \Bigl \lfloor \biggl \lfloor \Biggl \lfloor \dots \Biggr \rfloor \biggr \rfloor \Bigr \rfloor \bigr \rfloor \rceil \rceil \rceil</code>   | $[[[ \dots ]]]]$                                                                                              |
|                                                                                 | <code>\uparrow \big \uparrow \Big \uparrow \dots \Downarrow \big \Downarrow \Big \Downarrow</code>                                                                    | $\uparrow\uparrow\uparrow\uparrow \dots \Downarrow\Downarrow\Downarrow\Downarrow$                             |
|                                                                                 | <code>\updownarrow \big \updownarrow \Big \updownarrow \dots \Updownarrow \big \Updownarrow \Big \Updownarrow</code>                                                  | $\updownarrow\updownarrow\updownarrow \dots \Updownarrow\Updownarrow\Updownarrow\Updownarrow$                 |
|                                                                                 | <code>/ \big / \Big / \bigg / \Bigg / \dots \Bigg \backslash \bigg \backslash \Big \backslash \big \backslash</code>                                                  | $/// \dots \backslash \backslash \backslash$                                                                  |

### Equation numbering

The templates `\NumBlk` and `\EquationRef` can be used to number equations. The template `\EquationNote` can be used to refer to a numbered equation from surrounding text. For example, the following syntax:

```
\NumBlk|<math>x^2 + y^2 + z^2 = 1</math>|{\EquationRef{1}}|
```

produces the following result (note the equation number in the right margin):

$$x^2 + y^2 + z^2 = 1$$

(1)

Later on, the text can refer to this equation by its number using syntax like this:



2094

As seen in equation ([1](#)), blah blah blah...

The result looks like this:

As seen in equation (1), blah blah blah...

Note that the equation number produced by `{EquationNote}` is a link that the user can click to go immediately to the cited equation.

**Alphabets and typefaces**

*See also: [Wikipedia:LaTeX symbols § Fonts](#)*

Texvc cannot render arbitrary Unicode characters. Those it can handle can be entered by the expressions below. For others, such as Cyrillic, they can be entered as Unicode or HTML entities in running text, but cannot be used in displayed formulas.



2095

| Greek alphabet                                                                                                                                                             |                                                                                                                              |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <code>\Alpha \Beta \Gamma \Delta \Epsilon \Zeta \Eta \Theta</code>                                                                                                         | $\text{AB}\Gamma\Delta\text{E}\text{Z}\text{H}\Theta$                                                                        |
| <code>\Iota \Kappa \Lambda \Mu \Nu \Xi \Pi \Rho</code>                                                                                                                     | $\text{I}\text{K}\text{A}\text{M}\text{N}\text{E}\Pi\text{P}$                                                                |
| <code>\Sigma \Tau \Upsilon \Phi \Psi \Omega</code>                                                                                                                         | $\Sigma\text{T}\text{T}\Phi\text{X}\Psi\Omega$                                                                               |
| <code>\alpha \beta \gamma \delta \epsilon \zeta \eta \theta</code>                                                                                                         | $\alpha\beta\gamma\delta\epsilon\zeta\eta\theta$                                                                             |
| <code>\iota \kappa \lambda \mu \nu \xi \pi \rho</code>                                                                                                                     | $\iota\kappa\lambda\mu\nu\xi\pi\rho$                                                                                         |
| <code>\sigma \tau \upsilon \phi \chi \psi \omega</code>                                                                                                                    | $\sigma\tau\upsilon\phi\chi\psi\omega$                                                                                       |
| <code>\varepsilon \digamma \varkappa \varpi</code>                                                                                                                         | $\varepsilon\text{F}\kappa\omega$                                                                                            |
| <code>\varrho \varsigma \vartheta \varphi</code>                                                                                                                           | $\varrho\varsigma\vartheta\varphi$                                                                                           |
| Hebrew symbols                                                                                                                                                             |                                                                                                                              |
| <code>\aleph \beth \gimel \daleth</code>                                                                                                                                   | $\aleph\aleph\aleph$                                                                                                         |
| Blackboard bold/scripts                                                                                                                                                    |                                                                                                                              |
| <code>\mathbb{A} \mathbb{B} \mathbb{C} \mathbb{D} \mathbb{E} \mathbb{F} \mathbb{G} \mathbb{H} \mathbb{I}</code>                                                            | $\mathbb{A}\mathbb{B}\mathbb{C}\mathbb{D}\mathbb{E}\mathbb{F}\mathbb{G}\mathbb{H}\mathbb{I}$                                 |
| <code>\mathbb{J} \mathbb{K} \mathbb{L} \mathbb{M} \mathbb{N} \mathbb{O} \mathbb{P} \mathbb{Q} \mathbb{R}</code>                                                            | $\mathbb{J}\mathbb{K}\mathbb{L}\mathbb{M}\mathbb{N}\mathbb{O}\mathbb{P}\mathbb{Q}\mathbb{R}$                                 |
| <code>\mathbb{S} \mathbb{T} \mathbb{U} \mathbb{V} \mathbb{W} \mathbb{X} \mathbb{Y} \mathbb{Z}</code>                                                                       | $\mathbb{S}\mathbb{T}\mathbb{U}\mathbb{V}\mathbb{W}\mathbb{X}\mathbb{Y}\mathbb{Z}$                                           |
| Boldface                                                                                                                                                                   |                                                                                                                              |
| <code>\mathbf{A} \mathbf{B} \mathbf{C} \mathbf{D} \mathbf{E} \mathbf{F} \mathbf{G} \mathbf{H} \mathbf{I}</code>                                                            | <b><math>\text{A}\text{B}\text{C}\text{D}\text{E}\text{F}\text{G}\text{H}\text{I}</math></b>                                 |
| <code>\mathbf{J} \mathbf{K} \mathbf{L} \mathbf{M} \mathbf{N} \mathbf{O} \mathbf{P} \mathbf{Q} \mathbf{R}</code>                                                            | <b><math>\text{J}\text{K}\text{L}\text{M}\text{N}\text{O}\text{P}\text{Q}\text{R}</math></b>                                 |
| <code>\mathbf{S} \mathbf{T} \mathbf{U} \mathbf{V} \mathbf{W} \mathbf{X} \mathbf{Y} \mathbf{Z}</code>                                                                       | <b><math>\text{S}\text{T}\text{U}\text{V}\text{W}\text{X}\text{Y}\text{Z}</math></b>                                         |
| <code>\mathbf{a} \mathbf{b} \mathbf{c} \mathbf{d} \mathbf{e} \mathbf{f} \mathbf{g} \mathbf{h} \mathbf{i} \mathbf{j} \mathbf{k} \mathbf{l} \mathbf{m}</code>                | <b><math>\text{a}\text{b}\text{c}\text{d}\text{e}\text{f}\text{g}\text{h}\text{i}\text{j}\text{k}\text{l}\text{m}</math></b> |
| <code>\mathbf{n} \mathbf{o} \mathbf{p} \mathbf{q} \mathbf{r} \mathbf{s} \mathbf{t} \mathbf{u} \mathbf{v} \mathbf{w} \mathbf{x} \mathbf{y} \mathbf{z}</code>                | <b><math>\text{n}\text{o}\text{p}\text{q}\text{r}\text{s}\text{t}\text{u}\text{v}\text{w}\text{x}\text{y}\text{z}</math></b> |
| <code>\mathbf{0} \mathbf{1} \mathbf{2} \mathbf{3} \mathbf{4} \mathbf{5} \mathbf{6} \mathbf{7} \mathbf{8} \mathbf{9}</code>                                                 | <b><math>0123456789</math></b>                                                                                               |
| Boldface (Greek)                                                                                                                                                           |                                                                                                                              |
| <code>\boldsymbol{\Alpha} \boldsymbol{\Beta} \boldsymbol{\Gamma} \boldsymbol{\Delta} \boldsymbol{\Epsilon} \boldsymbol{\Zeta} \boldsymbol{\Eta} \boldsymbol{\Theta}</code> | <b><math>\text{A}\text{B}\Gamma\Delta\text{E}\text{Z}\text{H}\Theta</math></b>                                               |
| <code>\boldsymbol{\Iota} \boldsymbol{\Kappa} \boldsymbol{\Lambda} \boldsymbol{\Mu} \boldsymbol{\Nu} \boldsymbol{\Xi} \boldsymbol{\Pi} \boldsymbol{\Rho}</code>             | <b><math>\text{I}\text{K}\text{A}\text{M}\text{N}\text{E}\Pi\text{P}</math></b>                                              |
| <code>\boldsymbol{\Sigma} \boldsymbol{\Tau} \boldsymbol{\Upsilon} \boldsymbol{\Phi} \boldsymbol{\Psi} \boldsymbol{\Omega}</code>                                           | <b><math>\Sigma\text{T}\text{T}\Phi\text{X}\Psi\Omega</math></b>                                                             |
| <code>\boldsymbol{\alpha} \boldsymbol{\beta} \boldsymbol{\gamma} \boldsymbol{\delta} \boldsymbol{\epsilon} \boldsymbol{\zeta} \boldsymbol{\eta} \boldsymbol{\theta}</code> | <b><math>\alpha\beta\gamma\delta\epsilon\zeta\eta\theta</math></b>                                                           |
| <code>\boldsymbol{\iota} \boldsymbol{\kappa} \boldsymbol{\lambda} \boldsymbol{\mu} \boldsymbol{\nu} \boldsymbol{\xi} \boldsymbol{\pi} \boldsymbol{\rho}</code>             | <b><math>\iota\kappa\lambda\mu\nu\xi\pi\rho</math></b>                                                                       |
| <code>\boldsymbol{\sigma} \boldsymbol{\tau} \boldsymbol{\upsilon} \boldsymbol{\phi} \boldsymbol{\chi} \boldsymbol{\psi} \boldsymbol{\omega}</code>                         | <b><math>\sigma\tau\upsilon\phi\chi\psi\omega</math></b>                                                                     |
| <code>\boldsymbol{\varepsilon} \boldsymbol{\digamma} \boldsymbol{\varkappa} \boldsymbol{\varpi}</code>                                                                     | <b><math>\varepsilon\text{F}\kappa\omega</math></b>                                                                          |
| <code>\boldsymbol{\varrho} \boldsymbol{\varsigma} \boldsymbol{\vartheta} \boldsymbol{\varphi}</code>                                                                       | <b><math>\varrho\varsigma\vartheta\varphi</math></b>                                                                         |
| Italics (default for Latin alphabet)                                                                                                                                       |                                                                                                                              |
| <code>\mathit{0} \mathit{1} \mathit{2} \mathit{3} \mathit{4} \mathit{5} \mathit{6} \mathit{7} \mathit{8} \mathit{9}</code>                                                 | <i><math>0123456789</math></i>                                                                                               |
| Greek italics (default for lowercase Greek)                                                                                                                                |                                                                                                                              |
| <code>\mathit{\alpha} \mathit{\beta} \mathit{\gamma} \mathit{\delta} \mathit{\epsilon} \mathit{\zeta} \mathit{\eta} \mathit{\theta}</code>                                 | <i><math>\text{A}\text{B}\Gamma\Delta\text{E}\text{Z}\text{H}\Theta</math></i>                                               |
| <code>\mathit{\iota} \mathit{\kappa} \mathit{\lambda} \mathit{\mu} \mathit{\nu} \mathit{\xi} \mathit{\pi} \mathit{\rho}</code>                                             | <i><math>\text{I}\text{K}\text{A}\text{M}\text{N}\text{E}\Pi\text{P}</math></i>                                              |
| <code>\mathit{\sigma} \mathit{\tau} \mathit{\upsilon} \mathit{\phi} \mathit{\chi} \mathit{\psi} \mathit{\omega}</code>                                                     | <i><math>\Sigma\text{T}\text{T}\Phi\text{X}\Psi\Omega</math></i>                                                             |
| Roman typeface                                                                                                                                                             |                                                                                                                              |
| <code>\mathrm{A} \mathrm{B} \mathrm{C} \mathrm{D} \mathrm{E} \mathrm{F} \mathrm{G} \mathrm{H} \mathrm{I}</code>                                                            | $\text{A}\text{B}\text{C}\text{D}\text{E}\text{F}\text{G}\text{H}\text{I}$                                                   |
| <code>\mathrm{J} \mathrm{K} \mathrm{L} \mathrm{M} \mathrm{N} \mathrm{O} \mathrm{P} \mathrm{Q} \mathrm{R}</code>                                                            | $\text{J}\text{K}\text{L}\text{M}\text{N}\text{O}\text{P}\text{Q}\text{R}$                                                   |
| <code>\mathrm{S} \mathrm{T} \mathrm{U} \mathrm{V} \mathrm{W} \mathrm{X} \mathrm{Y} \mathrm{Z}</code>                                                                       | $\text{S}\text{T}\text{U}\text{V}\text{W}\text{X}\text{Y}\text{Z}$                                                           |
| <code>\mathrm{a} \mathrm{b} \mathrm{c} \mathrm{d} \mathrm{e} \mathrm{f} \mathrm{g} \mathrm{h} \mathrm{i} \mathrm{j} \mathrm{k} \mathrm{l} \mathrm{m}</code>                | $\text{a}\text{b}\text{c}\text{d}\text{e}\text{f}\text{g}\text{h}\text{i}\text{j}\text{k}\text{l}\text{m}$                   |
| <code>\mathrm{n} \mathrm{o} \mathrm{p} \mathrm{q} \mathrm{r} \mathrm{s} \mathrm{t} \mathrm{u} \mathrm{v} \mathrm{w} \mathrm{x} \mathrm{y} \mathrm{z}</code>                | $\text{n}\text{o}\text{p}\text{q}\text{r}\text{s}\text{t}\text{u}\text{v}\text{w}\text{x}\text{y}\text{z}$                   |
| <code>\mathrm{0} \mathrm{1} \mathrm{2} \mathrm{3} \mathrm{4} \mathrm{5} \mathrm{6} \mathrm{7} \mathrm{8} \mathrm{9}</code>                                                 | $0123456789$                                                                                                                 |
| Sans serif                                                                                                                                                                 |                                                                                                                              |
| <code>\mathsf{A} \mathsf{B} \mathsf{C} \mathsf{D} \mathsf{E} \mathsf{F} \mathsf{G} \mathsf{H} \mathsf{I}</code>                                                            | $\text{A}\text{B}\text{C}\text{D}\text{E}\text{F}\text{G}\text{H}\text{I}$                                                   |
| <code>\mathsf{J} \mathsf{K} \mathsf{L} \mathsf{M} \mathsf{N} \mathsf{O} \mathsf{P} \mathsf{Q} \mathsf{R}</code>                                                            | $\text{J}\text{K}\text{L}\text{M}\text{N}\text{O}\text{P}\text{Q}\text{R}$                                                   |
| <code>\mathsf{S} \mathsf{T} \mathsf{U} \mathsf{V} \mathsf{W} \mathsf{X} \mathsf{Y} \mathsf{Z}</code>                                                                       | $\text{S}\text{T}\text{U}\text{V}\text{W}\text{X}\text{Y}\text{Z}$                                                           |





2097

| Colors supported |                |             |               |
|------------------|----------------|-------------|---------------|
| Apricot          | Aquamarine     | Bittersweet | Black         |
| Blue             | BlueGreen      | BlueViolet  | BrickRed      |
| Brown            | BurntOrange    | CadetBlue   | CarnationPink |
| Cerulean         | CornflowerBlue | Cyan        | Dandelion     |
| DarkOrchid       | Emerald        | ForestGreen | Fuchsia       |
| Goldenrod        | Gray           | Green       | GreenYellow   |
| JungleGreen      | Lavender       | LimeGreen   | Magenta       |
| Mahogany         | Maroon         | Melon       | MidnightBlue  |
| Mulberry         | NavyBlue       | OliveGreen  | Orange        |
| OrangeRed        | Orchid         | Peach       | Periwinkle    |
| PineGreen        | Plum           | ProcessBlue | Purple        |
| RawSienna        | Red            | RedOrange   | RedViolet     |
| Rhodamine        | RoyalBlue      | RoyalPurple | RubineRed     |
| Salmon           | SeaGreen       | Sepia       | SkyBlue       |
| SpringGreen      | Tan            | TealBlue    | Thistle       |
| Turquoise        | Violet         | VioletRed   | White         |
| WildStrawberry   | Yellow         | YellowGreen | YellowOrange  |

Note that color should not be used as the *only* way to identify something, because it will become meaningless on black-and-white media or for color-blind people. See Wikipedia:Manual of Style (accessibility)#Color.

Latex does not have a command for setting the background color. The most effective of setting a background color is by setting a CSS styling rules for a table cell

```
| class="wikitable" align="center"
| style="background: gray;" | <math>\backslash\mathrm{pagecolor}\{Gray\}x^2</math>
| style="background: Goldenrod;" | <math>\backslash\mathrm{pagecolor}\{Goldenrod\}y^3</math>
|
```

Rendered as



The `\pagecolor{Goldenrod}` command is necessary for the Texvc renderer to use the correct anti-aliasing around the edges of the semi-transparent images. Without the command a default (white) background color is used — below are shown the results displayed on non-white background.

```
| class="wikitable" align="center"
| style="background: gray;" | <math>x^2</math>
| style="background: Goldenrod;" | <math>y^3</math>
|
```



Custom colours can be defined using

```
\definecolor{myorange}{rgb}{1,0.65,0.4}\color{myorange}e^{i\pi}\color{Black} + 1 = 0
```

$e^{i\pi} + 1 = 0$

Formatting issues

Spacing

Note that TeX handles most spacing automatically, but you may sometimes want manual control.





2098

| Feature                                        | Syntax                    | How it looks rendered          |
|------------------------------------------------|---------------------------|--------------------------------|
| double quad space                              | <code>a \qqquad b</code>  | <b><i>a      b</i></b>         |
| quad space                                     | <code>a \quad b</code>    | <b><i>a    b</i></b>           |
| text space                                     | <code>a\ b</code>         | <b><i>a b</i></b>              |
| text space without PNG conversion              | <code>a \mbox{ } b</code> | <b><i>a b</i></b>              |
| large space                                    | <code>a\;b</code>         | <b><i>a b</i></b>              |
| medium space                                   | <code>a\&lt;b</code>      | [not supported]                |
| small space                                    | <code>a\,b</code>         | <b><i>a b</i></b>              |
| tiny space (use for multiplication of factors) | <code>ab</code>           | <b><i>ab</i></b>               |
| tiny space (syntax space ignored)              | <code>a b</code>          | <b><i>ab</i></b>               |
| no space (use for multi-letter variables)      | <code>\mathit{ab}</code>  | <b><i>ab</i></b>               |
| small negative space                           | <code>a\!b</code>         | <b><i>a<b><i>b</i></b></i></b> |

Automatic spacing may be broken in very long expressions (because they produce an overfull hbox in TeX):

`0+1+2+3+4+5+6+7+8+9+10+11+12+13+14+15+16+17+18+19+20+\cdots`  
***0 + 1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 + 10 + 11 + 12 + 13 + 14 + 15 + 16 + 17 + 18 + 19 + 20 + ...***

This can be remedied by putting a pair of braces { } around the whole expression:

`{0+1+2+3+4+5+6+7+8+9+10+11+12+13+14+15+16+17+18+19+20+\cdots}`  
***0 + 1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 + 10 + 11 + 12 + 13 + 14 + 15 + 16 + 17 + 18 + 19 + 20 + ...***

Alignment with normal text flow

Because of the default CSS

```
\img.tex { vertical-align: middle; }
```

an inline expression like  $\int_{-N}^N e^x dx$  should look good.

If you need to align it otherwise, use `<math style="vertical-align:-100%;">...</math>` and play with the `vertical-align` argument until you get it right; however, how it looks may depend on the browser and the browser settings.

Also note that if you rely on this workaround, if/when the rendering on the server gets fixed in future releases, as a result of this extra manual offset your formulae will suddenly be aligned incorrectly. So use it sparingly, if at all.

Commutative diagrams

To make a commutative diagram, there are three steps:

- 1. write the diagram in TeX
- 2. convert to SVG
- 3. upload the file to Wikimedia Commons

Diagrams in TeX

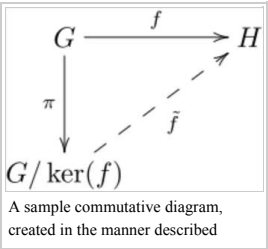
Xy-pic (<http://www.tug.org/applications/Xy-pic/>) (online manual (<http://tex.loria.fr/graph-pack/doc-xy-pic/xyguide-html/xyguide-html.html>)) is the most powerful and general-purpose diagram package in TeX. Diagrams created using it can be found at Commons: Category:Xy-pic diagrams.

Simpler packages include:

- AMS's amscd (<http://www.dante.de/CTAN/help/Catalogue/entries/amscd.html>)
- Paul Taylor's diagrams (<http://www.ctan.org/tex-archive/macros/generic/diagrams/taylor/>)
- François Borceux Diagrams (<http://www.ctan.org/tex-archive/help/Catalogue/entries/borceux.html>)

The following is a template for Xy-pic, together with a hack to increase the margins in dvips, so that the diagram is not truncated by over-eager cropping (suggested in TUGboat: TUGboat, Volume 17 1996, No. 3 (<http://www.tug.org/TUGboat/Articles/tb17-3/tb52rahtz.pdf>)):

```
\documentclass{amsart}
```



A sample commutative diagram, created in the manner described



2099

```
\usepackage{all, ps, dvips}(xy) % Loading the XY-Pic package
% Using postscript driver for smoother curves
\usepackage{color} % For invisible frame
\begin{document}
\thispagestyle{empty} % No page numbers
\SelectTips{eu}{} % Euler arrowheads (tips)
\setlength{\fboxsep}{0pt} % Frame box margin
\color{white}\framebox[{\color{black}$}]{ % Frame for margin
\begin{matrix}
% Diagram goes here %
\end{matrix}
\end{document}
```

Convert to SVG

Once you have produced your diagram in LaTeX (or TeX), you can convert it to an SVG file using the following sequence of commands:

```
pdflatex file.tex
pdfcrop --clip file.pdf tmp.pdf
pdf2svg tmp.pdf file.svg
rm tmp.pdf
```

The pdfcrop (<http://pdfcrop.sourceforge.net>) and pdf2svg (<http://www.cityinthesky.co.uk/opensource/pdf2svg/>) utilities are needed for this procedure. You can alternatively use pdf2svg (<http://www.pdftron.com/pdf2svg/>) from PDFTron for the last step.

If you do not have pdfTeX (which is unlikely) you can use the following commands to replace the first step (TeX → PDF):

```
latex file.tex
dvipdfm file.dvi
```

In general, you will not be able to get anywhere with diagrams without TeX and Ghostscript, and the `inkscape` program is a useful tool for creating or modifying your diagrams by hand. There is also a utility `pstoedit` which supports direct conversion from Postscript files to many vector graphics formats, but it requires a non-free plugin to convert to SVG, and regardless of the format, this editor has not been successful in using it to convert diagrams with diagonal arrows from TeX-created files.

These programs are:

- a working TeX distribution, such as TeX Live
- Ghostscript
- pstoedit
- Inkscape

Upload the file

See also: *commons:Commons:First steps/Upload form*  
See also: *Help:Contents/Images and media*

As the diagram is your own work, upload it to Wikimedia Commons, so that all projects (notably, all languages) can use it without having to copy it to their language's Wiki. (If you've previously uploaded a file to somewhere other than Commons, to Commons.)

Check size

Before uploading, check that the default size of the image is neither too large nor too small by opening in an SVG application and viewing at default size (100% scaling), otherwise adjust the `-y` option to `dvips`.

Name

Make sure the file has a meaningful name.

Upload

Login to Wikimedia Commons, then upload the file (<http://commons.wikimedia.org/w/index.php?title=Special:Upload&uselang=ownwork>); for the **Summary**, give a brief description.

Now go to the image page and add a description, including the **source code**, using this template:

```
[[Information
|description =
|en1= Description [[[:en:Link to WP page|topic]]
]]
|source = [[own]], created as per:
|[:en:Help:Displaying a formula#Commutative diagrams]];
|source code below.
|date = The Creation Date, like 1999-12-31
|author = [[User:YourUserName|Your Real Name]]
]]
```



```

%permission = {{self|PD-self (or other license)
%    |author = [[User:YourUserName|Your Real Name]]}}
%}}
%}}

==TeX source==

<source lang=latex>
% TeX source here
</source>

[[Category:Commutative diagrams]]
[[Category:Xy-pic diagrams]]
[[Category:Images with LaTeX source code]]

```

- Include the source code in the image page, in the `Source` section of the `{{Information}}` template, so that the diagram can be edited in future.
- Include the complete `.tex` file, not just the fragment, so future editors do not need to reconstruct a compilable file.
- You may optionally make the source code section collapsible, using the `{{cot}}/{{cob}}` templates.
- (Don't include it in the Summary section, which is just supposed to be a summary.)

The most common license for commutative diagrams is `PD-self`; some use `PD-ineligible`, especially for simple diagrams, or other licenses. Please *do not* use the GFDL (<http://www.gnu.org/copyleft/fdl.html>), as it requires the entire text of the GFDL to be attached to any document that uses the diagram.

If possible, link to a Wikipedia page relevant to the diagram. (The `1=` is necessary if you use nest templates within the description, and harmless otherwise.)

Include [[Category:Commutative diagrams]], so that it appears in commons:Category:Commutative diagrams. There are also subcategories, which you may choose to use.

Now include the image on the original page via `[[File:Diagram.svg]]`

A sample conforming diagram is commons:Image:PSU-PU.svg.

## \ojint and \oiint

These elements appear in many contexts: `\oiint` denotes a surface integral over the closed 2d boundary of a 3d region (which occurs in much of 3d vector calculus and physical applications – like Maxwell’s equations), likewise `\oiiint` denotes integration over the closed 3d boundary (surface volume) of a 4d region, and they would be strong candidates for the next TeX version. As such there are a lot of workarounds in the present version.

`\oint` looks like:

- [illegible]

`\oiint` (should also be preferably more tightly kerned) looks more or less like:



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- [illegible]

However, since no standardisation exists as yet, any workaround like this (which uses many `\!` symbols for backspacing) should be avoided, if possible. See below for a possibility using PNG image enforcement.

Note that `\iint` (the double integral) and `\iiint` (the triple integral) are still not kerned as they should preferably be, and are currently rendered as if they were successive `\int` symbols; this is not a major problem for reading the formulas, even if the integral symbols before the last one do not have bounds, so it's best to avoid backspacing "hacks" as they may be inconsistent with a possible future better implementation of integrals symbols (with more precisely computed kerning positions).

### `\oint` and `\oiint` as PNG images

These symbols are available as PNG images which are also integrated into two templates, `{oiint}` and `{oiint}`, which take care of the formatting around the symbols.

The templates have three parameters:

```
preintegral
```

the text or formula immediately before the integral

intsubscpt

the subscript below the integral

integrand

the text or formula immediately after the integral

### Examples

- Stokes' theorem:  $\int_{\text{oint}} \text{intsubsect} = \int \text{integrands} = \int \nabla \times \mathbf{F} \cdot d\mathbf{S} = \int \frac{\partial \mathbf{F}}{\partial x} \cdot d\mathbf{S}$

$$\oiint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S} = \oint_{\partial S} \mathbf{F} \cdot d\mathbf{e}$$

- Ampère's law + correction:  $\oint_{\text{oint } C} \boldsymbol{B} \cdot d\boldsymbol{s} = \mu_0 \int_{\text{intsubscpt}} \boldsymbol{S} \cdot d\boldsymbol{A} + \epsilon_0 \frac{\partial E}{\partial t} \cdot d\boldsymbol{s}$

$$\oint_{\partial S} \mathbf{B} \cdot d\boldsymbol{\ell} = \mu_0 \oiint_c \left( \mathbf{J} + \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right) \cdot d\mathbf{S}$$

- Continuity of 4-momentum flux (in general relativity):<sup>[1]</sup>

$$\int_{\text{intsubscript}} \partial_{\text{scriptstyle}} \Omega \int_{\text{integrand}} \boldsymbol{T} \cdot \boldsymbol{x}^{\text{d}} \boldsymbol{\Sigma} = 0$$

$$\mathbf{P} = \oint_{\partial\Omega} \mathbf{T} \cdot \mathbf{d}^3\Sigma = 0$$

### Oriented \oiint and \oiint as PNG images

Some variants of ∮ and ∯ have arrows on them to indicate the sense of integration, such as a line integral around a closed curve in the clockwise sense, and higher dimensional analogues. These are not implemented in TeX on Wikipedia either, although the template {\intorint} is available - see link for details.

 $\overline{\text{arc}}$ 

`\overarc` is not yet implemented to display the arc notation. However, there exists a workaround: use `\overset{\frown}{AB}`, which gives  $\overset{\frown}{AB}$



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`\dddot`

`\dddot` is not implemented in the TexVC renderer but does work in MathJax. For a workaround use `\overset{...}{x}`, which gives  $\ddot{x}$ .

### Syntax to avoid

The texvc processor accepts some non-standard syntax. These should be avoided as the MathJax based renderers do not support these syntax.

### Percentages

Texvc accepts `%` for representing percentages. This causes an error with MathJax and should be replaced with `\%` in all renderers.

### `\textrm`

In texvc spaces need to be represented inside the `\textrm` environment using `\,` `\` and normal spaces are ignored i.e. `\textrm{A\,B\,C}` would render as A BC. In mathjax `\textrm` is an alias for `\text` which renders its argument as normal text, hence `\textrm{A\,B\,C}` renders as A\,B C. To ensure compatibility between versions always use the `\text` environment: `\text{A B C}`.

### Unicode characters

Non-ASCII Unicode characters like  $\pi$  work in MathML, and MathJax but not in texvc so should be avoided.

## Chemistry

There are three ways to render chemical sum formulae as used in chemical equations:

- `<math chem>...</math>`
- `<ce>...</ce>`
- `{{chem}}`

`<ce>X</ce>` is short for `<math chem>\text{X}</math>`

(where x is a chemical sum formula)

Technically, `<math chem>...</math>` is a `math` tag with the extension `mhchem` enabled, according to the MathJax documentation (<http://mathjax.readthedocs.org/en/latest/tex.html#mhchem>).

Note, that the commands `\cee` and `\cef` are disabled, because they are marked as deprecated in the mhchem LaTeX package documentation (<http://www.ctan.org/pkg/mhchem>).

Please note that there are still major issues (<https://phabricator.wikimedia.org/T140217>) with mhchem support in MediaWiki.

### Molecular and Condensed formula

| mhchem                                    |                           | <code>{{chem}}</code> | Equivalent HTML |
|-------------------------------------------|---------------------------|-----------------------|-----------------|
| Markup                                    | Renders as                |                       |                 |
| <code>&lt;ce&gt;H2O&lt;/ce&gt;</code>     | $\text{H}_2\text{O}$      |                       |                 |
| <code>&lt;ce&gt;Sb2O3&lt;/ce&gt;</code>   | $\text{Sb}_2\text{O}_3$   |                       |                 |
| <code>&lt;ce&gt;(NH4)2S&lt;/ce&gt;</code> | $(\text{NH}_4)_2\text{S}$ |                       |                 |

### Bonds



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| mhchem                                           |                                   | Equivalent<br>{ { chem } }<br>and<br>HTML |
|--------------------------------------------------|-----------------------------------|-------------------------------------------|
| Markup                                           | Renders as                        |                                           |
| <code>&lt;ce&gt;C6H5-CHO&lt;/ce&gt;</code>       | $\text{C}_6\text{H}_5\text{-CHO}$ |                                           |
| <code>&lt;ce&gt;A-B=C{\equiv}D&lt;/ce&gt;</code> | $\text{A-B=C}\equiv\text{D}$      |                                           |

Charges

| mhchem                                                                                |                                      | { { chem } } | Equivalent<br>HTML |
|---------------------------------------------------------------------------------------|--------------------------------------|--------------|--------------------|
| Markup                                                                                | Renders as                           |              |                    |
| <code>&lt;ce&gt;H+&lt;/ce&gt;</code>                                                  | $\text{H}^+$                         |              |                    |
| <code>&lt;ce&gt;NO3-&lt;/ce&gt;</code>                                                | $\text{NO}_3^-$                      |              |                    |
| <code>&lt;ce&gt;CrO4^2-&lt;/ce&gt;</code>                                             | $\text{CrO}_4^{2-}$                  |              |                    |
| <code>&lt;ce&gt;AgCl2-&lt;/ce&gt;</code>                                              | $\text{AgCl}_2^-$                    |              |                    |
| <code>&lt;ce&gt;[AgCl2]-&lt;/ce&gt;</code>                                            | $[\text{AgCl}_2]^-$                  |              |                    |
| <code>&lt;ce&gt;Y^{99}&lt;/ce&gt;</code><br><code>&lt;ce&gt;Y^{99+}&lt;/ce&gt;</code> | $\text{Y}^{99+}$<br>$\text{Y}^{90+}$ |              |                    |

Addition Compounds and Stoichiometric Numbers

| mhchem                                                                        |                                                                                                                                              | { { chem } } |
|-------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| Markup                                                                        | Renders as                                                                                                                                   |              |
| <code>&lt;ce&gt;MgSO4.7H2O&lt;/ce&gt;</code>                                  | $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$                                                                                                    |              |
| <code>&lt;ce&gt;KCr(SO4)2*12H2O&lt;/ce&gt;</code>                             | $\text{KCr}(\text{SO}_4)_2 \cdot 12\text{H}_2\text{O}$                                                                                       |              |
| <code>&lt;ce&gt;[CaSO4.1/2H2O] + 1\!1/2H2O -&gt; CaSO4.2H2O&lt;/ce&gt;</code> | $\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O} + 1\frac{1}{2}\text{H}_2\text{O} \longrightarrow \text{CaSO}_4 \cdot 2\text{H}_2\text{O}$ |              |
| <code>&lt;ce&gt;(25/2O2) + C8H18 -&gt; (8CO2) + 9H2O&lt;/ce&gt;</code>        | $\frac{25}{2}\text{O}_2 + \text{C}_8\text{H}_{18} \longrightarrow 8\text{CO}_2 + 9\text{H}_2\text{O}$                                        |              |

(Italic) Math



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|        |            |                                                                                                                                 |
|--------|------------|---------------------------------------------------------------------------------------------------------------------------------|
| mhchem | Markup     | <code>&lt;ce&gt;[C_\mathit{x}]H_\mathit{y} + \mathit{z}O_2 -&gt; (\mathit{x}CO_2) + \frac{\mathit{y}}{2}H_2O&lt;/ce&gt;</code>  |
|        | Renders as | $C_xH_y + zO_2 \longrightarrow xCO_2 + \frac{y}{2}H_2O$                                                                         |
| {chem} | Markup     | <code>{{chem C 'x' H 'y' }} + 'z'{{chem O 2}} \hspace{0.5em}\rightarrow\hspace{0.5em} 'x'{{chem C O 2}} + {{frac 'y' 2}}</code> |
|        | Renders as | $C_xH_y + zO_2 \rightarrow xCO_2 + \frac{y}{2}H_2O$                                                                             |

Oxidation States

|                            |            |                                                                                        |
|----------------------------|------------|----------------------------------------------------------------------------------------|
| mhchem                     | Markup     | <code>&lt;ce&gt;Fe^{II}Fe^{III}2O4&lt;/ce&gt;</code>                                   |
|                            | Renders as | $Fe^{II}Fe_2^{III}O_4$                                                                 |
| {chem} with <sup>...</sup> | Markup     | <code>{{chem Fe &lt;sup&gt;II&lt;/sup&gt; Fe &lt;sup&gt;III&lt;/sup&gt; 2 O 4}}</code> |
|                            | Renders as | $Fe^{II}Fe_2^{III}O_4$                                                                 |

Greek characters

| mhchem                                                  |                         | Equivalent<br>{chem}<br>and<br>HTML |
|---------------------------------------------------------|-------------------------|-------------------------------------|
| Markup                                                  | Renders as              |                                     |
| <code>&lt;ce&gt;\mu-Cl&lt;/ce&gt;</code>                | $\mu-Cl$                |                                     |
| <code>&lt;ce&gt;[Fe(\eta^5-C_5H_5)_2]&lt;/ce&gt;</code> | $[Fe(\eta^5-C_5H_5)_2]$ |                                     |

Isotopes

| mhchem                                           |                          | Equivalent<br>{chem}<br>and<br>HTML |
|--------------------------------------------------|--------------------------|-------------------------------------|
| Markup                                           | Renders as               |                                     |
| <code>&lt;ce&gt;^{227}_{90}Th+&lt;/ce&gt;</code> | $^{227}_{90}\text{Th}^+$ |                                     |
| <code>&lt;ce&gt;^0_{-1}n-&lt;/ce&gt;</code>      | $^0_{-1}\text{n}^-$      |                                     |

States

States Subscripting is not IUPAC recommendation.



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| mhchem                                            |                               | {chem} |
|---------------------------------------------------|-------------------------------|--------|
| Markup                                            | Renders as                    |        |
| <code>&lt;ce&gt;H2_{(aq)}&lt;/ce&gt;</code>       | $\text{H}_{2(\text{aq})}$     |        |
| <code>&lt;ce&gt;CO3^{2-}_{(aq)}&lt;/ce&gt;</code> | $\text{CO}_3^{2-}(\text{aq})$ |        |

**Precipitate**

|                        |            |                                                                                                                                                     |
|------------------------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>mhchem</b>          | Markup     | <code>&lt;ce&gt;[Ba^2+] + SO4^{2-} -&gt; BaSO4 v&lt;/ce&gt;</code>                                                                                  |
|                        | Renders as | $\text{Ba}^{2+} + \text{SO}_4^{2-} \rightarrow \text{BaSO}_4\downarrow$                                                                             |
| {chem}                 | Markup     | <code>{{chem Ba 2+}} + {{chem S O 4 2-}} \xrightarrow{{chem Ba S O 4}} \&amp;darr;</code>                                                           |
|                        | Renders as | $\text{Ba}^{2+} + \text{SO}_4^{2-} \rightarrow \text{BaSO}_4\downarrow$                                                                             |
| <b>Equivalent HTML</b> | Markup     | <code>Ba&lt;sup&gt;2+&lt;/sup&gt; + SO&lt;sub&gt;4&lt;/sub&gt;&lt;sup&gt;2-&lt;/sup&gt; \xrightarrow{BaSO&lt;sub&gt;4&lt;/sub&gt;}&amp;darr;</code> |
|                        | Renders as | $\text{Ba}^{2+} + \text{SO}_4^{2-} \rightarrow \text{BaSO}_4\downarrow$                                                                             |

**Reaction Arrows**

| Markup                                                                            | Renders as                                                             |
|-----------------------------------------------------------------------------------|------------------------------------------------------------------------|
| <code>&lt;ce&gt;A -&gt;[x] B&lt;/ce&gt;</code>                                    | $\overset{x}{\text{A}} \rightarrow \text{B}$                           |
| <code>&lt;ce&gt;A -&gt;[\text{text above}][\text{text below}] B&lt;/ce&gt;</code> | $\text{A} \xrightarrow[\text{text below}]{\text{text above}} \text{B}$ |
| <code>&lt;ce&gt;A -&gt;[\text{ce} (+H2O)] B&lt;/ce&gt;</code>                     | $\text{A} \xrightarrow{+\text{H}_2\text{O}} \text{B}$                  |

**Comparison of arrow symbols**

| Markup                                                                               | Renders as                             |
|--------------------------------------------------------------------------------------|----------------------------------------|
| <code>&lt;math&gt;\rightarrow&lt;/math&gt;</code>                                    | $\rightarrow$                          |
| <code>&lt;math&gt;\rightleftarrows&lt;/math&gt;</code>                               | $\rightleftarrows$                     |
| <code>&lt;math&gt;\rightleftharpoons&lt;/math&gt;</code>                             | $\rightleftharpoons$                   |
| <code>&lt;math&gt;\leftrightarrow&lt;/math&gt;</code>                                | $\leftrightarrow$                      |
| <code>&lt;math&gt;\longrightarrow&lt;/math&gt;<br/>&lt;ce&gt;-&gt;&lt;/ce&gt;</code> | $\longrightarrow$<br>$\longrightarrow$ |
| <code>&lt;ce&gt;=&lt;/ce&gt;</code>                                                  | $\rightleftharpoons$                   |



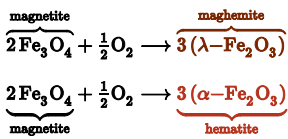


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|                       |                       |
|-----------------------|-----------------------|
| $\longleftrightarrow$ | $\longleftrightarrow$ |
| $\longleftrightarrow$ | $\longleftrightarrow$ |

Further Examples Using Ordinary LaTeX tags

```
<math chem>\begin{align}
\ce{\overbrace{2Fe3O4}^{magnetite} + (1/2O2) -> \color{Brown}\ce{\overbrace{3(\lambda-Fe2O3)}^{maghemite}}}\backslash
\ce{\underbrace{2Fe3O4}_{magnetite} + (1/2O2) -> \color{Red}\ce{\underbrace{3(\alpha-Fe2O3)}_{hematite}}}\backslash
\end{align}</math>
```



To align the equations or color them, use <math chem> and \ce.

Examples of implemented TeX formulas

Quadratic polynomial

|            |                                         |
|------------|-----------------------------------------|
| Markup     | $\text{<math>ax^2 + bx + c = 0</math>}$ |
| Renders as | $ax^2 + bx + c = 0$                     |

Quadratic formula

|            |                                                        |
|------------|--------------------------------------------------------|
| Markup     | $\text{<math>x=(-b\pm\sqrt{b^2-4ac})\over 2a}</math>}$ |
| Renders as | $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$               |

Tall parentheses and fractions

|            |                                                                           |
|------------|---------------------------------------------------------------------------|
| Markup     | $\text{<math>2 = \left(\frac{3-x}{3-x}\right)^2</math>}$                  |
| Renders as | $2 = \left(\frac{3-x}{3-x}\right)^2$                                      |
| Markup     | $\text{<math>S_{\text{new}} = S_{\text{old}} - \frac{(5-T)^2}{2}</math>}$ |
| Renders as | $S_{\text{new}} = S_{\text{old}} - \frac{(5-T)^2}{2}$                     |

Integrals

|            |                                                                                                   |
|------------|---------------------------------------------------------------------------------------------------|
| Markup     | $\text{<math>\int_a^s \int_a^s f(y) dy ds = \int_a^s f(y)(x-y) dy</math>}$                        |
| Renders as | $\int_a^s \int_a^s f(y) dy ds = \int_a^s f(y)(x-y) dy$                                            |
| Markup     | $\text{<math>\int_e^\infty \frac{1}{t(\ln t)^2} dt = \frac{-1}{\ln t} \Big _e^\infty = 1</math>}$ |
| Renders as | $\int_e^\infty \frac{1}{t(\ln t)^2} dt = \frac{-1}{\ln t} \Big _e^\infty = 1$                     |

Matrices and determinants

|            |                                                                 |
|------------|-----------------------------------------------------------------|
| Markup     | $\text{<math>\det(\mathbf{A} - \lambda \mathbf{I}) = 0</math>}$ |
| Renders as | $\det(\mathbf{A} - \lambda \mathbf{I}) = 0$                     |



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Summation

|            |                                                                                                                                                              |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Markup     | <code>&lt;math&gt;\sum_{i=0}^{n-1} i&lt;/math&gt;</code>                                                                                                     |
| Renders as | $\sum_{i=0}^{n-1} i$                                                                                                                                         |
| Markup     | <code>&lt;math&gt;\sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \frac{m^2 n}{3^m (m 3^n + n 3^m)} \{ 3^m \left( m \, 3^{n+n} \, 3^m \right) \}&lt;/math&gt;</code> |
| Renders as | $\sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \frac{m^2 n}{3^m (m 3^n + n 3^m)}$                                                                                  |

Differential equation

|            |                                                                                                  |
|------------|--------------------------------------------------------------------------------------------------|
| Markup     | <code>&lt;math&gt;u'' + p(x)u' + q(x)u=f(x), \quad \textcolor{green}{x}&gt;a&lt;/math&gt;</code> |
| Renders as | $u'' + p(x)u' + q(x)u = f(x), \quad x > a$                                                       |

Complex numbers

|            |                                                                                                                                                                                  |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Markup     | <div><code>&lt;math&gt; \textcolor{green}{\bar{z}}  =  z , \\  (\textcolor{green}{\bar{z}})^n  =  z ^n, \\ \textcolor{green}{\arg}(z^n) = n \, \arg(z)&lt;/math&gt;</code></div> |
| Renders as | $ \bar{z}  =  z ,  (\bar{z})^n  =  z ^n, \arg(z^n) = n \arg(z)$                                                                                                                  |

Limits

|            |                                                                              |
|------------|------------------------------------------------------------------------------|
| Markup     | <code>&lt;math&gt;\lim_{z \rightarrow z_0} f(z) = f(z_0)&lt;/math&gt;</code> |
| Renders as | $\lim_{z \rightarrow z_0} f(z) = f(z_0)$                                     |

Integral equation

|            |                                                                                                                                                                                                                                   |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Markup     | <div><code>&lt;math&gt;\phi_n(\kappa) = \\ \frac{1}{4\pi^2 \kappa^2} \int_0^\infty \frac{\sin(\kappa R)}{\kappa R} \frac{\partial}{\partial R} \left[ R^2 \frac{\partial D_n(R)}{\partial R} \right] dR&lt;/math&gt;</code></div> |
| Renders as | $\phi_n(\kappa) = \frac{1}{4\pi^2 \kappa^2} \int_0^\infty \frac{\sin(\kappa R)}{\kappa R} \frac{\partial}{\partial R} \left[ R^2 \frac{\partial D_n(R)}{\partial R} \right] dR$                                                   |

Example

|            |                                                                                                                                                   |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Markup     | <div><code>&lt;math&gt;\phi_n(\kappa) = \\ 0.033 C_n^2 \kappa^{-11/3}, \quad \frac{1}{L_0} \ll \kappa \ll \frac{1}{l_0}&lt;/math&gt;</code></div> |
| Renders as | $\phi_n(\kappa) = 0.033 C_n^2 \kappa^{-11/3}, \quad \frac{1}{L_0} \ll \kappa \ll \frac{1}{l_0}$                                                   |

Continuation and cases

|        |                                                                                                                                   |
|--------|-----------------------------------------------------------------------------------------------------------------------------------|
| Markup | <div><code>&lt;math&gt;f(x) = \\ \begin{cases} -1 &amp; x &lt; 0 \\ \frac{1}{2} &amp; x = 0 \end{cases}&lt;/math&gt;</code></div> |
|--------|-----------------------------------------------------------------------------------------------------------------------------------|



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```
1 - x^2 & \text{otherwise} \\ \end{cases}
```

|            |                                                                                      |
|------------|--------------------------------------------------------------------------------------|
| Renders as | $f(x) = \begin{cases} \frac{1}{2} & x = 0 \\ 1 - x^2 & \text{otherwise} \end{cases}$ |
|------------|--------------------------------------------------------------------------------------|

### Prefixed subscript

Markup

```
<math>\sum_{n=0}^{\infty} \frac{p_F q(a_1, \dots, a_p; c_1, \dots, c_q; z)^n}{n!} \frac{(c_1)_n \cdots (c_q)_n}{z^n}</math>
```

|            |                                                                                                                                           |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Renders as | ${}_pF_q(a_1, \dots, a_p; c_1, \dots, c_q; z) = \sum_{n=0}^{\infty} \frac{(a_1)_n \cdots (a_p)_n}{(c_1)_n \cdots (c_q)_n} \frac{z^n}{n!}$ |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------|

### Fraction and small fraction

|        |                                                                     |
|--------|---------------------------------------------------------------------|
| Markup | <code>&lt;math&gt;\frac{a}{b}\ \ \ \tfrac{a}{b}&lt;/math&gt;</code> |
|--------|---------------------------------------------------------------------|

|            |                             |
|------------|-----------------------------|
| Renders as | $\frac{a}{b}$ $\frac{a}{b}$ |
|------------|-----------------------------|

### Area of a quadrilateral

|        |                                                                                      |
|--------|--------------------------------------------------------------------------------------|
| Markup | <code>&lt;math&gt;S=dD\backslash,\backslash\sin\alpha\backslash!&lt;/math&gt;</code> |
|--------|--------------------------------------------------------------------------------------|

|            |                      |
|------------|----------------------|
| Renders as | $S = dD \sin \alpha$ |
|------------|----------------------|

### Volume of a sphere-stand

|        |                                                                                                                      |
|--------|----------------------------------------------------------------------------------------------------------------------|
| Markup | <code>&lt;math&gt; v = \frac{16 \pi h}{\left[ 3 \left( r^1{}^2 + r^2{}^2 \right) + h^2 \right]} &lt;/math&gt;</code> |
|--------|----------------------------------------------------------------------------------------------------------------------|

|            |                                                 |
|------------|-------------------------------------------------|
| Renders as | $V = \frac{1}{6}\pi h [3(r_1^2 + r_2^2) + h^2]$ |
|------------|-------------------------------------------------|

## Multiple equations

Markup

```
<math>\begin{align}<math>\begin{align}u &= \tfrac{1}{\sqrt{2}}(x+y) & \text{qqquad} x &= \tfrac{1}{\sqrt{2}}(u+v) \\v &= \tfrac{1}{\sqrt{2}}(x-y) & \text{qqquad} y &= \tfrac{1}{\sqrt{2}}(u-v)\end{align}</math>
```

|            |                                 |                                 |
|------------|---------------------------------|---------------------------------|
| Renders as | $u = \frac{1}{\sqrt{2}}(x + y)$ | $x = \frac{1}{\sqrt{2}}(u + v)$ |
|            | $v = \frac{1}{\sqrt{2}}(x - y)$ | $y = \frac{1}{\sqrt{2}}(u - v)$ |

## See also

- Typesetting of mathematical formulae
- Help:Score (a tag for tablatures, "sheet music") and Help:Musical symbols
- Table of mathematical symbols
- Wikipedia:Rendering math
- mw:Extension:Blahtex, or blahtex: a LaTeX to MathML converter for Wikipedia
- commons:Category:Images which should use TeX

## References

## Footnotes



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- a. Although, in all cases mentioned, TeX is generated by compilation, and not by an interpreter program, there is one essential difference between, e.g., Knuth's TeX or L<sup>A</sup>T<sub>E</sub>X and the present implementation: whereas in the first two cases the compiler typically generates an *all-in-one* printable output, which has the quality of a whole book with all chapters, sections and subsections, and where no line is "special", in the present case one has, typically, a mixture of TeX images (more precisely: PNG images) for the equations, embedded into usual text, and with short TeX elements usually replaced by HTML parts. As a consequence, in many cases TeX-elements, e.g. vector symbols, "stick out" below (or above) the text line. This "sticking out" is *not* the case in the above-mentioned original products, and the HTML-substitutes for small TeX additions to the text are often insufficient in quality for many readers. In spite of these shortcomings, the present product characterized by "many embedded PNG-images" should be preferred for small texts, where the equations do not dominate.
- b. This can cause difficulty with setting the baseline as vertical alignment with the surrounding text can also be a problem (see bug 32694)

### Citations

1. J. A. Wheeler; C. Misner; K. S. Thorne (1973). *Gravitation* (2nd ed.). W. H. Freeman & Co. ISBN 0-7167-0344-0.

### External links

- A LaTeX tutorial (<http://www.maths.tcd.ie/~dwilkins/LaTeXPrimer/>)
- LaTeX online editor (<http://www.codecogs.com/latex/eqneditor.php>)
- Doob, Michael, *A Gentle Introduction to TeX: A Manual for Self-study* (PDF). A paper introducing TeX — see page 39 onwards for a good introduction to the maths side of things.
- Oetiker, Tobias; Partl, Hubert; Hyna, Irene; Schlegl, Elisabeth (December 13, 2009), *The Not So Short Introduction to LaTeX 2<sub>ε</sub>* (PDF) (4.27 ed.). A paper introducing LaTeX — skip to page 49 for the math section. See page 63 for a complete reference list of symbols included in LaTeX and AMS-LaTeX.
- The Comprehensive LaTeX Symbol List (<http://tug.ctan.org/tex-archive/info/symbols/comprehensive/symbols-letter.pdf>) — symbols not found here may be documented there.
- Long list of many symbols (<http://www.tex.ac.uk/tex-archive/info/symbols/comprehensive/symbols-a4.pdf>)
- short list of common symbols (<http://amath.colorado.edu/documentation/LaTeX/Symbols.pdf>)
- The esint package for closed double integrals (<http://milde.users.sourceforge.net/LUCR/Math/mathpackages/esint-symbols.pdf>)
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- List of mathematical symbols with their Unicode characters and their LaTeX commands (<http://milde.users.sourceforge.net/LUCR/Math/unimathsymbols.xhtml>)
- MathML: A product of the W3C Math working group (<http://www.w3.org/Math/>), is a low-level specification for describing mathematics as a basis for machine to machine communication.



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## **Appendix H**

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## **IEEE EDITORIAL STYLE MANUAL**



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# IEEE EDITORIAL STYLE MANUAL

**IEEE Periodicals  
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445 Hoes Lane  
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V8 10-30-2014



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## I. INTRODUCTION

### A. Purpose of Manual

This style manual provides general editing guidelines for IEEE Transactions, Journals, and Letters. For guidance in grammar and usage not included in this manual, please consult *The Chicago Manual of Style*, published by the University of Chicago Press.

### B. IEEE Transactions Editing Philosophy

The IEEE's responsibility in editing papers for the Transactions is not to make any determination on or do any editing of the technical content of the papers we work with, but is instead to render the work as readable, grammatically correct, and as consistent with IEEE style as possible.

Since we are concerned with the IEEE house style, the author's style of writing is not changed. A mechanical edit to correct or question grammatical errors is done, obvious inconsistencies or omissions, spelling, and punctuation are fixed. Since we work with highly technical text, extensive formatting of mathematical material is also done.

Some manuscripts require closer editing than others. Some papers, for example, are from authors unfamiliar with the English language. Authors with questions or requiring assistance with the English language may visit [http://www.ieee.org/publications\\_standards/publications/authors/authors\\_journals.html](http://www.ieee.org/publications_standards/publications/authors/authors_journals.html). Often, an IEEE Staff Editor must determine how to correct a grammatical error or in decide what can be safely changed or corrected without altering the author's original meaning. Because of the highly technical nature of the material we deal with, and because of our often limited understanding of that material, it is especially important that Staff Editors do not risk making any unnecessary changes or any that may affect the author's meaning.

Sometimes there are cases where it is simply not possible to decipher an author's meaning or to find a way to correct a sentence. In these cases, a judgment is made either to query the author on the proof about the passage in question, to directly contact the author, or in rare cases, to work with the Transactions Editor or Guest Editor to clarify the material.

### C. Different Models of Editing

There are several different models of editing.

- *Fully edited articles*: These papers are edited and follow the IEEE Transactions/Journal style.
- *Moderately edited articles*: These articles are minimally edited. The abstract, first footnote, figure captions, and biographies are edited to style. The references are checked for accuracy and completion.
  - Excludes:
    - Editing text for grammar, punctuation, spelling or style
  - Includes Editing of:
    - Abstracts
    - Bios
    - Callouts & art captions
  - Ensures accuracy of:
    - Article metadata
    - Automated spell check
    - Reference validation
  - Also includes:
    - Author proofs & alterations

### D. Preprinting (Pre-edit Rapid Posting)

Preprinting is a term used to define the process of posting an author-submitted PDF of his/her manuscript online on the IEEEExplore site. This is done within a day or two of receipt at the IEEE. The author is required to include a signed copyright form with their submission package. If the form is not provided, the paper cannot be preprinted. On Xplore, it appears under "Early Access." This version of the paper has been accepted for publication by IEEE, but



has not yet been edited and may not have been assigned to a print issue. A paper that has been preprinted is considered published.

#### E. Rapid Posting (Post-edit Rapid Posting)

Rapid Posting is a term used to define the process of posting the author-approved edited version online. This is done within 3 weeks of receipt at the IEEE for a fully edited article, and within 2 weeks of receipt for a moderately edited article (see section I-C for explanation). The running head will contain only the publication title. The page numbers would contain generic numbers (e.g., 1 – 10). On IEEEExplore, the article appears under “Early Access” till it is assigned to an issue. Once the article is assigned to a print issue, the article is paginated, and the running head is “opened up” and will contain the volume, issue, month, and year.

#### F. Continuous Pagination

In a continuously paginated journal, each individual article goes through the entire workflow process, is assigned an issue, real-time page numbers, and finally posted to Xplore at the issue level. These articles may already be either pre-printed or rapid posted, not both. **Note:** Once the paginated article is on Xplore, no changes to the content or page layout may occur.

The running head should not indicate a month till the very end of the process. (Note to staff: The <proddate> tags for “first publication and current version...” are suppressed till author review, and unsuppressed prior to final posting to Xplore.)

- **Print Collections** — In addition, several journals have **Print Collections**. A print collection is a literal collection of online issues collected into one print edition. For this reason, additional concerns must be taken into considering when paginating. Each online issue will contain an Index of Contents listing of the papers in the issue. Due to postal requirements, in a print collection, a blank page **MUST** precede the Index of Contents in subsequent issues. The first article must begin on a verso page. Therefore, if the last page of one print collection ends on an even number (left-hand side), TWO blank pages must be left in order to start the next issue on the right-hand side.
- In Print Collections, the front cover will contain information reflecting the pages on which the Index of Contents will appear in each issue. Staff may refer to the “Table of Contents (ToC)” section for more information.
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#### I. Open Access

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## II. EDITING PRINCIPLES

The sections of a paper should generally be edited in the following order:

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- 3) Index Terms
- 4) Nomenclature (optional)
- 5) Introduction
- 6) Body of Paper
- 7) Conclusion
- 8) Appendix(es)
- 9) Acknowledgment
- 10) References
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- 12) Photos and Biographies

### A. Editing the Parts of a Paper

#### Paper Title

In the paper title, capitalize the first letter of the first and last word and all nouns, pronouns, adjectives, verbs, adverbs, and subordinating conjunctions (*If, Because, That, Which*). Capitalize abbreviations that are otherwise lower case (i.e., use DC, not dc or Dc) except for unit abbreviations and acronyms. Articles (*a, an, the*), coordinating conjunctions (*and, but, for, or, nor*), and most short prepositions are lower case unless they are the first or last word. Prepositions of more than three letters (*Before, Through, With, Versus, Among, Under, Between, Without*.) should be capitalized. Example:

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- Self-Pulsation in an InGaN Laser  $\frac{1}{m}$  Part I: Theory and Experiment

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## Shadow Codes and Weight Enumerators

Steven T. Dougherty, *Fellow, IEEE*

(*Invited Paper*)

Example in a table of contents:

Shadow Codes and Weight Enumerators (*Invited Paper*) . . . . . S. T. Dougherty 24

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IEEE TRANSACTIONS ON IMAGE PROCESSING, VOL. 23, NO. 3, MARCH 2014

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*For two authors:*

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The authors are with Fujitsu Laboratories Ltd., Atsugi, Kanagawa 243-01, Japan, and also with the Department of Electrical Engineering and the Electronics Research Laboratory, University of California at Berkeley, Berkeley, CA 94720 USA (e-mail: [corresponding-author@ieee.org](mailto:corresponding-author@ieee.org)).

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L. A. Tepper, retired, was with the Applied Research Laboratory, Bellcore, Morristown, NJ 07851 USA. He resides in Laguna Niguel, CA 92677 USA (e-mail: [retiredauthor@yahoo.com](mailto:retiredauthor@yahoo.com)).

*Deceased author:*

P. Dorigo, deceased, was with the Progetto di Intelligenza Artificiale e Robotica, Dipartimento di Elettronica e Informazione, Politecnico di Milano, 20133 Milano, Italy.

*Consultant:*

P. Leff Jr. was with the Department of Biomedical Engineering, University of Virginia, Charlottesville, VA 22908 USA. He resides in Charlottesville, VA 22908 USA.

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### Nomenclature

Nomenclature lists (lists of symbols and definitions) generally follow the Abstract and Index Terms and precede the Introduction. This type of list is characterized by the following.

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#### NOMENCLATURE

|             |                                   |
|-------------|-----------------------------------|
| <i>SPQ</i>  | Strictly proper pole constraints. |
| <i>M</i>    | Minimal weighted sensitivity.     |
| <i>P(s)</i> | Physical feedback.                |
| <i>W</i>    | Weighting.                        |
| <i>Q</i>    | $= P - 1$ . Improper function.    |
| <i>S, I</i> | Signal density, $= P, M$ .        |

**NOTE:** Acronyms defined in a Nomenclature list do not need to be defined again in the text. If the section headings are made up of only previously defined acronyms, we should continue to add the acronym in parentheses next to the definition, as it becomes unreadable otherwise.

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Standard specifications have been established for Transactions text section headings. There are four levels of section headings with established specs: primary (section), secondary (subsect1), tertiary (subsect2), and quaternary (subsect3) heads.



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#### I. INTRODUCTION

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##### A. Formal Frameworks

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1) *Sophisticated Local Control*: Sophisticated local control is applied when ...

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a) *Communication policies*: Policies developed to improve communication ...

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#### REFERENCES

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*Appendix headings* are a special case. The primary heading(s) in the Appendix or Appendixes (note spelling of plural) are set according to the usual style, except that there is flexibility in the enumeration of the heading. The author may use Roman numerals as heading numbers (Appendix I) or letters (Appendix A). Either is acceptable. The Appendix is not preceded by a Roman numeral. Follow the rules given earlier for labeling subsidiary heads. Note that if there is only one Appendix in the paper, leave the Appendix unnumbered and unnamed as is. (Appendix subheads should also not be enumerated in this case.) Examples:

#### APPENDIX

##### APPENDIX I

##### PROOF OF THEOREM

##### APPENDIX A

##### PROOF OF THEOREM

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#### Introduction

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If it is not possible to use the first word or character of the Introduction as an initial cap (i.e., if the paper begins with a quotation mark), try rewriting the sentence and query the author. See Section II-A of this guide for type specs of the initial cap.

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*Hyphens and Periods:* Hyphens and periods are accepted, if consistent in paper, e.g., (1a), (1.1), (1-1). This should be done consistently throughout the paper.

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Refer to the Appendix in text as "given in the Appendix." Note that the plural of Appendix is Appendixes. Also note that all figures and tables in the Appendixes must be labeled in consecutive order with the other figures in the paper. Never start a separate numbering system or group of numbers for the figures or tables in the Appendix section.

### Acknowledgment

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When citing names within the Acknowledgment, use first initials only, not full names. Drop Mr., Mrs., or Miss (list first initial and last name only). For Dr. or Prof., use the Dr. or Prof. title with each name separately; do not use plural Drs. or Profs. with lists of names.

All acknowledgments of financial support must be removed from the Acknowledgment section and placed in the first footnote/author affiliation.

Any acknowledgments of permission to publish and disclaimers to the content of the work made to/by the author's employer may be added as an Acknowledgment section.

Rewrite the Acknowledgment section to be read in the third person. Rewrite it even if the paper is given in the first person.

### References

A few guidelines related to the editing of references are summarized here. See Section V of this manual for a more complete discussion of reference style.

The numbering of references is employed by citing one reference per number. Every reference in a Transactions reference list should be a separate number entry. Use of one reference number to designate a group of references is not preferred, and is discouraged. If the author-supplied reference list is unnumbered, the Staff Editor must provide numbers, or if the list contains multiple references, these should be separated and renumbered by the Staff Editor. If numbering or renumbering is necessary, then in-text references to the reference list must be checked and renumbered by the Staff Editor.

Footnotes or other words and phrases that are part of the reference format do not belong on the reference list. These full footnotes or extraneous phrases must always be removed from the list, changed into text or footnotes on the appropriate page, and the references renumbered (renumber reference citation in text as well). Even the words "For example" should not introduce references in the actual list, but should instead be included in parentheses in text (or in a footnote), followed by the reference number, i.e., "For example, see [5]."

Do not say "in reference [1] ..."; rather, the text should be edited to read simply, "in [1] ..." The author's name should not be included in a text reference with a number (i.e., "In Smith [1]") and should be changed to "in [1]" except in such cases where the author's name is integral to the understanding of the sentence (e.g., "Smith [1] reduced calculated time ..."). Reference dates should not be used as reference identifiers and should be deleted in text except in rare cases where the date is somehow relevant to the paper's subject.

Sometimes an author will refer to a specific figure of a reference or to a specific page or equation from a reference. To avoid confusion, rewrite phrases such as "in Fig. 2 of reference [1]" to the IEEE cross-reference notation "in [1, Fig. 2]." Similarly, rewrite phrases such as "in equation (8) of reference [1]" to be [1, eq. (8)]. Other phrases may be rewritten as [1, Sec. IV], [1, Th. 4.2], or [1, Ch. 3].





If an author lists the same reference more than once on the reference list, giving a new reference number for each page or part of the same source that is cited, these separate references should all be made into one reference and the separate citations of pages, equations, etc., should be made in text using the notation explained in the previous paragraph.

If a reference author's name is mentioned in text, check its spelling against the reference list.

### Text Citation of Figures and Tables

All first citations of figures and tables in the paper must be in numerical order. If a figure is not mentioned or if the first text mentions are not in order, call or query the author and/or renumber the figures where necessary. Citations to figures in text always carry the abbreviation "Fig." followed by the figure number. The abbreviation is used even when it begins a sentence.

**Figures:** If labeled, parts of figures (callouts) should be 8-pt. lower case Roman letters within parentheses. Whenever possible, all caption parts shown on the figure must be removed and keyed along with the caption.

The general style for captions is such that each caption number should be cited with the abbreviation "Fig." and the number, followed by a period, an em space, and then the text of the caption. The first word of the caption should always be capitalized, regardless of any style that may be chosen to list caption parts (a), (b), etc., if included. In general, do not use A, An, or The at the beginning of a figure or table caption. Example:

Fig. 1. Theoretical measured values of  $n$ .

There are several acceptable styles for listing the parts of the figure in the caption. Be consistent within each paper, but otherwise use whichever style is most convenient for the figure. Regardless of which caption notation is used, the citation of (a), (b), etc., should always appear before the corresponding caption part. Examples:

Fig. 1. Intercomplex crosstalk characteristics. (a) Electrode transmission.  
(b) Interelectrode crosstalk.

Fig. 2. (a) Variation of effective mode index with time. (b) Step-index change.

Fig. 3. Output resistance as a function of channel doping for 1-m-long gate.  
(a) InGaAs and (b) InP JFETs with pinchoff voltage as a parameter.

Fig. 4. (a) and (b) Plain and side views, respectively, of the experimental setup used to measure the effective diffraction loss which can be achieved using the feedback technique.

Do not use:

Fig. 1. (a) Electrode transmission. (b) Interelectrode crosstalk.

If a figure after reduction will run more than one 21-pica column in width, the caption should be flush left on 43 picas.

If parts of a figure after reduction will run the length of more than one page, the full descriptive part of the caption should be cited with the first part of the figure followed by the corresponding caption for the part. On the subsequent pages, the word (*Continued.*) will be placed under the carryover parts of the figure followed by a repeat of the full descriptive part of the caption and the corresponding caption for the carryover parts.

**Captions for Landscape/broadside figures:** The text should appear below the figures and facing outward at all times. Example:

Fig. 6. True and estimated spectra for a real data sequence. (a) True spectrum.

Fig. 6. (*Continued.*) True and estimated spectra for a real data sequence. (b) Estimated with the periodogram.

**Tables:** The general style for table captions is such that each caption number should be centered above the table with the label TABLE (set in 8-pt. caps) and the enumeration given in Roman numerals. The descriptive text of the caption should be centered directly below the table number caption and is set in 8-pt. and 6-pt. caps. The captions are usually centered on 21 picas, unless the table will be wider than one column width, in which case the table caption should be centered on 43 picas.

The descriptive text of the table caption does not contain a period at the end of the caption, although punctuation may be necessary within the caption itself. In general, table captions should be set as an inverted pyramid.

As in figures, labeled parts of tables should be 8-pt. lower case Roman letters within parentheses. The style for listing the parts of a table in the caption and in text depends on whichever style is most convenient for the table. The most acceptable style is to follow the conventions for callouts of figures. Example:



TABLE I  
PARAMETER VALUES

TABLE II  
OPTIMAL WAVELENGTH AS A FUNCTION OF POLARIZER ANGLE. (a) WAVELENGTH  
FOR EXTERNAL CAVITY. (b) ESTIMATED WAVELENGTH FOR LASER DIODE

A single rule should be added above and below the table body. Use the **hrule** macro to create rules. The type specs for the text of a table is 8-pt. TR for full length papers, brief papers, and short papers.

The same rules as in figures apply for listing table part labels (callouts).

Table footnotes should be 8-pt. type and should be placed below the bottom rule of the table.

#### Obtaining permission to reuse copyrighted material

- 1) **Reusing IEEE graphics previously published in IEEE publications. Author should email IEEE Intellectual property department at: [copyrights@ieee.org](mailto:copyrights@ieee.org). In most cases, the only requirements will be to give full credit to the original source and to obtain the author's approval (as a courtesy to the author). At the end of the caption, add the reference number of the papers from which the graphics are being used.**
- 2) **Reusing graphics previously published in non-IEEE publications. Author must have obtained permission to republish from copyright holder (in most cases, this is the publishing house (not the author of the paper). The wording is provided by the author (usually supplied by the publishing house itself). This text is added at the end of the caption.**

#### Photos and Biographies

IEEE Transactions author biographies are generally divided into three paragraphs. However, if appropriate information for each paragraph is not provided by the author, the biography may be only one or two paragraphs.

The author's photograph is sized at 6 picas wide by 7.5 picas deep and is surrounded by the biography.

The biography begins with the author's full name and IEEE membership history as listed in the *IEEE Membership Directory*. The author's name appears in boldface type and must match the byline. A nickname may appear within parentheses, e.g., Sung-Mo (Steve) Kang, but not in the byline. The format for listing the IEEE membership history is to list each grade of membership attained followed by an apostrophe and the year it was attained, with each year and grade combination separated from the others by an en dash. Note that if an author attains the same membership grade in more than one year, list only the first year that it was reached. Check the current membership listed with the biography against the byline.

Abbreviations for IEEE membership grades are: S (Student Member), A (Associate Member), M (Member), SM (Senior Member), F (Fellow), LA (Life Associate Member), LM (Life Member), LSM (Life Senior Member), and LF (Life Fellow). Note that A stands for Associate, not Affiliate, Member. Affiliate memberships are not listed in the byline or biography membership history.

Delete all references to IEEE membership from the text of the biography.

**First Paragraph:** If provided by the author, the first paragraph may contain a place and/or date of birth (list place, then date). Next, the author's educational background is listed. When listing degrees earned, the biography should state "[S]he received the Ph.D. degree from ..." (not "[S]he received [her] his Ph.D. degree from ..."). Always add the word degree after a degree title if it is not included. Include the years degrees were received. If the author was educated overseas, the names of the degrees earned may not be familiar. Abbreviations for some common international and domestic degrees are:

Dipl.Ing., Diplom-Physiker, Dr. Ing., Dr. Phil., Dr. Eng., B.S., S.B., B.Sc.(Hons.), B.E.E., B.S.E., M.Eng., M.Sc.(tech.), M.S.E.E., M.S.E., Civilingenir, Lic.es Sci., Lic.es Lett.

Add the locations of universities and colleges the first time they are mentioned if not included (*refer to the University website for location*). For U.S. state-named universities, repeat the state name in the location, and included the country (e.g., University of Colorado, Boulder, CO, USA); but for city-named universities, repeat the name of the city when giving the location (e.g., University of Chicago, Chicago, IL, USA). For universities outside the U.S., give locations with the name of the city (postal abbreviations of Canadian Provinces, if used) and the country the first time.

Use lower case for the author's major field of study.

**Second Paragraph:** The second paragraph of the biography should list military and work experience, including summer and fellowship jobs and consultant positions. Job titles are capitalized. The current job must have a location; previous positions may be listed without one (retain if given). Do not abbreviate city names, Company, Laboratory, or Department. Use standard names for all countries. If there is space, information the author provides about



previous publications may be included at the end of this paragraph. Edit out long lists of published books or articles. Instead use the sentence s(he) “is the author of several books and many published articles.” The format for listing publishers of an author’s books within the biography is: *Title of the Book* (publisher name, year) similar to a reference. List author affiliations with non-IEEE journals. The author often notes current and sometimes previous research interests. If space is available, these may be retained; otherwise, edit out the prior interests and leave in the current. Any homepage of the author may be listed in the biography only.

Do not repeat the author’s name in the second paragraph; use “he” or “she.”

**Third Paragraph:** The third paragraph begins with the author’s title and last name (e.g., Dr. Smith, Prof. Jones, Mr. Kajor, Ms. Hunter). It lists the author’s memberships in professional societies other than the IEEE and his or her status as a Professional Engineer if given. Finally, list awards and work for IEEE committees and publications, affiliation with other professional societies, and symposia.

Personal notes such as hobbies should be deleted from the biography. Examples:

**Michael C. Author, Jr.** (S’87–A’89–SM’90–F’93) was born in New York, NY, USA, on March 2, 1969. He received the B.S. degree in applied mathematics from the University of Michigan, Ann Arbor, MI, USA, in 1989, the M.S. degree in mathematical physics from Stanford University, Stanford, CA, in 1991, and the Ph.D. degree in electrical engineering from the Massachusetts Institute of Technology, Cambridge, MA, USA, in 1995.

From 1993 to 1995, he was with the Raytheon Corporation, Bedford, MA, USA. From 1995 to 1996, he was with the General Electric Space Laboratory, Valley Forge, PA, USA. During 1996–1997, he was a Fulbright Lecturer at the University of Madrid, Madrid, Spain. He is currently an Associate Professor of Electrical Engineering at the University of Maryland, College Park, MD, USA. His research has been concerned with reentry plasma effects and microwave diagnostics of plasmas.

Dr. Author, Jr. is a Registered Professional Engineer in the State of Pennsylvania.

**Katsunari Okamoto** was born in Hiroshima Prefecture, Japan, on October 19, 1949. He received the B.S. degree from Rutgers University, New Brunswick, NJ, in 1979 and the M.S. degree from Monmouth University, Long Branch, NJ, USA, in 1984.

He was a Postdoctoral Fellow at the University of Tokyo in 1978. He joined the Ibaraki Electrical Communication Laboratory, N.T.T., Ibaraki-ken, Japan, in 1979, where he was engaged in research on the optimum waveguide structure of optical fibers. At present, he is a Member of Technical Staff at Bellcore, Red Bank, NJ, USA.

Dr. Okamoto is a member of the Institute of Electronics and Communication Engineers of Japan.

**NOTE:** If no photograph is available or the journal does not require them, the biography is set 8/9×21 picas.

### Squibs

If no biography or photograph is available, a squib is used. The phrase is run at 8/9 ×21 picas, flush left. Example:

**James A. Author, (S’87–A’89–SM’90–F’93) photograph and biography not available at the time of publication.**

### D. Other Text to Edit

#### Footnotes

Footnotes should be numbered in consecutive order throughout the text. In full length, brief, and short papers, they are 8/9 TR ×21. Each footnote should be a new paragraph. The footnote numbers are superscripts in text and in the actual footnotes. In text, place the superscript footnote numbers after punctuation such as periods, commas, and parentheses, quotation marks, but generally before dashes, colons, and semicolons in a compound sentence. The footnotes should be placed at the bottom of the text column in which they are cited.

#### Lists in Text

There are three types of lists in text: run-in lists, displayed lists, and where lists. The ordering of labeling for all lists is 1), 2), 3) followed by a), b), c), and then i), ii), iii). All are Roman; note single parenthesis. The order of indentation is 1 em, 2 ems, 3 ems.

**Run-In Lists:** Lists that run in with text must be grammatically correct. They must also be introduced by a colon, separated by semicolons, and have parallel construction. Example:

The carrier–phonon interaction matrices are given by: 1) polar optical phonons; 2) deformation potential optical phonons; and 3) piezoelectric acoustic phonons.

**Displayed Lists:** Lists that are displayed may be either incomplete sentence items or full sentence items. Incomplete sentence items contain a few items, are very short, are grammatically parallel, and are handled in two ways. If the items are not mentioned in the text or are less than three items, run in as shown in the example for run-in lists. If, however, the items are mentioned later in text, introduce the item with a colon, number the items, begin



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the entry with a lower case letter, and set block paragraph style. Use semicolons between items and a period at the end of the list. Example:

This operating scenario provides all of the contributors necessary to configure a resonant power distribution system:

- 1) the implementation of capacitor power factor correction on the power line;
- 2) the presence of nonlinear load;
- 3) the tuning of the power line by the load adjustments to a frequency present in the nonlinear generator.

Incomplete sentence items that are mentioned in text may also be formatted as shown in the example for full sentence items.

Full sentence items may be introduced by “that” or other words taking object and are rewritten to end with a period. If the items are introduced by a sentence ending with a colon, change the colon to a period. Number all items, start each entry with a capital letter, and end with a period. Example:

The synthesis is performed in three major steps.

- 1) Geometry is generated for the selected module variants.
- 2) Shape variants using different fold counts for resistors are generated for each module.
- 3) Routing and postprocessing complete the final layout.

*Where Lists:* Where lists define variables in the equations preceding the list. They are characterized by incomplete sentences and follow the same rules as *Nomenclature* lists, with the following exceptions.

- 1) There is no primary heading.
- 2) The left-hand side is indented one em space.
- 3) The first letter on the right-hand side is lower case.
- 4) Each item ends with a semicolon (except for the last item, which ends with a period).
- 5) The lists are at least three items long; if fewer than three items, the list is generally run in paragraph form.

Follow author preference for run-in or displayed lists. Example:

where

|              |                                              |
|--------------|----------------------------------------------|
| $\Delta V_S$ | $= \Delta V_S \cos(\omega't + \phi')$ ;      |
| $\Delta V_S$ | amplitude of supply voltage flicker;         |
| $\omega'$    | angular frequency of supply voltage flicker; |
| $V_{sf}$     | supply voltage amplitude;                    |
| $\omega$     | supply angular frequency.                    |

Note the alignment of the equal sign with the right-hand side.

Lists having mixed items (start with an incomplete item, then have a full sentence explanation) are treated as a full sentence item list.

#### Note Added in Proof

An author may wish to add a brief note in the proof stage, citing results obtained after acceptance of the paper or mentioning additional references that have come to the author's attention since acceptance. This added information is usually inserted at the end of the Conclusion section of the paper or in whatever section contains the last paragraph of the main body of the paper. As long as the note is not a major change to the paper or more than a few lines long, the addition generally does not require further review procedures. Use the tertiary heading “Note Added in Proof:” (run into text), but set in boldface italic with no enumeration and an em space indent. Example:

***Note Added in Proof:***

#### E. Other Types of Papers

##### Editorials

This category of papers includes the various types of introductory papers, such as Editorials, Guest Editorials, Forewords, Introductions, and Editorial Announcements that appear at the beginning of issues as non-technical introductory material. A discussion of the papers in an editorial should follow the order of the table of contents. The editorial may contain illustrations, citations, and references. Follow general rules for editing. An acknowledgment does not contain a heading. If a heading is required, set as a separate section and follow the primary heading specs without enumeration. *Note:* In the Editorial, the Acknowledgment does not need to be in third person.



Procedures and style for Editorials include the following.

*General Specs:* Type specs are the same as for full length papers. The initial cap remains the same. The title of the Editorial is set in 24 pt. as in a full length paper title. There is no Abstract. There is a rule above the DOI.

**NOTE:** Editorials generally do not carry a section heading above the title. Center the word “Editorial” in 24-pt. type above the title.

*Byline:* Note that the byline for the Editorial does NOT appear below the title as it does in a full length paper. The name of the author of the Editorial or Foreword (usually the Editor or Guest Editor) (called “signature”) appears at the end of the Editorial, 6 pts. below the end of text, in 10-pt. and 8-pt. caps. Stack and align the name or names with an identifier such as “Guest Editor” which should appear in italics next to the name. The affiliation should appear as a “list” under each name. The right edge of the longest of these aligned lines should then be flush right at the end of the last column of text. Example:

M. K. SAIN, *Guest Editor*  
Department of Electrical Engineering  
University of Illinois  
Urbana, IL 60617 USA

*Biographies and Photos:* Biographies and photographs that appear with Editorials are set differently from regular biographies and photos in the Transactions. They are, for example, not 8/9 type, but are the same type size as the text of the Editorial (normally 10/12). In addition, Editorial biographies are: first 13 lines  $\times 32$ , rest at 43 picas. The photos are reduced to  $9\frac{1}{2} \times 12$  picas.

*Copyright Line:* Run a copyright line for the Editorial, even if no copyright form is submitted by the Editor.

### Brief Papers

**Brief papers are set up like full length papers, except that the paper title is set in 16-pt. TR, centered on 43 picas. These papers do contain Abstracts and also take the initial cap. The byline includes the membership grade. See Section I-B. They do not contain biographies and photographs of the authors. Footnotes, references, and figure/table captions are 8/9 TR. The papers carry issue running heads on both left and right pages.**

### Short Papers, Correspondence, and Communications

Short papers are set up like full-length papers, except that usually they are 9/11 type and their titles and bylines are smaller type and run across only one column. Usually, short paper titles are 10/12 bold with bylines 9-pt. upper and lower case. These papers do contain Abstracts, but do not take the initial cap. The membership grade is not included in the byline. Author biographies and photos are not included. Footnotes, captions, references are 8/9 type.

### Comments and Replies

Comments are generally in response to a previously published paper. The Comments and Author(s) Reply are short papers published together in that the “Reply” is in response to the Comments. These short items may appear without Abstracts. A special format applies for Comments and Author(s) Reply. Begin the first sentence with “In the above paper [1], ...” Reference [1] is the commented paper’s citation, will appear as Reference [1] in the References section. Include a copyright line for Comments and Replies even if no new forms are required from the author(s). Some publications refer to these articles as Discussions and Closures. Index Terms are optional.

Example of the Comments:

*Title:* Comments on “Harmonics: The Effects on Power Quality and Transformers”

*Byline:* Keith H. Sueker

*Footnote:*

Manuscript received July 15, 1995.

The author is with the School of Engineering, Vanderbilt University, Nashville, TN 37235 USA (e-mail: k.sueker@ieee.org).

Digital Object Identifier 10.1109/JQE.2006.12345

**NOTE:** The footnote here relates back to the original paper being commented upon. The title is not repeated.

Example of the Reply:

*Title:* Authors’ Reply



*Byline:* Robert D. Henderson and Patrick J. Rose

*Footnote:*

Manuscript received October 3, 2009; accepted October 5, 2009. Date of publication November 2, 2009; date of current version November 25, 2009.

The authors are with RDH Consultants, Inc., Charlotte, NC 28241 USA.  
Digital Object Identifier 10.1109/JQE.2006.12348

### Corrections/Errata

The format for a Corrections is basically the same as for the Comments, except that a Corrections does not carry a Reply. Run a copyright line with a Corrections even if no new forms are received from the author(s). Corrections that has been generated in-house may be labeled “Erratum,” and should also follow the standard format, although the byline may be omitted because the IEEE Transactions/Journals Department assumes authorship of the Corrections. *Note:* The plural form of the word is used in the title, even if there may be only one correction. Example of a “Corrections” article:

*Title:* Corrections to “On the Exact Realization of LOG-Domain Elliptic Filters Using the Signal Flow Graph Approach”

*Byline:* Costas Psychalinos and Spiridon Vlassis

*Footnote:*

Manuscript received May 1, 2003.  
The authors are with the Physics Department, Electronics Laboratory, Aristotle University of Thessaloniki, GR-54124 Thessaloniki, Greece (e-mail: cpsychal@physics.auth.gr; svals@skiathos.physics.auth.gr).  
Digital Object Identifier 10.1109/TCSII.2003.814788

*Example of Errata:*

*Title:* Erratum

*Byline:* There is no byline for an erratum, as this is created if the department (staff/vendor) is responsible for the error.

*Footnote:*

Manuscript received January 20, 2004.  
Digital Object Identifier 10.1109/TVLSI.2004.830244

### Book Reviews

Some publications carry Book Reviews. The type specs of the text are the same as for a short paper or correspondence; however, the title runs additional information about the book that is being reviewed. The title is separated from the book’s author by an em dash. Included in parentheses is the city of publication, publisher, date of publication, the total number of pages of the book, and the price. Outside of the parentheses is the reviewer’s name in italics. Some Transactions carry a short biography of the reviewer under the title. Book Reviews appear in the table of contents with a listing for both the author of the book and the reviewer. Example:

*Title and Byline:*

**The Analysis and Design of Pneumatic Systems**—B. L. Andersen. (New York: Wiley, 1987, 302 pp., \$65.00.)  
*Reviewed by J. L. Shearer.*

*First Footnote:*

The reviewer is with the College of Engineering, Idaho State University, Pocatello, ID 83209 USA. Digital Identifier 0090-6778/TNN.2005.828433.

*Table of Contents:*

The Analysis and Design of Pneumatic Systems—B. L. Andersen . . . . . *Reviewed by J. L. Shearer* 123

### Obituaries/In Memoriam

Obituaries are usually run as the first page of an issue, like an Editorial. They are set up with the same specs as Editorials. Obituaries normally are formatted as one column, at 36 picas width. They may carry a photo of the person being memorialized, usually the same size as in Editorials (9-1/2 ×12 picas). The name appears above the photograph in 12-pts., boldface. The photograph is generally centered above the text. The years of birth and death are generally cited at the bottom of the photo in 12-pts., boldface type in parentheses. The obituary is normally set as one column, across a 36-pica width.



## F. Editing Style for Transactions

The following provides a summary of the most important style distinctions to be made in the final copy editing of a Transactions paper.

### Acronyms

Define acronyms the first time they appear in the Abstract as well as the first time they appear in the body of the paper, written out first as part of the sentence, followed by the acronym in parentheses. Widely used or familiar terms should be defined (see Section VIII-F of this manual for some terms that must be defined the first time they are used in text). Acronyms do not need to be defined in the text if mentioned in the Nomenclature. Coined plurals or plurals of acronyms do not take the apostrophe as per *Chicago Manual of Style*. Example: FET (singular); FETs (plural).

Indefinite articles are assigned to abbreviations to fit the sound of the first letter: an FCC regulation; a BRI.

### Spelling

Obviously, in reading and editing a paper, misspellings and typographical errors are top priority for correction. Note that IEEE Transactions use the first spellings indicated in our first reference, the most current edition of *Webster's New Collegiate Dictionary*.

**British Spellings and Terminology:** Change all British spellings to American spellings. In particular, watch for "our" endings in words like "behaviour" (change to "behavior") and "re" endings in words like "centre" (change to "center"). Also watch for the use of "s" rather than "z" in words like "polarisation" (change to "polarization"). See "Common Hyphenations and Misspellings" in Section VIII-E.

### Trademarks

The trademark symbol, <sup>TM</sup> and ® are no longer used. Capitalize the first letter in the trademark name only. Follow the author's notation. The symbols <sup>TM</sup> and ®, which often accompany registered trademark names on product packaging and in advertisements, need not be used in running text.

### Plurals

Plurals of units of measure take the "s." For example, the plural form of 3 mil is 3 mils; 3 bits/s instead of 3 bit/s. The plural of calendar years do not take the apostrophe before the "s." For example, the plural form of 1990 is 1990s.

### Hyphenation Rules

For hyphenation and spelling guidelines, IEEE style follows: 1) the list of preferred spellings and hyphenated words; 2) the guidelines discussed in the Grammar and Usage in Transactions section of this guide; and 3) the first version of the spelling given in *Webster's Tenth New Collegiate Dictionary*. Do not hyphenate most compound modifiers if they occur after the noun being modified, even if hyphenating them before the noun. "Except for *cooperate* and *coordinate*, use a hyphen if the prefix ends in a vowel and the word that follows begins with the same vowel."

*Example:*

**The plan was well prepared. The man was little known. The woman was better qualified. His boat was 42 feet long. He has a 42-feet-long boat. T was the data period of the 40-Gb/s data signal. The 160-GHz MLLD was a diode in which a 40-nm-long saturable absorber was located.**

Follow the author's preference if the result is consistent and clear. The most important hyphenation guideline is to be certain that the hyphenation for a particular word or group of adjectives is consistent within a particular paper.

### The En, Em, or Two-Em Dash

The en dash represents the words "to," "through," or "and." Use it between page numbers, reference numbers, figure citations, academic years, proper nouns, names, a range of values, or for opposites. Examples are: pp. 10–15, 1984–1990, Jones–Smith theorem, input–output, voltage–current curve, analog–digital converter, 10–20 cm. Also, use the en dash in chemical abbreviations such as Ni–Al–Si. When using the en dash to represent a range, if the word "from" occurs, the word "to" must be used rather than an en dash (ranges from 5 to 50 times).

The em dash is used in ordinary writing to mark a suspension of the sense. It is also used like parentheses, to mark a subordinate thought within a sentence.





### Grammar

Check closely for lapses of clarity, subject/verb agreement, and parallel clause construction. See samples below and a more detailed discussion in the Grammar and Usage in Transactions section of this guide.

#### Number:

A number of samples were taken ...

A number  $N$  expressing the relation  $x/y$  is chosen ...

#### Data:

The data were collected ... (always plural)

#### Series:

A series of tests was run ... (always singular with “a”)

#### Some, All, Half:

Some (all, half) of it is ...

Some of them are ...

#### Quantity:

Three volts was applied ...

Four grams was added ...

### Contractions

Contractions such as “don’t” and “can’t” are not used in technical text. Change to “do not” and “cannot.”

Note: “don’t care,” “best-case,” and “worst-case” are allowed and used often in journals like TCAD.

### Capitalization

In general, discourage capitalization in text except where absolutely necessary. For example, only proper names attached to the names of laws, principles, theorems, etc., get capitalized (Boyle’s law, Newton’s first law, etc.). Computer commands are in computer tags and remain small caps; most computer languages (Cobol, Java, LISP, PERL, etc.) are upper and lower case. In text, the names of IEEE publications are 10-pt. and 8-pt. caps: TRANSACTIONS, IEEE SPECTRUM, PROCEEDINGS OF THE IEEE.

### Math

Some brief guidelines for editing math are explained here. For further discussion, see Section IV of this guide.

- 1) Variables are set italic; vectors are usually boldface italic (if distinguished by the author).
- 2) Remove commas around variables in text.
- 3) If not included by the author, always add a zero before decimals, but do not add after (e.g., 0.25).
- 4) Set the use of the author’s parentheses and brackets (i.e., [0,1) may be correct).
- 5) Spell out units used in text without quantities (e.g., “where the noise is given in decibels”). For units appearing with quantities, use the standard abbreviations listed in Section VIII-G.
- 6) Numbers and units used as compound adjectives may be hyphenated only if needed for clarity: 10-kV voltage, 5-in-thick glass. Do not insert a hyphen when they are not used as adjectives: a current of 2 A, a line 4 in long, a length of 3.05 mm.
- 7) Use thin spaces instead of commas between numbers in tens or hundreds of thousands (e.g., 62 000, 100 000, but 4000).
- 8) Always change  $\mu$  to  $\mu\text{m}$ , “micron” to “micrometer,” “submicron” to submicrometer.” Always change cycle per second to hertz (Hz); cycle per second may not appear as cycle, cps, c/s, csec. See “Table of Units and Quantity Symbols” in Section VIII-G.
- 9) In text, break down (shill) multiline (built-up) fractions so they can be placed on one line. Sometimes parentheses may need to be added to distinguish between expressions, especially when a minus appears [e.g.,  $\frac{a}{b-c}$  becomes  $a/(b-c)$ ],  $\frac{c-d}{k+4}$  becomes  $[(c-d)/(k+4)]$ .
- 10) In exponential expressions [e.g.,  $e^{-(j\omega t)xyzk}$ ], there are sometimes long and complicated superscripts. These may be brought down on line with the substitution of “exp” for “e” and the addition of square brackets (e.g.,  $\exp[-(j\omega t)xyzk]$ ).
- 11) Distinguish between lower case italic “ell” or “oh” versus one and zero.
- 12) Always use numerals for numbers written with units. Otherwise, spell out numbers below 11, and use numerals for others unless they begin a sentence or are combined in a phrase (gives 7 to 13 times more).
- 13) Use zeroth, first,  $n$ th,  $(k+1)$ th, not 1st, 2nd,  $(k+1)$ st, etc.
- 14) Use the word “equation” at the start of a sentence, but in text, just use the number [e.g., in (1)].





15) Use the \$ symbol versus “dollars” in sums of money.

16) The slash (/) is acceptable in place of the word “per” when it lends to the clarity of the sentence. For example: “the ratio of 16 samples/s to 35 samples/s as compared to ...”

**Ellipses:** In mathematics an author may use dots (ellipses) to show continuation in an expression (e.g.,  $x_2, \dots, x_{16}$ ). The type of mathematical expression will determine whether the ellipses points are set on the baseline or centered. If commas or operational signs are present, they are placed after each term and after the three ellipses points (almost all expressions will use three points). If operational signs are used, the ellipses are centered on the operator. When commas are used the ellipses are on the baseline. Example:

$$x_1, x_2, \dots, x_n \text{ not } x_1, x_2 \dots x_n$$

$$x_1 + x_2 + \dots + x_n \text{ not } x_1 + x_2 + \dots x_n$$

$$y = 0, 1, 2, \dots \text{ not } y = 0, 1, 2 \dots$$

$$x_1 x_2 \dots a_n \text{ not } x_1 x_2 \dots a_n$$

**Conditions:** In displayed equations, there should be a comma or parentheses and a two-em space between the main expression and the condition following it. Example:

$$x = y n^{-2} \quad \forall n = 3$$

$$x = y n^{-2}, \quad \text{if } n = 3 - y^{-4}.$$

$$x = y n^{-2}, \quad y = 3, \dots, m$$

**NOTE:** There is no comma before a for all “ $\forall$ ” symbol.

**Compound Units:** Compound units should be separated by a multidot (e.g.,  $4 \text{ V} \cdot \text{s}$ ), but leave the slash if the author uses it since this has a different meaning (for instance,  $6 \text{ V/s}$  means volts per second). It is also possible to use a negative power to put a unit in the denominator:  $\text{cm/s}^2 = \text{cm} \cdot \text{s}^{-2}$ . Parentheses may be used to clarify a unit:  $\text{g}/(\text{cm} \cdot \text{s})$  or  $\text{g} \cdot \text{cm}^{-1} \cdot \text{s}^{-1}$ .

**Use of Periods and Commas:** Equations which conclude a sentence should end with a period. The only time punctuation is used to lead into an equation is when the lead-in text is a complete sentence. Example:  
where we had the following:

$$x = Y + Z.$$

or where, i.e.,

$$x = Y + Z.$$

Commas appearing at the ends of equations are deleted unless they are critical to the punctuation of the sentence containing the equation.

### Equation Numbers

Check that equation numbering is consecutive, that it appears flush right on line with the last line of an equation, that there are no repeats or missing numbers, and that a correct numbering style has been used.

### Displayed Equations

Material in displayed equations is automatically italic unless otherwise indicated by the author. Some simple general rules apply. All variables are italic. Function names and abbreviations are Roman, as are units, unit abbreviations, complete words, and abbreviations of words. Superscripts and subscripts follow this same formula: when they are variables, they are italic; when they are abbreviations of words (such as “in” and “out” for input and output), they are Roman. Single-letter superscripts and subscripts may be italic even if they are abbreviations, unless this leads to inconsistency between italic and Roman characters for similar types of subscripts.

### Typical Problems

Which does the author mean: zero or “oh”? one or “ell”? subscript variable or on line? A general guideline to help resolve these questions before querying the author is to read carefully through the paper—does the author mention “O” for output or use a series of numbers like 0, 1, 2,?—and look through the illustrations—does  $V_s$  appear in the figures or is it  $V_s$ ? This may provide clues.

**G. General Layout Rules**

- 1) Normal page depth for a Transactions is 60 picas (called even).
- 2) Pages may run one line long (61 picas) or short (59 picas), but facing pages (left and right) must be the same depth.
- 3) Transactions papers are set in a two-column format. Each column is 21 picas wide, with a 1-pica space between the two columns, giving a total page width of 43 picas.
- 4) Specifics of type area spacing are approximately 18 pts. between text and footnotes or figures and text, 6 pts. above and below equations and lists, 12 pts. above primary heads, at least 6 pts. above secondary heads, and 3 picas between biographies.
- 5) Figures and tables are placed at the tops of columns as close to their first mention as possible, but preferably after the mention.
- 6) Figures and tables progress vertically, not horizontally, on pages.
- 7) Footnotes must appear at the bottom of the column where they are first mentioned.
- 8) There must be at least two or three lines of text under a head at the bottom of a column.
- 9) Never leave widows at the tops of columns when breaking text. (A “widow” is any single last line of a paragraph, even if it is of full column width.) The exceptions are when widows are used to introduce equations or when they are in the Reference section.
- 10) Avoid breaking multiline equations so that one line appears at the bottom of a column and the others at the top of the next column.
- 11) The starting page number is determined by checking the previous issue—it is the next page number after the last page of the preceding issue, including any fillers. Issues beginning a new calendar year always start with page 1.
- 12) Obituaries/In Memoriam(s) are articles formatted on 36-pica width.



### III. GRAMMAR AND USAGE IN TRANSACTIONS

#### A. Rules of Grammar

The principles of style given below aim to concentrate on fundamentals of modern usage. Particular emphasis is given to the rules most commonly violated.

- 1) **Form the possessive singular of nouns by adding's** (*Avoquadro's theorem*).
- 2) **In a series of three or more terms, use a comma immediately before the coordinating conjunction** (*usually and, or, or nor*).
- 3) **Enclose parenthetical expressions between commas.** (*Improvement, as shown in Fig. 1, is attained by the addition of the cogeneration.*).
- 4) **Use the semicolon, not the comma, to separate two complete sentences which form a compound sentence.**
- 5) **Use a colon after an independent clause to introduce a list.**
- 6) **Punctuation always goes inside quotation marks, except for the colon and semicolon.** Use single quotation marks around quotes within quotes. Quotes may be used around a new or special usage of a term the first time only, but use of quotes in this manner should be kept to a minimum.
- 7) **Do not use double parentheses in text expressions, but keep them in math.** For example, (see (10)) should become [see (10)].
- 8) **All acronyms and numerical plurals do not use apostrophes, i.e., FETs, 1980s.**
- 9) **Compound nouns made from a one-syllable verb and a short adverb are one word when found that way in the dictionary** (setup, takeoff, breakup). Compound nouns are likely to be two words, without a hyphen, or one word (bandwidth, bypass, flowchart, phase shift, sideband, standing wave). Compound nouns of more than two words can be hyphenated.
- 10) **A pair of words, modifying a third word separately, does not get a hyphen** (a tall water tower, a hot metal cylinder). If the first word modifies the second, and the pair together modify the third, there is a hyphen between the pair (a high-frequency signal, a second-order equation). The exception to this is the adverb ending in "ly," which needs no hyphen to join it to the next word.
- 11) **A hyphen is not used after the comparative or the superlative** (a higher order equation, a worst case value, nearest neighbor method). Do not hyphenate chemical compounds (sodium chloride crystals). Alloys and mixtures take the en dash (Ni-Co, He-Ne laser).
- 12) **Do not use commas between adjectives** (a planar equiangular spiral antenna).
- 13) **Do not hyphenate predicate adjectives** (... is well known, ...is second order).
- 14) **Compound verbs are generally hyphenated** (arc-weld, freeze-dry). Keep the hyphen when using the participles of such verbs as adjectives (freeze-dried, arc-welded). However, verbs with up, out, down, off, on, etc., do not have a hyphen, although the nouns formed from them may be hyphenated or one word (Verb: set up, break down, read out; Noun: setup, breakdown, readout).

#### B. Words Often Confused

Affect: to change or modify (verb).  
Effect: result (noun); cause (verb).

Alternate: a substitute.  
Alternative: a matter of choice.

Among: involves more than two things.  
Between: involves more than two things, but considers each individually.

Compare to: point out resemblances between different objects.  
Compare with: point out differences between same objects.

Compose: a set composed of members.  
Comprise: a set comprising members; members comprising a set.

Farther: distance.  
Further: quantity.

Fewer: modifies plural nouns specifying countable units, e.g., fewer tubes.  
Less: modifies singular mass nouns and singular abstract nouns, e.g., less air.

Imply: something suggested though not expressed.



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Infer: something deduced from evidence.

Number: a large number of people.

Amount: a large amount of water.

Principal: chief, main, most important (adjective).

Principle: a rule (noun).

Precede: come before.

Proceed: continue, advance.

That: (defining, restrictive).

Which: (nondefining, nonrestrictive)



## IV. EDITING MATHEMATICS

### A. The Language of Math

When editing technical publications it is important to remember that the mathematics often carries as much if not more meaning than the body of text itself. Therefore, it is critical that the grammar of an equation be taken into account when editing.

Most equations should read like a sentence. They should contain a noun and a verb and often contain adjectives, prepositional phrases, conjunctions, and conditions. Equations also contain punctuation. When math occurs along with text it shares the grammatical characteristics of the text. A displayed expression may be a main or subordinate clause, an expression in apposition, a direct object, an item in a list, or the object of a preposition. **Use comma at end of introductory sentences after: i.e., e.g., “Hence” or “That is.” Use a colon after words such as “following” or “as follows.”** There should be no punctuation after forms of the verb to be, or between a verb and its object or a preposition and its object. IEEE style dictates that the only punctuation used at the end of an equation is a period. There is, however, other punctuation permitted in the equation itself and between an equation and its condition. This interior punctuation contains mathematical meaning and must not be changed.

Some examples of interior punctuation are as follows.

*Mathematical Ellipses:*

$$I = 1, 2, 3, \dots, n$$

**NOTE:** Only three dots are used and they are enclosed by commas and are on the baseline.

*Matrix:*

$$C_{\text{Eopt}} = \begin{bmatrix} -4.65 E^{+0} & -1.07 E^{-1} & -1.42 E^{-1} & -9.50 E^{-4} & 2.52 E^{+1} & 3.36 E^{+0} \\ 1.97 E^{+0} & 1.44 E^{-1} & 8.80 E^{+0} & 5.88 E^{-2} & 2.14 E^{+1} & 1.46 E^{+0} \\ -1.62 E^{+0} & -1.10 E^{-1} & 1.01 E^{+1} & 6.27 E^{-2} & -1.92 E^{+1} & -1.37 E^{+0} \end{bmatrix} \quad (1)$$

**NOTE:** There is a centered operator, equation number, and period.

*Parenthetic Statement:*

$$v(t) = u(t), \quad t = 1, 2, \dots, m.$$

**NOTE:** There is a 2em space after the comma and before the condition  $t = 1, 2, \dots, m$ . Multiple conditions should be separated with a semicolon, with a comma at the end of the equation, a 2em space, and the condition aligned on the operator.

### B. In-Line Equations and Expressions

An inline equation is an equation within text or part of a paragraph. It is not displayed.

**Rule 1:** Equations appearing in text should be broken after a verb or an operator, meaning, if at all possible, the verb or operator should remain on the top line of text.

**Rule 2:** Fractions should not appear stacked in line.  $\frac{(xy + 6\alpha)}{xy}$  should be written as  $(xy + 6\alpha)/(xy)$ .

**Rule 3:** Collective signs should not appear with limits to top and bottom, but to the side instead.

$\sum_{i=0}^{i=\infty}$  should be written as  $\sum_{i=0}^{i=\infty}$ .

**Rule 4:** Use Roman function exp instead of  $e$  followed by a lengthy superscript.  $e^{(zx^2+y)(\alpha-2yx)+zx}$  should be written as  $\exp[(zx^2+y)(\alpha-2yx)+zx]$ .

**Rule 5 (optional):** Avoid square roots (radical signs) having long bars.  $\sqrt{(x+\alpha)}$  should be rewritten as  $(x+\alpha)^{1/2}$ .

### C. Break/Alignment Rules

**Rule 1:** Break equations at verbs and align on same when possible for a displayed equation.

$$\begin{aligned} A &= (5\alpha + x) + (10y + \beta)^2 \\ &\geq (5x - \alpha + y + x^2) \\ &\equiv B^2 \end{aligned}$$



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## Appendix I

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## IEEE CITATION REFERENCE



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## IEEE Citation Reference

IEEE Publications uses *Webster's College Dictionary*, 4th Edition. For guidance on grammar and usage not included in this manual, please consult *The Chicago Manual of Style*, published by the University of Chicago Press.

Citation standards in this reference are provided for:

|                               |                                         |
|-------------------------------|-----------------------------------------|
| Books                         | Online Sources                          |
| Handbooks                     | Patents, Standards, Theses, Unpublished |
| Reports                       | Periodicals                             |
| Conference Technical Articles | References                              |

## Books

*Basic Format:*

- [1] J. K. Author, "Title of chapter in the book," in *Title of His Published Book*, xth ed. City of Publisher, Country if not USA: Abbrev. of Publisher, year, ch. x, sec. x, pp. xxx-xxx.

**NOTE:** Use *et al.* when three or more names are given.

*Examples:*

- [1] B. Klaus and P. Horn, *Robot Vision*. Cambridge, MA: MIT Press, 1986.
- [2] L. Stein, "Random patterns," in *Computers and You*, J. S. Brake, Ed. New York: Wiley, 1994, pp. 55-70.
- [3] R. L. Myer, "Parametric oscillators and nonlinear materials," in *Nonlinear Optics*, vol. 4, P. G. Harper and B. S. Wherret, Eds. San Francisco, CA: Academic, 1977, pp. 47-160.
- [4] M. Abramowitz and I. A. Stegun, Eds., *Handbook of Mathematical Functions* (Applied Mathematics Series 55). Washington, DC: NBS, 1964, pp. 32-33.
- [5] E. F. Moore, "Gedanken-experiments on sequential machines," in *Automata Studies* (Ann. of Mathematical Studies, no. 1), C. E. Shannon and J. McCarthy, Eds. Princeton, NJ: Princeton Univ. Press, 1965, pp. 129-153.
- [6] Westinghouse Electric Corporation (Staff of Technology and Science, Aerospace Div.), *Integrated Electronic Systems*. Englewood Cliffs, NJ: Prentice-Hall, 1970.
- [7] M. Gorkii, "Optimal design," *Dokl. Akad. Nauk SSSR*, vol. 12, pp. 111-122, 1961 (Transl.: in L. Pontryagin, Ed., *The Mathematical Theory of Optimal Processes*. New York: Interscience, 1962, ch. 2, sec. 3, pp. 127-135).
- [8] G. O. Young, "Synthetic structure of industrial plastics," in *Plastics*, vol. 3, *Polymers of Hexadromicon*, J. Peters, Ed., 2nd ed. New York: McGraw-Hill, 1964, pp. 15-64.

**Handbooks**

*Basic Format:* [1] *Name of Manual/Handbook*, x ed., Abbrev. Name of Co., City of Co., Abbrev. State, year, pp. xx-xx.

*Examples:*

- [1] *Transmission Systems for Communications*, 3rd ed., Western Electric Co., Winston-Salem, NC, 1985, pp. 44-60.
- [2] *Motorola Semiconductor Data Manual*, Motorola Semiconductor Products Inc., Phoenix, AZ, 1989.
- [3] *RCA Receiving Tube Manual*, Radio Corp. of America, Electronic Components and Devices, Harrison, NJ, Tech. Ser. RC-23, 1992.



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TR-0200 (4230-46)-3, Nov. 1988.

**Reports**

The general form for citing technical reports is to place the name and location of the company or institution after the author and title and to give the report number and date at the end of the reference.

*Basic Format:*

- [1] J. K. Author, "Title of report," Abbrev. Name of Co., City of Co., Abbrev. State, Rep. xxx, year.

*Examples:*

- [1] E. E. Reber *et al.*, "Oxygen absorption in the earth's atmosphere," Aerospace Corp., Los Angeles, CA, Tech. Rep. Angeles, CA, Tech. Rep. TR-0200 (4230-46)-3, Nov. 1988.  
 [2] J. H. Davis and J. R. Cogdell, "Calibration program for the 16-foot antenna," Elect. Eng. Res. Lab., Univ. Texas, Austin, Tech. Memo. NGL-006-69-3, Nov. 15, 1987.  
 [3] R. E. Haskell and C. T. Case, "Transient signal propagation in lossless isotropic plasmas," USAF Cambridge Res. Labs., Cambridge, MA, Rep. ARCRL-66-234 (II), 1994, vol. 2.  
 [4] M. A. Brusberg and E. N. Clark, "Installation, operation, and data evaluation of an oblique-incidence ionosphere sounder system," in "Radio Propagation Characteristics of the Washington-Honolulu Path," Stanford Res. Inst., Stanford, CA, Contract NOBSR-87615, Final Rep., Feb. 1995, vol. 1.  
 [5] P. Diamant and W. L. Lupatkin, "V-line surface-wave radiation and scanning," Dept. Elect. Eng., Columbia Univ., New York, Sci. Rep. 85, Aug. 1991.

**Conference Technical Articles**

The general form for citing technical articles published in conference proceedings is to list the author/s and title of the paper, followed by the name (and location, if given) of the conference publication *in italics* using these standard abbreviations.

| <i>When the word below appears in the conference publication title,</i> | <i>abbreviate to</i> | <i>When the word below appears in the conference publication title,</i> | <i>abbreviate to</i> |
|-------------------------------------------------------------------------|----------------------|-------------------------------------------------------------------------|----------------------|
| Annals                                                                  | Ann.                 | Proceedings                                                             | Proc.                |
| Annual                                                                  | Annu.                | Record                                                                  | Rec.                 |
| Colloquium                                                              | Colloq.              | Symposium                                                               | Symp.                |
| Conference                                                              | Conf.                | Technical Digest                                                        | Tech. Dig.           |
| Congress                                                                | Congr.               | Technical Paper                                                         | Tech. Paper          |
| Convention                                                              | Conv.                | First                                                                   | 1st                  |
| Digest                                                                  | Dig.                 | Second                                                                  | 2nd                  |
| Exposition                                                              | Expo.                | Third                                                                   | 3rd                  |
| International                                                           | Int.                 | Fourth/nth ...                                                          | 4th/nth...           |
| National                                                                | Nat.                 |                                                                         |                      |

Write out all the remaining words, but omit most articles and prepositions like "of the" and "on." That is, *Proceedings of the 1996 Robotics and Automation Conference* becomes *Proc. 1996 Robotics and Automation Conf.*

*Basic Format:*

- [1] J. K. Author, "Title of paper," in *Unabbreviated Name of Conf.*, City of Conf., Abbrev. State (if given), year, pp. xxx-xxx.

*For an electronic conference article when there are no page numbers:*

- [1] J. K. Author [two authors: J. K. Author and A. N. Writer ] [three or more authors: J. K. Author *et al.*], "Title of Article," in [Title of Conf. Record as it appears on the copyright page], [copyright year] © [IEEE or applicable copyright holder of the Conference Record]. doi: [DOI number]

*For an unpublished paper presented at a conference:*

- [1] J. K. Author, "Title of paper," presented at the Unabbrev. Name of Conf., City of Conf., Abbrev. State, year.





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## Online Sources

The basic guideline for citing online sources is to follow the standard citation for the source given previously and add the Digital Object Identifier (DOI) at the end of the citation, or add the DOI in place of page numbers if the source is not paginated. The DOI for each IEEE conference article is assigned when the article is processed for inclusion in the IEEE Xplore digital library and is included with the reference data of the article in Xplore. See The DOI System for more information about the benefits of DOI referencing.

The following sources are unique in that they are electronic only sources.

*FTP**Basic Format:*

- [1] J. K. Author. (year). *Title* (edition) [Type of medium]. Available FTP: Directory: File:

*Example:*

- [1] R. J. Vidmar. (1994). *On the use of atmospheric plasmas as electromagnetic reflectors* [Online]. Available FTP: atmnext.usc.edu Directory: pub/etext/1994 File: atmosplasma.txt

*WWW**Basic Format:*

- [1] J. K. Author. (year, month day). *Title* (edition) [Type of medium]. Available: http://www.(URL)

*Example:*

- [1] J. Jones. (1991, May 10). *Networks (2nd ed.)* [Online]. Available: http://www.atm.com

*E-Mail**Basic Format:*

- [1] J. K. Author. (year, month day). *Title* (edition) [Type of medium]. Available e-mail: Message:

*Example:*

- [1] S. H. Gold. (1995, Oct. 10). *Inter-Network Talk* [Online]. Available e-mail: COMSERVE@RPIECS Message: Get NETWORK TALK

*Telnet**Basic Format:*

- [1] J. K. Author. (year, month day). *Title* (edition) [Type of medium]. Available Telnet: Directory: File:

*Example:*

- [1] V. Meligna. (1993, June 11). *Periodic table of elements* [Online]. Available Telnet: Library.CMU.edu Directory: Libraries/Reference Works File: Periodic Table of Elements



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## Patents, Standards, Theses, Unpublished

**Patents***Basic Format:*

- [1] J. K. Author, "Title of patent," U.S. Patent x xxx xxx, Abbrev. Month, day, year.

*Example:*

- [1] J. P. Wilkinson, "Nonlinear resonant circuit devices," U.S. Patent 3 624 125, July 16, 1990.

**NOTE:** Use "issued date" if several dates are given.

**Standards***Basic Format:*

- [1] *Title of Standard*, Standard number, date.

*Examples:*

- [1] *IEEE Criteria for Class IE Electric Systems*, IEEE Standard 308, 1969.  
 [2] *Letter Symbols for Quantities*, ANSI Standard Y10.5-1968.

**Theses (M.S.) and Dissertations (Ph.D.)***Basic Format:*

- [1] J. K. Author, "Title of thesis," M.S. thesis, Abbrev. Dept., Abbrev. Univ., City of Univ., Abbrev. State, year.  
 [2] J. K. Author, "Title of dissertation," Ph.D. dissertation, Abbrev. Dept., Abbrev. Univ., City of Univ., Abbrev. State, year.

*Examples:*

- [1] J. O. Williams, "Narrow-band analyzer," Ph.D. dissertation, Dept. Elect. Eng., Harvard Univ., Cambridge, MA, 1993.  
 [2] N. Kawasaki, "Parametric study of thermal and chemical nonequilibrium nozzle flow," M.S. thesis, Dept. Electron. Eng., Osaka Univ., Osaka, Japan, 1993.  
 [3] N. M. Amer, "The effects of homogeneous magnetic fields on developments of tribolium confusum," Ph.D. dissertation, Radiation Lab., Univ. California, Berkeley, Tech. Rep. 16854, 1995. \*\*\* *The state abbreviation is omitted if the name of the university includes the state name, i.e., "Univ. California, Berkeley."*\*\*\*  
 [4] C. Beclé, These de doctoral d'état, Univ. Grenoble, Grenoble, France, 1968.

**Unpublished**

These are the two most common types of unpublished references.

*Basic Format :*

- [1] J. K. Author, private communication, Abbrev. Month, year.  
 [2] J. K. Author, "Title of paper," unpublished.

*Examples:*

- [1] A. Harrison, private communication, May 1995.  
 [2] B. Smith, "An approach to graphs of linear forms," unpublished.  
 [3] A. Brahms, "Representation error for real numbers in binary computer arithmetic," IEEE Computer Group Repository, Paper R-67-85.



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## Periodicals

**NOTE:** When referencing IEEE Transactions, the issue number should be deleted and month carried.

*Basic Format:*

- [1] J. K. Author, "Name of paper," *Abbrev. Title of Periodical*, vol. x, no. x, pp. xxx-xxx, Abbrev. Month, year.

*Examples:*

- [1] R. E. Kalman, "New results in linear filtering and prediction theory," *J. Basic Eng.*, ser. D, vol. 83, pp. 95-108, Mar. 1961.
- [2] Ye. V. Lavrova, "Geographic distribution of ionospheric disturbances in the F2 layer," *Tr. IZMIRAN*, vol. 19, no. 29, pp. 31-43, 1961 (Transl.: E. R. Hope, Directorate of Scientific Information Services, Defence Research Board of Canada, Rep. T384R, Apr. 1963).
- [3] E. P. Wigner, "On a modification of the Rayleigh-Schrodinger perturbation theory," (in German), *Math. Naturwiss. Anz. Ungar. Akad. Wiss.*, vol. 53, p. 475, 1935.
- [4] E. H. Miller, "A note on reflector arrays," *IEEE Trans. Antennas Propag.*..., to be published.\*\*
- [5] C. K. Kim, "Effect of gamma rays on plasma," submitted for publication. \*\*
- [6] W. Rafferty, "Ground antennas in NASA's deep space telecommunications," *Proc. IEEE* vol. 82, pp. 636-640, May 1994.

\*\* Always use this style when the paper has not yet been accepted or scheduled for publication. Do not use "to appear in."

*Abbreviations for IEEE Periodicals*

*Proceedings of the IEEE* abbreviates to: Proc. IEEE

*Proceedings of the IRE* abbreviates to: Proc. IRE (until 1962)

|                |                                           |                                            |
|----------------|-------------------------------------------|--------------------------------------------|
| IEEE Journals  | IEEE J. Comput. Aid. Des.                 | IEEE J. Solid-State Circuits               |
|                | IEEE J. Ocean. Eng.                       | IEEE Sensors J.                            |
|                | IEEE J. Quantum Electron.                 | IEEE Syst. J.                              |
|                | IEEE J. Sel. Areas Commun.                | IEEE Transl. J. Magn. Jpn.                 |
|                | IEEE J. Sel. Topics Signal Process.       | J. Lightw. Technol.                        |
| IEEE Letters   | IEEE J. Sel. Topics. Quantum Electron.    | J. Microelectromech. Syst.                 |
|                | IEEE Antennas Wireless Propag. Lett.      | IEEE Photonics Technol. Lett.              |
|                | IEEE Commun. Lett.                        | IEEE Power Electron. Lett. (until 2005)    |
|                | IEEE Electron Device Lett.                | IEEE Signal Process. Lett.                 |
|                |                                           |                                            |
| IEEE Magazines | IEEE Aerosp. Electron. Syst. Mag.         | IEEE Ind. Appl. Mag.                       |
|                | IEEE Annals Hist. Comput.                 | IEEE Instrum. Meas. Mag.                   |
|                | IEEE Antennas Propagat. Mag.              | IEEE Intell. Syst.                         |
|                | IEEE ASSP Mag. (1984-1990)                | IEEE Internet Comput.                      |
|                | IEEE Circuits Devices Mag. (1985-present) | IEEE IT Prof.                              |
|                | IEEE Circuits Syst. Mag. (1979-1984)      | IEEE Micro                                 |
|                | IEEE Commun. Mag. (1979-present)          | IEEE Microwave                             |
|                | IEEE Commun. Soc. Mag. (until 1978)       | IEEE Multimedia                            |
|                | IEEE Comput. Appl. Power                  | IEEE Nanotechnol. Mag.                     |
|                | IEEE Comput. Graph. Appl.                 | IEEE Network                               |
|                | IEEE Comput. Intell. Mag.                 | IEEE Pers. Commun.                         |
|                | IEEE Comput. Sci. Eng. Mag.               | IEEE Potentials                            |
|                | IEEE Computer                             | IEEE Power Eng. Rev.                       |
|                | IEEE Concurrency                          | IEEE Robot. Automat. Mag.                  |
|                | IEEE Control. Syst. Mag.                  | IEEE Signal Processing Mag. (1991-present) |
|                | IEEE Des. Test Comput.                    | IEEE Softw.                                |
|                | IEEE Electr. Insul. Mag.                  | IEEE Spectr.                               |
|                | IEEE Eng. Manag. Rev.                     | IEEE Technol. Soc. Mag.                    |
|                | IEEE Eng. Med. Biol. Mag.                 | IEEE Veh. Technol. Mag.                    |
|                | IEEE Expert (until 1997)                  | Today's Eng.                               |
|                |                                           |                                            |
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## IEEE Transactions abbreviations

IEEE Adv. Packag.  
 IEEE/ACM Trans. Netw.  
 IEEE Human–Factors Electron. (*until 1968*)  
 IEEE Man–Mach. Syst. (*until 1970*)  
 IEEE Trans. Acoust., Speech, Signal Process. (*1975–1990*)  
 IEEE Trans. Aeronaut. Navig. Electron.  
 IEEE Trans. Aerosp.  
 IEEE Trans. Aerosp. Electron. Syst.  
 IEEE Trans. Aerosp. Navig. Electron.  
 IEEE Trans. Airborne Electron.  
 IEEE Trans. Antennas Propag.  
 IEEE Trans. Appl. Supercond.  
 IEEE Trans. Audio Electroacoust. (*until 1974*)  
 IEEE Trans. Autom. Control  
 IEEE Trans. Biomed. Circuits Syst.  
 IEEE Trans. Biomed. Eng.  
 IEEE Trans. Broadcast.  
 IEEE Trans. Broadcast. Technol.  
 IEEE Trans. Circuit Theory (*until 1973*)  
 IEEE Trans. Circuits Syst. (*1974–1992*)  
 IEEE Trans. Circuits Syst. I, Fundam. Theory Appl. (*until 2003*)  
 IEEE Trans. Circuits Syst. I, Reg. Papers  
 IEEE Trans. Circuits Syst. II, Analog Digit. Signal Process. (*until 2003*)  
 IEEE Trans. Circuits Syst. II, Exp. Briefs  
 IEEE Trans. Circuits Syst. Video Technol.  
 IEEE Trans. Commun.  
 IEEE Trans. Commun. Technol. (*until 1971*)  
 IEEE Trans. Compon. Hybrids, Manuf. Technol. (*1978–1993*)  
 IEEE Trans. Compon. Packag. Manuf. Technol. A (*1994–1998*)  
 IEEE Trans. Compon. Packag. Manuf. Technol. B (*1994–1998*)  
 IEEE Trans. Compon. Packag. Manuf. Technol. C (*1996–1998*)  
 IEEE Trans. Compon. Packag. Technol.  
 IEEE Trans. Comput.  
 IEEE Trans. Comput.-Aided Des. Integr. Circuits Syst.  
 IEEE Trans. Consum. Electron.  
 IEEE Trans. Control Syst. Technol.  
 IEEE Trans. Dev. Mat. Rel.  
 IEEE Trans. Dielectr. Electr. Insul.  
 IEEE Trans. Edu.  
 IEEE Trans. Electromagn. Compat.  
 IEEE Trans. Electron Devices  
 IEEE Trans. Electron. Packag. Manuf.  
 IEEE Trans. Energy Convers.  
 IEEE Trans. Eng. Manag.  
 IEEE Trans. Evol. Comput.  
 IEEE Trans. Fuzzy Syst.  
 IEEE Trans. Geosci. Electron. (*1962–1979*)  
 IEEE Trans. Geosci. Remote Sens.  
 IEEE Trans. Image Process.  
 IEEE Trans. Ind. Appl.  
 IEEE Trans. Ind. Electron.  
 IEEE Trans. Ind. Informat.  
 IEEE Trans. Inf. Forens. Security  
 IEEE Trans. Inf. Technol. Biomed.  
 IEEE Trans. Inf. Theory  
 IEEE Trans. Instrum.  
 IEEE Trans. Instrum. Meas.  
 IEEE Trans. Intell. Transp. Syst.  
 IEEE Trans. Knowl. Data Eng.  
 IEEE Trans. Magn.  
 IEEE Trans. Manuf. Technol. (*1972–1977*)  
 IEEE Trans. Mechatron.  
 IEEE Trans. Med. Imag.  
 IEEE Trans. Microw. Guid. Wave Lett. (*1987–1999*)  
 IEEE Trans. Microw. Theory Tech.  
 IEEE Trans. Microw. Wireless Compon. Lett. (*until 2004*)  
 IEEE Trans. Mil. Electron.  
 IEEE Trans. Multimedia  
 IEEE Trans. Nanotechnol.  
 IEEE Trans. Neural Netw.  
 IEEE Trans. Neural Syst. Rehabil. Eng.  
 IEEE Trans. Nucl. Sci.  
 IEEE Trans. Parallel Distrib. Syst.  
 IEEE Trans. Parts, Hybrids, Packag. Technol. (*June 1971–1977*)  
 IEEE Trans. Parts, Mater. Packag.  
 IEEE Trans. Pattern Anal. Mach. Intell.  
 IEEE Trans. Plasma Sci.  
 IEEE Trans. Power App. Syst. (*until 1985*)  
 IEEE Trans. Power Del.  
 IEEE Trans. Power Electron.  
 IEEE Trans. Power Syst.  
 IEEE Trans. Prof. Commun.  
 IEEE Trans. Rehabil. Eng. (*until 2000*)  
 IEEE Trans. Reliab.  
 IEEE Trans. Robot. Autom.  
 IEEE Trans. Semicond. Manuf.  
 IEEE Trans. Signal Process.  
 IEEE Trans. Softw. Eng.  
 IEEE Trans. Sonics Ultrason. (*until 1985*)  
 IEEE Trans. Speech Audio Process.  
 IEEE Trans. Syst. Man Cybern. (*1971–1995*)  
 IEEE Trans. Syst. Man Cybern. A., Syst. Humans  
 IEEE Trans. Syst. Man Cybern. B, Cybern.  
 IEEE Trans. Syst. Man Cybern. C, Appl. Rev.  
 IEEE Trans. Ultrason. Eng.  
 IEEE Trans. Ultrason. Ferroelectr. Freq. Control  
 IEEE Trans. Veh. Technol.  
 IEEE Trans. Very Large Scale Integr. (VLSI) Syst.  
 IEEE Trans. Vis. Comput. Graphics  
 IEEE Trans. Wireless Commun.



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**NOTE:** Use *et al.* when three or more names are given.

*References in Text:*

References need not be cited in the text. When they are, they appear on the line, in square brackets, *inside the punctuation*. Grammatically, they may be treated as if they were footnote numbers, e.g.,

as shown by Brown [4], [5]; as mentioned earlier [2], [4]–[7], [9]; Smith [4] and Brown and Jones [5]; Wood et al. [7]

or as nouns:

as demonstrated in [3]; according to [4] and [6]–[9].

*References Within a Reference:*

Check the reference list for *ibid.* or *op. cit.* These refer to a previous reference and should be eliminated from the reference section. In text, repeat the earlier reference number and renumber the reference section accordingly. If the *ibid.* gives a new page number, or other information, use the following forms:

[3, Th. 1]; [3, Lemma 2]; [3, pp. 5–10]; [3, eq. (2)]; [3, Fig. 1]; [3, Appendix I]; [3, Sec. 4.5]; [3, Ch. 2, pp. 5–10]; [3, Algorithm 5].

**NOTE:** Editing of references may entail careful renumbering of references, as well as the citations in text.

*Style*

Reference numbers are set flush left and form a column of their own, hanging out beyond the body of the reference. The reference numbers are on the line, enclosed in square brackets. In all references, the given name of the author or editor is abbreviated to the initial only and precedes the last name. Use commas around Jr., Sr., and III in names. If there are many names, use *et al.* Note that when citing IEEE Transactions, if the month is not available, the number may be kept, although it is normally deleted. Keep the day of the month when referencing a patent. References may not include all information; please obtain and include relevant information. Do not combine references. There must be only one reference with each number. If there is a URL included with the print reference, it can be included at the end of the reference.

When the word below appears in the reference, *abbreviate to*

|                |            |               |           |                   |             |
|----------------|------------|---------------|-----------|-------------------|-------------|
| Acoustics      | Acoust.    | Electrical    | Elect.    | Nuclear           | Nucl.       |
| Administration | Admin.     | Electronic    | Electron. | Occupation        | Occupat.    |
| Administrative | Administ.  | Engineering   | Eng.      | Philosophical     | Philosph.   |
| American       | Amer.      | Ergonomics    | Ergonom.  | Proceedings       | Proc.       |
| Analysis       | Anal.      | Evolutionary  | Evol.     | Processing        | Process.    |
| Annals         | Ann.       | Foundation    | Found.    | Production        | Prod.       |
| Annual         | Annu.      | Geoscience    | Geosci.   | Productivity      | Productiv.  |
| Apparatus      | App.       | Graphics      | Graph.    | Quarterly         | Quart.      |
| Applications   | Applicat.  | Industrial    | Ind.      | Record            | Rec.        |
| Applied        | Appl.      | Industry      | Ind.      | Reliability       | Rel.        |
| Association    | Assoc.     | Information   | Inform.   | Report            | Rep.        |
| Automatic      | Automat.   | Institute     | Inst.     | Royal             | Roy.        |
| Broadcasting   | Broadcast. | Intelligence  | Intell.   | Science           | Sci.        |
| Business       | Bus.       | International | Int.      | Selected          | Select.     |
| Communications | Commun.    | Journal       | J.        | Society           | Soc.        |
| Computer(s)    | Comput.    | Letter(s)     | Lett.     | Sociological      | Sociol.     |
| Congress       | Congr.     | Machine       | Mach.     | Statistics        | Stat.       |
| Convention     | Conv.      | Magazine      | Mag.      | Studies           | Stud.       |
| Correspondence | Corresp.   | Management    | Manage.   | Supplement        | Suppl.      |
| Cybernetics    | Cybern.    | Managing      | Manag.    | Symposium         | Symp.       |
| Department     | Dept.      | Mathematic(s) | Math.     | Systems           | Syst.       |
| Development    | Develop.   | Mathematical  | Math.     | Technical         | Tech.       |
| Digest         | Dig.       | Mechanical    | Mech.     | Telecommunication | Telecommun. |
| Economic(s)    | Econ.      | National      | Nat.      | Transactions      | Trans.      |
| Education      | Educ.      | Newsletter    | Newslett. | Vehicular         | Veh.        |
|                |            |               |           | Working           | Work.       |



De La Salle University

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## Appendix J

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## IEEE PUBLICATION ABBREVIATIONS



## IEEE Abbreviations for Transactions, Journals, Letters, and Magazines

NOTE: \* denotes past acronyms/abbreviations of journals (used for pre-1988 publications).

### List of IEEE Transactions, Journals, and Letters

| Publication                                                                        | Acronym | Reference Abbreviation                                                          |
|------------------------------------------------------------------------------------|---------|---------------------------------------------------------------------------------|
| IEEE TRANSACTIONS ON AEROSPACE AND ELECTRONIC SYSTEMS                              | AES     | <i>IEEE Trans. Aerosp. Electron. Syst.</i>                                      |
|                                                                                    | ANE*    | <i>IEEE Trans. Aeronaut. Navig. Electron.*</i>                                  |
|                                                                                    | ANE*    | <i>IEEE Trans. Aerosp. Navig. Electron.*</i>                                    |
|                                                                                    | AS*     | <i>IEEE Trans. Aerosp.*</i>                                                     |
|                                                                                    | MIL*    | <i>IEEE Trans. Mil. Electron.*</i>                                              |
|                                                                                    | AE*     | <i>IEEE Trans. Airborne Electron.*</i>                                          |
| IEEE TRANSACTIONS ON ANTENNAS AND PROPAGATION                                      | AP      | <i>IEEE Trans. Antennas Propag.</i>                                             |
| IEEE ANTENNAS AND WIRELESS PROPAGATION LETTERS                                     | LAWP    | <i>IEEE Antennas Wireless Propag. Lett.</i>                                     |
| IEEE TRANSACTIONS ON APPLIED SUPERCONDUCTIVITY                                     | ASC     | <i>IEEE Trans. Appl. Supercond.</i>                                             |
| IEEE/ACM TRANSACTIONS ON AUDIO, SPEECH, AND LANGUAGE PROCESSING                    | ASLP    | <i>IEEE/ACM Trans. Audio, Speech, Language Process.</i>                         |
| IEEE TRANSACTIONS ON AUDIO, SPEECH, AND LANGUAGE PROCESSING                        | ASL     | <i>IEEE Trans. Audio, Speech, Language Process.</i><br>(2006–2013)              |
|                                                                                    | SAP*    | <i>IEEE Speech Audio Process. (1993–2005)</i>                                   |
| IEEE TRANSACTIONS ON AUTOMATIC CONTROL                                             | AC      | <i>IEEE Trans. Autom. Control</i>                                               |
| IEEE TRANSACTIONS ON AUTOMATION SCIENCE AND ENGINEERING                            | ASE     | <i>IEEE Trans. Autom. Sci. Eng. (from July 2004)</i>                            |
| IEEE TRANSACTIONS ON AUTONOMOUS MENTAL DEVELOPMENT                                 | AMD     | <i>IEEE Trans. Auton. Mental Develop.</i>                                       |
| IEEE TRANSACTIONS ON BIG DATA                                                      | BD      | <i>IEEE Trans. Big Data</i>                                                     |
| IEEE JOURNAL OF BIOMEDICAL AND HEALTH INFORMATICS                                  | BHI     | <i>IEEE J. Biomed. Health Inform.</i>                                           |
|                                                                                    | ITB     | <i>IEEE Trans. Inf. Technol. Biomed. (1995–2012)</i>                            |
| IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS                               | BCAS    | <i>IEEE Trans. Biomed. Circuits Syst.</i>                                       |
| IEEE REVIEWS IN BIOMEDICAL ENGINEERING                                             | RBME    | <i>IEEE Rev. Biomed. Eng.</i>                                                   |
| IEEE TRANSACTIONS ON BIOMEDICAL ENGINEERING                                        | BME     | <i>IEEE Trans. Biomed. Eng.</i>                                                 |
|                                                                                    | BME*    | <i>IEEE Trans. Bio Med. Eng.*</i>                                               |
|                                                                                    | BME*    | <i>IEEE Trans. Bio Med. Electron.*</i>                                          |
|                                                                                    | PGME*   | <i>IEEE Trans. Med. Electron.*</i>                                              |
| IEEE TRANSACTIONS ON BROADCASTING                                                  | BC      | <i>IEEE Trans. Broadcast.</i>                                                   |
| IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS—I: REGULAR PAPERS                        | CSI     | <i>IEEE Trans. Circuits Syst. I, Reg. Papers</i>                                |
| IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS—II: EXPRESS BRIEFS                       | CSII    | <i>IEEE Trans. Circuits Syst. II, Exp. Briefs</i>                               |
| IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS—I: FUNDAMENTAL THEORY AND APPLICATIONS   | CAS1*   | <i>IEEE Trans. Circuits Syst. I, Fundam. Theory Appl. (1993–2003)</i>           |
| IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS—II: ANALOG AND DIGITAL SIGNAL PROCESSING | CAS2*   | <i>IEEE Trans. Circuits Syst. II, Analog Digit. Signal Process. (1993–2003)</i> |
|                                                                                    | CAS*    | <i>IEEE Trans. Circuits Syst.* (1974–1992)</i>                                  |
|                                                                                    | CT*     | <i>IEEE Trans. Circuit Theory* (until 1973)</i>                                 |
| IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS FOR VIDEO TECHNOLOGY                     | CSVT    | <i>IEEE Trans. Circuits Syst. Video Technol.</i>                                |
| IEEE TRANSACTIONS ON CLOUD COMPUTING                                               | CC      | <i>IEEE Trans. Cloud Comput.</i>                                                |
| IEEE TRANSACTIONS ON COGNITIVE COMMUNICATIONS AND NETWORKING                       | CCN     | <i>IEEE Trans. Cogn. Commun. Netw.</i>                                          |
| IEEE TRANSACTIONS ON COMMUNICATIONS                                                | COM     | <i>IEEE Trans. Commun.</i>                                                      |
|                                                                                    | COM*    | <i>IEEE Trans. Commun. Technol.* (until 1971)</i>                               |



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### IEEE Abbreviations for Transactions, Journals, Letters, and Magazines (ctd.)

| Publication                                                                   | Acronym | Reference Abbreviation                                             |
|-------------------------------------------------------------------------------|---------|--------------------------------------------------------------------|
| IEEE COMMUNICATIONS LETTERS                                                   | COMML   | <i>IEEE Commun. Lett.</i>                                          |
| IEEE TRANSACTIONS ON COMPONENTS, PACKAGING AND MANUFACTURING TECHNOLOGY       | CPMT    | <i>IEEE Trans. Compon. Packag. Manuf. Technol.</i>                 |
|                                                                               | CAPT    | <i>IEEE Trans. Compon. Packag. Technol.</i> (1999–2010)            |
|                                                                               | CPMTA   | <i>IEEE Trans. Compon., Packag., Manuf. Technol. A</i> (1994–1998) |
|                                                                               | CHMT*   | <i>IEEE Trans. Compon., Hybrids, Manuf. Technol.*</i> (1978–1993)  |
|                                                                               | MFT*    | <i>IEEE Trans. Manuf. Technol.*</i> (1972–1977)                    |
|                                                                               | PHP*    | <i>IEEE Trans. Parts, Hybrids, Packag.*</i> (June 1971–1977)       |
|                                                                               | PMP*    | <i>IEEE Trans. Parts, Mater., Packag.*</i> (1965–1971)             |
|                                                                               | ADVP    | <i>IEEE Trans. Adv. Packag.</i> (1999–2010)                        |
|                                                                               | CPMTB   | <i>IEEE Trans. Compon., Packag., Manuf. Technol. B</i> (1994–1998) |
|                                                                               | EPM     | <i>IEEE Trans. Electron. Packag. Manuf.</i> (1999–2010)            |
|                                                                               | CPMTC   | <i>IEEE Trans. Compon., Packag., Manuf. Technol. C</i> (1996–1998) |
| IEEE/ACM TRANSACTIONS ON COMPUTATIONAL BIOLOGY AND BIOINFORMATICS             | CBB     | <i>IEEE/ACM Trans. Comput. Biol. Bioinf.</i>                       |
| IEEE TRANSACTIONS ON COMPUTATIONAL INTELLIGENCE AND AI IN GAMES               | CIAIG   | <i>IEEE Trans. Comput. Intell. AI in Games</i>                     |
| IEEE TRANSACTIONS ON COMPUTATIONAL SOCIAL SYSTEMS                             | CSS     | <i>IEEE Trans. Comput. Social Syst.</i>                            |
| IEEE TRANSACTIONS ON COMPUTERS                                                | C       | <i>IEEE Trans. Comput.</i>                                         |
| IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS | CAD     | <i>IEEE Trans. Comput.-Aided Design Integr. Circuits Syst.</i>     |
| IEEE COMPUTER ARCHITECTURAL LETTERS                                           | CAL     | <i>IEEE Comput. Archit. Lett.</i>                                  |
| IEEE TRANSACTIONS ON CONSUMER ELECTRONICS                                     | CE      | <i>IEEE Trans. Consum. Electron.</i>                               |
|                                                                               | BTR     | <i>IEEE Trans. Broadcast. Telev. Receiv.</i> (1963–1974)           |
| IEEE TRANSACTIONS ON CONTROL SYSTEMS TECHNOLOGY                               | CST     | <i>IEEE Trans. Control Syst. Technol.</i>                          |
| IEEE TRANSACTIONS ON CONTROL OF NETWORK SYSTEMS                               | CNS     | <i>IEEE Trans. Control Netw. Syst.</i>                             |
| IEEE TRANSACTIONS ON CYBERNETICS                                              | CYB     | <i>IEEE Trans. Cybern.</i>                                         |
|                                                                               | SMCB*   | <i>IEEE Trans. Syst. Man, Cybern. B, Cybern.</i> (1995–2012)       |
| IEEE TRANSACTIONS ON DEVICE AND MATERIALS RELIABILITY                         | DMR     | <i>IEEE Trans. Device Mater. Rel.</i>                              |
| IEEE TRANSACTIONS ON DIELECTRICS AND ELECTRICAL INSULATION                    | DEI     | <i>IEEE Trans. Dielectr. Electr. Insul.</i>                        |
|                                                                               | EI*     | <i>IEEE Trans. Electr. Insul.*</i> (through 1993)                  |
| IEEE/OSA JOURNAL OF DISPLAY TECHNOLOGY                                        | DT      | <i>J. Display Technol.</i>                                         |
| IEEE TRANSACTIONS ON EDUCATION                                                | E       | <i>IEEE Trans. Edu.</i>                                            |
| IEEE TRANSACTIONS ON ELECTROMAGNETIC COMPATIBILITY                            | EMC     | <i>IEEE Trans. Electromagn. Compat.</i>                            |
|                                                                               | RFI*    | <i>IEEE Trans. Radio Freq. Interference*</i>                       |
| IEEE TRANSACTIONS ON ELECTRON DEVICES                                         | ED      | <i>IEEE Trans. Electron Devices</i>                                |
| IEEE JOURNAL OF ELECTRON DEVICES SOCIETY                                      | EDS     | <i>IEEE J. Electron Devices Soc.</i>                               |
| IEEE ELECTRON DEVICE LETTERS                                                  | EDL     | <i>IEEE Electron Device Lett.</i>                                  |
| IEEE TRANSACTIONS ON ELECTRONICS PACKAGING MANUFACTURING                      | EPM     | <i>IEEE Trans. Electron. Packag. Manuf.</i> (1999–2010)            |
| IEEE EMBEDDED SYSTEMS LETTERS                                                 | ES      | <i>IEEE Embedded Syst. Lett.</i>                                   |
| IEEE TRANSACTIONS ON EMERGING TOPICS IN COMPUTING                             | ETC     | <i>IEEE Trans. Emerg. Topics Comput.</i>                           |
| IEEE TRANSACTIONS ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS     | ETCAS   | <i>IEEE Trans. Emerg. Sel. Topics Circuits Syst.</i>               |
| IEEE TRANSACTIONS ON EMERGING AND SELECTED TOPICS IN POWER ELECTRONICS        | ESTPE   | <i>IEEE Trans. Emerg. Sel. Topics Power Electron.</i>              |
| IEEE TRANSACTIONS ON ENERGY CONVERSION                                        | EC      | <i>IEEE Trans. Energy Convers.</i>                                 |





### IEEE Abbreviations for Transactions, Journals, Letters, and Magazines (ctd.)

| Publication                                                                | Acronym | Reference Abbreviation                                           |
|----------------------------------------------------------------------------|---------|------------------------------------------------------------------|
| IEEE TRANSACTIONS ON ENGINEERING MANAGEMENT                                | EM      | <i>IEEE Trans. Eng. Manag.</i>                                   |
| IEEE TRANSACTIONS ON EVOLUTIONARY COMPUTATION                              | EVC     | <i>IEEE Trans. Evol. Comput.</i>                                 |
| IEEE JOURNAL ON EXPLORATORY SOLID-STATE COMPUTATIONAL DEVICES AND CIRCUITS | XCDC    | <i>IEEE J. Explor. Solid-State Computat. Devices Circuits</i>    |
| IEEE TRANSACTIONS ON FUZZY SYSTEMS                                         | FUZZ    | <i>IEEE Trans. Fuzzy Syst.</i>                                   |
| IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING                         | GRS     | <i>IEEE Trans. Geosci. Remote Sens.</i>                          |
| IEEE GEOSCIENCE AND REMOTE SENSING LETTERS                                 | GRSL    | <i>IEEE Geosci. Remote Sens. Lett.</i>                           |
| IEEE TRANSACTIONS ON HUMAN-MACHINE SYSTEMS                                 | HMS*    | <i>IEEE Trans. Human-Mach. Syst.</i>                             |
|                                                                            | SMCC*   | <i>IEEE Trans. Syst., Man, Cybern. C, Appl. Rev.</i> (1995–2012) |
|                                                                            | SMC*    | <i>IEEE Trans. Syst., Man, Cybern.*</i> (1971–1995)              |
|                                                                            | SSC*    | <i>IEEE Trans. Syst. Sci. Cybern.*</i> (through 1970)            |
| IEEE TRANSACTIONS ON IMAGE PROCESSING                                      | IP      | <i>IEEE Trans. Image Process.</i>                                |
| IEEE TRANSACTIONS ON INDUSTRIAL ELECTRONICS                                | IE      | <i>IEEE Trans. Ind. Electron.</i>                                |
| IEEE TRANSACTIONS ON INDUSTRIAL INFORMATICS                                | II      | <i>IEEE Trans. Ind. Informat.</i>                                |
| IEEE TRANSACTIONS ON INDUSTRY APPLICATIONS                                 | IA      | <i>IEEE Trans. Ind. Appl.</i>                                    |
| IEEE TRANSACTIONS ON INFORMATION FORENSICS AND SECURITY                    | IFS     | <i>IEEE Trans. Inf. Forensics Security</i>                       |
| IEEE TRANSACTIONS ON INFORMATION THEORY                                    | IT      | <i>IEEE Trans. Inf. Theory</i>                                   |
| IEEE TRANSACTIONS ON INSTRUMENTATION AND MEASUREMENT                       | IM      | <i>IEEE Trans. Instrum. Meas.</i>                                |
|                                                                            | I, PGI* | <i>IEEE Trans. Instrum.*</i>                                     |
| IEEE TRANSACTIONS ON INTELLIGENT TRANSPORTATION SYSTEMS                    | ITS     | <i>IEEE Trans. Intell. Transp. Syst.</i>                         |
| IEEE INTERNET OF THINGS JOURNAL                                            | IoT     | <i>IEEE Internet Things J.</i>                                   |
| IEEE TRANSACTIONS ON KNOWLEDGE AND DATA ENGINEERING                        | KDE     | <i>IEEE Trans. Knowl. Data Eng.</i>                              |
| IEEE LIFE SCIENCES LETTERS                                                 | LS      | <i>IEEE Life Sci. Lett.</i>                                      |
| IEEE/OSA JOURNAL OF LIGHTWAVE TECHNOLOGY                                   | LT      | <i>J. Lightw. Technol.</i>                                       |
| IEEE TRANSACTIONS ON MAGNETICS                                             | MAG     | <i>IEEE Trans. Magn.</i>                                         |
| IEEE MAGNETICS LETTERS                                                     | MAGL    | <i>IEEE Magn. Lett.</i>                                          |
| IEEE/ASME TRANSACTIONS ON MECHATRONICS                                     | MECH    | <i>IEEE/ASME Trans. Mechatronics</i>                             |
| IEEE TRANSACTIONS ON MEDICAL IMAGING                                       | MI      | <i>IEEE Trans. Med. Imag.</i>                                    |
| IEEE JOURNAL OF MICROELECTROMECHANICAL SYSTEMS                             | MEMS    | <i>J. Microelectromech. Syst.</i>                                |
| IEEE/ASME JOURNAL OF MICROELECTROMECHANICAL SYSTEMS                        | MEMS    | <i>J. Microelectromech. Syst.</i> (1992–2013)                    |
| IEEE MICROWAVE AND WIRELESS COMPONENTS LETTERS                             | MWCL    | <i>IEEE Microw. Compon. Lett.</i>                                |
|                                                                            | MGWL    | <i>IEEE Microw. Guided Wave Lett.</i> (1991–2000)                |
| IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES                       | MTT     | <i>IEEE Trans. Microw. Theory Techn.</i>                         |
| IEEE TRANSACTIONS ON MOBILE COMPUTING                                      | MC      | <i>IEEE Trans. Mobile Comput.</i>                                |
| IEEE TRANSACTIONS ON MOLECULAR, BIOLOGICAL AND MULTI-SCALE COMMUNICATIONS  | MBSC    | <i>IEEE Trans. Mol. Biol. Multi-Scale Commun.</i>                |
| IEEE TRANSACTIONS ON MULTIMEDIA                                            | MM      | <i>IEEE Trans. Multimedia</i>                                    |
| IEEE TRANSACTIONS ON MULTI-SCALE COMPUTING SYSTEMS                         | MSCS    | <i>IEEE Trans. Multi-Scale Comput. Syst.</i>                     |
| IEEE TRANSACTIONS ON NANOBIOSCIENCE                                        | NB      | <i>IEEE Trans. Nanobiosci.</i>                                   |
| IEEE TRANSACTIONS ON NANOTECHNOLOGY                                        | NANO    | <i>IEEE Trans. Nanotechnol.</i>                                  |
| IEEE NANOTECHNOLOGY EXPRESS                                                | ENANO   | <i>IEEE Nanotechnol. Express</i>                                 |
| IEEE/ACM TRANSACTIONS ON NETWORKING                                        | NET     | <i>IEEE/ACM Trans. Netw.</i>                                     |
| IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS                  | NNLS    | <i>IEEE Trans. Neural Netw. Learn. Syst.</i>                     |
|                                                                            | NN      | <i>IEEE Trans. Neural Netw.</i> (1990–2011)                      |
| IEEE TRANSACTIONS ON NUCLEAR SCIENCE                                       | NS      | <i>IEEE Trans. Nucl. Sci.</i>                                    |



### IEEE Abbreviations for Transactions, Journals, Letters, and Magazines (ctd.)

| Publication                                                                     | Acronym | Reference Abbreviation                                              |
|---------------------------------------------------------------------------------|---------|---------------------------------------------------------------------|
| IEEE TRANSACTIONS ON NEURAL SYSTEMS AND REHABILITATION ENGINEERING              | NSRE    | <i>IEEE Trans. Neural Syst. Rehabil. Eng.</i>                       |
| IEEE JOURNAL OF OCEANIC ENGINEERING                                             | RE*     | <i>IEEE Trans. Rehabil. Eng.* (1993–2000)</i>                       |
| IEEE JOURNAL OF OPTICAL COMMUNICATIONS AND NETWORKING                           | OE      | <i>IEEE J. Ocean. Eng.</i>                                          |
| IEEE TRANSACTIONS ON PARALLEL AND DISTRIBUTED SYSTEMS                           | OCN     | <i>IEEE J. Opt. Commun. Netw.</i>                                   |
| IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE                  | PDS     | <i>IEEE Trans. Parallel Distrib. Syst.</i>                          |
| IEEE PHOTONICS TECHNOLOGY LETTERS                                               | PAMI    | <i>IEEE Trans. Pattern Anal. Mach. Intell.</i>                      |
| IEEE PHOTONICS JOURNAL                                                          | PTL     | <i>IEEE Photon. Technol. Lett.</i>                                  |
| IEEE JOURNAL OF PHOTOVOLTAICS                                                   | PJ      | <i>IEEE Photon. J.</i>                                              |
| IEEE TRANSACTIONS ON PLASMA SCIENCE                                             | PHOT    | <i>IEEE J. Photovolt.</i>                                           |
| IEEE TRANSACTIONS ON POWER APPARATUS AND SYSTEMS                                | PS      | <i>IEEE Trans. Plasma Sci.</i>                                      |
| IEEE TRANSACTIONS ON POWER DELIVERY                                             | PAS*    | <i>IEEE Trans. Power App. Syst.* (through 1985)</i>                 |
| IEEE TRANSACTIONS ON POWER ELECTRONICS                                          | PWRD    | <i>IEEE Trans. Power Del.</i>                                       |
| IEEE POWER ELECTRONICS LETTERS                                                  | PEL     | <i>IEEE Trans. Power Electron.</i>                                  |
| IEEE TRANSACTIONS ON POWER SYSTEMS                                              | LPEL    | <i>IEEE Power Electron Lett. (2003–2005; abolished)</i>             |
| IEEE JOURNAL OF PRODUCT SAFETY ENGINEERING                                      | PWRS    | <i>IEEE Trans. Power Syst.</i>                                      |
| IEEE POWER AND ENERGY TECHNOLOGY SYSTEMS JOURNAL                                | PSE     | <i>IEEE J. Product Safety Eng.</i>                                  |
| IEEE TRANSACTIONS ON PROFESSIONAL COMMUNICATION                                 | PETS    | <i>IEEE Power Energy Technol. Syst. J.</i>                          |
| IEEE JOURNAL OF QUANTUM ELECTRONICS                                             | PC      | <i>IEEE Trans. Prof. Commun.</i>                                    |
| IEEE RFIC JOURNAL                                                               | QE      | <i>IEEE J. Quantum Electron.</i>                                    |
| IEEE RFID JOURNAL                                                               | RFIC    | <i>IEEE RFIC J.</i>                                                 |
| IEEE TRANSACTIONS ON RELIABILITY                                                | RFID    | <i>IEEE RFID J.</i>                                                 |
| IEEE TRANSACTIONS ON ROBOTICS                                                   | R       | <i>IEEE Trans. Rel.</i>                                             |
| IEEE TRANSACTIONS ON ROBOTICS AND AUTOMATION                                    | RO      | <i>IEEE Trans. Robot.</i>                                           |
|                                                                                 | RA*     | <i>IEEE Trans. Robot. Autom. (1989–June 2004)</i>                   |
|                                                                                 | RA*     | <i>IEEE J. Robot. Autom.* (1985–1988)</i>                           |
| IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS IN REMOTE SENSING | STARS   | <i>IEEE J. Sel. Topics Appl. Earth Observ. in Remote Sens.</i>      |
| IEEE JOURNAL ON SELECTED AREAS IN COMMUNICATIONS                                | SAC     | <i>IEEE J. Sel. Areas Commun.</i>                                   |
| IEEE JOURNAL OF SELECTED TOPICS IN QUANTUM ELECTRONICS                          | STQE    | <i>IEEE J. Sel. Topics Quantum Electron.</i>                        |
| IEEE JOURNAL OF SELECTED TOPICS IN SIGNAL PROCESSING                            | STSP    | <i>IEEE J. Sel. Topics Signal Process.</i>                          |
| IEEE TRANSACTIONS ON SEMICONDUCTOR MANUFACTURING                                | SM      | <i>IEEE Trans. Semicond. Manuf.</i>                                 |
| IEEE SENSORS JOURNAL                                                            | SEN     | <i>IEEE Sensors J.</i>                                              |
| IEEE TRANSACTIONS ON SIGNAL PROCESSING                                          | SP      | <i>IEEE Trans. Signal Process.</i>                                  |
|                                                                                 | ASSP*   | <i>IEEE Trans. Acoust., Speech, Signal Process. * (1975–1990)</i>   |
| IEEE SIGNAL PROCESSING LETTERS                                                  | AU*     | <i>IEEE Trans. Audio Electroacoust. (until 1974)</i>                |
| IEEE TRANSACTIONS ON SMART GRID                                                 | SPL     | <i>IEEE Signal Process. Lett.</i>                                   |
| IEEE TRANSACTIONS ON SUSTAINABLE ENERGY                                         | SG      | <i>IEEE Trans. Smart Grid</i>                                       |
| IEEE SYSTEMS JOURNAL                                                            | STE     | <i>IEEE Trans. Sustain. Energy</i>                                  |
| IEEE TRANSACTIONS ON SOFTWARE ENGINEERING                                       | SJ      | <i>IEEE Syst. J.</i>                                                |
| IEEE JOURNAL OF SOLID-STATE CIRCUITS                                            | SE      | <i>IEEE Trans. Softw. Eng.</i>                                      |
| IEEE TRANSACTIONS ON SYSTEMS, MAN, AND CYBERNETICS: SYSTEMS                     | SSC     | <i>IEEE J. Solid-State Circuits</i>                                 |
|                                                                                 | SMC     | <i>IEEE Trans. Syst., Man, Cybern., Syst.</i>                       |
|                                                                                 | SMCA*   | <i>IEEE Trans. Syst., Man, Cybern. A, Syst., Humans (1995–2012)</i> |
|                                                                                 | MMS*    | <i>IEEE Trans. Man-Mach. Syst.* (through 1970)</i>                  |
|                                                                                 | HFE*    | <i>Hum. Factors Electron.* (through 1968)</i>                       |



### IEEE Abbreviations for Transactions, Journals, Letters, and Magazines (ctd.)

| Publication                                                             | Acronym | Reference Abbreviation                                   |
|-------------------------------------------------------------------------|---------|----------------------------------------------------------|
| IEEE JOURNAL OF TRANSLATIONAL ENGINEERING IN HEALTH AND MEDICINE        | TEHM    | <i>IEEE J. Transl. Eng. Health Med.</i>                  |
| IEEE TRANSLATION JOURNAL ON MAGNETICS IN JAPAN                          | TJMJ    | <i>IEEE Transl. J. Magn. Jpn. (through 2010)</i>         |
| IEEE JOURNAL ON TECHNOLOGY IN COMPUTER AIDED DESIGN                     | TCAD    | <i>IEEE J. Technol. Computer Aided Des.</i>              |
| IEEE TRANSACTIONS ON TERAHERTZ SCIENCE AND TECHNOLOGY                   | THz     | <i>IEEE Trans. THz Sci. Technol.</i>                     |
| IEEE TRANSACTIONS ON TRANSPORTATION ELECTRIFICATION                     |         | <i>IEEE Trans. Transport. Electrification.</i>           |
| IEEE TRANSACTIONS ON ULTRASONICS, FERROELECTRICS, AND FREQUENCY CONTROL | UFFC    | <i>IEEE Trans. Ultrason., Ferroelect., Freq. Control</i> |
|                                                                         | SU*     | <i>IEEE Trans. Sonics Ultrason.* (through 1985)</i>      |
|                                                                         | UE*     | <i>IEEE Trans. Ultrason. Eng.*</i>                       |
|                                                                         | PGUE*   | <i>IEEE Trans. Ultrason. Eng.*</i>                       |
| IEEE TRANSACTIONS ON VEHICULAR TECHNOLOGY                               | VT      | <i>IEEE Trans. Veh. Technol.</i>                         |
|                                                                         | VC*     | <i>IEEE Trans. Veh. Commun.*</i>                         |
| IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS        | VLSI    | <i>IEEE Trans. Very Large Scale Integr. (VLSI) Syst.</i> |
| IEEE TRANSACTIONS ON VISUALIZATION AND COMPUTER GRAPHICS                | VCG     | <i>IEEE Trans. Vis. Comput. Graphics</i>                 |
| IEEE TRANSACTIONS ON WIRELESS COMMUNICATIONS                            | WC      | <i>IEEE Trans. Wireless Commun.</i>                      |
| PROCEEDINGS OF THE IEEE                                                 |         | <i>Proc. IEEE</i>                                        |
|                                                                         |         | <i>Proc. IRE* (through 1962)</i>                         |



## List of IEEE Magazines

### List of IEEE Magazines

| Magazine                                            | Reference Abbreviation                                      |
|-----------------------------------------------------|-------------------------------------------------------------|
| IEEE Aerospace and Electronics Systems Magazine     | <i>IEEE Aerosp. Electron. Syst. Mag.</i>                    |
| IEEE Annals of the History of Computing             | <i>IEEE Ann. Hist. Comput.</i>                              |
| IEEE Antennas and Propagation Magazine              | <i>IEEE Antennas Propag. Mag.</i>                           |
| IEEE ASSP Magazine (1984–1990)                      | <i>IEEE ASSP Mag.</i>                                       |
| IEEE Circuits and Systems Magazine                  | <i>IEEE Circuits Syst. Mag.</i>                             |
| IEEE Circuits and Devices Magazine (1988–2006)      | <i>IEEE Circuits Devices Mag.</i>                           |
| IEEE Communications Society Magazine (through 1978) | <i>IEEE Commun. Soc. Mag.</i>                               |
| IEEE Communications Magazine (1979–present)         | <i>IEEE Commun. Mag.</i>                                    |
| IEEE Computational Intelligence Magazine            | <i>IEEE Comput. Intell. Mag.</i>                            |
| IEEE Computing in Science and Engineering Magazine  | <i>IEEE Comput. Sci. Eng.</i>                               |
| IEEE Computer Applications in Power                 | <i>IEEE Comput. Appl. Power</i>                             |
| IEEE Computer Graphics and Applications Magazine    | <i>IEEE Comput. Graph. Appl. Mag.</i>                       |
| IEEE Concurrency                                    | <i>IEEE Concurrency</i>                                     |
| IEEE Consumer Electronics Magazine                  | <i>IEEE Consum. Electron. Mag.</i>                          |
| IEEE Control Systems Magazine                       | <i>IEEE Control Syst. Mag.</i>                              |
| IEEE Design & Test                                  | <i>IEEE Des. Test.</i>                                      |
|                                                     | <i>IEEE Des. Test. Comput.* (through 2012)</i>              |
| IEEE Electrical Insulation Magazine                 | <i>IEEE Elect. Insul. Mag.</i>                              |
| IEEE Electromagnetic Compatibility Magazine         | <i>IEEE Electromagn. Compat.</i>                            |
| IEEE Electrification Magazine                       | <i>IEEE Electrific. Mag.</i>                                |
| IEEE ElectroTechnology Review                       | <i>IEEE ElectroTechnol. Rev.</i>                            |
|                                                     | <i>IEEE Eng. Med. Biol. Mag. (1982–2009)</i>                |
| IEEE Engineering Management Review                  | <i>IEEE Eng. Manag. Rev.</i>                                |
| IEEE Expert (through 1997)                          | <i>IEEE Expert</i>                                          |
| IEEE Geoscience and Remote Sensing Magazine         | <i>IEEE Geosci. Remote Sens. Mag. (replaces Newsletter)</i> |
| IEEE Industrial Electronics Magazine                | <i>IEEE Ind. Electron. Mag.</i>                             |
| IEEE Industry Applications Magazine                 | <i>IEEE Ind. Appl. Mag.</i>                                 |
| IEEE Instrumentation and Measurement Magazine       | <i>IEEE Instrum. Meas. Mag.</i>                             |
| IEEE Intelligent Systems (formerly IEEE Expert)     | <i>IEEE Intell. Syst.</i>                                   |
| IEEE Intelligent Transportation Systems Magazine    | <i>IEEE Intell. Transp. Syst. Mag.</i>                      |
| IEEE Internet Computing Magazine                    | <i>IEEE Internet Comput.</i>                                |
| IEEE IT Professional                                | <i>IEEE IT Prof.</i>                                        |
| IEEE Micro Magazine                                 | <i>IEEE Micro</i>                                           |
| IEEE Microwave Magazine                             | <i>IEEE Microw. Mag.</i>                                    |
| IEEE MultiMedia                                     | <i>IEEE Multimedia Mag.</i>                                 |
| IEEE Nanotechnology Magazine                        | <i>IEEE Nanotechnol. Mag.</i>                               |
| IEEE Network                                        | <i>IEEE Netw.</i>                                           |
| IEEE Personal Communications                        | <i>IEEE Pers. Commun.</i>                                   |
| IEEE Potentials                                     | <i>IEEE Potentials</i>                                      |
| IEEE Power Electronics Magazine                     | <i>IEEE Power Electron. Mag.</i>                            |
| IEEE Power and Energy Magazine                      | <i>IEEE Power Energy Mag.</i>                               |
| IEEE Power Engineering Review                       | <i>IEEE Power Eng. Rev.</i>                                 |
| IEEE Pulse                                          | <i>IEEE Pulse</i>                                           |
| IEEE Robotics and Automation Magazine               | <i>IEEE Robot. Autom. Mag.</i>                              |
| IEEE Signal Processing Magazine (1991–present)      | <i>IEEE Signal Process. Mag.</i>                            |



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## List of IEEE Magazines

### Magazine

IEEE Solid-State Circuits Magazine  
 IEEE Security and Privacy  
 IEEE Software  
 IEEE Spectrum  
 IEEE Technology and Society Magazine  
 IEEE Vehicular Technology Magazine  
 China Communications Magazine  
 Communications Surveys and Tutorials  
 Computer Magazine  
 Internet Computing  
 Pervasive Computing  
 Today's Engineer  
 Wireless Communications

### Reference Abbreviation

*IEEE Solid State Circuits Mag.*  
*IEEE Security Privacy*  
*IEEE Softw.*  
*IEEE Spectr.*  
*IEEE Technol. Soc. Mag.*  
*IEEE Veh. Technol. Mag.*  
*China Commun.*  
*Commun. Surveys Tuts.*  
*Computer*  
*Internet Comput.*  
*Pervasive Comput.*  
*Today's Engineer*  
*Wireless Commun.*



De La Salle University

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## **Appendix K**

### **IEEE INDEX TERMS**



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**2014 IEEE  
Taxonomy**

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**Version  
1.0**

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**Created by  
The Institute  
of Electrical  
and  
Electronics  
Engineers  
(IEEE)**

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### IEEE Taxonomy: A Subset Hierarchical Display of IEEE Thesaurus Terms

The IEEE Taxonomy comprises the first three hierarchical 'levels' under each term-family (or branch) that is formed from the top-most terms of the IEEE Thesaurus. In this document these term-families are arranged alphabetically and denoted by **boldface** type. Each term family's hierarchy goes to no more than three sublevels, denoted by indents (in groups of four dots) preceding the next level terms. A term can appear in more than one hierarchical branch and can appear more than once in any particular hierarchy. The 2014 IEEE Taxonomy is defined in this way so that it is always a subset of the 2014 IEEE Thesaurus.

#### **Aerospace and electronic systems**

- ....Aerospace control
  - .....Air traffic control
  - .....Attitude control
  - .....Ground support
  - .....Aerospace engineering
    - .....Aerospace biophysics
    - .....Aerospace electronics
    - .....Aerospace safety
      - .....Air safety
    - .....Aerospace simulation
    - .....Aerospace testing
    - .....Satellites
      - .....Artificial satellites
      - .....Earth Observing System
      - .....Low earth orbit satellites
      - .....Moon
      - .....Space stations
      - .....Space technology
      - .....Space exploration
    - .....Aerospace materials
    - .....Aerospace components
    - ....Aircraft manufacture
    - ....Aircraft navigation
    - ....Aircraft propulsion
      - .....Propellers
    - ....Command and control systems
    - ....Electronic warfare
      - .....Electronic countermeasures
      - .....Jamming
    - .....Radar countermeasures
    - ....Military equipment
      - .....Military aircraft
        - .....Payloads
      - .....Military satellites
      - .....Weapons
        - .....Guns
        - .....Missiles
        - .....Nuclear weapons
        - .....Projectiles
    - ....Radar
      - .....Airborne radar
      - .....Bistatic radar
      - .....Doppler radar
      - .....Ground penetrating radar
      - .....Laser radar
      - .....Meteorological radar
      - .....Millimeter wave radar
      - .....Multistatic radar
      - .....MIMO radar
      - .....Passive radar
      - .....Radar applications
        - .....Radar countermeasures
        - .....Radar detection
        - .....Radar imaging
        - .....Radar measurements
        - .....Radar polarimetry
        - .....Radar remote sensing
        - .....Radar tracking
        - .....Radar clutter
        - .....Radar cross-sections
        - .....Radar equipment
        - .....Radar theory
        - .....Spaceborne radar
        - .....Spread spectrum radar
        - .....Synthetic aperture radar
          - .....Inverse synthetic aperture radar
          - .....Polarimetric synthetic aperture radar
        - .....Ultra wideband radar
      - .....Sensor systems
        - .....Gunshot detection systems
      - ....Sonar
        - .....Sonar applications
          - .....Sonar detection
          - .....Sonar measurements
          - .....Sonar equipment
        - .....Synthetic aperture sonar
      - ....Telemetry
        - .....Biomedical telemetry





**Antennas and propagation**

....Antennas  
 .....Antenna accessories  
 .....Antenna arrays  
 .....Adaptive arrays  
 .....Butler matrices  
 .....Linear antenna arrays  
 .....Log periodic antennas  
 .....Microstrip antenna arrays  
 .....Microwave antenna arrays  
 .....Phased arrays  
 .....Planar arrays  
 .....Antenna radiation patterns  
 .....Near-field radiation pattern  
 .....Antenna theory  
 .....Frequency selective surfaces  
 .....Apertures  
 .....Aperture antennas  
 .....Aperture coupled antennas  
 .....Broadband antennas  
 .....Ultra wideband antennas  
 .....Vivaldi antennas  
 .....Dielectric resonator antennas  
 .....Dipole antennas  
 .....Directional antennas  
 .....Directive antennas  
 .....Feeds  
 .....Antenna feeds  
 .....Fractal antennas  
 .....Helical antennas  
 .....Horn antennas  
 .....Leaky wave antennas  
 .....Loaded antennas  
 .....Log-periodic dipole antennas  
 .....Microstrip antennas  
 .....Microwave antennas  
 .....Mobile antennas  
 .....Multifrequency antennas  
 .....Omnidirectional antennas  
 .....Patch antennas  
 .....Radar antennas  
 .....Receiving antennas  
 .....Rectennas  
 .....Reflector antennas  
 .....Satellite antennas  
 .....Slot antennas  
 .....Transmission line antennas  
 .....Transmitting antennas  
 .....UHF antennas  
 .....Yagi-Uda antennas  
 ....Electromagnetic propagation

.....Electromagnetic diffraction  
 .....Optical diffraction  
 .....Physical theory of diffraction  
 .....X-ray diffraction  
 .....Electromagnetic propagation in absorbing media  
 .....Electromagnetic reflection  
 .....Optical reflection  
 .....Microwave propagation  
 .....Millimeter wave propagation  
 .....Optical propagation  
 .....Optical surface waves  
 .....Optical waveguides  
 .....Propagation constant  
 .....Propagation losses  
 .....Radio propagation  
 .....Radiowave propagation  
 .....Submillimeter wave propagation  
 .....UHF propagation  
 ....Radio astronomy

**Broadcast technology**

....Broadcasting  
 .....Digital audio broadcasting  
 .....Digital audio players  
 .....Digital Radio Mondiale  
 .....Digital multimedia broadcasting  
 .....Digital video broadcasting  
 .....Radio broadcasting  
 .....Frequency modulation  
 .....Radio networks  
 .....Satellite broadcasting  
 .....TV broadcasting

**Circuits and systems**

....Circuits  
 .....Active circuits  
 .....Active inductors  
 .....Gyrators  
 .....Operational amplifiers  
 .....Adders  
 .....Analog circuits  
 .....Analog integrated circuits  
 .....Analog processing circuits  
 .....Application specific integrated circuits  
 .....System-on-chip  
 .....Asynchronous circuits  
 .....Bipolar integrated circuits  
 .....BiCMOS integrated circuits





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2014 IEEE Taxonomy

|                                          |                                          |
|------------------------------------------|------------------------------------------|
| .....Bipolar transistor circuits         | .....Integrated circuit modeling         |
| .....Bipolar integrated circuits         | .....Integrated circuit noise            |
| .....Bistable circuits                   | .....Integrated circuit synthesis        |
| .....Latches                             | .....Large scale integration             |
| .....Bridge circuits                     | .....MESFET integrated circuits          |
| .....Charge pumps                        | .....Microprocessors                     |
| .....Circuit analysis                    | .....Microwave integrated circuits       |
| .....Circuit analysis computing          | .....Millimeter wave integrated circuits |
| .....Coupled mode analysis               | .....Mixed analog digital integrated     |
| .....Nonlinear network analysis          | circuits                                 |
| .....Circuit faults                      | .....Monolithic integrated circuits      |
| .....Electrical fault detection          | .....Photonic integrated circuits        |
| .....Circuit noise                       | .....Power integrated circuits           |
| .....Thermal noise                       | .....Radiofrequency integrated circuits  |
| .....Circuit simulation                  | .....Submillimeter wave integrated       |
| .....Circuit synthesis                   | circuits                                 |
| .....High level synthesis                | .....Superconducting integrated circuits |
| .....Integrated circuit synthesis        | .....Thick film circuits                 |
| .....Coprocessors                        | .....Thin film circuits                  |
| .....Counting circuits                   | .....Three-dimensional integrated        |
| .....Coupling circuits                   | circuits                                 |
| .....Digital circuits                    | .....Through-silicon vias                |
| .....Circuit topology                    | .....UHF integrated circuits             |
| .....Digital integrated circuits         | .....Ultra large scale integration       |
| .....Digital signal processors           | .....Very high speed integrated circuits |
| .....Distributed parameter circuits      | .....Very large scale integration        |
| .....Driver circuits                     | .....Wafer scale integration             |
| .....Electronic circuits                 | .....Isolators                           |
| .....Breadboard circuit                  | .....Large scale integration             |
| .....Central Processing Unit             | .....Ultra large scale integration       |
| .....Stripboard circuit                  | .....Very large scale integration        |
| .....Equivalent circuits                 | .....Wafer scale integration             |
| .....Feedback                            | .....Linear circuits                     |
| .....Feedback circuits                   | .....Logic arrays                        |
| .....Negative feedback                   | .....Programmable logic arrays           |
| .....Neurofeedback                       | .....Logic circuits                      |
| .....Hybrid integrated circuits          | .....Combinational circuits              |
| .....Integrated circuits                 | .....Logic arrays                        |
| .....Analog-digital integrated circuits  | .....Programmable logic arrays           |
| .....Analog integrated circuits          | .....Superconducting logic circuits      |
| .....Application specific integrated     | .....Magnetic circuits                   |
| circuits                                 | .....Microprocessors                     |
| .....Bipolar integrated circuits         | .....Automatic logic units               |
| .....CMOS integrated circuits            | .....Biomimetics                         |
| .....Coprocessors                        | .....Coprocessors                        |
| .....Current-mode circuits               | .....Microcontrollers                    |
| .....Digital integrated circuits         | .....Microprocessor chips                |
| .....FET integrated circuits             | .....Vector processors                   |
| .....Field programmable gate arrays      | .....Microwave circuits                  |
| .....Hybrid integrated circuits          | .....Millimeter wave circuits            |
| .....Integrated circuit interconnections | .....Millimeter wave integrated circuits |





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2014 IEEE Taxonomy

|                                             |                                    |
|---------------------------------------------|------------------------------------|
| .....Millimeter wave integrated circuits    | .....Thin film circuits            |
| .....MIMICs                                 | .....Thyristor circuits            |
| .....Monolithic integrated circuits         | .....Time varying circuits         |
| .....MIMICs                                 | .....Trigger circuits              |
| .....MMICs                                  | .....UHF circuits                  |
| .....MOSFET circuits                        | .....UHF integrated circuits       |
| .....CMOSFET circuits                       | .....UHF integrated circuits       |
| .....MOS integrated circuits                | .....Ultra large scale integration |
| .....Power MOSFET                           | .....Very large scale integration  |
| .....Multiplying circuits                   | .....Neuromorphics                 |
| .....Nonlinear circuits                     | .....Wafer scale integration       |
| .....Nonlinear network analysis             | .....VHF circuits                  |
| .....Passive circuits                       | .....Wafer scale integration       |
| .....Phase shifters                         | ....Contacts                       |
| .....Phase transformers                     | .....Brushes                       |
| .....Power dissipation                      | .....Contact resistance            |
| .....Power integrated circuits              | .....Ohmic contacts                |
| .....Printed circuits                       | ....Filtering                      |
| .....Flexible printed circuits              | .....Filters                       |
| .....Programmable circuits                  | .....Active filters                |
| .....Field programmable analog arrays       | .....Anisotropic                   |
| .....Programmable logic arrays              | .....Bragg gratings                |
| .....Programmable logic devices             | .....Channel bank filters          |
| .....Programmable logic arrays              | .....Digital filters               |
| .....Programmable logic devices             | .....Equalizers                    |
| .....Pulse circuits                         | .....Filtering theory              |
| .....Flip-flops                             | .....Gabor filters                 |
| .....Radiation detector circuits            | .....Harmonic filters              |
| .....Rail to rail operation                 | .....IIR filters                   |
| .....Rail to rail amplifiers                | .....Kalman filters                |
| .....Rail to rail inputs                    | .....Low-pass filters              |
| .....Rail to rail outputs                   | .....Matched filters               |
| .....Rectifiers                             | .....Microstrip filters            |
| .....RLC circuits                           | .....Nonlinear filters             |
| .....Sampled data circuits                  | .....Particle filters              |
| .....Sequential circuits                    | .....Power filters                 |
| .....Silicon-on-insulator                   | .....Resonator filters             |
| .....Silicon on sapphire                    | .....Spatial filters               |
| .....Submillimeter wave circuits            | .....Superconducting filters       |
| .....Submillimeter wave integrated circuits | .....Transversal filters           |
| .....Summing circuits                       | .....Information filtering         |
| .....Switched circuits                      | .....Information filters           |
| .....Switched capacitor circuits            | .....Recommender systems           |
| .....Switching circuits                     | .....Integrated circuit technology |
| .....Choppers (circuits)                    | .....CMOS technology               |
| .....Logic circuits                         | .....CMOS process                  |
| .....Switching converters                   | .....Silicon on sapphire           |
| .....Zero current switching                 | .....Moore's Law                   |
| .....Zero voltage switching                 | ....Logic devices                  |
| .....Thick film circuits                    | .....Logic gates                   |
|                                             | .....Programmable logic devices    |



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2014 IEEE Taxonomy

....Oscillators  
 .....Digital-controlled oscillators  
 .....Injection-locked oscillators  
 .....Local oscillators  
 .....Microwave oscillators  
 .....Phase noise  
 .....Ring oscillators  
 .....Voltage-controlled oscillators  
 ....Single electron devices  
 .....Single electron memory  
 .....Hetero-nanocrystal memory  
 .....Single electron transistors  
 ....Tunable circuits and devices  
 .....RLC circuits  
 .....Tuned circuits

### **Communications technology**

....Communication equipment  
 .....Auditory displays  
 .....Codecs  
 .....Speech codecs  
 .....Video codecs  
 .....Modems  
 .....Optical communication equipment  
 .....Optical transmitters  
 .....Radio communication equipment  
 .....Base stations  
 .....Ham radios  
 .....Land mobile radio equipment  
 .....Radio transceivers  
 .....Transponders  
 .....Receivers  
 .....Optical receivers  
 .....RAKE receivers  
 .....Receiving antennas  
 .....Repeaters  
 .....Speech codecs  
 .....Telephone equipment  
 .....Cellular phones  
 .....Telephone sets  
 .....Vocoders  
 .....Transceivers  
 .....Radio transceivers  
 .....Transmitters  
 .....Auxiliary transmitters  
 .....Diversity methods  
 .....Neurotransmitters  
 .....Optical transmitters  
 .....Radio transmitters  
 .....Transmitting antennas

.....Transponders  
 .....TV equipment  
 .....Large screen displays  
 .....TV receivers  
 .....Video codecs  
 .....Video equipment  
 .....Video codecs  
 .....Vocoders  
 .....Communication switching  
 .....Code division multiplexing  
 .....Electronic switching systems  
 .....Frame relay  
 .....Handover  
 .....Multiprotocol label switching  
 .....Packet switching  
 .....Burst switching  
 .....Frame relay  
 .....Multiprotocol label switching  
 .....Packet loss  
 ....Communication systems  
 .....ARPANET  
 .....Biomedical communication  
 .....Biomedical telemetry  
 .....Telemedicine  
 .....Broadband communication  
 .....B-ISDN  
 .....Broadband amplifiers  
 .....Communication networks  
 .....Central office  
 .....Cyberspace  
 .....Industrial communication  
 .....Relay networks  
 (telecommunications)  
 .....Software defined networking  
 .....Communication system control  
 .....Telecommunication control  
 .....Communication system security  
 .....Radio communication  
 countermeasures  
 .....Communication system signaling  
 .....Communication system software  
 .....Streaming media  
 .....Communication system traffic  
 .....Communication system traffic control  
 .....Computer networks  
 .....Ad hoc networks  
 .....Computer network management  
 .....Content distribution networks  
 .....Cyberspace  
 .....Diffserv networks  
 .....Domain Name System



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|                                              |                                                              |
|----------------------------------------------|--------------------------------------------------------------|
| .....Ethernet networks                       | .....Internet topology                                       |
| .....Google                                  | .....Middleboxes                                             |
| .....Internet                                | .....Semantic Web                                            |
| .....Intserv networks                        | .....Social computing                                        |
| .....IP networks                             | .....Web 2.0                                                 |
| .....Metropolitan area networks              | .....Web services                                            |
| .....Multiprocessor interconnection networks | .....IP networks                                             |
| .....Network servers                         | .....TCPIP                                                   |
| .....Next generation networking              | .....ISDN                                                    |
| .....Overlay networks                        | .....B-ISDN                                                  |
| .....Peer-to-peer computing                  | .....Land mobile radio cellular systems                      |
| .....Software defined networking             | .....Cellular networks                                       |
| .....Storage area networks                   | .....Paging strategies                                       |
| .....Token networks                          | .....Local area networks                                     |
| .....Unicast                                 | .....Wireless LAN                                            |
| .....Virtual private networks                | .....Machine-to-machine communications                       |
| .....Wide area networks                      | .....Metropolitan area networks                              |
| .....Cross layer design                      | .....Microwave communication                                 |
| .....Data buses                              | .....Rectennas                                               |
| .....Backplanes                              | .....Military communication                                  |
| .....Data communication                      | .....Reconnaissance                                          |
| .....Asynchronous communication              | .....Millimeter wave communication                           |
| .....Asynchronous transfer mode              | .....MIMO                                                    |
| .....Data buses                              | .....Rician channels                                         |
| .....Data transfer                           | .....Mobile communication                                    |
| .....Telecommunication buffers               | .....3G mobile communication                                 |
| .....Telemetry                               | .....4G mobile communication                                 |
| .....Teleprinting                            | .....Ambient networks                                        |
| .....Digital communication                   | .....Dual band                                               |
| .....Baseband                                | .....Land mobile radio                                       |
| .....DICOM                                   | .....Land mobile radio cellular systems                      |
| .....Digital audio broadcasting              | .....Mobile nodes                                            |
| .....Digital images                          | .....Mobile radio mobility management                        |
| .....Digital multimedia broadcasting         | .....Software radio                                          |
| .....Digital video broadcasting              | .....Molecular communication                                 |
| .....DSL                                     | .....Multiaccess communication                               |
| .....ISDN                                    | .....Direct-sequence code-division multiple access           |
| .....Passband                                | .....Frequency division multiaccess                          |
| .....Portable media players                  | .....Multicarrier code division multiple access              |
| .....SONET                                   | .....Subscriber loops                                        |
| .....Spread spectrum communication           | .....Time division multiple access                           |
| .....Facsimile                               | .....Time division synchronous code division multiple access |
| .....FDDI                                    | .....Multicast communication                                 |
| .....Indoor communication                    | .....Multicast VPN                                           |
| .....Indoor environments                     | .....Multimedia communication                                |
| .....Internet                                | .....Narrowband                                              |
| .....Crowdsourcing                           | .....Optical fiber communication                             |
| .....Instant messaging                       | .....FDDI                                                    |
| .....Internet of Things                      |                                                              |
| .....Internet telephony                      |                                                              |



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|                                         |                                       |
|-----------------------------------------|---------------------------------------|
| .....Optical buffering                  | .....Submillimeter wave communication |
| .....Optical fiber networks             | .....Subscriber loops                 |
| .....Optical fiber subscriber loops     | .....Switching systems                |
| .....Optical interconnections           | .....Electronic switching systems     |
| .....Optical packet switching           | .....Switching frequency              |
| .....Optical wavelength conversion      | .....Switching loss                   |
| .....Scheduling algorithms              | .....Telecommunication switching      |
| .....SONET                              | .....Synchronous digital hierarchy    |
| .....Personal communication networks    | .....Telecommunications               |
| .....Protocols                          | .....Ambient intelligence             |
| .....Access protocols                   | .....Feedback communications          |
| .....Asynchronous transfer mode         | .....IP networks                      |
| .....Cryptographic protocols            | .....Radio access networks            |
| .....Master-slave                       | .....Railway communication            |
| .....Multicast protocols                | .....Telecommunication computing      |
| .....Multiprotocol label switching      | .....Telecommunication network        |
| .....Routing protocols                  | topology                              |
| .....Transport protocols                | .....Telecommunication services       |
| .....Wireless application protocol      | .....Telematics                       |
| .....Quality of service                 | .....Teleconferencing                 |
| .....Admission control                  | .....Telegraphy                       |
| .....Radio communication                | .....Telephony                        |
| .....Baseband                           | .....Teleprinting                     |
| .....Bluetooth                          | .....Teletext                         |
| .....Indoor radio communication         | .....Token networks                   |
| .....Land mobile radio                  | .....UHF communication                |
| .....Land mobile radio cellular systems | .....Underwater communication         |
| .....Packet radio networks              | .....Videophone systems               |
| .....Passband                           | .....Videotex                         |
| .....Personal area networks             | .....Visual communication             |
| .....Radio broadcasting                 | .....Wide area networks               |
| .....Radio communication                | .....Wideband                         |
| countermeasures                         | .....Wireless communication           |
| .....Radio frequency                    | .....Cognitive radio                  |
| .....Radio link                         | .....Cooperative communication        |
| .....Radio spectrum management          | .....GSM                              |
| .....Satellite communication            | .....Open wireless architecture       |
| .....Satellite ground stations          | .....Roaming                          |
| .....Software radio                     | .....Spatial diversity                |
| .....Zigbee                             | .....WiMAX                            |
| .....Routing                            | .....Wireless application protocol    |
| .....Wavelength routing                 | .....Wireless networks                |
| .....Satellite communication            | .....Wireless mesh networks           |
| .....Downlink                           | .....Wireless sensor networks         |
| .....Satellite broadcasting             | .....Body sensor networks             |
| .....Satellite ground stations          | .....Event detection                  |
| .....Uplink                             | .....Couplers                         |
| .....Satellite ground stations          | .....Directional couplers             |
| .....SIMO                               | .....High-speed electronics           |
| .....SISO                               | .....High-speed integrated circuits   |
| .....Spatial diversity                  | .....High-speed networks              |





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.....Ultrafast electronics  
 ....Image communication  
 .....Facsimile  
 .....Picture archiving and communication systems  
 ....Message systems  
 .....Electronic mail  
 .....Unified messaging  
 .....Unsolicited electronic mail  
 .....Electronic messaging  
 .....Instant messaging  
 .....Unified messaging  
 .....Postal services  
 .....Publish subscribe systems  
 .....Voice mail  
 ....Modulation  
 .....Amplitude modulation  
 .....Amplitude shift keying  
 .....Quadrature amplitude modulation  
 .....Chirp modulation  
 .....Demodulation  
 .....Digital modulation  
 .....Constellation diagram  
 .....Partial response signaling  
 .....Frequency modulation  
 .....Frequency shift keying  
 .....Magnetic modulators  
 .....Modulation coding  
 .....Interleaved codes  
 .....Optical modulation  
 .....Electrooptic modulators  
 .....Intensity modulation  
 .....Phase modulation  
 .....Continuous phase modulation  
 .....Differential phase shift keying  
 .....Phase shift keying  
 .....Pulse modulation  
 .....Pulse width modulation  
 .....Pulse width modulation inverters  
 .....Space vector pulse width modulation  
 ....Multiplexing  
 .....Code division multiplexing  
 .....Demultiplexing  
 .....Frequency division multiplexing  
 .....Multiplexing equipment  
 .....Add-drop multiplexers  
 .....OFDM  
 .....Multiple access interference  
 .....OFDM modulation  
 .....Partial transmit sequences

.....Peak to average power ratio  
 .....Time division multiplexing  
 .....Wavelength division multiplexing  
 .....WDM networks  
 ....Network topology  
 .....Complex networks  
 .....Computer network reliability  
 .....Presence network agents  
 ....TV  
 .....Cable TV  
 .....Digital TV  
 .....Analog TV  
 .....HDTV  
 .....IPTV  
 .....Mobile TV  
 .....Three-dimensional television  
 ....UHF technology  
 .....UHF antennas  
 .....UHF circuits  
 .....UHF integrated circuits  
 .....UHF communication  
 .....UHF devices  
 .....UHF integrated circuits  
 ....Ultra wideband technology  
 .....Ultra wideband antennas  
 .....Ultra wideband communication  
 .....Ultra wideband radar  
 ....VHF devices

#### **Components, packaging, and manufacturing technology**

....Component architectures  
 ....Electronic components  
 .....Capacitors  
 .....Power capacitors  
 .....Varactors  
 .....Coils  
 .....Superconducting coils  
 .....Connectors  
 .....Plugs  
 .....Sockets  
 .....Diodes  
 .....Diode lasers  
 .....Electrodes  
 .....Anodes  
 .....Cathodes  
 .....Microelectrodes  
 .....Fuses  
 .....Inductors  
 .....Active inductors



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- .....Thick film inductors
- .....Thin film inductors
- .....Resistors
- .....Memristors
- .....Switched capacitor networks
- .....Varistors
- .....Structural plates
- .....Switches
- .....Contactors
- .....Microswitches
- .....Optical switches
- .....Transducers
- .....Acoustic transducers
- .....Biomedical transducers
- .....Chemical transducers
- .....Piezoelectric transducers
- .....Ultrasonic transducer arrays
- ....Electronic equipment manufacture
- .....Damascene integration
- .....Micromachining
- .....Radiation hardening (electronics)
- .....Semiconductor device manufacture
- .....Diffusion processes
- .....Flip-chip devices
- .....High-K gate dielectrics
- .....Quasi-doping
- .....Semiconductor device doping
- .....Semiconductor epitaxial layers
- .....Semiconductor growth
- .....Silicidation
- .....Wafer bonding
- ....Electronics packaging
- .....Chip scale packaging
- ....Environmentally friendly manufacturing techniques
- ....Integrated circuit manufacture
- .....Surface-mount technology
- ....Integrated circuit packaging
- .....Multichip modules
- .....Plastic integrated circuit packaging
- ....Semiconductor device packaging
- ....Thermal management of electronics
- .....Electronic packaging thermal management
- .....Electronics cooling
- Computational and artificial intelligence**
- ....Artificial intelligence
- .....Context awareness
- .....Cooperative systems
- .....Decision support systems
- .....Intelligent systems
- .....Intelligent robots
- .....Knowledge based systems
- .....Expert systems
- .....Mobile agents
- .....Knowledge engineering
- .....Inference mechanisms
- .....Knowledge acquisition
- .....Knowledge discovery
- .....Knowledge representation
- .....Learning (artificial intelligence)
- .....Distance learning
- .....Electronic learning
- .....Learning systems
- .....Backpropagation
- .....Learning automata
- .....Semisupervised learning
- .....Supervised learning
- .....Unsupervised learning
- .....Machine learning
- .....Boosting
- .....Statistical learning
- .....Prediction methods
- .....Linear predictive coding
- .....Predictive coding
- .....Predictive encoding
- .....Predictive models
- ....Autonomous mental development
- ....Computational intelligence
- .....Computation theory
- .....Computational complexity
- .....Concurrent computing
- .....Greedy algorithms
- .....Support vector machines
- .....Evolutionary computation
- .....Particle swarm optimization
- .....Fuzzy systems
- .....Fuzzy control
- .....Fuzzy neural networks
- .....Hybrid intelligent systems
- .....Genetic algorithms
- ....Logic
- .....Fuzzy logic
- .....Fuzzy cognitive maps
- .....Takagi-Sugeno model
- .....Multivalued logic
- .....Probabilistic logic
- .....Sufficient conditions
- ....Machine intelligence
- .....Pattern analysis







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....Neural networks  
 .....Artificial neural networks  
 .....Hebbian theory  
 .....Self-organizing feature maps  
 .....Biological neural networks  
 .....Cellular neural networks  
 .....Feedforward neural networks  
 .....Multilayer perceptrons  
 .....Multi-layer neural network  
 .....Neural network hardware  
 .....Radial basis function networks  
 .....Recurrent neural networks  
 .....Hopfield neural networks

#### Computers and information processing

....Computer applications  
 .....Affective computing  
 .....Application virtualization  
 .....Computer aided analysis  
 .....Computer aided engineering  
 .....Computer aided instruction  
 .....Computer generated music  
 .....Computer integrated manufacturing  
 .....Control engineering computing  
 .....Green computing  
 .....High energy physics instrumentation  
 computing  
 .....Linear particle accelerator  
 .....Knowledge management  
 .....Knowledge transfer  
 .....Medical information systems  
 .....Electronic medical records  
 .....Military computing  
 .....Physics computing  
 .....Power engineering computing  
 .....Power system analysis computing  
 .....Publishing  
 .....Bibliometrics  
 .....Company reports  
 .....Desktop publishing  
 .....Electronic publishing  
 .....Open Access  
 .....Scientific publishing  
 .....Scientific computing  
 .....Telecommunication computing  
 .....Internetworking  
 .....Soft switching  
 .....Virtual enterprises  
 .....Virtual manufacturing  
 .....Virtual machining

.....Web sites  
 .....Facebook  
 .....MySpace  
 .....Uniform resource locators  
 .....Web design  
 .....YouTube  
 .....World Wide Web  
 .....Mashups  
 ....Computer architecture  
 .....Accelerator architectures  
 .....Data structures  
 .....Arrays  
 .....Binary decision diagrams  
 .....Null value  
 .....Octrees  
 .....Table lookup  
 .....Tree data structures  
 .....Dynamic voltage scaling  
 .....Memory architecture  
 .....Memory management  
 .....Multiprocessor interconnection  
 .....Hypercubes  
 .....Parallel architectures  
 .....Multicore processing  
 .....Reconfigurable architectures  
 ....Computer interfaces  
 .....Application programming interfaces  
 .....WebRTC  
 .....Browsers  
 .....Field buses  
 .....Firewire  
 .....Haptic interfaces  
 .....Data gloves  
 .....Force feedback  
 .....Grasping  
 .....Hypertext systems  
 .....Interface phenomena  
 .....Network interfaces  
 .....Interface states  
 .....Musical instrument digital interfaces  
 .....Ports (Computers)  
 .....System buses  
 ....Computer networks  
 .....Ad hoc networks  
 .....AODV  
 .....Mesh networks  
 .....Mobile ad hoc networks  
 .....Vehicular ad hoc networks  
 .....Computer network management  
 .....Computer network reliability  
 .....Disruption tolerant networking



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|                                     |                                     |
|-------------------------------------|-------------------------------------|
| .....Management information base    | .....Difference engines             |
| .....Middleboxes                    | .....Microcomputers                 |
| .....Network address translation    | .....Portable computers             |
| .....Network synthesis              | .....Workstations                   |
| .....Content distribution networks  | .....Parallel machines              |
| .....Cyberspace                     | .....Supercomputers                 |
| .....Diffserv networks              | .....Tablet computers               |
| .....Domain Name System             | .....Wearable computers             |
| .....Ethernet networks              | .....Computer science               |
| .....EPON                           | .....Formal languages               |
| .....Google                         | .....Computer languages             |
| .....Internet                       | .....Runtime library                |
| .....Crowdsourcing                  | .....Network theory (graphs)        |
| .....Instant messaging              | .....Programming                    |
| .....Internet of Things             | .....Augmented reality              |
| .....Internet telephony             | .....Automatic programming          |
| .....Internet topology              | .....Concatenated codes             |
| .....Middleboxes                    | .....Functional programming         |
| .....Semantic Web                   | .....Granular computing             |
| .....Social computing               | .....Integer linear programming     |
| .....Web 2.0                        | .....Logic programming              |
| .....Web services                   | .....Microprogramming               |
| .....Intserv networks               | .....Object oriented methods        |
| .....IP networks                    | .....Object oriented programming    |
| .....TCPIP                          | .....Opportunistic software systems |
| .....Metropolitan area networks     | development                         |
| .....Multiprocessor interconnection | .....Parallel programming           |
| networks                            | .....Performance analysis           |
| .....Network servers                | .....Programming profession         |
| .....Next generation networking     | .....Robot programming              |
| .....Overlay networks               | .....Concurrency control            |
| .....Peer-to-peer computing         | .....Processor scheduling           |
| .....Software defined networking    | .....Scheduling algorithms          |
| .....Storage area networks          | .....Database machines              |
| .....Token networks                 | .....Data systems                   |
| .....Unicast                        | .....Data acquisition               |
| .....Virtual private networks       | .....Fastbus                        |
| .....Extranets                      | .....User-generated content         |
| .....Wide area networks             | .....Data compression               |
| .....Computer performance           | .....Adaptive coding                |
| .....Computer errors                | .....Audio compression              |
| .....Computer crashes               | .....Huffman coding                 |
| .....Performance loss               | .....Source coding                  |
| .....Computer peripherals           | .....Test data compression          |
| .....Disk drives                    | .....Transform coding               |
| .....Keyboards                      | .....Data conversion                |
| .....Modems                         | .....Analog-digital conversion      |
| .....Printers                       | .....Digital-analog conversion      |
| .....Computers                      | .....Data engineering               |
| .....Analog computers               | .....Data handling                  |
| .....Calculators                    | .....Data assimilation              |



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|                                      |                                   |
|--------------------------------------|-----------------------------------|
| .....Data encapsulation              | .....Internet of Things           |
| .....Document handling               | .....Internet telephony           |
| .....Merging                         | .....Internet topology            |
| .....Sorting                         | .....Middleboxes                  |
| .....Data processing                 | .....Semantic Web                 |
| .....Associative processing          | .....Social computing             |
| .....Business data processing        | .....Web 2.0                      |
| .....Data analysis                   | .....Web services                 |
| .....Data collection                 | .....Metacomputing                |
| .....Data integration                | .....Grid computing               |
| .....Data preprocessing              | .....Peer-to-peer computing       |
| .....Data transfer                   | ....DNA computing                 |
| .....Information exchange            | ....File servers                  |
| .....Spreadsheet programs            | ....Hardware                      |
| .....Text processing                 | .....Open source hardware         |
| .....Virtual enterprises             | ....High performance computing    |
| .....Data storage systems            | ....Image processing              |
| .....Data warehouses                 | .....Active shape model           |
| ....Digital systems                  | .....Feature extraction           |
| .....Internet                        | .....Geophysical image processing |
| .....Crowdsourcing                   | .....Gray-scale                   |
| .....Instant messaging               | .....Image analysis               |
| .....Internet of Things              | .....Image classification         |
| .....Internet telephony              | .....Image motion analysis        |
| .....Internet topology               | .....Image quality                |
| .....Middleboxes                     | .....Image sequence analysis      |
| .....Semantic Web                    | .....Image texture analysis       |
| .....Social computing                | .....Object detection             |
| .....Web 2.0                         | .....Subtraction techniques       |
| .....Web services                    | .....Image coding                 |
| .....ISDN                            | .....Image color analysis         |
| .....B-ISDN                          | .....Image decomposition          |
| .....Local area networks             | .....Image denoising              |
| .....Wireless LAN                    | .....Image enhancement            |
| .....Metropolitan area networks      | .....Image fusion                 |
| .....Token networks                  | .....Image generation             |
| ....Distributed computing            | .....Plasma displays              |
| .....Client-server systems           | .....Visual effects               |
| .....Middleware                      | .....Image recognition            |
| .....Servers                         | .....Image edge detection         |
| .....Collaborative work              | .....Image reconstruction         |
| .....Cooperative communication       | .....Image registration           |
| .....Crowdsourcing                   | .....Image representation         |
| .....Social computing                | .....Image resolution             |
| .....Diffserv networks               | .....High-resolution imaging      |
| .....Distributed databases           | .....Spatial resolution           |
| .....Distributed information systems | .....Image restoration            |
| .....Publish-subscribe               | .....Image sampling               |
| .....Internet                        | .....Image segmentation           |
| .....Crowdsourcing                   | .....Image sequences              |
| .....Instant messaging               | .....Image texture                |



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|                                         |                                             |
|-----------------------------------------|---------------------------------------------|
| .....Machine vision                     | .....Multiprocessing systems                |
| .....Object recognition                 | .....Data flow computing                    |
| .....Object segmentation                | .....Processor scheduling                   |
| .....Morphological operations           | .....Systolic arrays                        |
| .....Optical feedback                   | .....Multithreading                         |
| .....Smart pixels                       | .....Parallel algorithms                    |
| .....Spatial coherence                  | .....Pipeline processing                    |
| .....Table lookup                       | ....Pattern recognition                     |
| ....Memory                              | .....Active shape model                     |
| .....Analog memory                      | .....Character recognition                  |
| .....Associative memory                 | .....Clustering methods                     |
| .....Buffer storage                     | .....Pattern clustering                     |
| .....Computer buffers                   | .....Data mining                            |
| .....Cache memory                       | .....Association rules                      |
| .....Cache storage                      | .....Data privacy                           |
| .....Content addressable storage        | .....Text analysis                          |
| .....Flash memories                     | .....Text mining                            |
| .....Flash memory cells                 | .....Web mining                             |
| .....Magnetic memory                    | .....Face recognition                       |
| .....Floppy disks                       | .....Fingerprint recognition                |
| .....Hard disks                         | .....Gesture recognition                    |
| .....Memory management                  | .....Sign language                          |
| .....Nonvolatile memory                 | .....Handwriting recognition                |
| .....Nonvolatile single electron memory | .....Forgery                                |
| .....Phase change memory                | .....Pattern matching                       |
| .....Phase change random access         | .....Image matching                         |
| memory                                  | .....Speech recognition                     |
| .....Random access memory               | .....Automatic speech recognition           |
| .....DRAM chips                         | .....Speech analysis                        |
| .....Phase change random access         | .....Text recognition                       |
| memory                                  | ....Pervasive computing                     |
| .....SDRAM                              | .....Ubiquitous computing                   |
| .....SRAM cells                         | .....Context-aware services                 |
| .....SRAM chips                         | .....Wearable computers                     |
| .....Read only memory                   | ....Petascale computing                     |
| .....PROM                               | ....Platform virtualization                 |
| .....Read-write memory                  | ....Quantum computing                       |
| .....Registers                          | .....Quantum cellular automata              |
| .....Shift registers                    | ....Real-time systems                       |
| .....Scanning probe data storage        | .....WebRTC                                 |
| .....Semiconductor memory               | ....Software                                |
| ....Mobile computing                    | .....Application software                   |
| ....Molecular computing                 | .....Embedded software                      |
| ....Multitasking                        | .....Middleware                             |
| .....Parametric study                   | .....Mediation                              |
| ....Open systems                        | .....Message-oriented middleware            |
| .....Open Access                        | .....Web services                           |
| .....Public domain software             | .....Open source software                   |
| .....Physical layer                     | .....Optical character recognition software |
| ....Optical computing                   | .....Public domain software                 |
| ....Parallel processing                 | .....Software agents                        |



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.....Autonomous agents  
 .....Intelligent agents  
 .....Software as a service  
 .....Software debugging  
 .....Software design  
 .....Software maintenance  
 .....Software packages  
 .....EMTDC  
 .....MATLAB  
 .....PSCAD  
 .....SPICE  
 .....Software performance  
 .....Software quality  
 .....Software reusability  
 .....Software safety  
 .....Software systems  
 .....Software tools  
 .....Authoring systems  
 .....System software  
 .....File systems  
 .....Operating systems  
 .....Program processors  
 .....Utility programs  
 ....Software engineering  
 .....Capability maturity model  
 .....Computer aided software engineering  
 .....Formal verification  
 .....Programming environments  
 .....Reasoning about programs  
 .....Runtime  
 .....Dynamic compiler  
 .....Runtime environment  
 .....Software architecture  
 .....Client-server systems  
 .....Microarchitecture  
 .....Representational state transfer  
 .....Software libraries  
 ....System recovery  
 .....Checkpointing  
 .....Core dumps  
 .....Debugging  
 ....Time sharing computer systems  
 ....Virtual machine monitors

#### Consumer electronics

....Ambient intelligence  
 ....Audio systems  
 .....Audio-visual systems  
 .....Auditory displays  
 .....Headphones

.....Loudspeakers  
 .....Microphones  
 .....Microphone arrays  
 .....Portable media players  
 .....Sonification  
 ....Home automation  
 .....Portable media players  
 .....Refrigerators  
 .....Smart homes  
 .....Washing machines  
 ....Home computing  
 ....Low-power electronics  
 ....Microwave ovens  
 ....Multimedia systems  
 .....Multimedia communication  
 .....Multimedia computing  
 .....Multimedia databases

#### Control systems

....Automatic control  
 .....Power generation control  
 ....Automatic generation control  
 ....Bidirectional control  
 ....CAMAC  
 ....Centralized control  
 ....Closed loop systems  
 ....Control design  
 ....Control engineering  
 ....Control equipment  
 .....Actuators  
 .....Electrostatic actuators  
 .....Hydraulic actuators  
 .....Intelligent actuators  
 .....Microactuators  
 .....Piezoelectric actuators  
 .....Pneumatic actuators  
 .....Fasteners  
 .....Microcontrollers  
 .....Regulators  
 .....Servosystems  
 .....Servomotors  
 .....Switches  
 .....Contactors  
 .....Microswitches  
 .....Optical switches  
 .....Switchgear  
 .....Circuit breakers  
 .....Interrupters  
 .....Relays  
 .....Telecontrol equipment



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.....Thermostats  
 ....Controllability  
 ....Control system synthesis  
 ....Decentralized control  
 .....Distributed parameter systems  
 ....Delay systems  
 .....Added delay  
 .....Delay lines  
 ....Digital control  
 .....Programmable control  
 .....Flow graphs  
 ....Feedback  
 .....Feedback circuits  
 .....Output feedback  
 .....Negative feedback  
 .....Neurofeedback  
 ....Fluid flow control  
 ....Fluidics  
 .....Microfluidics  
 .....Nanofluidics  
 ....Linear feedback control systems  
 .....Frequency locked loops  
 .....Phase locked loops  
 .....State feedback  
 .....Tracking loops  
 ....Magnetic variables control  
 ....Mechanical variables control  
 .....Displacement control  
 .....Force control  
 .....Level control  
 .....Gyroscopes  
 .....Motion control  
 .....Collision avoidance  
 .....Collision mitigation  
 .....Kinetic theory  
 .....Motion planning  
 .....Path planning  
 .....Visual servoing  
 .....Position control  
 .....Nanopositioning  
 .....Shape control  
 .....Size control  
 .....Strain control  
 .....Stress control  
 .....Thickness control  
 .....Torque control  
 .....Velocity control  
 .....Angular velocity control  
 .....Vibration control  
 .....Weight control  
 ....Medical control systems

....Moisture control  
 .....Humidity control  
 ....Motion compensation  
 ....Networked control systems  
 ....Nonlinear control systems  
 ....Open loop systems  
 ....Optical control  
 .....Lighting control  
 .....Optical variables control  
 ....Optimal control  
 .....Bang-bang control  
 .....Infinite horizon  
 ....PD control  
 ....Pi control  
 ....Pneumatic systems  
 ....Pressure control  
 ....Proportional control  
 ....Radio control  
 ....Robot control  
 .....Robot motion  
 ....SCADA systems  
 ....Sensorless control  
 ....Sliding mode control  
 ....Supervisory control  
 .....SCADA systems  
 ....Thermal variables control  
 .....Temperature control  
 .....Cooling  
 .....Heating  
 .....Thermal analysis  
 .....Thermomechanical processes  
 ....Traffic control  
 .....Queueing analysis  
 .....Vehicle routing

#### **Dielectrics and electrical insulation**

....Dielectrics  
 .....Dielectric constant  
 .....High-K gate dielectrics  
 .....Dielectric devices  
 .....Capacitors  
 .....Ferroelectric devices  
 .....Piezoelectric devices  
 .....Pyroelectric devices  
 .....Dielectric losses  
 .....Dielectric substrates  
 .....Dielectrophoresis  
 .....Electrohydrodynamics  
 .....Electrokinetics  
 .....Electrostriction





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- ....Electric breakdown
  - .....Avalanche breakdown
  - .....Corona
  - .....Dielectric breakdown
    - .....Arc discharges
    - .....Discharges (electric)
    - .....Electrostatic discharges
    - .....Flashover
    - .....Glow discharges
    - .....Partial discharges
    - .....Surface discharges
    - .....Vacuum breakdown
  - .....Sparks
- ....Insulation
  - .....Cable insulation
    - .....Power cable insulation
  - .....Ceramics
    - .....Porcelain
  - .....Gas insulation
    - .....Sulfur hexafluoride
  - .....Insulators
    - .....Metal-insulator structures
    - .....Plastic insulators
    - .....Rubber
    - .....Topological insulators
  - .....Trees - insulation
  - .....Isolation technology
  - .....Oil insulation
    - .....Oil filled cables
  - .....Plastic insulation

#### Education

- ....Computer science education
- ....Continuing education
- ....Education courses
- ....Educational institutions
- ....Educational technology
  - .....Computer aided instruction
  - .....Courseware
  - .....Electronic learning
- ....Engineering education
  - .....Biomedical engineering education
  - .....Communication engineering education
  - .....Control engineering education
  - .....Electrical engineering education
  - .....Electronics engineering education
  - .....Engineering students
  - .....Power engineering education
  - .....Student experiments
  - .....Systems engineering education

- ....Physics education
- ....Power engineering education
- ....Qualifications
- ....Training
  - .....Industrial training
  - .....Management training
  - .....On the job training
  - .....Vocational training

#### Electromagnetic compatibility and interference

- ....Electromagnetic compatibility
  - .....Immunity testing
  - .....Reverberation chambers
- ....Electromagnetics
  - .....Electromagnetic analysis
    - .....Air gaps
    - .....Computational electromagnetics
    - .....Delay effects
    - .....Electromagnetic fields
    - .....Electromagnetic forces
    - .....Electromagnetic refraction
    - .....Permeability
    - .....Spark gaps
    - .....Time-domain analysis
  - .....Electromagnetic coupling
    - .....Mutual coupling
    - .....Optical coupling
  - .....Electromagnetic devices
  - .....Electromagnetic induction
    - .....Eddy currents
  - .....Inductive power transmission
  - .....Electromagnetic metamaterials
  - .....Electromagnetic radiation
    - .....Correlators
    - .....Electromagnetic wave absorption
    - .....Frequency
    - .....Gamma-rays
    - .....Line-of-sight propagation
  - .....Electromagnetic shielding
    - .....Cable shielding
    - .....Magnetic shielding
  - .....Electromagnetic transients
    - .....EMP radiation effects
    - .....EMTDC
    - .....EMTP
    - .....Power system transients
    - .....Surges
    - .....Proximity effects
  - .....Interference



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.....Clutter  
 .....Crosstalk  
 .....Diffraction  
 .....Echo interference  
 .....Electromagnetic interference  
 .....Radiofrequency interference  
 .....Specific absorption rate  
 .....Electromagnetic radiative interference  
 .....Electrostatic interference  
 .....Immunity testing  
 .....Interchannel interference  
 .....Interference cancellation  
 .....Interference channels  
 .....Interference constraints  
 .....Interference elimination  
 .....Interference suppression  
 .....Intersymbol interference  
 .....Rain fading  
 .....Terrain factors  
 .....TV interference

**Electron devices**

....Cathode ray tubes  
 ....Electron guns  
 ....Electron multipliers  
 ....Electron tubes  
 .....Field emitter arrays  
 .....Klystrons  
 .....Magnetrons  
 .....Thyratrons  
 ....Mechatronics  
 .....Biomechatronics  
 ....Microelectromechanical systems  
 .....Microelectromechanical devices  
 .....Microactuators  
 .....Micromotors  
 .....Micropumps  
 .....Microvalves  
 .....Radiofrequency  
 microelectromechanical systems  
 ....Microfluidics  
 ....Micromechanical devices  
 .....Biomedical microelectromechanical  
 systems  
 .....Fluidic microsystems  
 .....Microfabrication  
 ....Photoelectricity  
 .....Photovoltaic effects  
 .....Shunts (electrical)  
 ....Photovoltaic cells

.....Light trapping  
 ....Quantum computing  
 .....Quantum cellular automata  
 ....Quantum well devices  
 .....Quantum well lasers  
 .....Quantum cascade lasers  
 .....Quantum wells  
 .....Two dimensional hole gas  
 ....Semiconductivity  
 ....Semiconductor devices  
 .....Flip-chip devices  
 .....Gunn devices  
 .....Hall effect devices  
 .....Junctions  
 .....Heterojunctions  
 .....Hybrid junctions  
 .....P-n junctions  
 .....Waveguide junctions  
 .....MIS devices  
 .....Charge coupled devices  
 .....MOS devices  
 .....MONOS devices  
 .....Piezoresistive devices  
 .....P-i-n diodes  
 .....Power semiconductor devices  
 .....Power transistors  
 .....Power semiconductor switches  
 .....Bipolar transistors  
 .....Thyristors  
 .....Quantum dots  
 .....Quantum well lasers  
 .....Quantum cascade lasers  
 .....Schottky diodes  
 .....Semiconductor counters  
 .....Semiconductor detectors  
 .....Semiconductor device modeling  
 .....Semiconductor device noise  
 .....Semiconductor diodes  
 .....P-i-n diodes  
 .....Schottky diodes  
 .....Semiconductor-metal interfaces  
 .....Superluminescent diodes  
 .....Varactors  
 .....Semiconductor-insulator interfaces  
 .....Semiconductor lasers  
 .....Laser tuning  
 .....Quantum dot lasers  
 .....Quantum well lasers  
 .....Semiconductor laser arrays  
 .....Semiconductor optical amplifiers  
 .....Surface emitting lasers







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.....Semiconductor waveguides  
 .....Silicon devices  
 .....SONOS devices  
 .....Superluminescent diodes  
 .....Surface emitting lasers  
 .....Vertical cavity surface emitting lasers  
 .....Thermistors  
 .....Transistors  
 .....Field effect transistors  
 .....Heterojunction bipolar transistors  
 .....Millimeter wave transistors  
 .....Phototransistors  
 ....Single electron devices  
 ....Single electron memory  
 .....Hetero-nanocrystal memory  
 ....Single electron transistors  
 ....Thick film devices  
 .....Thick film inductors  
 ....Thin film devices  
 .....Film bulk acoustic resonators  
 .....Thin film inductors  
 .....Thin film transistors  
 .....Organic thin film transistors  
 ....Tunneling  
 .....Gate leakage  
 .....Josephson effect  
 .....Magnetic tunneling  
 .....Resonant tunneling devices  
 .....Tunneling magnetoresistance  
 ....Vacuum technology  
 .....Photomultipliers  
 .....Vacuum systems  
 .....Gettering

#### Electronic design automation and methodology

....Design automation  
 .....CAD/CAM  
 .....Logic design  
 .....Reconfigurable logic  
 .....PSCAD  
 ....Design methodology  
 .....Design for disassembly  
 .....Design for experiments  
 .....Design for manufacture  
 .....Design for quality  
 .....Design for testability  
 .....Graphics  
 .....Animation

.....Art  
 .....Character generation  
 .....Computer graphics  
 .....Engineering drawings  
 .....Layout  
 .....Shape  
 .....Symbols  
 .....Virtual reality  
 .....Visualization  
 .....Green design  
 .....Ecodesign  
 .....Green computing  
 .....Process design  
 .....Pattern formation  
 .....Product design  
 .....Prototypes  
 .....Technical drawing  
 .....Time to market  
 .....User centered design  
 .....Virtual prototyping

#### Engineering - general

....Acoustical engineering  
 ....Agricultural engineering  
 ....Chemical engineering  
 ....Civil engineering  
 .....Railway engineering  
 .....Railway safety  
 .....Structural engineering  
 .....Offshore installations  
 ....Concurrent engineering  
 ....Design engineering  
 ....Electrical engineering  
 .....Electrical engineering computing  
 ....Engineering profession  
 ....Maintenance engineering  
 .....Predictive maintenance  
 .....Preventive maintenance  
 .....Condition monitoring  
 ....Mechanical engineering  
 .....Mechanical power transmission  
 .....Torque converters  
 .....Mechanical systems  
 .....Mechanical energy  
 .....Micromechanical devices  
 ....Precision engineering  
 ....Production engineering  
 .....Production planning  
 .....Capacity planning  
 .....Materials requirements planning



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.....Process planning  
 ....Research and development  
 ....Reverse engineering  
 ....Sanitary engineering  
 ....Standardization  
 .....Formal specifications  
 .....Guidelines  
 .....Standards  
 .....ANSI standards  
 .....Code standards  
 .....Communication standards  
 .....IEC standards  
 .....IEEE standards  
 .....ISO standards  
 .....Measurement standards  
 .....Military standards  
 .....Software standards  
 .....Standards activities board  
 .....Standards organizations  
 .....Telecommunication standards  
 .....Universal Serial Bus  
 ....Thermal engineering

#### Engineering in medicine and biology

....Bioinformatics  
 ....Biology  
 .....Biochemistry  
 .....Amino acids  
 .....Biochemical analysis  
 .....Peptides  
 .....Proteins  
 .....Biodiversity  
 .....Biogeography  
 .....Bioelectric phenomena  
 .....Electric shock  
 .....Biological cells  
 .....Cells (biology)  
 .....Chromosome mapping  
 .....Fibroblasts  
 .....RNA  
 .....Stem cells  
 .....Biological information theory  
 .....Biological processes  
 .....Biological interactions  
 .....Chronobiology  
 .....Circadian rhythm  
 .....Coagulation  
 .....Symbiosis  
 .....Biological system modeling  
 .....Biological systems

.....Anatomy  
 .....Molecular communication  
 .....Organisms  
 .....Biology computing  
 .....Biophotonics  
 .....Biophysics  
 .....Aerospace biophysics  
 .....Biomagnetics  
 .....Cellular biophysics  
 .....Molecular biophysics  
 .....Evolution (biology)  
 .....Memetics  
 .....Phylogeny  
 .....Genetics  
 .....DNA  
 .....Gene therapy  
 .....Genetic communication  
 .....Genetic expression  
 .....Genetic programming  
 .....Genomics  
 .....Microinjection  
 .....Nanobioscience  
 .....DNA computing  
 .....Nanobiotechnology  
 .....Physiology  
 .....Predator prey systems  
 .....Synthetic biology  
 .....Systematics  
 .....Systems biology  
 .....Vegetation  
 .....Crops  
 .....Marine vegetation  
 .....Zoology  
 .....Animals  
 ....Biomedical communication  
 ....Biomedical telemetry  
 ....Telemedicine  
 ....Biomedical computing  
 ....Biomedical informatics  
 ....Medical expert systems  
 ....Medical information systems  
 ....Electronic medical records  
 ....Biomedical engineering  
 ....Bioimpedance  
 ....Biological techniques  
 ....Biomedical applications of radiation  
 ....Biomedical electronics  
 ....Biomedical signal processing  
 ....Biomedical image processing  
 ....Biotechnology  
 ....Cloning





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|                                     |                                    |
|-------------------------------------|------------------------------------|
| .....Drug delivery                  | .....Computational biochemistry    |
| .....Targeted drug delivery         | .....Computational biophysics      |
| .....Neural engineering             | .....Computational systems biology |
| .....Neural microtechnology         | ....Genetic engineering            |
| .....Neural nanotechnology          | ....Medical services               |
| .....Neural prosthesis              | .....Assisted living               |
| .....Protein engineering            | .....Catheterization               |
| .....Tissue engineering             | .....Clinical diagnosis            |
| .....Regeneration engineering       | .....Cybercare                     |
| ....Biomedical equipment            | .....Health information management |
| .....Assistive technology           | .....Hospitals                     |
| .....Assistive devices              | .....In vitro                      |
| .....Wheelchairs                    | .....In vitro fertilization        |
| .....Biomedical electrodes          | .....In vivo                       |
| .....Biomedical telemetry           | .....Medical conditions            |
| .....Biomedical transducers         | .....Aneurysm                      |
| .....Catheters                      | .....Arteriosclerosis              |
| .....Cybercare                      | .....Arthritis                     |
| .....Endoscopes                     | .....Atrophy                       |
| .....Gerontechnology                | .....Blindness                     |
| .....Hypodermic needles             | .....Cancer                        |
| .....Implantable biomedical devices | .....Deafness                      |
| .....Implants                       | .....Diabetes                      |
| .....Auditory implants              | .....Diseases                      |
| .....Brainstem implants             | .....Epilepsy                      |
| .....Cochlear implants              | .....Hemorrhaging                  |
| .....Microelectronic implants       | .....Hypertension                  |
| .....Intracranial pressure sensors  | .....Hyperthermia                  |
| .....Lithotriptors                  | .....Influenza                     |
| .....Pacemakers                     | .....Injuries                      |
| .....Stethoscope                    | .....Pregnancy                     |
| .....Surgical instruments           | .....Retinopathy                   |
| .....Laparoscopes                   | .....Sleep apnea                   |
| ....Biomedical imaging              | .....Thrombosis                    |
| .....Angiocardiology                | .....Tumors                        |
| .....Angiography                    | .....Medical diagnosis             |
| .....Biomedical optical imaging     | .....Autopsy                       |
| .....Cardiology                     | .....Bronchoscopy                  |
| .....Echocardiography               | .....Colonography                  |
| .....Electrocardiography            | .....Computer aided diagnosis      |
| .....Phonocardiography              | .....Medical signal detection      |
| .....DICOM                          | .....Nanomedicine                  |
| .....Encephalography                | .....Plethysmography               |
| .....Mammography                    | .....Sensitivity and specificity   |
| .....Medical diagnostic imaging     | .....Medical tests                 |
| .....Anatomical structure           | .....Amniocentesis                 |
| .....Molecular imaging              | .....Biopsy                        |
| .....Phantoms                       | .....Cancer detection              |
| ....Bionanotechnology               | .....Colonoscopy                   |
| ....Bioterrorism                    | .....Pregnancy test                |
| ....Computational biology           | .....Medical treatment             |



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.....Anesthesia  
 .....Angioplasty  
 .....Brachytherapy  
 .....Brain stimulation  
 .....Cardiology  
 .....Chemotherapy  
 .....Clinical trials  
 .....Defibrillation  
 .....Dentistry  
 .....Electrical stimulation  
 .....Electronic medical prescriptions  
 .....Embolization  
 .....Fibrillation  
 .....Gastroenterology  
 .....Gerontology  
 .....Gynecology  
 .....Hepatectomy  
 .....Hospitals  
 .....Hyperthermia  
 .....Lithotripsy  
 .....Magnetic stimulation  
 .....Neonatology  
 .....Neuromuscular stimulation  
 .....Neutron capture therapy  
 .....Noninvasive treatment  
 .....Oncology  
 .....Orthopedic procedures  
 .....Orthotics  
 .....Pathology  
 .....Patient rehabilitation  
 .....Pediatrics  
 .....Pharmaceuticals  
 .....Surgery  
 .....Occupational medicine  
 .....Prosthetics  
 .....Artificial biological organs  
 .....Artificial limbs  
 .....Prosthetic hand  
 .....Prosthetic limbs  
 .....Visual prosthesis  
 .....Public healthcare  
 .....Sensory aids  
 .....Hearing aids  
 .....Vaccines  
 .....X-rays  
 .....X-ray applications  
 .....X-ray detection  
 .....X-ray scattering  
 .....X-ray tomography  
 .....Nuclear medicine  
 .....Synthetic biology

**Engineering management**

....Business  
 .....Business data processing  
 .....Industrial relations  
 .....Management  
 .....Asset management  
 .....Best practices  
 .....Business continuity  
 .....Business process re-engineering  
 .....Communication system operations  
 and management  
 .....Content management  
 .....Contingency management  
 .....Contracts  
 .....Customer relationship management  
 .....Decision making  
 .....Enterprise resource planning  
 .....Facilities management  
 .....Financial management  
 .....Governmental factors  
 .....Human resource management  
 .....Information management  
 .....International collaboration  
 .....Knowledge management  
 .....Marketing management  
 .....Organizational aspects  
 .....Outsourcing  
 .....Process planning  
 .....Production management  
 .....Project management  
 .....Public relations  
 .....Quality management  
 .....Research and development  
 management  
 .....Resource management  
 .....Risk analysis  
 .....Storage management  
 .....Supply chain management  
 .....Operations research  
 .....Inventory control  
 .....Virtual enterprises  
 .....Organizations  
 .....BNSC  
 .....Companies  
 .....Government  
 .....Sociotechnical systems  
 ....Commercialization  
 .....Economics  
 .....Costs  
 .....Cost benefit analysis  
 .....Econometrics



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.....Economic forecasting  
 .....Economic indicators  
 .....Share prices  
 .....Electronic commerce  
 .....Environmental economics  
 .....Carbon tax  
 .....Exchange rates  
 .....Fuel economy  
 .....International trade  
 .....Macroeconomics  
 .....Privatization  
 .....Microeconomics  
 .....Economies of scale  
 .....Industrial economics  
 .....Monopoly  
 .....Oligopoly  
 .....Power generation economics  
 .....Electricity supply industry  
 deregulation  
 .....Profitability  
 .....Stock markets  
 .....Supply and demand  
 .....Trade agreements  
 .....Venture capital  
 .....Virtual enterprises  
 ....Innovation management  
 ....Legal factors  
 .....Copyright protection  
 .....Software protection  
 .....Law  
 .....Censorship  
 .....Commercial law  
 .....Consumer protection  
 .....Contract law  
 .....Criminal law  
 .....Employment law  
 .....Forensics  
 .....Law enforcement  
 .....Patent law  
 .....Trademarks  
 .....Law enforcement  
 .....Patents  
 .....Product liability  
 .....Warranties  
 .....Software protection  
 .....Trademarks  
 ....Market research  
 ....Product development  
 .....Graphical user interfaces  
 .....Avatars  
 .....Product customization

.....Product life cycle management  
 .....Prognostics and health  
 management  
 .....Time to market  
 ....Project engineering  
 .....Scheduling  
 .....Adaptive scheduling  
 .....Dynamic scheduling  
 .....Job shop scheduling  
 .....Single machine scheduling  
 ....Research and development  
 management  
 .....Innovation management  
 ....Research initiatives  
 ....Software development management  
 .....Agile software development  
 .....Scrum (Software development)  
 ....Technology management

#### **Geoscience and remote sensing**

....Environmental factors  
 .....Biosphere  
 .....Ecosystems  
 .....Environmental economics  
 .....Carbon tax  
 .....Environmental monitoring  
 .....Global warming  
 .....Green products  
 .....Green buildings  
 .....Green cleaning  
 .....Pollution  
 .....Air pollution  
 .....Industrial pollution  
 .....Land pollution  
 .....Oil pollution  
 .....Radioactive pollution  
 .....Thermal pollution  
 .....Urban pollution  
 .....Water pollution  
 ....Geographic information systems  
 .....Geospatial analysis  
 ....Gunshot detection systems  
 ....Geophysical measurements  
 .....Geodesy  
 .....Level measurement  
 .....Sea measurements  
 .....Geoacoustic inversion  
 .....Seismic measurements  
 ....Geophysical measurement techniques  
 ....Geophysical signal processing



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|                                 |                                            |
|---------------------------------|--------------------------------------------|
| ....Geoscience                  | .....Ocean salinity                        |
| .....Antarctica                 | .....Ocean temperature                     |
| .....South Pole                 | .....Sea coast                             |
| .....Arctic                     | .....Sea floor                             |
| .....North Pole                 | .....Sea level                             |
| .....Atmosphere                 | .....Sea surface                           |
| .....Atmospheric modeling       | .....Tides                                 |
| .....Atmospheric waves          | .....Rivers                                |
| .....Biosphere                  | .....Sediments                             |
| .....Continents                 | .....Soil                                  |
| .....Africa                     | .....Soil moisture                         |
| .....Asia                       | .....Soil properties                       |
| .....Australia                  | .....Soil texture                          |
| .....Europe                     | .....Tornadoes                             |
| .....North America              | .....Tsunami                               |
| .....South America              | .....Volcanoes                             |
| .....Cyclones                   | .....Planetary volcanoes                   |
| .....Hurricanes                 | .....Volcanic activity                     |
| .....Tropical cyclones          | .....Volcanic ash                          |
| .....Earth                      | ....Land surface temperature               |
| .....Earthquakes                | ....Photometry                             |
| .....Earthquake engineering     | ....Radar                                  |
| .....Forestry                   | .....Airborne radar                        |
| .....Geoengineering             | .....Bistatic radar                        |
| .....Geography                  | .....Doppler radar                         |
| .....Cities and towns           | .....Ground penetrating radar              |
| .....Rural areas                | .....Laser radar                           |
| .....Urban areas                | .....Meteorological radar                  |
| .....Geology                    | .....Millimeter wave radar                 |
| .....Minerals                   | .....Multistatic radar                     |
| .....Rocks                      | .....MIMO radar                            |
| .....Geophysics                 | .....Passive radar                         |
| .....EMTDC                      | .....Radar applications                    |
| .....Extraterrestrial phenomena | .....Radar countermeasures                 |
| .....Geodynamics                | .....Radar detection                       |
| .....Geophysics computing       | .....Radar imaging                         |
| .....Meteorology                | .....Radar measurements                    |
| .....Moisture                   | .....Radar polarimetry                     |
| .....Seismology                 | .....Radar remote sensing                  |
| .....Surface waves              | .....Radar tracking                        |
| .....Well logging               | .....Radar clutter                         |
| .....Ice                        | .....Radar cross-sections                  |
| .....Ice shelf                  | .....Radar equipment                       |
| .....Ice surface                | .....Radar theory                          |
| .....Ice thickness              | .....Spaceborne radar                      |
| .....Sea ice                    | .....Spread spectrum radar                 |
| .....Lakes                      | .....Synthetic aperture radar              |
| .....Land surface               | .....Inverse synthetic aperture radar      |
| .....Levee                      | .....Polarimetric synthetic aperture radar |
| .....Meteorological factors     | .....Ultra wideband radar                  |
| .....Oceans                     |                                            |





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....Radiometry  
 .....Microwave radiometry  
 .....Radiometers  
 .....Spectroradiometers  
 ....Remote sensing  
 .....Hyperspectral sensors  
 .....Hyperspectral imaging  
 .....Passive microwave remote sensing  
 .....Remote monitoring  
 ....Terrain mapping  
 .....Digital elevation models  
 ....Terrestrial atmosphere  
 .....Clouds  
 .....Global warming  
 .....Ionosphere  
 .....Magnetosphere  
 ....Vegetation mapping

#### IEEE organizational topics

....IEEE activities  
 .....Awards activities  
 .....Corporate recognition awards  
 .....External awards  
 .....Honorary membership  
 .....Medals  
 .....Prize paper awards  
 .....Scholarships  
 .....Service awards  
 .....Student awards  
 .....Technical field awards  
 .....Conferences  
 .....Corporate activities  
 .....Calendars  
 .....Ethics  
 .....Finance  
 .....Legislation  
 .....Meetings  
 .....Member relations  
 .....Membership development  
 .....Motion-planning  
 .....Planning  
 .....Public relations  
 .....Strategic planning  
 .....Technology planning  
 .....Educational activities  
 .....Accreditation  
 .....Career development  
 .....Continuing education  
 .....Curriculum development  
 .....Educational programs

.....Scholarships  
 .....Intersociety activities  
 .....Local activities  
 .....Member and Geographic Activities  
 .....Conferences  
 .....Meetings  
 .....Nominations and elections  
 .....Organizing  
 .....Professional activities  
 .....Career development  
 .....Certification  
 .....Consortia  
 .....Continuing education  
 .....Employment  
 .....Ethics  
 .....Intellectual property  
 .....Legislation  
 .....Meetings  
 .....Professional aspects  
 .....Public policy  
 .....Publishing activities  
 .....Books  
 .....CD-ROMs  
 .....Conference proceedings  
 .....Indexes  
 .....Standards publication  
 .....Standards activities  
 .....Standards development  
 .....Standards publication  
 .....Student activities  
 .....Technical activities  
 .....Conferences  
 .....Meetings  
 .....Technical Activities Guide - TAG  
 .....United States activities  
 .....Career development  
 .....Continuing education  
 .....Employment  
 .....Ethics  
 .....Intellectual property  
 .....Legislation  
 .....PACE network  
 .....Public policy  
 .....Volunteer activities  
 .....Audit Committee  
 .....Board of Directors Awards Board  
 Committee  
 .....Credentials Committee  
 .....Ethics Committee  
 .....Executive Committee  
 .....Fellow Committee



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|                                                       |                                                                       |
|-------------------------------------------------------|-----------------------------------------------------------------------|
| .....Life Members Committee                           | .....Operations Council                                               |
| .....Member Conduct Committee                         | .....Outreach Council                                                 |
| .....Nominations and elections                        | .....Professional Activities Council                                  |
| .....Strategic Planning Committee                     | .....Systems Council                                                  |
| .....Tellers Committee                                | .....Technical Councils                                               |
| .....Women in Engineering Committee                   | .....Technical Field Awards Council                                   |
| ....IEEE entities                                     | .....Technology Policy Council                                        |
| .....Boards                                           | .....IEEE Computer Society Press                                      |
| .....Board of Directors                               | .....IEEE Foundation                                                  |
| .....Educational Activities Board                     | .....IEEE Press                                                       |
| .....IEEE Press Editorial Board                       | .....Regions                                                          |
| .....IEEE Spectrum Editorial Board                    | .....Chapters                                                         |
| .....Member and Geographic Activities Board           | .....Region 1                                                         |
| .....Proceedings Editorial Board                      | .....Region 10                                                        |
| .....Publications Board                               | .....Region 2                                                         |
| .....Standards Board                                  | .....Region 3                                                         |
| .....Technical Activities Board                       | .....Region 4                                                         |
| .....The Institute Editorial Board                    | .....Region 5                                                         |
| .....United States Activities Board                   | .....Region 6                                                         |
| .....Center for the History of Electrical Engineering | .....Region 7                                                         |
| .....History                                          | .....Region 8                                                         |
| .....Chapters                                         | .....Region 9                                                         |
| .....Student Chapters                                 | .....Sections                                                         |
| .....Committees                                       | .....Student Chapters                                                 |
| .....Awards committees                                | .....Sections                                                         |
| .....Board committees                                 | .....Chapters                                                         |
| .....Communities                                      | .....Student Chapters                                                 |
| .....New Technology Connections Portal                | .....Societies                                                        |
| .....Online Communities/Technical Collaboration       | .....IEEE Aerospace and Electronic Systems Society                    |
| .....Standards Working Groups                         | .....IEEE Antennas and Propagation Society                            |
| .....Councils                                         | .....IEEE Broadcast Technology Society                                |
| .....Accreditation Policy Council                     | .....IEEE Circuits and Systems Society                                |
| .....Career Policy Council                            | .....IEEE Communications Society                                      |
| .....Geographic Councils                              | .....IEEE Components, Packaging, and Manufacturing Technology Society |
| .....IEEE Biometrics Council                          | .....IEEE Computational Intelligence Society                          |
| .....IEEE Council on Electronic Design Automation     | .....IEEE Computer Society                                            |
| .....IEEE Council on Superconductivity                | .....IEEE Consumer Electronics Society                                |
| .....IEEE Nanotechnology Council                      | .....IEEE Control Systems Society                                     |
| .....IEEE Sensors Council                             | .....IEEE Dielectrics and Electrical Insulation Society               |
| .....IEEE Systems Council                             | .....IEEE Education Society                                           |
| .....IEEE Technology Management Council               | .....IEEE Electromagnetic Compatibility Society                       |
| .....Lifelong Learning Council                        | .....IEEE Electron Devices Society                                    |
| .....Member Activities Council                        | .....IEEE Engineering in Medicine and Biology Society                 |
| .....Metropolitan Councils                            |                                                                       |
| .....Nanotechnology Council                           |                                                                       |



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|                                                                      |                                                                   |
|----------------------------------------------------------------------|-------------------------------------------------------------------|
| .....IEEE Engineering Management Society                             | .....Student members                                              |
| .....IEEE Geoscience and Remote Sensing Society                      | ....IEEE news                                                     |
| .....IEEE Industrial Electronics Society                             | .....Chapter news                                                 |
| .....IEEE Industry Applications Society                              | .....Region news                                                  |
| .....IEEE Information Theory Society                                 | .....Section news                                                 |
| .....IEEE Instrumentation and Measurement Society                    | .....Society news                                                 |
| .....IEEE Intelligent Transportation Systems Society                 | ....IEEE products                                                 |
| .....IEEE Lasers and Electro-Optics Society                          | .....Audio tapes                                                  |
| .....IEEE Magnetics Society                                          | .....Catalogs                                                     |
| .....IEEE Microwave Theory and Techniques Society                    | .....Educational Activities Product Catalog                       |
| .....IEEE Nuclear and Plasma Sciences Society                        | .....IEEE catalog                                                 |
| .....IEEE Oceanic Engineering Society                                | .....IEEE Electronic catalog                                      |
| .....IEEE Photonics Society                                          | .....IEEE standards catalog                                       |
| .....IEEE Power Electronics Society                                  | .....New products catalog                                         |
| .....IEEE Power & Energy Society                                     | .....Conference proceedings                                       |
| .....IEEE Reliability Society                                        | .....Educational products                                         |
| .....IEEE Robotics and Automation Society                            | .....Reading series                                               |
| .....IEEE Signal Processing Society                                  | .....Self-study courses                                           |
| .....IEEE Society on Social Implications of Technology               | .....Videos                                                       |
| .....IEEE Solid-State Circuits Society                               | .....IEEE standards                                               |
| .....IEEE Systems, Man, and Cybernetics Society                      | .....IEEE 1394 Standard                                           |
| .....IEEE Technology Management Council                              | .....IEEE 802.11 Standards                                        |
| .....IEEE Ultrasonics, Ferroelectrics, and Frequency Control Society | .....IEEE 802.15 Standards                                        |
| .....IEEE Vehicular Technology Society                               | .....IEEE 802.16 Standards                                        |
| .....Student Chapters                                                | .....IEEE 802.3 Standards                                         |
| ....IEEE governance                                                  | .....IEEE Xplore                                                  |
| .....Bylaws                                                          | .....IEL                                                          |
| .....Constitution                                                    | .....Merchandise                                                  |
| .....IEEE Policy and Procedures                                      | .....Reading series                                               |
| .....IEEE Staff                                                      | .....Self-study courses                                           |
| .....Mission and Vision                                              | .....Videos                                                       |
| .....Organization Charts                                             | ....IEEE publications                                             |
| ....IEEE members                                                     | .....IEEE conference proceedings                                  |
| .....Associate members                                               | .....IEEE directories                                             |
| .....Fellows                                                         | .....IEEE Membership Directory                                    |
| .....Joining IEEE                                                    | .....IEEE Staff Directory                                         |
| .....Signup web site                                                 | .....IEEE indexing                                                |
| .....Life members                                                    | .....Awards                                                       |
| .....Senior members                                                  | .....Book reviews                                                 |
|                                                                      | .....CD-ROM reviews                                               |
|                                                                      | .....Editorials                                                   |
|                                                                      | .....Interviews                                                   |
|                                                                      | .....Obituaries                                                   |
|                                                                      | .....Software reviews                                             |
|                                                                      | .....Special issues and sections                                  |
|                                                                      | .....Tutorials                                                    |
|                                                                      | .....Video reviews                                                |
|                                                                      | .....IEEE journals                                                |
|                                                                      | .....IEEE Canadian Journal of Electrical and Computer Engineering |



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|--------------------------------------------------------------------------------------|------------------------------------------------------------|
| .....IEEE Communications Letters                                                     | .....IEEE Annals of the History of Computing               |
| .....IEEE Communications Surveys & Tutorials                                         | .....IEEE Antennas and Propagation Magazine                |
| .....IEEE Computer Architecture Letters                                              | .....IEEE Circuits and Devices                             |
| .....IEEE Electrochemical and Solid-State Letters                                    | .....IEEE Communications Magazine                          |
| .....IEEE Electron Device Letters                                                    | .....IEEE Computational Intelligence                       |
| .....IEEE Embedded Systems Letters                                                   | .....IEEE Computational Science and Engineering            |
| .....IEEE Journal of Microelectromechanical Systems                                  | .....IEEE Computer Applications in Power                   |
| .....IEEE Journal of Oceanic Engineering                                             | .....IEEE Computer Graphics and Applications               |
| .....IEEE Journal of Quantum Electronics                                             | .....IEEE Computer Magazine                                |
| .....IEEE Journal of Robotics and Automation                                         | .....IEEE Concurrency                                      |
| .....IEEE Journal of Selected Topics in Applied Earth Observation and Remote Sensing | .....IEEE Control Systems                                  |
| .....IEEE Journal of Selected Topics in Quantum Electronics                          | .....IEEE Design and Test of Computers                     |
| .....IEEE Journal of Selected Topics in Signal Processing                            | .....IEEE Electrical Insulation Magazine                   |
| .....IEEE Journal of Solid-State Circuits                                            | .....IEEE Engineering in Medicine and Biology Magazine     |
| .....IEEE Journal of Technology                                                      | .....IEEE Engineering Management Review                    |
| .....IEEE Journal of Technology Computer Aided Design                                | .....IEEE Industrial Electronics Magazine                  |
| .....IEEE Journal on Selected Areas in Communications                                | .....IEEE Industry Applications Magazine                   |
| .....IEEE Latin America Learning Technologies Journal [IEEE-RITA]                    | .....IEEE Instrumentation and Measurement Magazine         |
| .....IEEE Learning Technology                                                        | .....IEEE Intelligent Systems and their Applications       |
| .....IEEE Magnetism Letters                                                          | .....IEEE Intelligent Transportation Systems Magazine      |
| .....IEEE Microwave and Guided Wave Letters                                          | .....IEEE Internet Computing                               |
| .....IEEE/OSA Journal of Display Technology                                          | .....IEEE Micro                                            |
| .....IEEE/OSA Journal of Lightwave Technology                                        | .....IEEE Multidisciplinary Engineering Education Magazine |
| .....IEEE/OSA Journal of Optical Communications and Networking                       | .....IEEE Multimedia                                       |
| .....IEEE Photonics Journal                                                          | .....IEEE Nanotechnology Magazine                          |
| .....IEEE Photonics Technology Letters                                               | .....IEEE Network                                          |
| .....IEEE Reviews in Biomedical Engineering                                          | .....IEEE Personal Communications                          |
| .....IEEE Signal Processing Letters                                                  | .....IEEE Potentials                                       |
| .....IEEE Systems Journal                                                            | .....IEEE Power Engineering Review                         |
| .....Proceedings of the IEEE                                                         | .....IEEE Robotics and Automation Magazine                 |
| .....IEEE magazines                                                                  | .....IEEE Signal Processing Magazine                       |
| .....IEEE Aerospace and Electronics Society Magazine                                 | .....IEEE Software                                         |
|                                                                                      | .....IEEE Solid-State Circuits Magazine                    |
|                                                                                      | .....IEEE Spectrum                                         |
|                                                                                      | .....IEEE Technology and Society Magazine                  |



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- .....IEEE-USA Today's Engineer  
 .....IEEE newsletters  
 .....Broadcast Technology Society  
 Newsletter  
 .....Center for the History of Electrical  
 Engineering Newsletter  
 .....Circuits and Systems Society  
 Newsletter  
 .....Components, Packaging, and  
 Manufacturing Technology Society  
 Newsletter  
 .....Consumer Electronics Society  
 Newsletter  
 .....Education Society Newsletter  
 .....Electromagnetic Compatibility  
 Society Newsletter  
 .....Electron Devices Society  
 Newsletter  
 .....Electronics and the Environment  
 Newsletter  
 .....Engineering Management Society  
 Newsletter  
 .....Geoscience and Remote Sensing  
 Society Newsletter  
 .....IEEE Circuitboard  
 .....IEEE Looking Forward  
 .....IEEE Publications Bulletin  
 .....Industrial Electronics Society  
 Newsletter  
 .....Information Theory Society  
 Newsletter  
 .....Instrumentation and Measurement  
 Society Newsletter  
 .....Lasers and Electro-Optics Society  
 Newsletter  
 .....Magnetics Society Newsletter  
 .....Microwave Theory and Techniques  
 Society Newsletter  
 .....Nuclear and Plasma Sciences  
 Society Newsletter  
 .....Oceanic Engineering Society  
 Newsletter  
 .....Power Electronics Society  
 Newsletter  
 .....Professional Communication  
 Society Newsletter  
 .....Reliability Society Newsletter  
 .....Systems, Man and Cybernetics  
 Society Newsletter  
 .....The Institute  
 .....The Staff Circuit  
 .....Ultrasonics, Ferroelectrics, and  
 Frequency Control Society Newsletter  
 .....Vehicular Technology Society  
 Newsletter  
 .....IEEE online publications  
 .....IEEE Bibliographies On-line  
 .....IEEE Circuitboard  
 .....IEEE Communications Interactive  
 .....IEEE Communications Surveys &  
 Tutorials  
 .....IEEE Distributed Systems Online  
 .....IEEE Electrochemical and Solid-  
 State Letters  
 .....IEEE Electronic catalog  
 .....IEEE Journal of Technology  
 Computer Aided Design  
 .....IEEE Journals and Transactions  
 On-LINE - OpeRA  
 .....IEEE Latin America Learning  
 Technologies Journal [IEEE-RITA]  
 .....IEEE Latin America Transactions  
 [Revista IEEE America Latina]  
 .....IEEE Learning Technology  
 .....IEEE Looking Forward  
 .....IEEE Multidisciplinary Engineering  
 Education Magazine  
 .....IEEE Network Interactive  
 .....IEEE Personal Communications  
 Interactive  
 .....IEEE Photonics Journal  
 .....IEEE Transactions on  
 Computational Intelligence and AI in  
 Games  
 .....IEEE Transactions on Learning  
 Technologies  
 .....IEEE Transactions on Network and  
 Service Management  
 .....IEEE Transactions on Services  
 Computing  
 .....IEEE standard glossaries  
 .....IEEE transactions  
 .....IEEE/ACM Transactions on  
 Networking  
 .....IEEE Biometrics Compendium  
 .....IEEE Latin America Transactions  
 [Revista IEEE America Latina]  
 .....IEEE Transactions on Aerospace  
 and Electronic Systems  
 .....IEEE Transactions on Affective  
 Computing





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|                                                                                         |                                                                     |
|-----------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| .....IEEE Transactions on Antennas and Propagation                                      | .....IEEE Transactions on Electromagnetic Compatibility             |
| .....IEEE Transactions on Applied Superconductivity                                     | .....IEEE Transactions on Electron Devices                          |
| .....IEEE Transactions on Audio, Speech, and Language Processing                        | .....IEEE Transactions on Energy Conversion                         |
| .....IEEE Transactions on Automatic Control                                             | .....IEEE Transactions on Engineering Management                    |
| .....IEEE Transactions on Automation Science and Engineering                            | .....IEEE Transactions on Evolutionary Computation                  |
| .....IEEE Transactions on Autonomous Mental Development                                 | .....IEEE Transactions on Fuzzy Systems                             |
| .....IEEE Transactions on Biomedical Circuits and Systems                               | .....IEEE Transactions on Geoscience and Remote Sensing             |
| .....IEEE Transactions on Biomedical Engineering                                        | .....IEEE Transactions on Haptics                                   |
| .....IEEE Transactions on Broadcasting                                                  | .....IEEE Transactions on Image Processing                          |
| .....IEEE Transactions on Circuits and Systems for Video Technology                     | .....IEEE Transactions on Industrial Electronics                    |
| .....IEEE Transactions on Circuits and Systems I: Fundamental Theory and Applications   | .....IEEE Transactions on Industry Applications                     |
| .....IEEE Transactions on Circuits and Systems II: Analog and Digital Signal Processing | .....IEEE Transactions on Information Forensics and Security        |
| .....IEEE Transactions on Communications                                                | .....IEEE Transactions on Information Technology in Biomedicine     |
| .....IEEE Transactions on Components, Packaging, and Manufacturing Technology Part A    | .....IEEE Transactions on Information Theory                        |
| .....IEEE Transactions on Components, Packaging, and Manufacturing Technology Part B    | .....IEEE Transactions on Instrumentation and Measurement           |
| .....IEEE Transactions on Components, Packaging, and Manufacturing Technology Part C    | .....IEEE Transactions on Knowledge and Data Engineering            |
| .....IEEE Transactions on Computational Intelligence and AI in Games                    | .....IEEE Transactions on Learning Technologies                     |
| .....IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems      | .....IEEE Transactions on Magnetism                                 |
| .....IEEE Transactions on Computers                                                     | .....IEEE Transactions on Mechatronics                              |
| .....IEEE Transactions on Consumer Electronics                                          | .....IEEE Transactions on Medical Imaging                           |
| .....IEEE Transactions on Control Systems Technology                                    | .....IEEE Transactions on Microwave Theory and Techniques           |
| .....IEEE Transactions on Dielectrics and Electrical Insulation                         | .....IEEE Transactions on Nanotechnology                            |
| .....IEEE Transactions on Education                                                     | .....IEEE Transactions on Network and Service Management            |
|                                                                                         | .....IEEE Transactions on Neural Networks                           |
|                                                                                         | .....IEEE Transactions on Nuclear Science                           |
|                                                                                         | .....IEEE Transactions on Pattern Analysis and Machine Intelligence |
|                                                                                         | .....IEEE Transactions on Plasma Science                            |



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.....IEEE Transactions on Power  
Delivery  
.....IEEE Transactions on Power  
Electronics  
.....IEEE Transactions on Power  
Systems  
.....IEEE Transactions on Professional  
Communication  
.....IEEE Transactions on Rehabilitation  
Engineering  
.....IEEE Transactions on Reliability  
.....IEEE Transactions on Robotics  
.....IEEE Transactions on Robotics and  
Automation  
.....IEEE Transactions on  
Semiconductor Manufacturing  
.....IEEE Transactions on Services  
Computing  
.....IEEE Transactions on Signal  
Processing  
.....IEEE Transactions on Smart Grid  
.....IEEE Transactions on Software  
Engineering  
.....IEEE Transactions on Speech and  
Audio Processing  
.....IEEE Transactions on Sustainable  
Energy  
.....IEEE Transactions on Systems,  
Man, and Cybernetics Part A: Systems and  
Humans  
.....IEEE Transactions on Systems,  
Man, and Cybernetics Part B: Cybernetics  
.....IEEE Transactions on Systems,  
Man, and Cybernetics Part C: Applications  
and Reviews  
.....IEEE Transactions on Ultrasonics,  
Ferroelectrics and Frequency Control  
.....IEEE Transactions on Vehicular  
Technology  
.....IEEE Transactions on Very Large  
Scale Integration - VLSI  
.....IEEE Transactions on Visualization  
and Computer Graphics  
.....IEEE Women in Engineering  
.....Notice of Violation  
.....IEEE services  
.....Ask IEEE  
.....Conference management  
.....Meeting services  
.....Member services  
.....Career development

.....Electronic mail  
.....Financial advantage program  
.....IEEE Bibliographies On-line  
.....IEEE Electronic catalog  
.....Job listing service  
.....Membership renewal  
.....Travel services  
.....Web and internet services  
.....Subscriptions  
.....Web and internet services  
.....Electronic mail  
.....IEEE Electronic catalog  
.....IEEE Journals and Transactions  
On-LINE - OpeRA  
.....Online banking  
.....IEEE web sites  
.....Society home pages  
.....Web page design

### Imaging

....Biomedical imaging  
.....Angiocardiology  
.....Angiography  
.....Biomedical optical imaging  
.....Cardiology  
.....Echocardiography  
.....Electrocardiography  
.....Phonocardiography  
.....DICOM  
.....Encephalography  
.....Mammography  
.....Medical diagnostic imaging  
.....Anatomical structure  
.....Molecular imaging  
.....Phantoms  
....Cameras  
.....Digital cameras  
.....Webcams  
....Focusing  
....Ground penetrating radar  
....Holography  
....Image converters  
.....Image intensifiers  
....Image sensors  
.....Active pixel sensors  
.....CCD image sensors  
.....Charge-coupled image sensors  
.....CMOS image sensors  
.....Infrared image sensors  
....Image storage



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- ....Infrared imaging
- .....Night vision
- ....Magnetic resonance imaging
- .....Diffusion tensor imaging
- ....Magneto electrical resistivity imaging technique
- ....Microscopy
- .....Atomic force microscopy
- .....Electron microscopy
- .....Photoelectron microscopy
- .....Scanning electron microscopy
- .....Transmission electron microscopy
- .....Scanning probe microscopy
- ....Microwave imaging
- ....Motion pictures
- ....Multispectral imaging
- ....Nuclear imaging
- .....Energy resolution
- ....Optical imaging
- .....Talbot effect
- .....Thermoreflectance imaging
- ....Photography
- .....Cinematography
- .....Digital photography
- .....Image forensics
- .....Photomicrography
- ....Radiation imaging
- ....Radiography
- .....Diagnostic radiography
- ....Stereo vision
- .....Stereo image processing
- ....Tomography
- .....Computed tomography
- .....Electrical capacitance tomography
- .....Positron emission tomography
- .....Whole-body PET
- .....Reconstruction algorithms
- .....Single photon emission computed tomography
- .....Predictive control
- .....Three-term control
- .....Two-term control
- .....Production control
- .....Continuous production
- .....Lot sizing
- .....Optimized production technology
- .....Scheduling
- .....Integrated manufacturing systems
- ....Machine control
- .....Machine vector control
- ....Manufacturing automation
- .....Computer aided manufacturing
- .....CAD/CAM
- .....Silicon compiler
- .....Computer integrated manufacturing
- .....Computer numerical control
- .....Flexible manufacturing systems
- ....Testing
- .....Aerospace testing
- .....Automatic testing
- .....Automatic test pattern generation
- .....Ring generators
- .....Benchmark testing
- .....Built-in self-test
- .....Circuit testing
- .....Integrated circuit measurements
- .....Electronic equipment testing
- .....Immunity testing
- .....Error analysis
- .....Bit error rate
- .....Finite wordlength effects
- .....Error-free operations
- .....Failure analysis
- .....Equipment failure
- .....Semiconductor device breakdown
- .....Frequency response
- .....Impulse testing
- .....Insulator testing
- .....Insulation testing
- .....Integrated circuit testing
- .....Integrated circuit yield
- .....Logic testing
- .....Life testing
- .....Materials testing
- .....Accelerated aging
- .....Acoustic testing
- .....Adhesive strength
- .....Bonding forces
- .....Delamination
- .....Elastic recovery

#### Industrial electronics

- ....Assembly systems
- .....Flexible electronics
- .....Robotic assembly
- ....Computer aided manufacturing
- .....CAD/CAM
- .....Silicon compiler
- ....Cryogenic electronics
- ....Industrial control
- .....Process control



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.....Nondestructive testing  
 .....Optical fiber testing  
 .....Remaining life assessment  
 .....Ring generators  
 .....Semiconductor device testing  
 .....Software testing  
 .....System testing  
 .....Model checking  
 .....Test equipment  
 .....Automatic test equipment  
 .....Test facilities  
 .....Anechoic chambers  
 .....Laboratories  
 .....Large Hadron Collider  
 .....Open area test sites  
 .....TEM cells

#### Industry applications

....Accident prevention  
 .....Accidents  
 .....Aerospace accidents  
 .....Electrical accidents  
 .....Industrial accidents  
 .....Marine accidents  
 .....Railway accidents  
 .....Road accidents  
 ....Chemical technology  
 .....Chemical reactors  
 .....Bioreactors  
 .....Continuous-stirred tank reactor  
 .....Ignition  
 .....Chemical sensors  
 .....Crystallizers  
 .....Distillation equipment  
 .....Fluidization  
 .....Pharmaceutical technology  
 .....Vitrification  
 ....Cryogenics  
 ....Electrochemical devices  
 .....Amperometric sensors  
 .....Batteries  
 .....Lithium batteries  
 .....Battery management systems  
 .....Fuel cells  
 .....Supercapacitors  
 ....Electrochemical processes  
 ....Electromechanical systems  
 .....Electromechanical devices  
 .....Armature  
 .....SAW filters

....Electrostatic devices  
 ....Electrostatic precipitators  
 ....Electrostatic processes  
 .....Aerosols  
 .....Electrophotography  
 .....Electrostatic analysis  
 .....Electrostatic induction  
 .....Electrostatics  
 .....Electrostatic levitation  
 .....Particle charging  
 .....Particle production  
 .....Space charge  
 .....Surface charging  
 .....Triboelectricity  
 .....Triboelectricity  
 ....Engines  
 .....Heat engines  
 .....Steam engines  
 .....Stirling engines  
 .....Internal combustion engines  
 .....Diesel engines  
 .....Ignition  
 .....Jet engines  
 ....Environmental management  
 .....Biodegradation  
 .....Biodegradable materials  
 .....Land use planning  
 .....Pest control  
 .....Pollution control  
 .....Recycling  
 .....Renewable energy sources  
 .....Biomass  
 .....Sustainable development  
 .....Waste management  
 .....Waste disposal  
 .....Waste handling  
 .....Waste recovery  
 .....Waste reduction  
 .....Water conservation  
 .....Desalination  
 .....Water resources  
 .....Desalination  
 .....Reservoirs  
 ....Food technology  
 .....Food preservation  
 ....High-temperature techniques  
 .....Rapid thermal processing  
 ....Industrial engineering  
 .....Industrial communication  
 ....Industries  
 .....Agriculture



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|                                      |                                   |
|--------------------------------------|-----------------------------------|
| .....Agricultural products           | .....Electricity supply industry  |
| .....Aquaculture                     | .....Nuclear facility regulation  |
| .....Fertilizers                     | .....Power system interconnection |
| .....Greenhouses                     | .....Sugar industry               |
| .....Irrigation                      | .....Sugar refining               |
| .....Architecture                    | .....Textile technology           |
| .....Banking                         | .....Spinning                     |
| .....Beverage industry               | .....Weaving                      |
| .....Chemical industry               | .....Toy industry                 |
| .....Coal industry                   | .....Wood industry                |
| .....Communication industry          | .....Inspection                   |
| .....Computer industry               | .....Automatic optical inspection |
| .....Construction                    | .....Machinery                    |
| .....Buildings                       | .....Agricultural machinery       |
| .....Green buildings                 | .....Ball bearings                |
| .....Modular construction            | .....Belts                        |
| .....Prefabricated construction      | .....Drives                       |
| .....Construction industry           | .....Hydraulic drives             |
| .....Prefabricated construction      | .....Motor drives                 |
| .....Defense industry                | .....Variable speed drives        |
| .....Entertainment industry          | .....Electric machines            |
| .....Gas industry                    | .....AC machines                  |
| .....Manufacturing industries        | .....Alternators                  |
| .....Aerospace industry              | .....Brushless machines           |
| .....Cement industry                 | .....Compressors                  |
| .....Ceramics industry               | .....Conductors                   |
| .....Clothing industry               | .....DC machines                  |
| .....Electrical products industry    | .....Electric fences              |
| .....Electronics industry            | .....Generators                   |
| .....Food industry                   | .....Permanent magnet machines    |
| .....Footwear industry               | .....Rotating machines            |
| .....Fuel processing industries      | .....Rotors                       |
| .....Glass industry                  | .....Stators                      |
| .....Machinery production industries | .....Washing machines             |
| .....Metal product industries        | .....Fans                         |
| .....Plastics industry               | .....Furnaces                     |
| .....Pulp and paper industry         | .....Blast furnaces               |
| .....Rubber industry                 | .....Kilns                        |
| .....Shipbuilding industry           | .....Gears                        |
| .....Textile industry                | .....Hydraulic systems            |
| .....Toy manufacturing industry      | .....Electrohydraulics            |
| .....Metals industry                 | .....Hydraulic equipment          |
| .....Mining industry                 | .....Hydraulic fluids             |
| .....Coal mining                     | .....Machine components           |
| .....Natural gas industry            | .....Air cleaners                 |
| .....Petroleum industry              | .....Belts                        |
| .....Oil drilling                    | .....Cams                         |
| .....Oil refineries                  | .....Engine cylinders             |
| .....Well logging                    | .....Exhaust systems              |
| .....Power industry                  | .....Impellers                    |
| .....Electrical equipment industry   | .....Intake systems               |



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|                                  |                                        |
|----------------------------------|----------------------------------------|
| .....Manifolds                   | .....Chemical products                 |
| .....Mechanical splines          | .....Consumer products                 |
| .....Pistons                     | .....Electrical products               |
| .....Rotors                      | .....Food products                     |
| .....Shafts                      | .....Fuels                             |
| .....Valves                      | .....Glass products                    |
| .....Motors                      | .....Mechanical products               |
| .....AC motors                   | .....Metal products                    |
| .....Brushless motors            | .....Paper products                    |
| .....Commutation                 | .....Paper pulp                        |
| .....DC motors                   | .....Plastic products                  |
| .....Electric motors             | .....Rubber products                   |
| .....Hysteresis motors           | .....Sports equipment                  |
| .....Induction motors            | .....Textile products                  |
| .....Micromotors                 | .....Windows                           |
| .....Permanent magnet motors     | .....Manufacturing systems             |
| .....Servomotors                 | .....Agile manufacturing               |
| .....Traction motors             | .....Automobile manufacture            |
| .....Universal motors            | .....Batch production systems          |
| .....Printing machinery          | .....Blanking                          |
| .....Pumps                       | .....Cellular manufacturing            |
| .....Fuel pumps                  | .....Flow production systems           |
| .....Heat pumps                  | .....Food manufacturing                |
| .....Micropumps                  | .....Forging                           |
| .....Textile machinery           | .....Glass manufacturing               |
| .....Spinning machines           | .....Integrated manufacturing systems  |
| .....Manufacturing               | .....Intelligent manufacturing systems |
| .....Assembly                    | .....Job production systems            |
| .....Fitting                     | .....Joining processes                 |
| .....Microassembly               | .....Layered manufacturing             |
| .....Preforms                    | .....Lean production                   |
| .....Soldering                   | .....Manufacturing processes           |
| .....Assembly systems            | .....Mass production                   |
| .....Flexible electronics        | .....Melt processing                   |
| .....Robotic assembly            | .....Pulp manufacturing                |
| .....Embossing                   | .....Sheet metal processing            |
| .....Fabrication                 | .....Thermoforming                     |
| .....Bonding processes           | .....Mass customization                |
| .....Microfabrication            | .....Tolerance analysis                |
| .....Optical device fabrication  | .....Packaging                         |
| .....Soldering                   | .....Bagging                           |
| .....Welding                     | .....Bottling                          |
| .....Lithography                 | .....Canning                           |
| .....Colloidal lithography       | .....Encapsulation                     |
| .....Interferometric lithography | .....Labeling                          |
| .....Nanolithography             | .....Multichip modules                 |
| .....Soft lithography            | .....Plastic packaging                 |
| .....Stereolithography           | .....Wrapping                          |
| .....X-ray lithography           | .....Paper technology                  |
| .....Manufactured products       | .....Production                        |
| .....Ceramic products            | .....Ball milling                      |



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|                                  |                                        |
|----------------------------------|----------------------------------------|
| .....Compression molding         | .....Process planning                  |
| .....Embossing                   | .....Cause effect analysis             |
| .....Food products               | .....Production control                |
| .....Dairy products              | .....Continuous production             |
| .....Fats                        | .....Lot sizing                        |
| .....Sugar                       | .....Optimized production technology   |
| .....Group technology            | .....Scheduling                        |
| .....Injection molding           | .....Production engineering            |
| .....Materials processing        | .....Production planning               |
| .....Annealing                   | .....Production equipment              |
| .....Bleaching                   | .....Applicators                       |
| .....Casting                     | .....Clamps                            |
| .....Coatings                    | .....Cutting tools                     |
| .....Curing                      | .....Fixtures                          |
| .....Etching                     | .....Machine tools                     |
| .....Heat treatment              | .....Mining equipment                  |
| .....Joining processes           | .....Molding equipment                 |
| .....Lamination                  | .....Packaging machines                |
| .....Machining                   | .....Paper making machines             |
| .....Melt processing             | .....Polishing machines                |
| .....Plasma materials processing | .....Soldering equipment               |
| .....Pressing                    | .....Production facilities             |
| .....Punching                    | .....Foundries                         |
| .....Refining                    | .....Greenhouses                       |
| .....Shearing                    | .....Industrial plants                 |
| .....Smelting                    | .....Machine shops                     |
| .....Softening                   | .....Paper mills                       |
| .....Swaging                     | .....Production management             |
| .....Mechanical products         | .....Control charts                    |
| .....Automotive components       | .....Inventory management              |
| .....Axles                       | .....Lead time reduction               |
| .....Bellows                     | .....Logistics                         |
| .....Blades                      | .....Process planning                  |
| .....Couplings                   | .....Production planning               |
| .....Fasteners                   | .....Production materials              |
| .....Flanges                     | .....Abrasives                         |
| .....Gears                       | .....Aerospace materials               |
| .....Hoses                       | .....Automotive materials              |
| .....Machine components          | .....Inhibitors                        |
| .....Mechanical guides           | .....Ink                               |
| .....Needles                     | .....Joining materials                 |
| .....Orifices                    | .....Lubricants                        |
| .....Pistons                     | .....Retardants                        |
| .....Seals                       | .....Production systems                |
| .....Springs                     | .....Assembly systems                  |
| .....Steering systems            | .....Exhaust systems                   |
| .....Structural shapes           | .....Intelligent manufacturing systems |
| .....Suspensions                 | .....Lean production                   |
| .....Tires                       | .....Manufacturing systems             |
| .....Vents                       | .....Steering systems                  |
| .....Wheels                      | .....Productivity                      |



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.....Shafts  
 .....Camshafts  
 .....Springs  
 .....Suspensions  
 .....Shock absorbers  
 .....Transfer molding  
 ....Safety  
 .....Aerospace safety  
 .....Air safety  
 .....Domestic safety  
 .....Emergency services  
 .....Explosion protection  
 .....Hazards  
 .....Biohazards  
 .....Chemical hazards  
 .....Explosions  
 .....Fires  
 .....Flammability  
 .....Floods  
 .....Hazardous areas  
 .....Hazardous materials  
 .....Toxicology  
 .....Health and safety  
 .....Occupational health  
 .....Occupational safety  
 .....Marine safety  
 .....Product safety  
 .....Protection  
 .....Explosion protection  
 .....Lightning protection  
 .....Radiation safety  
 .....Safety devices  
 .....Eye protection  
 .....Protective clothing  
 .....Vehicle safety  
 ....Security  
 .....Access control  
 .....Authorization  
 .....Alarm systems  
 .....Smoke detectors  
 .....Computer security  
 .....Authentication  
 .....Computer crime  
 .....Computer hacking  
 .....Firewalls (computing)  
 .....Identity management systems  
 .....Invasive software  
 .....Permission  
 .....Cryptography  
 .....Ciphers  
 .....Encryption

.....Public key  
 .....Random number generation  
 .....Data security  
 .....Cryptography  
 .....Message authentication  
 .....Digital signatures  
 .....Information security  
 .....Intrusion detection  
 .....Power system security  
 .....Reconnaissance  
 .....Terrorism  
 .....Bioterrorism  
 .....National security  
 .....Watermarking  
 ....Wine industry  
 .....Wineries

#### Information theory

....Audio coding  
 ....Biological information theory  
 ....Channel coding  
 .....Block codes  
 .....Linear codes  
 .....Combined source-channel coding  
 .....Turbo codes  
 ....Codes  
 .....Binary codes  
 .....Reflective binary codes  
 .....Convolutional codes  
 .....Cyclic redundancy check codes  
 .....Error correction codes  
 .....Reed-Solomon codes  
 .....Parity check codes  
 .....Iterative decoding  
 .....Product codes  
 .....Bar codes  
 .....Space-time codes  
 ....Communication channels  
 .....Channel allocation  
 .....Channel capacity  
 .....Channel estimation  
 .....Channel models  
 .....Channel spacing  
 .....Channel state information  
 .....Gaussian channels  
 .....AWGN channels  
 .....Multipath channels  
 .....Multiuser channels  
 .....Partial response channels  
 .....Throughput



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.....Time-varying channels  
 ....Decoding  
 .....Maximum likelihood decoding  
 ....Encoding  
 .....Audio coding  
 .....Channel coding  
 .....Block codes  
 .....Combined source-channel coding  
 .....Turbo codes  
 .....Entropy coding  
 .....Huffman coding  
 .....Source coding  
 .....Speech coding  
 .....Transcoding  
 ....Error compensation  
 ....Genetic communication  
 ....Hamming distance  
 ....Hamming weight  
 ....Information entropy  
 ....Mutual information  
 ....Network coding  
 ....Rate-distortion  
 ....Rate distortion theory  
 .....Channel rate control  
 ....Source coding  
 ....Speech coding

#### Instrumentation and measurement

....Computerized instrumentation  
 ....Electric variables  
 .....Admittance  
 .....Capacitance  
 .....Parasitic capacitance  
 .....Quantum capacitance  
 .....Capacitance-voltage characteristics  
 .....Conductivity  
 .....Photoconductivity  
 .....Semiconductivity  
 .....Transconductance  
 .....Current  
 .....Bioimpedance  
 .....Current slump  
 .....Dark current  
 .....Fault currents  
 .....Leakage currents  
 .....Persistent currents  
 .....Short-circuit currents  
 .....Threshold current  
 .....Current-voltage characteristics  
 .....Electric potential

.....Gain  
 .....Impedance  
 .....Impedance matching  
 .....Inductance  
 .....Permittivity  
 .....Piezoresistance  
 .....Q-factor  
 .....Resistance  
 .....Electric resistance  
 .....Piezoresistance  
 .....Surface resistance  
 .....Thermal resistance  
 .....Viscosity  
 .....Voltage  
 .....Breakdown voltage  
 .....Dynamic voltage scaling  
 .....Threshold voltage  
 .....Voltage fluctuations  
 .....Wiring  
 ....High energy physics instrumentation  
 computing  
 .....Linear particle accelerator  
 ....Instruments  
 .....Compass  
 .....Goniometers  
 .....Microscopy  
 .....Atomic force microscopy  
 .....Electron microscopy  
 .....Scanning probe microscopy  
 .....Oscilloscopes  
 .....Potentiometers  
 .....Pressure gauges  
 .....Probes  
 .....Radiometers  
 .....Spectroradiometers  
 .....Telescopes  
 .....Theodolites  
 .....Tuners  
 .....Vibrometers  
 .....Voltmeters  
 .....Watt-hour meters  
 .....Wattmeters  
 ....Measurement  
 .....Accelerometers  
 .....Acoustic measurements  
 .....Antenna measurements  
 .....Anthropometry  
 .....Area measurement  
 .....Atmospheric measurements  
 .....Atomic measurements  
 .....Biomedical measurement





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|                                        |                                       |
|----------------------------------------|---------------------------------------|
| .....Biomarkers                        | .....Frequency-domain analysis        |
| .....Biomedical monitoring             | .....Frequency estimation             |
| .....Electroencephalography            | .....Gain measurement                 |
| .....Electromyography                  | .....Gas chromatography               |
| .....Electrooculography                | .....Geologic measurements            |
| .....Electrophysiology                 | .....Geophysical image processing     |
| .....Photoplethysmography              | .....Geophysical measurements         |
| .....Reproducibility of results        | .....Geodesy                          |
| .....Sensitivity and specificity       | .....Sea measurements                 |
| .....Calorimetry                       | .....Seismic measurements             |
| .....Coordinate measuring machines     | .....Interferometry                   |
| .....Density measurement               | .....Fabry-Perot                      |
| .....Hydrometers                       | .....Interferometers                  |
| .....Distance measurement              | .....Optical interferometry           |
| .....Euclidean distance                | .....Phase shifting interferometry    |
| .....Distortion measurement            | .....Radar interferometry             |
| .....Total harmonic distortion         | .....Radio interferometry             |
| .....Doppler measurement               | .....Sagnac interferometers           |
| .....Dosimetry                         | .....Length measurement               |
| .....Dynamic range                     | .....Lifetime estimation              |
| .....Electric variables measurement    | .....Loss measurement                 |
| .....Admittance measurement            | .....Packet loss                      |
| .....Ammeters                          | .....Magnetic variables measurement   |
| .....Attenuation measurement           | .....Magnetic field measurement       |
| .....Capacitance measurement           | .....Magnetometers                    |
| .....Conductivity measurement          | .....Permeability measurement         |
| .....Current measurement               | .....Measurement by laser beam        |
| .....Dielectric measurement            | .....Laser velocimetry                |
| .....Electrical resistance measurement | .....Measurement techniques           |
| .....Electrostatic measurements        | .....Calibration                      |
| .....Energy measurement                | .....Dynamic equilibrium              |
| .....Impedance measurement             | .....Measurement uncertainty          |
| .....Inductance measurement            | .....Measurement units                |
| .....Partial discharge measurement     | .....Nanometers                       |
| .....Phasor measurement units          | .....Mechanical variables measurement |
| .....Power measurement                 | .....Angular velocity                 |
| .....Q measurement                     | .....Displacement measurement         |
| .....Transmission line measurements    | .....Force measurement                |
| .....Voltage measurement               | .....Motion measurement               |
| .....Electromagnetic measurements      | .....Position measurement             |
| .....Electromagnetic modeling          | .....Rotation measurement             |
| .....Linearity                         | .....Strain measurement               |
| .....Microwave measurement             | .....Stress measurement               |
| .....Millimeter wave measurements      | .....Thickness measurement            |
| .....Parameter extraction              | .....Torque measurement               |
| .....Polarimetry                       | .....Velocity measurement             |
| .....Radiometry                        | .....Vibration measurement            |
| .....Submillimeter wave measurements   | .....Volume measurement               |
| .....Extraterrestrial measurements     | .....Weight measurement               |
| .....Fluid flow measurement            | .....Moisture measurement             |
| .....Frequency measurement             | .....Humidity measurement             |



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|                                             |                                        |
|---------------------------------------------|----------------------------------------|
| .....Noise measurement                      | .....UHF measurements                  |
| .....Multiple signal classification         | .....Ultrasonic variables measurement  |
| .....Noise figure                           | .....Viscosity                         |
| .....Noise shaping                          | .....Wavelength measurement            |
| .....Nuclear measurements                   | .....Wide area measurements            |
| .....Particle tracking                      | .....Monitoring                        |
| .....Optical variables measurement          | .....Computerized monitoring           |
| .....Ellipsometry                           | .....Environmental monitoring          |
| .....Photometry                             | .....Patient monitoring                |
| .....Reflection coefficient                 | .....Radiation monitoring              |
| .....Refractive index                       | .....Radiation dosage                  |
| .....Particle beam measurements             | .....Remote monitoring                 |
| .....Particle measurements                  | .....Surveillance                      |
| .....Performance evaluation                 | .....Infrared surveillance             |
| .....Phase measurement                      | .....Video surveillance                |
| .....pH measurement                         | .....Testing                           |
| .....Plasma measurements                    | .....Aerospace testing                 |
| .....Plethysmography                        | .....Automatic testing                 |
| .....Pollution measurement                  | .....Automatic test pattern generation |
| .....Pressure measurement                   | .....Ring generators                   |
| .....Altimetry                              | .....Benchmark testing                 |
| .....Tire pressure                          | .....Built-in self-test                |
| .....Pulse measurements                     | .....Circuit testing                   |
| .....Reflectometry                          | .....Integrated circuit measurements   |
| .....Reproducibility of results             | .....Electronic equipment testing      |
| .....Scintillation counters                 | .....Immunity testing                  |
| .....Solid scintillation detectors          | .....Error analysis                    |
| .....Sea state                              | .....Bit error rate                    |
| .....Semiconductor device measurement       | .....Finite wordlength effects         |
| .....Sensitivity                            | .....Error-free operations             |
| .....Sensitivity analysis                   | .....Failure analysis                  |
| .....Shape measurement                      | .....Equipment failure                 |
| .....Size measurement                       | .....Semiconductor device breakdown    |
| .....Software measurement                   | .....Frequency response                |
| .....Software metrics                       | .....Impulse testing                   |
| .....Soil measurements                      | .....Insulator testing                 |
| .....Spectroscopy                           | .....Insulation testing                |
| .....Electrochemical impedance spectroscopy | .....Integrated circuit testing        |
| .....Kirchhoff's Law                        | .....Integrated circuit yield          |
| .....Mass spectroscopy                      | .....Logic testing                     |
| .....MERIS                                  | .....Life testing                      |
| .....Neutron spin echo                      | .....Materials testing                 |
| .....Photoacoustic effects                  | .....Accelerated aging                 |
| .....Resonance light scattering             | .....Acoustic testing                  |
| .....Thermal variables measurement          | .....Adhesive strength                 |
| .....Temperature measurement                | .....Bonding forces                    |
| .....Time measurement                       | .....Delamination                      |
| .....Clocks                                 | .....Elastic recovery                  |
| .....Time dissemination                     | .....Nondestructive testing            |
| .....Timing                                 | .....Optical fiber testing             |
|                                             | .....Remaining life assessment         |





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.....Ring generators  
 .....Semiconductor device testing  
 .....Software testing  
 .....System testing  
 .....Model checking  
 .....Test equipment  
 .....Automatic test equipment  
 .....Test facilities  
 .....Anechoic chambers  
 .....Laboratories  
 .....Large Hadron Collider  
 .....Open area test sites  
 .....TEM cells

#### Intelligent transportation systems

....Automated highways  
 ....Geographic information systems  
 .....Geospatial analysis  
 .....Gunshot detection systems  
 ....Intelligent vehicles  
 .....Vehicle routing  
 ....Navigation  
 .....Aircraft navigation  
 .....Course correction  
 .....Dead reckoning  
 .....Inertial navigation  
 .....Marine navigation  
 .....Radio navigation  
 .....Satellite navigation systems  
 .....Global Positioning System  
 .....Satellite constellations  
 .....Sonar navigation  
 ....Transportation  
 .....Air transportation  
 .....Aircraft  
 .....Airports  
 .....Land transportation  
 .....Rail transportation  
 .....Road transportation  
 .....Vehicles  
 .....Land vehicles  
 .....Remotely operated vehicles  
 .....Space vehicles

#### Lasers and electrooptics

....Electrooptic devices  
 .....Electrochromic devices  
 .....Electrooptic deflectors  
 .....Electrooptic modulators

....Electrooptic effects  
 .....Electrochromism  
 .....Kerr effect  
 .....Optical bistability  
 .....Stark effect  
 ....Lasers  
 .....Atom lasers  
 .....Chemical lasers  
 .....Diode lasers  
 .....Free electron lasers  
 .....Gas lasers  
 .....Laser applications  
 .....Dark states  
 .....Distributed feedback devices  
 .....Laser ablation  
 .....Laser beam cutting  
 .....Laser fusion  
 .....Laser theory  
 .....Magnetooptic recording  
 .....Laser excitation  
 .....Optical pumping  
 .....Laser modes  
 .....Laser mode locking  
 .....Laser stability  
 .....Laser transitions  
 .....Power lasers  
 .....Pump lasers  
 .....Quantum well lasers  
 .....Quantum cascade lasers  
 .....Ring lasers  
 .....Fiber lasers  
 .....Semiconductor lasers  
 .....Laser tuning  
 .....Quantum dot lasers  
 .....Quantum well lasers  
 .....Semiconductor laser arrays  
 .....Semiconductor optical amplifiers  
 .....Surface emitting lasers  
 .....Solid lasers  
 .....Microchip lasers  
 .....Quantum well lasers  
 .....Semiconductor lasers  
 .....Surface emitting lasers  
 .....Surface emitting lasers  
 .....Vertical cavity surface emitting lasers  
 .....X-ray lasers  
 ....Optics  
 .....Adaptive optics  
 .....Birefringence  
 .....Brightness



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|                                       |                                      |
|---------------------------------------|--------------------------------------|
| .....Brightness temperature           | .....Holographic optical components  |
| .....Color                            | .....Lenses                          |
| .....Pigmentation                     | .....Light deflectors                |
| .....Electron optics                  | .....Lighting                        |
| .....Extinction coefficients          | .....Luminescent devices             |
| .....Extinction ratio                 | .....Mirrors                         |
| .....Fiber optics                     | .....Optical arrays                  |
| .....Fiber nonlinear optics           | .....Optical attenuators             |
| .....Optical fibers                   | .....Optical collimators             |
| .....Fluorescence                     | .....Optical device fabrication      |
| .....Four-wave mixing                 | .....Optical filters                 |
| .....Geometrical optics               | .....Optical resonators              |
| .....Ray tracing                      | .....Optical sensors                 |
| .....Integrated optics                | .....Thermooptical devices           |
| .....Light sources                    | .....Optical distortion              |
| .....Electroluminescent devices       | .....Optical fiber applications      |
| .....Fast light                       | .....Optical fiber devices           |
| .....Luminescent devices              | .....Optical harmonic generation     |
| .....Phosphors                        | .....Optical losses                  |
| .....Slow light                       | .....Optical microscopy              |
| .....Stray light                      | .....Optical mixing                  |
| .....Superluminescent diodes          | .....Multiwave mixing                |
| .....Ultraviolet sources              | .....Optical polarization            |
| .....Luminescence                     | .....Polarization shift keying       |
| .....Bioluminescence                  | .....Stokes parameters               |
| .....Electroluminescence              | .....Optical pulses                  |
| .....Fluorescence                     | .....Optical retarders               |
| .....Phosphorescence                  | .....Optical saturation              |
| .....Photoluminescence                | .....Optical solitons                |
| .....Thermoluminescence               | .....Optical tuning                  |
| .....Microoptics                      | .....Particle beam optics            |
| .....Micromirrors                     | .....Atom optics                     |
| .....Nonlinear optics                 | .....Electron optics                 |
| .....Fiber nonlinear optics           | .....Stimulated emission             |
| .....Nonlinear optical devices        | .....Photoluminescence               |
| .....Optical mixing                   | .....Physical optics                 |
| .....Optical saturation               | .....Optical refraction              |
| .....Photorefractive effect           | .....Optical vortices                |
| .....Raman scattering                 | .....Ray tracing                     |
| .....Supercontinuum generation        | .....Stray light                     |
| .....Optical amplifiers               | .....Ultrafast optics                |
| .....Doped fiber amplifiers           | .....Whispering gallery modes        |
| .....Erbium-doped fiber amplifiers    | .....Optoelectronic devices          |
| .....Semiconductor optical amplifiers | .....Charge-coupled image sensors    |
| .....Optical crosstalk                | .....Integrated optoelectronics      |
| .....Optical design                   | .....Light emitting diodes           |
| .....Optical design techniques        | .....Inorganic light emitting diodes |
| .....Optical devices                  | .....LED lamps                       |
| .....Bragg gratings                   | .....Organic light emitting diodes   |
| .....Collimators                      | .....Superluminescent diodes         |
| .....Displays                         | .....Photoconducting devices         |



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.....Electrophotography  
 .....Photodetectors  
 .....Photodiodes  
 .....Phototransistors  
 .....Superconducting photodetectors  
 .....Superluminescent diodes  
 ....Photonics  
 .....Biophotonics  
 .....Microwave photonics  
 .....Nanophotonics  
 .....Photochromism  
 .....Photothermal effects  
 .....Silicon photonics  
 .....Spontaneous emission  
 .....Radiative recombination

### Magnetics

....Biomagnetics  
 .....Magnetoencephalography  
 ....Demagnetization  
 ....Gyromagnetism  
 ....Magnetic analysis  
 .....Magnetization  
 ....Magnetic anisotropy  
 .....Magnetic domains  
 .....Magnetic domain walls  
 .....Magnetic moments  
 .....Perpendicular magnetic anisotropy  
 ....Magnetic devices  
 .....Accelerator magnets  
 .....Ferrite devices  
 .....Circulators  
 .....Magnetic cores  
 .....Transformer cores  
 .....Magnetic heads  
 .....Magnetic memory  
 .....Floppy disks  
 .....Hard disks  
 .....Magnetic modulators  
 .....Magneto optic devices  
 .....Magnetoresistive devices  
 .....Magnetostrictive devices  
 .....Solenoids  
 .....Transformer cores  
 .....Undulators  
 ....Magnetic fields  
 .....Geomagnetism  
 .....Magnetic reconnection  
 .....Magnetic separation  
 .....Magnetostatics

.....Toroidal magnetic fields  
 ....Magnetic flux  
 .....Flux pinning  
 .....Magnetic flux density  
 .....Magnetic flux leakage  
 ....Magnetic force microscopy  
 ....Magnetic forces  
 .....Coercive force  
 ....Magnetic hysteresis  
 ....Magnetic levitation  
 ....Magnetic losses  
 ....Magnetic materials  
 .....Amorphous magnetic materials  
 .....Antiferromagnetic materials  
 .....Diamagnetic materials  
 .....Ferrimagnetic films  
 .....Ferrite films  
 .....Garnet films  
 .....Ferrimagnetic materials  
 .....Ferrimagnetic films  
 .....Ferrite films  
 .....Ferrites  
 .....Garnet films  
 .....Garnets  
 .....Ferrite films  
 .....Ferrites  
 .....Ferrite films  
 .....Garnet films  
 .....Garnets  
 .....Garnet films  
 .....Magnetic films  
 .....Ferrimagnetic films  
 .....Ferrite films  
 .....Garnet films  
 .....Magnetic liquids  
 .....Magnetic semiconductors  
 .....Magnetic superlattices  
 .....Paramagnetic materials  
 .....Soft magnetic materials  
 ....Magnetic multilayers  
 ....Magnetic particles  
 ....Magnetic properties  
 ....Magnetic sensors  
 .....Spin valves  
 ....Magnetic susceptibility  
 ....Magnetic switching  
 ....Magnetization processes  
 .....Magnetization reversal  
 .....Saturation magnetization  
 ....Magnetoacoustic effects  
 ....Magnetoelectric effects



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.....Hall effect  
 .....Magnetic tunneling  
 .....Magnetoelectronics  
 .....Spin polarized transport  
 .....Magnetoresistance  
 .....Anisotropic magnetoresistance  
 .....Ballistic magnetoresistance  
 .....Colossal magnetoresistance  
 .....Enhanced magnetoresistance  
 .....Extraordinary magnetoresistance  
 .....Giant magnetoresistance  
 .....Ordinary magnetoresistance  
 .....Tunneling magnetoresistance  
 ....Magnetomechanical effects  
 .....Magnetic field induced strain  
 .....Magnetoelasticity  
 .....Magnetostriction  
 .....Magnetostriction  
 ....Magneto optic effects  
 .....Faraday effect  
 .....Gyrotropism  
 ....Magnets  
 .....Electromagnets  
 .....Superconducting magnets  
 .....Micromagnetics  
 .....Permanent magnets  
 ....Microwave magnetics  
 ....Nonlinear magnetics  
 ....Remanence

#### Materials, elements, and compounds

....Chemical elements  
 .....Boron  
 .....Boron alloys  
 .....Carbon  
 .....Cerium  
 .....Darmstadtium  
 .....Helium  
 .....Hydrogen  
 .....Deuterium  
 .....Isotopes  
 .....Lutetium  
 .....Nitrogen  
 .....Silicon nitride  
 .....Oxygen  
 .....Roentgenium  
 .....Tellurium  
 .....Titanium  
 .....Titanium alloys  
 .....Titanium compounds

.....Ytterbium  
 .....Zirconium  
 ....Compounds  
 .....Bismuth compounds  
 .....Gallium compounds  
 .....Aluminum gallium nitride  
 .....Gallium arsenide  
 .....Gallium nitride  
 .....Indium gallium arsenide  
 .....Indium gallium nitride  
 .....Indium compounds  
 .....Indium gallium arsenide  
 .....Indium tin oxide  
 .....Inorganic compounds  
 .....Lead compounds  
 .....Organic compounds  
 .....Carbon compounds  
 .....Organic semiconductors  
 .....Volatile organic compounds  
 .....Silicon compounds  
 .....Silicides  
 .....Silicon carbide  
 .....Silicon nitride  
 ....Materials  
 .....Acoustic materials  
 .....Additives  
 .....Aggregates  
 .....Amorphous materials  
 .....Diamond-like carbon  
 .....Glass  
 .....Auxetic materials  
 .....Biological materials  
 .....Biomedical materials  
 .....Bioceramics  
 .....Biomembranes  
 .....Building materials  
 .....Asphalt  
 .....Concrete  
 .....Floors  
 .....Mortar  
 .....Tiles  
 .....Windows  
 .....Ceramics  
 .....Porcelain  
 .....Composite materials  
 .....Conducting materials  
 .....Corrosion inhibitors  
 .....Crystalline materials  
 .....Nanocrystals  
 .....Superlattices  
 .....Crystals



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|                                   |                                         |
|-----------------------------------|-----------------------------------------|
| .....Colloidal crystals           | .....Magnetic superlattices             |
| .....Crystallography              | .....Paramagnetic materials             |
| .....Crystal microstructure       | .....Soft magnetic materials            |
| .....Grain boundaries             | .....Material properties                |
| .....Grain size                   | .....Creep                              |
| .....Liquid crystals              | .....Elasticity                         |
| .....Dielectric materials         | .....Resilience                         |
| .....Dielectric films             | .....Media                              |
| .....Dielectric liquids           | .....Nonhomogeneous media               |
| .....Electrets                    | .....Random media                       |
| .....Epoxy resins                 | .....Mesoporous materials               |
| .....High K dielectric materials  | .....Metal foam                         |
| .....Piezoelectric materials      | .....Metamaterials                      |
| .....Films                        | .....Electromagnetic metamaterials      |
| .....Conductive films             | .....Optical cloaking                   |
| .....Dielectric films             | .....Optical metamaterials              |
| .....Epitaxial layers             | .....Nanostructured materials           |
| .....Ferrimagnetic films          | .....Nanocomposites                     |
| .....Ferrite films                | .....Nanoporous materials               |
| .....Garnet films                 | .....Oils                               |
| .....Magnetic films               | .....Lubricating oils                   |
| .....Optical films                | .....Vegetable oils                     |
| .....Piezoelectric films          | .....Optical materials                  |
| .....Plastic films                | .....Optical cloaking                   |
| .....Polymer films                | .....Optical polymers                   |
| .....Semiconductor films          | .....Optical retarders                  |
| .....Thick films                  | .....Optical superlattices              |
| .....Thin films                   | .....Photorefractive materials          |
| .....Fluids                       | .....Organic inorganic hybrid materials |
| .....Fluid dynamics               | .....Organic materials                  |
| .....Gases                        | .....Paints                             |
| .....Hydraulic fluids             | .....Paper pulp                         |
| .....Liquids                      | .....Petrochemicals                     |
| .....Viscosity                    | .....Phase change materials             |
| .....Hazardous materials          | .....Photoconducting materials          |
| .....Inorganic materials          | .....Plastics                           |
| .....Lacquers                     | .....Epoxy resins                       |
| .....Laminates                    | .....Fiber reinforced plastics          |
| .....Magnetic materials           | .....Plastic films                      |
| .....Amorphous magnetic materials | .....Plastic optical fiber              |
| .....Antiferromagnetic materials  | .....Polymer foams                      |
| .....Diamagnetic materials        | .....Polymer gels                       |
| .....Ferrimagnetic films          | .....Polymers                           |
| .....Ferrimagnetic materials      | .....Liquid crystal polymers            |
| .....Ferrite films                | .....Optical polymers                   |
| .....Ferrites                     | .....Polyethylene                       |
| .....Garnet films                 | .....Polyimides                         |
| .....Garnets                      | .....Production materials               |
| .....Magnetic films               | .....Abrasives                          |
| .....Magnetic liquids             | .....Aerospace materials                |
| .....Magnetic semiconductors      | .....Automotive materials               |



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|                                       |                                       |
|---------------------------------------|---------------------------------------|
| .....Inhibitors                       | .....Wire                             |
| .....Ink                              | .....Materials science and technology |
| .....Joining materials                | .....Absorption                       |
| .....Lubricants                       | .....Aging                            |
| .....Retardants                       | .....Accelerated aging                |
| .....Radioactive materials            | .....Chemical analysis                |
| .....Nuclear fuels                    | .....Activation analysis              |
| .....Radioactive decay                | .....Chemical processes               |
| .....Radioactive waste                | .....Chemicals                        |
| .....Raw materials                    | .....Electronic noses                 |
| .....Resins                           | .....pH measurement                   |
| .....Epoxy resins                     | .....Contamination                    |
| .....Resists                          | .....Surface contamination            |
| .....Semiconductor materials          | .....Degradation                      |
| .....Amorphous semiconductors         | .....Filtration                       |
| .....Elemental semiconductors         | .....Microfiltration                  |
| .....Gallium                          | .....Hysteresis                       |
| .....Gallium arsenide                 | .....Impurities                       |
| .....Germanium                        | .....Semiconductor impurities         |
| .....III-V semiconductor materials    | .....Materials handling               |
| .....II-VI semiconductor materials    | .....Cleaning                         |
| .....Indium gallium arsenide          | .....Decontamination                  |
| .....Indium phosphide                 | .....Freight handling                 |
| .....Magnetic semiconductors          | .....Materials handling equipment     |
| .....Organic semiconductors           | .....Remote handling                  |
| .....Semiconductor superlattices      | .....Materials preparation            |
| .....Silicon                          | .....Doping                           |
| .....Silicon germanium                | .....Firing                           |
| .....Substrates                       | .....Ion implantation                 |
| .....Wide band gap semiconductors     | .....Laser sintering                  |
| .....Sheet materials                  | .....Sputtering                       |
| .....Solids                           | .....Materials reliability            |
| .....Young's modulus                  | .....Materials testing                |
| .....Superconducting materials        | .....Accelerated aging                |
| .....Granular superconductors         | .....Acoustic testing                 |
| .....High-temperature superconductors | .....Adhesive strength                |
| .....Multifilamentary superconductors | .....Bonding forces                   |
| .....Niobium-tin                      | .....Delamination                     |
| .....Type II superconductors          | .....Elastic recovery                 |
| .....Textiles                         | .....Nondestructive testing           |
| .....Cotton                           | .....Microstructure                   |
| .....Fabrics                          | .....Periodic structures              |
| .....Textile fibers                   | .....Gratings                         |
| .....Wool                             | .....Photonic crystals                |
| .....Waste materials                  | .....Pigmentation                     |
| .....Effluents                        | .....Pigments                         |
| .....Electronic waste                 | .....Separation processes             |
| .....Industrial waste                 | .....Fractionation                    |
| .....Radioactive waste                | .....Particle separators              |
| .....Slurries                         | .....Surface engineering              |
| .....Wastewater                       | .....Surfaces                         |



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|                          |                                       |
|--------------------------|---------------------------------------|
| .....Corrosion           | .....Germanium                        |
| .....Corrugated surfaces | .....Germanium alloys                 |
| .....Rough surfaces      | .....Gold                             |
| .....Surface impedance   | .....Gold alloys                      |
| .....Surface morphology  | .....Hafnium                          |
| .....Surface resistance  | .....Hafnium compounds                |
| .....Surface roughness   | .....Indium                           |
| .....Surface soil        | .....Iron                             |
| .....Surface structures  | .....Cast iron                        |
| .....Surface tension     | .....Iron alloys                      |
| .....Surface texture     | .....Lanthanum                        |
| .....Surface topography  | .....Lanthanum compounds              |
| .....Surface treatment   | .....Lead                             |
| ....Material storage     | .....Lead isotopes                    |
| .....Bulk storage        | .....Lithium                          |
| .....Containers          | .....Lithium compounds                |
| .....Freight containers  | .....Magnesium                        |
| .....Fuel storage        | .....Magnesium compounds              |
| .....Secure storage      | .....Manganese                        |
| .....Stacking            | .....Manganese alloys                 |
| .....Storage automation  | .....Mercury (metals)                 |
| .....Warehousing         | .....Metallization                    |
| .....Water storage       | .....Integrated circuit metallization |
| .....Reservoirs          | .....Neodymium                        |
| ....Metals               | .....Neodymium alloys                 |
| .....Alloying            | .....Neodymium compounds              |
| .....Intermetallic       | .....Nickel                           |
| .....Shape memory alloys | .....Nickel alloys                    |
| .....Aluminum            | .....Niobium                          |
| .....Aluminum alloys     | .....Niobium alloys                   |
| .....Aluminum compounds  | .....Niobium compounds                |
| .....Barium              | .....Palladium                        |
| .....Barium compounds    | .....Platinum                         |
| .....Bismuth             | .....Platinum alloys                  |
| .....Boron               | .....Rare earth metals                |
| .....Boron alloys        | .....Samarium                         |
| .....Cadmium             | .....Samarium alloys                  |
| .....Cadmium compounds   | .....Silver                           |
| .....Calcium             | .....Steel                            |
| .....Calcium compounds   | .....Strontium                        |
| .....Chromium            | .....Strontium compounds              |
| .....Chromium alloys     | .....Tin                              |
| .....Cobalt              | .....Tin alloys                       |
| .....Cobalt alloys       | .....Tin compounds                    |
| .....Copper              | .....Titanium                         |
| .....Copper alloys       | .....Titanium alloys                  |
| .....Copper compounds    | .....Titanium compounds               |
| .....Digital alloys      | .....Tungsten                         |
| .....Erbium              | .....Yttrium                          |
| .....Gallium             | .....Yttrium compounds                |
| .....Gallium alloys      | .....Zinc                             |



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.....Zinc compounds

**Mathematics**

....Accuracy  
 ....Algebra  
 .....Abstract algebra  
 .....Galois fields  
 .....Modules (abstract algebra)  
 .....Boolean algebra  
 .....Boolean functions  
 .....Linear algebra  
 .....Linear programming  
 .....Matrices  
 .....Vectors  
 .....Set theory  
 .....Fuzzy sets  
 .....Fuzzy set theory  
 .....Rough sets  
 ....Algorithms  
 .....Adaptive algorithms  
 .....Adaptation models  
 .....Algorithm design and analysis  
 .....Approximation algorithms  
 .....Backpropagation algorithms  
 .....Basis algorithms  
 .....Change detection algorithms  
 .....Classification algorithms  
 .....Clustering algorithms  
 .....Compression algorithms  
 .....Density estimation robust algorithm  
 .....Detection algorithms  
 .....Distributed algorithms  
 .....Dynamic programming  
 .....Filtering algorithms  
 .....Genetic algorithms  
 .....Heuristic algorithms  
 .....Inference algorithms  
 .....Least mean square algorithms  
 .....Machine learning algorithms  
 .....Matching pursuit algorithms  
 .....Maximum likelihood detection  
 .....MLFMA  
 .....Multicast algorithms  
 .....Parallel algorithms  
 .....Partitioning algorithms  
 .....Prediction algorithms  
 .....Projection algorithms  
 .....Pursuit algorithms  
 .....Signal processing algorithms  
 .....Software algorithms

.....Viterbi algorithm  
 ....Arithmetic  
 .....Digital arithmetic  
 .....Fixed-point arithmetic  
 .....Floating-point arithmetic  
 ....Azimuth  
 .....Azimuthal angle  
 .....Azimuthal component  
 .....Azimuthal current  
 .....Azimuthal harmonics  
 .....Azimuthal plane  
 ....Boundary value problems  
 .....Boundary conditions  
 .....Upper bound  
 ....Calculus  
 .....Differential equations  
 .....Differential algebraic equations  
 .....Navier-Stokes equations  
 .....Partial differential equations  
 .....Transfer functions  
 .....Integral equations  
 .....Probability density function  
 ....Level set  
 ....Closed-form solutions  
 ....Combinatorial mathematics  
 .....Graph theory  
 .....Bipartite graph  
 .....Optimal matching  
 .....Reachability analysis  
 .....Shortest path problem  
 .....Tree graphs  
 .....Steiner trees  
 ....Computational efficiency  
 ....Conformal mapping  
 ....Convergence  
 ....Convex functions  
 ....Cyclic redundancy check  
 .....Cyclic redundancy check codes  
 ....Eigenvalues and eigenfunctions  
 ....Equations  
 .....Boltzmann equation  
 .....Difference equations  
 .....Integrodifferential equations  
 .....Maxwell equations  
 .....Nonlinear equations  
 .....Bifurcation  
 .....Polynomials  
 .....Riccati equations  
 ....Estimation  
 .....Estimation error  
 .....Estimation theory





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.....Cramer-Rao bounds  
 .....Maximum a posteriori estimation  
 .....Life estimation  
 .....Maximum likelihood estimation  
 .....State estimation  
 .....Observers  
 .....Yield estimation  
 ....Euclidean distance  
 .....Hilbert space  
 ....Finite difference methods  
 ....Finite element analysis  
 ....Fourier series  
 ....Functional analysis  
 ....Geometry  
 .....Computational geometry  
 .....Fractals  
 .....Elliptic curves  
 .....Elliptic design  
 .....Ellipsoids  
 .....Information geometry  
 .....Surface topography  
 .....Nanotopography  
 ....Gradient methods  
 ....Graph theory  
 .....Bipartite graph  
 .....Optimal matching  
 .....Reachability analysis  
 .....Shortest path problem  
 .....Tree graphs  
 ....Harmonic analysis  
 ....Iterative methods  
 .....Expectation-maximization algorithms  
 .....Iterative algorithms  
 .....Belief propagation  
 .....Iterative closest point algorithm  
 .....Sum product algorithm  
 ....Kernel  
 .....Null space  
 ....Laplace equations  
 ....Lattices  
 .....Lattice Boltzmann methods  
 ....Limit-cycles  
 ....Linearization techniques  
 ....Linear matrix inequalities  
 ....Linear systems  
 ....Mathematical model  
 .....Mathematical analysis  
 .....Formal concept analysis  
 .....Fractional calculus  
 .....Modal analysis  
 ....Mathematical programming  
 ....Method of moments  
 ....Minimization  
 .....Minimization methods  
 ....Mode matching methods  
 ....Network theory (graphs)  
 ....Nonlinear equations  
 .....Bifurcation  
 ....Nonlinear systems  
 .....Chaos  
 .....Chaotic communication  
 .....Complexity theory  
 .....Spatiotemporal phenomena  
 .....Nonlinear dynamical systems  
 ....Numerical analysis  
 .....Adaptive mesh refinement  
 .....Approximation methods  
 .....Approximation error  
 .....Chebyshev approximation  
 .....Curve fitting  
 .....Extrapolation  
 .....Function approximation  
 .....Interpolation  
 .....Least squares approximations  
 .....Linear approximation  
 .....Perturbation methods  
 ....Convergence of numerical methods  
 .....Finite difference methods  
 .....Finite element analysis  
 .....Finite volume methods  
 .....Gradient methods  
 .....Independent component analysis  
 .....Iterative methods  
 .....Expectation-maximization algorithms  
 .....Iterative algorithms  
 ....Method of moments  
 .....Mode matching methods  
 .....Multigrid methods  
 .....Newton method  
 .....Numerical simulation  
 .....Numerical stability  
 .....Relaxation methods  
 .....Sparse matrices  
 .....Splines (mathematics)  
 .....Surface fitting  
 .....Response surface methodology  
 .....Symmetric matrices  
 .....Transmission line matrix methods  
 ....Optimization  
 .....Cost function  
 .....Optimal scheduling



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|                                      |                                    |
|--------------------------------------|------------------------------------|
| .....Optimization methods            | ....Spirals                        |
| .....Circuit optimization            | ....Statistics                     |
| .....Design optimization             | .....Adaptive estimation           |
| .....Gradient methods                | .....Autoregressive processes      |
| .....H infinity control              | .....Boltzmann distribution        |
| .....Mathematical programming        | .....Lattice Boltzmann methods     |
| .....Optimized production technology | .....Correlation                   |
| .....Pareto optimization             | .....Autocorrelation               |
| .....Quadratic programming           | .....Correlation coefficient       |
| .....Simulated annealing             | .....Covariance matrices           |
| ....Piecewise linear techniques      | .....Gaussian mixture model        |
| .....Piecewise linear approximation  | .....Higher order statistics       |
| ....Predator prey systems            | .....Histograms                    |
| ....Probability                      | .....Least squares methods         |
| .....Ant colony optimization         | .....Least mean squares methods    |
| .....Bayes methods                   | .....Least squares approximations  |
| .....Recursive estimation            | .....Linear discriminant analysis  |
| .....Error probability               | .....Maximum likelihood estimation |
| .....Forecasting                     | .....Mean square error methods     |
| .....Demand forecasting              | .....Minimax techniques            |
| .....Economic forecasting            | .....Parametric statistics         |
| .....Forecast uncertainty            | .....Prediction theory             |
| .....Technology forecasting          | .....Ranking (statistics)          |
| .....Memoryless systems              | .....Root mean square              |
| .....Pairwise error probability      | .....Sampling methods              |
| .....Possibility theory              | .....Compressed sensing            |
| .....Probability distribution        | .....Nonuniform sampling           |
| .....Exponential distribution        | .....Statistical analysis          |
| .....Log-normal distribution         | .....Analysis of variance          |
| .....Maxwell-Boltzmann distribution  | .....Mode matching methods         |
| .....Nakagami distribution           | .....Monte Carlo methods           |
| .....Random variables                | .....Parameter estimation          |
| .....Statistical distributions       | .....Pareto analysis               |
| .....Distribution functions          | .....Principal component analysis  |
| .....Gaussian distribution           | .....Regression analysis           |
| .....Weibull distribution            | .....Time series analysis          |
| .....Uncertainty                     | ....Stochastic processes           |
| .....Forecast uncertainty            | .....Gaussian processes            |
| ....Quaternions                      | .....Gaussian mixture model        |
| ....Random processes                 | .....Markov processes              |
| .....Brownian motion                 | .....Markov random fields          |
| ....Root mean square                 | ....Taylor series                  |
| ....Sequences                        | ....Topology                       |
| .....Binary sequences                | ....Transforms                     |
| .....Random sequences                | .....Discrete transforms           |
| ....Set theory                       | .....Discrete cosine transforms    |
| .....Fuzzy sets                      | .....Empirical mode decomposition  |
| .....Fuzzy set theory                | .....Fourier transforms            |
| .....Rough sets                      | .....Discrete Fourier transforms   |
| ....Simulated annealing              | .....Fast Fourier transforms       |
| ....Smoothing methods                | .....Karhunen-Loeve transforms     |







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.....Poincare invariance  
 .....Wavelet transforms  
 .....Biorthogonal modulation  
 .....Continuous wavelet transforms  
 .....Discrete wavelet transforms  
 .....Wavelet coefficients  
 .....Wavelet packets  
 ....Transmission line matrix methods  
 ....Uncertain systems  
 ....Utility theory

#### **Microwave theory and techniques**

....Microwave technology  
 .....Beam steering  
 .....Circulators  
 .....Masers  
 .....Gyrotrons  
 .....Microwave bands  
 .....C-band  
 .....K-band  
 .....L-band  
 .....Microwave circuits  
 .....Microwave communication  
 .....Rectennas  
 .....Microwave devices  
 .....Masers  
 .....Microwave amplifiers  
 .....Microwave filters  
 .....Microwave transistors  
 .....Microwave generation  
 .....High power microwave generation  
 .....Microwave photonics  
 .....Microwave sensors  
 ....Millimeter wave technology  
 .....Millimeter wave circuits  
 .....Millimeter wave integrated circuits  
 .....Millimeter wave communication  
 .....Millimeter wave devices  
 .....Millimeter wave transistors  
 .....Millimeter wave integrated circuits  
 .....MIMICs  
 .....Millimeter wave radar  
 ....Submillimeter wave technology  
 .....Submillimeter wave circuits  
 .....Submillimeter wave integrated circuits  
 .....Submillimeter wave communication  
 .....Submillimeter wave devices  
 .....Submillimeter wave filters  
 .....Submillimeter wave integrated circuits

#### **Nanotechnology**

....Bionanotechnology  
 ....Casimir effect  
 ....Molecular computing  
 ....Molecular electronics  
 ....Nanobioscience  
 .....DNA computing  
 .....Nanobiotechnology  
 ....Nanoelectromechanical systems  
 ....Nanoelectronics  
 ....Nanofabrication  
 ....Nanofluidics  
 ....Nanolithography  
 ....Nanomaterials  
 ....Nanopatterning  
 .....Colloidal lithography  
 ....Nanophotonics  
 ....Nanopositioning  
 ....Nanoscale devices  
 .....Nanocontacts  
 ....Nanotube devices  
 ....Nanosensors  
 ....Nanostructured materials  
 .....Nanocomposites  
 .....Nanoporous materials  
 ....Nanostructures  
 .....Nanoparticles  
 .....Nanocrystals  
 .....Nanotubes  
 .....Carbon nanotubes  
 .....Semiconductor nanotubes  
 .....Nanowires  
 ....Semiconductor nanostructures  
 ....Self-assembly  
 .....Electrostatic self-assembly  
 ....Self-replicating machines

#### **Nuclear and plasma sciences**

....Biomedical applications of radiation  
 ....Colliding beam devices  
 .....Colliding beam accelerators  
 .....Muon colliders  
 ....Electron emission  
 .....Ballistic transport  
 .....Electronic ballasts  
 ....Elementary particles  
 .....Charge carriers  
 .....Charge carrier density  
 .....Charge carrier lifetime  
 .....Charge carrier mobility



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|                                                   |                                        |
|---------------------------------------------------|----------------------------------------|
| .....Charge carrier processes                     | .....Nuclear phase transformations     |
| .....Hot carriers                                 | .....Nuclear thermodynamics            |
| .....Electrons                                    | .....Relativistic effects              |
| .....Electron sources                             | ....Particle accelerators              |
| .....Quantum wells                                | .....Accelerator magnets               |
| .....Trions                                       | .....Colliding beam accelerators       |
| .....Elementary particle exchange interactions    | .....Cyclotrons                        |
| .....Elementary particle vacuum                   | .....Electron accelerators             |
| .....Ions                                         | .....Ion accelerators                  |
| .....Ionization                                   | .....Linear accelerators               |
| .....Ion sources                                  | .....Photon collider                   |
| .....Mesons                                       | .....Plasma accelerators               |
| .....Neutrino sources                             | .....Proton accelerators               |
| .....Neutrons                                     | .....Storage rings                     |
| .....Particle beams                               | .....Synchrocyclotrons                 |
| .....Atomic beams                                 | .....Synchrotrons                      |
| .....Electron beams                               | .....Synchrotron radiation             |
| .....Ion beams                                    | .....Undulators                        |
| .....Particle collisions                          | ....Particle beam handling             |
| .....Phonons                                      | ....Particle beam injection            |
| .....Positrons                                    | ....Plasmas                            |
| .....Protons                                      | .....Atmospheric-pressure plasmas      |
| ....Fusion power generation                       | .....Plasma applications               |
| ....Fusion reactors                               | .....Plasma devices                    |
| .....Fusion reactor design                        | .....Plasma immersion ion implantation |
| .....Tokamaks                                     | .....Plasma welding                    |
| .....Tokamak devices                              | .....Tokamaks                          |
| ....Gamma-rays                                    | .....Plasma confinement                |
| .....Gamma-ray bursts                             | .....Inertial confinement              |
| .....Gamma-ray detection                          | .....Magnetic confinement              |
| .....Gamma-ray effects                            | .....Plasma diagnostics                |
| ....Gas discharge devices                         | .....Plasma properties                 |
| .....Glow discharge devices                       | .....Dusty plasmas                     |
| ....High energy physics instrumentation computing | .....Plasma chemistry                  |
| .....Linear particle accelerator                  | .....Plasma density                    |
| ....Ion beam applications                         | .....Plasma sheaths                    |
| .....Ion implantation                             | .....Plasma stability                  |
| .....Plasma immersion ion implantation            | .....Plasma temperature                |
| ....Ion emission                                  | .....Plasmons                          |
| ....Nuclear electronics                           | .....Plasma simulation                 |
| .....Nuclear imaging                              | .....Plasma sources                    |
| .....Energy resolution                            | .....Plasma transport processes        |
| ....Nuclear medicine                              | ....Radiation effects                  |
| .....Nuclear physics                              | .....Biological effects of radiation   |
| .....Alpha particles                              | .....Gamma-ray effects                 |
| .....Beta rays                                    | .....Ion radiation effects             |
| .....Ignition                                     | .....Neutron radiation effects         |
| .....Ion sources                                  | ....Radiation hardening (electronics)  |
| .....Isotopes                                     | .....Radiation monitoring              |
|                                                   | .....Radiation dosage                  |
|                                                   | ....Radiation safety                   |



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....Reactor instrumentation  
 ....Scintillation counters  
 .....Solid scintillation detectors  
 ....Thermionic emission

#### **Oceanic engineering and marine technology**

....Marine navigation  
 ....Marine technology  
 .....Marine equipment  
 .....Marine transportation  
 .....Marine vehicles  
 .....Underwater cables  
 .....Underwater communication  
 .....Underwater equipment  
 .....Rebreathing equipment  
 .....Underwater structures  
 .....Underwater technology  
 .....Underwater communication  
 .....Underwater equipment  
 .....Underwater structures  
 ....Oceanographic techniques  
 ....Ocean temperature  
 ....Water pollution  
 .....Marine pollution

#### **Power electronics**

....Converters  
 .....AC-AC converters  
 .....DC-AC power converters  
 .....Digital-to-frequency converters  
 .....Frequency conversion  
 .....Mixers  
 .....Optical frequency conversion  
 .....Power conversion  
 .....AC-AC converters  
 .....AC-DC power converters  
 .....DC-AC power converters  
 .....DC-DC power converters  
 .....Matrix converters  
 .....Power conversion harmonics  
 .....Pulse width modulation converters  
 .....Static power converters  
 .....Wavelength converters  
 ....Current limiters  
 .....Fault current limiters  
 ....Inverters  
 .....Pulse inverters  
 .....Resonant inverters

....Phase control  
 ....Power conditioning  
 .....Power smoothing  
 ....Power semiconductor devices  
 .....Power transistors  
 ....Power semiconductor switches  
 .....Bipolar transistors  
 .....Insulated gate bipolar transistors  
 .....Kirk field collapse effect  
 .....Thyristors  
 .....Photothyristors  
 ....Snubbers  
 ....Three-phase electric power

#### **Power engineering and energy**

....Electric variables control  
 .....Current control  
 .....Electrical ballasts  
 .....Electric current control  
 .....Gain control  
 .....Power control  
 .....Power system control  
 .....Bidirectional power flow  
 .....Load flow control  
 .....SCADA systems  
 .....Reactive power control  
 .....Voltage control  
 .....Automatic voltage control  
 ....Energy  
 .....Energy barrier  
 .....Energy capture  
 .....Energy consumption  
 .....Energy conversion  
 .....Batteries  
 .....Fuel cells  
 .....Motors  
 .....Photovoltaic cells  
 .....Potential well  
 .....Solar heating  
 .....Thermoelectricity  
 .....Waste heat  
 .....Energy dissipation  
 .....Energy exchange  
 .....Inductive charging  
 .....Energy harvesting  
 .....Energy management  
 .....Energy conservation  
 .....Energy efficiency  
 .....Load management  
 .....Energy resources



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|                                              |                                       |
|----------------------------------------------|---------------------------------------|
| .....Fuels                                   | ....Power systems                     |
| .....Geothermal energy                       | .....Hybrid power systems             |
| .....Nuclear fuels                           | .....Industrial power systems         |
| .....Solar energy                            | .....Power distribution               |
| .....Wave power                              | .....Power distribution faults        |
| .....Wind energy                             | .....Power distribution lines         |
| .....Wind farms                              | .....Power grids                      |
| .....Energy states                           | .....Microgrids                       |
| .....Effective mass                          | .....Smart grids                      |
| .....Orbital calculations                    | .....Power supplies                   |
| .....Energy storage                          | .....Battery chargers                 |
| .....Batteries                               | .....Charging stations                |
| .....Flywheels                               | .....Current supplies                 |
| .....Fuel cells                              | .....Emergency power supplies         |
| .....Hydrogen storage                        | .....Inductive charging               |
| .....Supercapacitors                         | .....Islanding                        |
| .....Superconducting magnetic energy storage | .....Power demand                     |
| ....Power engineering                        | .....Power quality                    |
| .....Ferroresonance                          | .....Power system restoration         |
| .....High-voltage techniques                 | .....Switched-mode power supply       |
| .....Power engineering computing             | .....Traction power supplies          |
| .....Power system simulation                 | .....Umbilical cable                  |
| ....Power generation                         | .....Power system analysis computing  |
| .....Automatic generation control            | .....Power system dynamics            |
| .....Cogeneration                            | .....Power system economics           |
| .....Distributed power generation            | .....Power system faults              |
| .....Geothermal power generation             | .....Power system harmonics           |
| .....Hydroelectric power generation          | .....Power harmonic filters           |
| .....Hydroelectric-thermal power generation  | .....Power system management          |
| .....Microhydro power                        | .....Load flow                        |
| .....Picohydro power                         | .....Power system measurements        |
| .....Magnetohydrodynamic power generation    | .....Meter reading                    |
| .....Nuclear power generation                | .....Power system planning            |
| .....Fission reactors                        | .....Power demand                     |
| .....Fusion power generation                 | .....Power system protection          |
| .....Power generation control                | .....Electrical safety                |
| .....Power generation dispatch               | .....Substation protection            |
| .....Power generation planning               | .....Surge protection                 |
| .....Solar power generation                  | .....Power system reliability         |
| .....Maximum power point trackers            | .....Power system stability           |
| .....Photovoltaic systems                    | .....Power transmission               |
| .....Trigeneration                           | .....Flexible AC transmission systems |
| .....Turbomachinery                          | .....HVDC transmission                |
| .....Turbines                                | .....Inductive power transmission     |
| .....Turbogenerators                         | .....Static VAR compensators          |
| .....Wind energy generation                  | .....Transmission lines               |
| .....Wind energy integration                 | .....PSCAD                            |
| .....Wind power generation                   | .....Pulse power systems              |
|                                              | .....Pulsed power supplies            |
|                                              | .....Reactive power                   |
|                                              | .....Substations                      |





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.....Substation automation  
 .....Substation protection  
 .....Transformers  
 .....Current transformers  
 .....Flyback transformers  
 .....Instrument transformers  
 .....Phase transformers  
 .....Power transformers  
 .....Pulse transformers  
 .....Uninterruptible power systems  
 .....Wind energy integration

#### **Product safety engineering**

....Consumer protection  
 ....Power system protection  
 .....Electrical safety  
 .....Fault protection  
 .....Grounding  
 .....Substation protection  
 .....Surge protection  
 .....Arresters  
 ....Safety  
 .....Aerospace safety  
 .....Air safety  
 .....Domestic safety  
 .....Emergency services  
 .....Explosion protection  
 .....Hazards  
 .....Biohazards  
 .....Chemical hazards  
 .....Explosions  
 .....Fires  
 .....Flammability  
 .....Floods  
 .....Hazardous areas  
 .....Hazardous materials  
 .....Toxicology  
 .....Health and safety  
 .....Occupational health  
 .....Occupational safety  
 .....Marine safety  
 .....Product safety  
 .....Protection  
 .....Explosion protection  
 .....Lightning protection  
 .....Radiation safety  
 .....Safety devices  
 .....Eye protection  
 .....Protective clothing  
 .....Vehicle safety

....Vehicle crash testing

#### **Professional communication**

....Collaboration  
 .....Collaborative tools  
 .....Call conference  
 .....Collaborative software  
 .....Videoconferences  
 .....Discussion forums  
 .....Teamwork  
 .....Virtual groups  
 ....Communication aids  
 ....Communication effectiveness  
 ....Communication symbols  
 .....Semiotics  
 .....Pragmatics  
 .....Semantics  
 .....Syntactics  
 ....Context  
 ....Databases  
 .....Database systems  
 .....Audio databases  
 .....Deductive databases  
 .....Image databases  
 .....Indexes  
 .....Multimedia databases  
 .....Object oriented databases  
 .....Query processing  
 .....Deductive databases  
 .....Distributed databases  
 .....Image databases  
 .....Image retrieval  
 .....Multimedia databases  
 .....Object oriented databases  
 .....Relational databases  
 .....Spatial databases  
 .....Transaction databases  
 .....Itemsets  
 .....Visual databases  
 ....Global communication  
 .....Cross-cultural communication  
 .....Geographic information systems  
 .....Geospatial analysis  
 .....Gunshot detection systems  
 ....Grammar  
 ....Information analysis  
 .....Indexing  
 ....Information resources  
 ....Information retrieval  
 ....Blogs



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|                                      |                                      |
|--------------------------------------|--------------------------------------|
| .....Content-based retrieval         | ....Information systems              |
| .....Hypertext systems               | .....Database systems                |
| .....Information filtering           | .....Audio databases                 |
| .....Information filters             | .....Deductive databases             |
| .....Recommender systems             | .....Image databases                 |
| .....Information rates               | .....Indexes                         |
| .....Music information retrieval     | .....Multimedia databases            |
| .....Online services                 | .....Object oriented databases       |
| .....Search engines                  | .....Query processing                |
| .....Search methods                  | .....Data systems                    |
| .....Keyword search                  | .....Data acquisition                |
| .....Metasearch                      | .....Data compression                |
| .....Nearest neighbor searches       | .....Data conversion                 |
| .....Search problems                 | .....Data engineering                |
| .....Web search                      | .....Data handling                   |
| .....Social network services         | .....Data processing                 |
| .....Computer mediated communication | .....Data storage systems            |
| .....Facebook                        | .....Data warehouses                 |
| .....LinkedIn                        | .....Distributed information systems |
| .....MySpace                         | .....Publish-subscribe               |
| .....Second Life                     | .....Identity management systems     |
| .....Twitter                         | .....Informatics                     |
| .....YouTube                         | .....Biomedical informatics          |
| .....Tagging                         | .....Cognitive informatics           |
| .....Tag clouds                      | .....Information architecture        |
| .....Taxonomy                        | .....Information management          |
| .....Terminology                     | .....Competitive intelligence        |
| .....Dictionaries                    | .....Document handling               |
| .....Video sharing                   | .....Information security            |
| .....Facebook                        | .....Information sharing             |
| .....MySpace                         | .....Knowledge transfer              |
| .....YouTube                         | .....Information processing          |
| .....Vocabulary                      | .....Informatics                     |
| .....Web sites                       | .....Information exchange            |
| .....Facebook                        | .....Sonification                    |
| .....MySpace                         | .....Management information systems  |
| .....Uniform resource locators       | .....Portals                         |
| .....Web design                      | .....Medical information systems     |
| .....YouTube                         | .....Electronic medical records      |
| ....Information science              | ....Information technology           |
| ....Information services             | .....Information representation      |
| .....Ask IEEE                        | .....Printing                        |
| .....Dictionaries                    | .....Digital printing                |
| .....Document delivery               | .....Teleprinting                    |
| .....Ask IEEE                        | .....Service computing               |
| .....Encyclopedias                   | .....Telematics                      |
| .....Libraries                       | .....Universal Serial Bus            |
| .....Software libraries              | ....Manuals                          |
| .....Teletext                        | ....Oral communication               |
| .....Videotex                        | .....Public speaking                 |
| .....Wikipedia                       | .....Speech                          |



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....Plagiarism  
 ....Portfolios  
 ....Professional societies  
 ....Public speaking  
 ....Rhetoric  
 ....Writing  
 .....Abstracts  
 .....Bibliographies  
 .....Biographies  
 .....Autobiographies  
 .....Dictionaries  
 .....Documentation  
 .....Grammar  
 .....Readability metrics  
 .....Resumes  
 .....Reviews  
 .....Thesauri

**Reliability**

....Availability  
 ....Fault diagnosis  
 .....Dissolved gas analysis  
 .....Fault location  
 ....Fault tolerance  
 .....Redundancy  
 ....Fluctuations  
 ....Integrated circuit reliability  
 ....Maintenance  
 ....Maldistribution  
 ....Materials reliability  
 ....Reliability engineering  
 ....Reliability theory  
 ....Robustness  
 ....Semiconductor device reliability  
 ....Software reliability  
 ....Stability  
 .....Circuit stability  
 .....Robust stability  
 .....Stability analysis  
 .....Stability criteria  
 ....Thermal stability  
 ....Telecommunication network reliability

**Resonance**

....Ferroresonance  
 ....Magnetic resonance  
 .....Nuclear magnetic resonance  
 .....Paramagnetic resonance  
 ....Resonance light scattering

....Stochastic resonance

**Robotics and automation**

....Animatronics  
 ....Automation  
 .....Automated highways  
 .....Automatic generation control  
 .....Automatic testing  
 .....Automatic test pattern generation  
 .....Ring generators  
 .....Building automation  
 .....Manufacturing automation  
 .....Computer aided manufacturing  
 .....Computer integrated manufacturing  
 .....Computer numerical control  
 .....Flexible manufacturing systems  
 .....Office automation  
 .....Workflow management software  
 .....Storage automation  
 ....Multi-robot systems  
 ....Robots  
 .....Androids  
 .....Aquatic robots  
 .....Automata  
 .....Turing machines  
 .....Cognitive robotics  
 .....Computer vision  
 .....Active appearance model  
 .....Face detection  
 .....Smart cameras  
 .....Educational robots  
 .....Humanoid robots  
 .....Intelligent robots  
 .....Manipulators  
 .....End effectors  
 .....Manipulator dynamics  
 .....Micromanipulators  
 .....Medical robotics  
 .....Rehabilitation robotics  
 .....Mobile robots  
 .....Climbing robots  
 .....Legged locomotion  
 .....Orbital robotics  
 .....Parallel robots  
 .....Robot control  
 .....Robot motion  
 .....Robot kinematics  
 .....Motion analysis  
 .....Robot programming  
 .....Robot sensing systems  
 .....Robot vision systems



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.....Simultaneous localization and mapping  
 .....Tactile sensors  
 .....Service robots  
 .....Telerobotics  
 .....Teleoperators

**Science - general**

....Astronomy  
 .....Astrophysics  
 .....Observatories  
 .....Orbits (stellar)  
 .....Planets  
 .....Earth  
 .....Extrasolar planets  
 .....Jupiter  
 .....Mars  
 .....Mercury (planets)  
 .....Pluto  
 .....Saturn  
 .....Sun  
 .....Venus  
 .....Radio astronomy  
 .....Solar system  
 .....Kuiper belt  
 .....Stellar dynamics  
 .....Stellar motion  
 ...Biology  
 .....Biochemistry  
 .....Amino acids  
 .....Biochemical analysis  
 .....Peptides  
 .....Proteins  
 .....Biodiversity  
 .....Biogeography  
 .....Bioelectric phenomena  
 .....Electric shock  
 .....Biological cells  
 .....Cells (biology)  
 .....Chromosome mapping  
 .....Fibroblasts  
 .....RNA  
 .....Stem cells  
 .....Biological information theory  
 .....Biological processes  
 .....Biological interactions  
 .....Chronobiology  
 .....Circadian rhythm  
 .....Coagulation  
 .....Symbiosis

.....Biological system modeling  
 .....Biological systems  
 .....Anatomy  
 .....Molecular communication  
 .....Organisms  
 .....Biology computing  
 .....Biophotonics  
 .....Biophysics  
 .....Aerospace biophysics  
 .....Biomagnetics  
 .....Cellular biophysics  
 .....Molecular biophysics  
 .....Evolution (biology)  
 .....Memetics  
 .....Phylogeny  
 .....Genetics  
 .....DNA  
 .....Gene therapy  
 .....Genetic communication  
 .....Genetic expression  
 .....Genetic programming  
 .....Genomics  
 .....Microinjection  
 .....Nanobioscience  
 .....DNA computing  
 .....Nanobiotechnology  
 .....Physiology  
 .....Predator prey systems  
 .....Synthetic biology  
 .....Systematics  
 .....Systems biology  
 .....Vegetation  
 .....Crops  
 .....Marine vegetation  
 .....Zoology  
 .....Animals  
 ...Chemistry  
 .....Astrochemistry  
 .....Biochemistry  
 .....Amino acids  
 .....Biochemical analysis  
 .....Peptides  
 .....Proteins  
 .....Chemical analysis  
 .....Activation analysis  
 .....Chemical processes  
 .....Chemicals  
 .....Electronic noses  
 .....pH measurement  
 .....Chemical compounds  
 .....Anti-freeze







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|                                 |                                 |
|---------------------------------|---------------------------------|
| .....Ethanol                    | .....EMTDC                      |
| .....Methanol                   | .....Extraterrestrial phenomena |
| .....Inorganic chemicals        | .....Geodynamics                |
| .....Interstellar chemistry     | .....Geophysics computing       |
| .....Organic chemicals          | .....Meteorology                |
| .....Hydrocarbons               | .....Moisture                   |
| .....Photochemistry             | .....Seismology                 |
| .....Photobleaching             | .....Surface waves              |
| ....Electricity                 | .....Well logging               |
| .....Photoelectricity           | .....Ice                        |
| .....Photovoltaic effects       | .....Ice shelf                  |
| .....Piezoelectricity           | .....Ice surface                |
| .....Piezoelectric effect       | .....Ice thickness              |
| .....Piezoelectric polarization | .....Sea ice                    |
| .....Pyroelectricity            | .....Lakes                      |
| .....Thermoelectricity          | .....Land surface               |
| .....Electrothermal effects     | .....Levee                      |
| .....Thermoelectric devices     | .....Meteorological factors     |
| .....Triboelectricity           | .....Oceans                     |
| ....Geoscience                  | .....Ocean salinity             |
| .....Antarctica                 | .....Ocean temperature          |
| .....South Pole                 | .....Sea coast                  |
| .....Arctic                     | .....Sea floor                  |
| .....North Pole                 | .....Sea level                  |
| .....Atmosphere                 | .....Sea surface                |
| .....Atmospheric modeling       | .....Tides                      |
| .....Atmospheric waves          | .....Rivers                     |
| .....Biosphere                  | .....Sediments                  |
| .....Continents                 | .....Soil                       |
| .....Africa                     | .....Soil moisture              |
| .....Asia                       | .....Soil properties            |
| .....Australia                  | .....Soil texture               |
| .....Europe                     | .....Tornadoes                  |
| .....North America              | .....Tsunami                    |
| .....South America              | .....Volcanoes                  |
| .....Cyclones                   | .....Planetary volcanoes        |
| .....Hurricanes                 | .....Volcanic activity          |
| .....Tropical cyclones          | .....Volcanic ash               |
| .....Earth                      | ....Metrology                   |
| .....Earthquakes                | ....Physics                     |
| .....Earthquake engineering     | .....Acoustics                  |
| .....Forestry                   | .....Acoustic applications      |
| .....Geoengineering             | .....Acoustic devices           |
| .....Geography                  | .....Acoustic emission          |
| .....Cities and towns           | .....Acoustic noise             |
| .....Rural areas                | .....Acoustic propagation       |
| .....Urban areas                | .....Acoustic pulses            |
| .....Geology                    | .....Acoustic waves             |
| .....Minerals                   | .....Acoustooptic effects       |
| .....Rocks                      | .....Biomedical acoustics       |
| .....Geophysics                 | .....Cepstral analysis          |



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|                                 |                                     |
|---------------------------------|-------------------------------------|
| .....Music                      | .....Hydrodynamics                  |
| .....Nonlinear acoustics        | .....Kinematics                     |
| .....Psychoacoustics            | .....Lubrication                    |
| .....Reverberation              | .....Magnetohydrodynamics           |
| .....Spectral shape             | .....Photoelasticity                |
| .....Underwater acoustics       | .....Pressure effects               |
| .....Astrophysics               | .....Shock (mechanics)              |
| .....Beams                      | .....Strain                         |
| .....Acoustic beams             | .....Stress                         |
| .....Laser beams                | .....Surface cracks                 |
| .....Molecular beams            | .....Torque                         |
| .....Optical beams              | .....Vibrations                     |
| .....Particle beams             | .....Volume relaxation              |
| .....Biophysics                 | .....Workability                    |
| .....Aerospace biophysics       | .....Network theory (graphs)        |
| .....Biomagnetics               | .....Orbits                         |
| .....Cellular biophysics        | .....Physics education              |
| .....Molecular biophysics       | .....Quantum mechanics              |
| .....Dark energy                | .....Density functional theory      |
| .....Entropy                    | .....Lagrangian functions           |
| .....Fluid flow                 | .....Proton effects                 |
| .....Fluid dynamics             | .....Quantum capacitance            |
| .....Hydraulic diameter         | .....Quantum entanglement           |
| .....Hydrology                  | .....Relativistic quantum mechanics |
| .....Pipelines                  | .....Schrodinger equation           |
| .....Valves                     | .....Stationary state               |
| .....Geophysics                 | .....Teleportation                  |
| .....EMTDC                      | .....Tunneling                      |
| .....Extraterrestrial phenomena | .....String theory                  |
| .....Geodynamics                | .....Thermal factors                |
| .....Geophysics computing       | .....Temperature                    |
| .....Meteorology                | .....Temperature dependence         |
| .....Moisture                   | .....Thermal conductivity           |
| .....Seismology                 | .....Thermal expansion              |
| .....Surface waves              | .....Thermal management             |
| .....Well logging               | .....Thermal stresses               |
| .....Kinetic theory             | .....Thermoelasticity               |
| .....Kinetic energy             | .....Thermoelectricity              |
| .....Levitation                 | .....Thermolysis                    |
| .....Electrostatic levitation   | .....Thermooptic effects            |
| .....Magnetic levitation        | .....Thermoresistivity              |
| .....Lorentz covariance         | .....Waves                          |
| .....Mechanical factors         | .....Atmospheric waves              |
| .....Acceleration               | .....Berry phase                    |
| .....Aerodynamics               | .....Doppler effect                 |
| .....Biomechanics               | .....Electrodynamics                |
| .....Damping                    | .....Magnetostatic waves            |
| .....Dynamics                   | .....Matter waves                   |
| .....Fatigue                    | .....Plasma waves                   |
| .....Force                      | .....Propagation                    |
| .....Friction                   | .....Reflectivity                   |



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.....Seismic waves  
 .....Shock waves  
 .....Solitons  
 .....Surface acoustic waves  
 .....Wave functions  
 ....Sociology  
 .....Digital divide  
 ....Thermodynamics  
 .....Isobaric  
 .....Isothermal processes

**Sensors**

....Acoustic sensors  
 ....Chemical and biological sensors  
 .....Biosensors  
 .....Gas detectors  
 .....Amperometric sensors  
 ....Electromechanical sensors  
 .....Microsensors  
 ....Force sensors  
 ....Infrared sensors  
 ....Intelligent sensors  
 ....Intracranial pressure sensors  
 ....Ionizing radiation sensors  
 .....Position sensitive particle detectors  
 .....Radiation detectors  
 .....Bolometers  
 .....Gamma-ray detectors  
 .....Infrared detectors  
 .....Photodetectors  
 .....Semiconductor radiation detectors  
 .....Silicon radiation detectors  
 .....X-ray detectors  
 ....Magnetic sensors  
 ....Spin valves  
 ....Mechanical sensors  
 .....Capacitive sensors  
 ....Multimodal sensors  
 ....Nanosensors  
 ....Optical sensors  
 .....Optical detectors  
 .....Bar codes  
 .....Optical fiber sensors  
 ....Optoelectronic and photonic sensors  
 ....Sensor phenomena and characterization  
 ....Sensor systems and applications  
 .....Detectors  
 .....Envelope detectors  
 .....Semiconductor detectors  
 .....Electric sensing devices

.....Leak detection  
 .....Radiofrequency identification  
 .....RFID tags  
 .....Robot sensing systems  
 .....Robot vision systems  
 .....Simultaneous localization and mapping  
 .....Tactile sensors  
 .....Sensor arrays  
 .....Sensor fusion  
 .....Sensor systems  
 .....Gunshot detection systems  
 ....Thermal sensors  
 .....Temperature sensors  
 ....Thick film sensors  
 ....Thin film sensors  
 ....Wearable sensors

**Signal processing**

....Acoustic signal processing  
 .....Active noise reduction  
 .....Echo cancellers  
 .....Speech processing  
 .....Human voice  
 .....Speech enhancement  
 .....Speech synthesis  
 ....Adaptive signal processing  
 .....Adaptive filters  
 .....Adaptive signal detection  
 ....Amplifiers  
 .....Broadband amplifiers  
 .....Cavity resonators  
 .....Laser cavity resonators  
 .....Differential amplifiers  
 .....Distributed amplifiers  
 .....Low-noise amplifiers  
 .....Operational amplifiers  
 .....Feedback amplifier  
 .....Power amplifiers  
 .....High power amplifiers  
 .....Predistortion  
 .....Preamplifiers  
 .....Pulse amplifiers  
 .....Radiofrequency amplifiers  
 ....Array signal processing  
 ....Attenuators  
 .....Optical attenuators  
 ....Chirp  
 ....Convolution  
 .....Convolver



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- ....Decorrelation
- ....Digital signal processing
- .....Delta modulation
- .....Delta-sigma modulation
- .....Sigma-delta modulation
- .....Digital signal processing chips
- ....Dispersion
- .....Chromatic dispersion
- .....Optical fiber dispersion
- ....Distortion
- .....Acoustic distortion
- .....Four-wave mixing
- .....Jitter
- .....Timing jitter
- .....Nonlinear distortion
- .....Harmonic distortion
- .....Intermodulation distortion
- .....Phase distortion
- ....Error correction
- .....Forward error correction
- ....Fading
- .....Frequency-selective fading channels
- .....Rayleigh channels
- .....Weibull fading channels
- ....Filters
- .....Active filters
- .....Band-pass filters
- .....Anisotropic
- .....Bragg gratings
- .....Fiber gratings
- .....Channel bank filters
- .....Digital filters
- .....Finite impulse response filters
- .....Equalizers
- .....Adaptive equalizers
- .....Blind equalizers
- .....Decision feedback equalizers
- .....Filtering theory
- .....Gabor filters
- .....Harmonic filters
- .....IIR filters
- .....Kalman filters
- .....Low-pass filters
- .....Matched filters
- .....Microstrip filters
- .....Nonlinear filters
- .....Particle filters
- .....Power filters
- .....Spurline
- .....Resonator filters
- .....Spatial filters
- .....Superconducting filters
- .....Transversal filters
- ....Frequency locked loops
- ....Geophysical signal processing
- ....Limiting
- ....Modulation
- .....Amplitude modulation
- .....Amplitude shift keying
- .....Quadrature amplitude modulation
- .....Chirp modulation
- .....Demodulation
- .....Digital modulation
- .....Constellation diagram
- .....Partial response signaling
- .....Frequency modulation
- .....Frequency shift keying
- .....Magnetic modulators
- .....Modulation coding
- .....Interleaved codes
- .....Optical modulation
- .....Electrooptic modulators
- .....Intensity modulation
- .....Phase modulation
- .....Continuous phase modulation
- .....Differential phase shift keying
- .....Phase shift keying
- .....Pulse modulation
- .....Pulse width modulation
- .....Pulse width modulation inverters
- .....Space vector pulse width modulation
- ....Multidimensional signal processing
- ....Video signal processing
- .....Video coding
- .....Video compression
- ....Noise
- .....1f noise
- .....Additive noise
- .....Additive white noise
- .....AWGN
- .....Colored noise
- .....Gaussian noise
- .....AWGN
- .....Laser noise
- .....Laser feedback
- .....Low-frequency noise
- .....Noise cancellation
- .....Phase noise
- .....Signal to noise ratio
- .....PSNR
- .....Superconducting device noise





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.....White noise  
 .....AWGN  
 ....Optical signal processing  
 .....Laser noise  
 .....Laser feedback  
 ....Optical wavelength conversion  
 ....Phase locked loops  
 ....Pulse compression methods  
 .....Optical pulse compression  
 ....Pulse shaping methods  
 .....Optical pulse shaping  
 ....Quantization (signal)  
 .....Vector quantization  
 ....Radar signal processing  
 ....Recording  
 .....Audio recording  
 .....Digital recording  
 .....Disk recording  
 .....Magnetic recording  
 .....Digital magnetic recording  
 .....Heat-assisted magnetic recording  
 .....Magnetic noise  
 .....Magneto-optic recording  
 .....Microwave-assisted magnetic recording  
 .....Perpendicular magnetic recording  
 .....Optical recording  
 .....CD recording  
 .....Video recording  
 .....High definition video  
 .....Webcams  
 ....RF signals  
 ....Signal analysis  
 .....Discrete-event systems  
 .....Harmonic analysis  
 .....Parameter estimation  
 .....Amplitude estimation  
 .....Direction-of-arrival estimation  
 .....Frequency estimation  
 .....Motion estimation  
 .....Phase estimation  
 .....Time of arrival estimation  
 .....Signal mapping  
 .....Spectral analysis  
 .....Infrared spectra  
 .....Judd-Ofelt theory  
 .....Spectroradiometers  
 ....Signal design  
 ....Signal detection  
 .....Acoustic signal detection  
 .....Sonar detection

.....Motion detection  
 .....Multiuser detection  
 .....Optical signal detection  
 .....Phase detection  
 .....Phase frequency detector  
 .....Radar detection  
 ....Signal generators  
 .....Noise generators  
 .....Pulse generation  
 .....Optical pulse generation  
 ....Signal reconstruction  
 .....Signal denoising  
 ....Signal resolution  
 .....Diversity reception  
 ....Signal restoration  
 ....Signal sampling  
 ....Signal synthesis  
 ....Source separation  
 .....Blind source separation  
 ....Spectrogram  
 ....Tracking loops

#### **Social implications of technology**

....Cultural differences  
 ....Environmental factors  
 .....Biosphere  
 .....Ecosystems  
 .....Environmental economics  
 .....Carbon tax  
 .....Environmental monitoring  
 .....Global warming  
 .....Green products  
 .....Green buildings  
 .....Green cleaning  
 .....Pollution  
 .....Air pollution  
 .....Industrial pollution  
 .....Land pollution  
 .....Oil pollution  
 .....Radioactive pollution  
 .....Thermal pollution  
 .....Urban pollution  
 .....Water pollution  
 ....Ethical aspects  
 ....Globalization  
 ....International relations  
 ....Peace technology  
 ....Philosophical considerations  
 ....Social factors  
 .....Demography



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.....Technology social factors  
 .....Privacy  
 ....Sustainable development  
 ....Technology  
 .....Appropriate technology  
 .....Technological innovation  
 .....Technology social factors  
 .....Privacy  
 .....Technology transfer  
 .....Small business technology transfer

### **Solid state circuits**

....Circuit subsystems  
 ....Circuit theory  
 ....FET circuits  
 .....FET integrated circuits  
 .....Field effect MMIC  
 .....MESFET integrated circuits  
 .....JFET circuits  
 .....JFET integrated circuits  
 .....MESFET circuits  
 .....MESFET integrated circuits  
 .....MODFET circuits  
 .....MODFET integrated circuits  
 .....MOSFET circuits  
 .....CMOSFET circuits  
 .....MOS integrated circuits  
 .....Power MOSFET  
 ....Gate leakage  
 ....Solid state circuit design  
 ....Transistors  
 .....Field effect transistors  
 .....CNTFETs  
 .....Double-gate FETs  
 .....HEMTs  
 .....JFETs  
 .....MESFETs  
 .....MISFETs  
 .....MODFETs  
 .....MOSFET  
 .....MOSHFETs  
 .....OFETs  
 .....Schottky gate field effect transistors  
 .....Thin film transistors  
 .....Heterojunction bipolar transistors  
 .....Double heterojunction bipolar transistors  
 .....Millimeter wave transistors  
 .....Phototransistors

### **Superconductivity**

....Bean model  
 ....Critical current density (superconductivity)  
 .....Critical current density  
 ....Flux pinning  
 ....Superconducting devices  
 .....Josephson junctions  
 .....SQUIDs  
 .....Superconducting coils  
 .....Superconducting magnets  
 .....Superconducting microwave devices  
 .....Superconducting photodetectors  
 ....Superconducting filaments and wires  
 ....Superconducting films  
 .....Superconducting thin films  
 ....Superconducting integrated circuits  
 ....Superconducting magnetic energy storage  
 ....Superconducting materials  
 .....Granular superconductors  
 .....High-temperature superconductors  
 .....Yttrium barium copper oxide  
 .....Multifilamentary superconductors  
 .....Niobium-tin  
 .....Type II superconductors  
 ....Superconducting transition temperature  
 Systems engineering and theory  
 ....Adaptive systems  
 .....Adaptive control  
 .....Line enhancers  
 .....Multi-agent systems  
 .....Variable structure systems  
 ....Hierarchical systems  
 .....Multilevel systems  
 ....Modeling  
 .....Analytical models  
 .....Atmospheric modeling  
 .....Brain modeling  
 .....Computational modeling  
 .....Computational cultural modeling  
 .....Context modeling  
 .....Data models  
 .....Deformable models  
 .....Digital elevation models  
 .....Emulation  
 .....Graphical models  
 .....Green's function methods  
 .....Hidden Markov models  
 .....Input variables  
 .....Integrated circuit modeling



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.....Cutoff frequency  
 .....Inverse problems  
 .....Deconvolution  
 .....Load modeling  
 .....Metamodeling  
 .....Numerical models  
 .....Object oriented modeling  
 .....Power system modeling  
 .....Load modeling  
 .....Semiconductor device modeling  
 .....Semiconductor process modeling  
 .....Signal representation  
 .....Simulation  
 .....Computer simulation  
 .....Digital simulation  
 .....Medical simulation  
 .....Solid modeling  
 .....System identification  
 ....Multidimensional systems  
 ....Reduced order systems  
 ....Stochastic systems  
 ....System analysis and design  
 .....Asymptotic stability  
 .....Control system analysis  
 .....State-space methods  
 .....Diakoptics  
 .....Distributed processing  
 .....Message passing  
 .....Distributed vision networks  
 .....Fault detection  
 .....Fault tolerant systems  
 .....Interconnected systems  
 .....Large-scale systems  
 .....Lyapunov methods  
 .....Open systems  
 .....Open Access  
 .....Physical layer  
 .....Petri nets  
 .....Robust control  
 .....Scalability  
 .....Scattering parameters  
 .....Sequential analysis  
 .....Sequential diagnosis  
 .....Software prototyping  
 .....System-level design  
 .....System performance  
 .....Cooperative caching  
 .....Time factors  
 .....Continuous time systems  
 .....Discrete-time systems  
 .....Time invariant systems

.....Time-varying systems  
 ....Systems engineering education

### **Systems, man, and cybernetics**

....Behavioral science  
 .....Animal behavior  
 .....Cognition  
 .....Consumer behavior  
 .....Psychiatry  
 .....Mental disorders  
 .....Psychology  
 .....Industrial psychology  
 .....Mood  
 .....Psychometric testing  
 ....Biological control systems  
 .....Biomarkers  
 .....Molecular biomarkers  
 ....Computational linguistics  
 .....Sentiment analysis  
 ....Cybernetics  
 .....Adaptive systems  
 .....Adaptive control  
 .....Line enhancers  
 .....Multi-agent systems  
 .....Variable structure systems  
 .....Cognitive informatics  
 .....Cognitive science  
 .....Problem-solving  
 .....Control theory  
 .....Control nonlinearities  
 .....Observability  
 .....Decision theory  
 .....Decision trees  
 .....Econophysics  
 .....Emergent phenomena  
 .....Intelligent control  
 .....Feedforward systems  
 .....Neurocontrollers  
 .....Linear feedback control systems  
 .....Frequency locked loops  
 .....Phase locked loops  
 .....State feedback  
 .....Tracking loops  
 ....Ergonomics  
 .....Job design  
 .....Human factors  
 .....Affective computing  
 .....Anthropomorphism  
 ....Identification of persons  
 .....Biometrics (access control)



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.....Gait recognition  
 .....Iris recognition  
 .....Face recognition  
 .....Fingerprint recognition  
 .....Handwriting recognition  
 .....Forgery  
 .....Speaker recognition  
 .....Speech recognition  
 .....Automatic speech recognition  
 .....Speech analysis  
 ....Man machine systems  
 .....Interactive systems  
 ....Natural languages  
 .....Natural language processing  
 .....Morphology  
 .....Sentiment analysis  
 ....Pervasive computing  
 .....Ubiquitous computing  
 .....Context-aware services  
 .....Wearable computers  
 ....Posthuman  
 ....Teleworking  
 ....Transhuman  
 ....User interfaces  
 .....Audio user interfaces  
 .....Brain-computer interfaces  
 .....Data visualization  
 .....Isosurfaces  
 .....Emotion recognition  
 .....Exoskeletons  
 .....Graphical user interfaces  
 .....Avatars  
 .....Human computer interaction  
 .....Human-robot interaction  
 .....Smart cards

#### **Ultrasonics, ferroelectrics, and frequency control**

....Ferroelectric materials  
 .....Ferroelectric films  
 .....Relaxor ferroelectrics  
 ....Frequency control  
 .....Automatic frequency control  
 .....Tunable circuits and devices  
 .....RLC circuits  
 .....Tuned circuits  
 .....Tuning  
 .....Laser tuning  
 .....Optical tuning  
 .....Tuners

....Piezoelectricity  
 .....Piezoelectric effect  
 .....Piezoelectric polarization  
 ....Pyroelectricity  
 ....Ultrasonic imaging  
 .....Ultrasonography  
 .....Sonogram  
 ....Ultrasonic transducers

#### **Vehicular and wireless technologies**

....Automotive engineering  
 .....Automotive applications  
 .....Automotive electronics  
 .....Power steering  
 .....Vehicle crash testing  
 .....Vehicle detection  
 .....Vehicle driving  
 .....Vehicle dynamics  
 .....Vehicle safety  
 ....Land mobile radio equipment  
 .....Mobile antennas  
 ....Navigation  
 .....Aircraft navigation  
 .....Course correction  
 .....Dead reckoning  
 .....Inertial navigation  
 .....Marine navigation  
 .....Radio navigation  
 .....Satellite navigation systems  
 .....Global Positioning System  
 .....Satellite constellations  
 .....Sonar navigation  
 ....Propulsion  
 .....Aircraft propulsion  
 .....Propellers  
 .....Electromagnetic launching  
 .....Coilguns  
 .....Railguns  
 .....Electrothermal launching  
 .....Rockets  
 ....Vehicles  
 .....Land vehicles  
 .....Bicycles  
 .....Electric vehicles  
 .....Road vehicles  
 .....Remotely operated vehicles  
 .....Unmanned aerial vehicles  
 .....Space vehicles  
 .....Space shuttles  
 ....Wireless sensor networks



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.....Body sensor networks  
.....Event detection

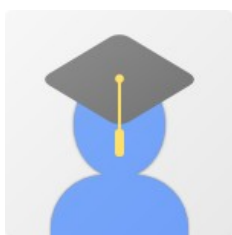


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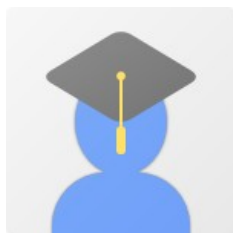
Page 67



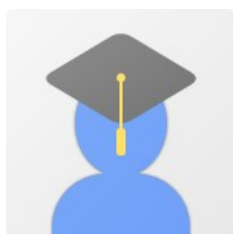
## Appendix L VITA



Juan Z. dela Cruz received the B.Sc., M.Sc., and Ph.D. degrees in chemistry all from the Pamantasan ng Pilipinas, San Juan, Metro Manila, Philippines, in 2020, 2022 and 2025 respectively. He is currently taking up his B.Sc. Electronics and Communications Engineering studies. He has developed several high-speed packet-switched network systems and node modules. His research interests include high-speed packet-switched networks, high speed radio interface design, discrete simulation and statistical models for packet switches.



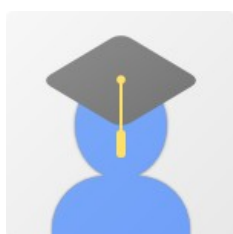
Nat Y. Franco received the B.Sc., M.Sc., and Ph.D. degrees in chemistry all from the Pamantasan ng Pilipinas, San Juan, Metro Manila, Philippines, in 2020, 2022 and 2025 respectively. He is currently taking up his B.Sc. Electronics and Communications Engineering studies. He has developed several high-speed packet-switched network systems and node modules. His research interests include high-speed packet-switched networks, high speed radio interface design, discrete simulation and statistical models for packet switches.



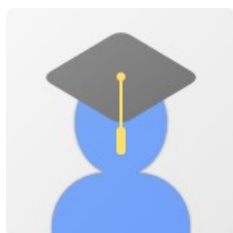
Sebastian X. Garcia received the B.Sc., M.Sc., and Ph.D. degrees in chemistry all from the Pamantasan ng Pilipinas, San Juan, Metro Manila, Philippines, in 2020, 2022 and 2025 respectively. He is currently taking up his B.Sc. Electronics



2250 and Communications Engineering studies. He has developed several high-speed packet-  
 2251 switched network systems and node modules. His research interests include high-speed  
 2252 packet-switched networks, high speed radio interface design, discrete simulation and  
 2253 statistical models for packet switches.



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 2258 switched network systems and node modules. His research interests include high-speed  
 2259 packet-switched networks, high speed radio interface design, discrete simulation and  
 2260 statistical models for packet switches.



2261 Max V. Rianzares received the B.Sc., M.Sc., and Ph.D. degrees  
 2262 in chemistry all from the Pamantasan ng Pilipinas, San Juan, Metro Manila, Philippines,  
 2263 in 2020, 2022 and 2025 respectively. He is currently taking up his B.Sc. Electronics  
 2264 and Communications Engineering studies. He has developed several high-speed packet-  
 2265 switched network systems and node modules. His research interests include high-speed  
 2266 packet-switched networks, high speed radio interface design, discrete simulation and  
 2267 statistical models for packet switches.



De La Salle University

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## Appendix M

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## ARTICLE PAPER(S)

# Article/Forum Paper Format (IEEE LaTeX format)

Michael Shell, *Member, IEEE*, John Doe, *Fellow, OSA*, and Jane Doe, *Life Fellow, IEEE*

2270

**Abstract**—The abstract goes here. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.

**Index Terms**—Computer Society, IEEE, IEEEtran, journal, LaTeX, paper, template.

## I. INTRODUCTION

THIS demo file is intended to serve as a “starter file” for IEEE article papers produced under LaTeX using IEEEtran.cls version 1.8b and later. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.

### A. Subsection Heading Here

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M. Shell was with the Department of Electrical and Computer Engineering, Georgia Institute of Technology, Atlanta, GA, 30332.

E-mail: see <http://www.michaelshell.org/contact.html>

J. Doe and J. Doe are with Anonymous University.



Fig. 1. Simulation results for the network.

TABLE I  
AN EXAMPLE OF A TABLE

|       |      |
|-------|------|
| One   | Two  |
| Three | Four |

Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.

### 1) Subsubsection Heading Here: Subsubsection text here.

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## II. CONCLUSION

The conclusion goes here.

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(a) Case I



(b) Case II

Fig. 2. Simulation results for the network.

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#### APPENDIX A

##### PROOF OF THE FIRST ZONKLAR EQUATION

Appendix one text goes here.

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#### APPENDIX B

Appendix two text goes here. [1].

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#### ACKNOWLEDGMENT

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#### REFERENCES

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